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A review of convolutional neural network based methods for medical image classification

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ABSTRACT

This study systematically reviews CNN-based medical image classification methods. We surveyed 149 of the latest and most important papers published to date and conducted an in-depth analysis of the methods used therein. Based on the selected literature, we organized this review systematically. First, the development and evolution of CNN in the field of medical image classification are analyzed. Subsequently, we provide an in-depth overview of the main techniques of CNN applied to medical image classification, which is also the current research focus in this field, including data preprocessing, transfer learning, CNN architectures, and explainability, and their role in improving classification accuracy and efficiency. In addition, this overview summarizes the main public datasets for various diseases. Although CNN has great potential in medical image classification tasks and has achieved good results, clinical application is still difficult. Therefore, we conclude by discussing the main challenges faced by CNNs in medical image analysis and pointing out future research directions to address these challenges. This review will help researchers with their future studies and can promote the successful integration of deep learning into clinical practice and smart medical systems.

1. Introduction

Medical imaging is an important auxiliary means of clinical diagnosis, and it plays an important role in life science research. Medical images include X-ray, Ultrasound (US), Computed Tomography (CT), Magnetic Resonance Imaging (MRI), Positron Emission Tomography (PET), etc. These medical imaging data are huge and difficult to process in clinical diagnosis, so auto-detection of diseases from medical images has become a prime issue in medical sciences [1]. In recent years, as deep learning has made remarkable achievements in the field of computer vision, it has become a current research hot spot to utilize deep learning technology to establish computer-aided diagnosis framework [2], then achieve intelligent diagnosis, intelligent clinical decision-making and prognosis treatment based on image data.

Medical image classification plays an essential role in clinical treatment, because clinicians often need to use medical images to help diagnose whether there are diseases in the human body, and to quantify the severity of the diseases [3]. Before the emergence of deep learning, handcrafted features such as image color, texture, shape, etc., were widely used in medical image classification [4]. After feature extraction and selection, traditional machine learning methods such as Support Vector Machines (SVM), Logistic Regression (LR), Random Forests (RF), etc., were used for classification [5]. However, extracting and selecting features manually requires a lot of time and effort. Moreover, the artificial feature-based method has reached its ceiling on performance [6]. The deep learning technology, especially convolutional neural network (CNN) is an emerging machine learning method that has proven its potential for different classification tasks [7]. It automatically extracts and selects features and classifies them, which provide radiologists with quicker and more accurate findings in real time. Actually, as an important branch of deep learning, CNN is increasingly used in various computer vision problems [8], especially in medical imaging domain. Therefore, in the following, we will review the applications of CNN in medical image processing.

CNN is a hierarchical and cascade model, which starting from the lowest level of pixel information of input image matrix, to extract the important feature information layer by layer. CNN architecture consists of 3 important parts: convolution layer, pooling layer, and fully connected layer [9]. With the convolution layer, local features will be extracted by convolutional operation of filters on the input image. Then the obtained feature maps will be transferred to the next convolution layer to extract higher-level features. The pooling layer, also called the down sampling layer, is used to reduce the image dimension after

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Nomenclature

ANN	Artificial Neural Network
AUC	Area Under the Curve
CAM	Class Activation Mapping
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CT	Computed Tomography
DBN	Deep Belief Network
DC	Deep Cascade
DCNN	Deep Convolutional Neural Network
DR	Diabetic Retinopathy
DT	Decision Tree
ELM	Extreme Learning Machine
GA	Genetic Algorithm
GAN	Generative Adversarial Network
GB	Guided Backpropagation
GNB	Gaussian Naive Bayes
GP	Gaussian Process
Grad-CAM	Gradient Weighted Class Activation Mapping
HE	Histogram Equalization
HSNN	Hierarchical Spiking Neural Network
KNN	K Nearest Neighbor
LDA	Linear Discriminant Analysis
LR	Logistic Regression
LSTM	Long Short-Term Memory
LSVM	Linear Support Vector Machine
MC-SVM	Multiclass Support Vector Machine
MLP	Multilayer Perceptron
MRI	Magnetic Resonance Imaging
NAS	Neural Architecture Search
NB	Naive Bayes
OSL	Orthogonal SoftMax Layer
PDR	Proliferative Diabetic Retinopathy
PET	Positron Emission Tomography
PSO	Particle Swarm Optimization
QFFA	Quantum Fruit Fly Algorithm technique
RBF	Radial Basis Function
RF	Random Forest
RPN	Region Proposal Network
RNN	Recurrent Neural Network
SVM	Support Vector Machine
T-SNE	T-distributed Stochastic Neighbor Embedding
UMAP	Uniform Manifold Approximation and Projection
US	Ultrasound
WOA	Whale Optimization Algorithm
WoS	Web of Science

convolutional layer. Max-pooling is often used as the down sampling operation because it reduces the dimension while retaining the dominant feature in the image. The fully connected layer is located after the last convolutional layer, flattening the feature map into a vector. Usually a softmax function is used at the output layer to normalize the output so that the final output is in probability form. This probability indicates that the image is classified into a certain category. During the training phase, CNN usually uses a gradient descent algorithm to update the weights of the convolution kernel and fully connected layers. The learning algorithm takes a classification loss as input and back propagates the loss into the network to update the weights until convergence.

Since Alex Krizhevsk et al. [10] proposed AlexNet in 2012, many classic CNN structures have emerged, such as VGGNet [11], GoogLeNet (Inception) [12], ResNet [13], DenseNet [14], MobileNet [15] and EfficientNet [16], etc. most of which are widely used in medical image classification. Although CNN has been successful applied in many computer vision tasks, its application in medical image classification still faces many difficulties, such as unbalanced sample categories, small datasets, small structural lesions, poor model generalization ability, lack of explainability, etc. Facing with these challenges, the research community is constantly exploring related solutions, aiming to build artificial intelligence models that are more consistent with medical image classification. Therefore, it is necessary to summarize the existing research works and conduct a comprehensive comparison and analysis of the methods used in this field, so as to provide a reference basis for further research. Prior to our review, there have been several review articles that have explored and studied medical image classification methods.

Each research reviewed the current literature from a different perspective. For example, Muhammad et al. [2] covered all existing CNN-based brain tumor classification methods, investigated and fine-tuned several pre-trained CNN models for brain tumor classification using two different datasets with and without data augmentation, and then discussed their achievements and limitations. Subramanian et al. [17] conducted a detailed survey of the available deep learning approaches for the detection of coronavirus infection from images such as CT scans or X-ray images. This review summarized existing COVID-19 detection methods from three aspects: transfer learning, novel structures and other methods. Kumar et al. [18] focused on medical image classification using deep learning and studied a total of 156 research papers. Medical image classification methods are reviewed from 5 perspectives: CNN, transfer learning model, CNN and Recurrent Neural Network (RNN), CNN and auto encoder, and CNN and Generative Adversarial Network (GAN). Additionally, the advantages and limitations of these methods were discussed. Hassan et al. [19] reviewed and categorized COVID-19 CT-based diagnosis methods in terms of computer vision tasks, including classification, segmentation, and detection or localization. Majtner et al. [20] compared multiple methods, including transfer learning and the CNNs as feature extractors, used in medical and biomedical images recognition.

Existing reviews analyzed medical image classification methods based on a specific disease or imaging modality, or just reviewed part of the technologies, and readers can gain substantial knowledge from these studies. This article, on the other hand, aims to conduct a comprehensive review of medical image classification methods, including all diseases and various parts of the body. By collecting relevant literature in the past 5 years, explore the main issues and coping methods of current research in this field, this article provides researchers with a reliable reference. Based on the hot issues in these research, we will start comprehensively summarize existing methods from data preprocessing, transfer learning, model structure, and model explainability. The main contributions of this review are as follows.

(1) First, compared with previous review articles, this review innovatively uses the PRISMA criteria for systematic literature search and screening. It includes medical images of all diseases and various parts of the human body, covers almost all CNN-based medical image classification methods.
(2) Second, it classifies and analyses CNN architectures for medical image classification from four aspects: existing models with transfer learning, variants, novel structures, and ensemble strategy. This is different from methods in other reviews. This classification method allows readers to have a comprehensive and clearer understanding of the application of CNN and its optimization strategies in medical image classification.

(3) Third, this review not only explores existing technologies, but also conducts an in-depth analysis on how to improve the interpretability of CNN models.
(4) Fourth, we summarize 38 public datasets, which is the largest summary compared with other reviews, covering different diseases and different parts. It will provide a reference for subsequent related research.
(5) Fifth, this review not only summarizes the current challenges of CNN in medical image classification, but also proposes comprehensive future research directions, especially in the fields of small datasets, data imbalance, automatic machine learning technology, unsupervised learning, and multi-modal data fusion. These research directions provide important references for future research.

The remainder of this survey is divided into seven sections. In section 2, literature research strategies are introduced. In section 3, the evolution of CNN used in medical image classification are explored. In section 4, CNN architecture and relevant optimization technology are analyzed and categorized in terms of data preprocessing, transfer learning, model architectures, and explainability. The main public datasets for various disease are listed in section 5. Section 6 discusses the challenges faced. In section 7, the future research directions are summarized. Conclusions are discussed in section 8.

2. Research methodology

The pertinent studies on the given topics, medical image classification and convolutional neural network, are mainly obtained from systematic literature search in Web of Science (WoS), ScienceDirect, Scopus. The search encompass papers published in the past five years. First, the search is augmented with filtering criteria, such as article type, research field, etc., to reduce the number of search results. Next, the duplicate articles are excluded. Subsequently, the records are screened based on title and abstract, as well as our study selection criteria. Then, 37 studies that cannot retrieved are excluded. A total of 161 studies are assessed for full-text screening. Finally, 149 studies that meet our criteria are included for systematic review. Fig. 1 shows the flowchart of the systematic literature review with the PRISMA guideline [21]. The distribution of these studies by year is shown in Fig. 2(a). All these studies focus on using CNN to classify medical image, involving diseases of various parts of the body, as shown in Fig. 2(b).

Based on Fig. 1, the search strings for WoS Databases are "(TS=(medical image classification)) AND (TS=(CNN) OR TS=(deep learning) OR TS=(convolutional neutral network*) OR TS=(deep-learning))". The search strategy for ScienceDirect Databases is "Find articles with these terms:medical image classification, Year:2019–2023, Title, abstract or author-specified keywords:CNN, Title:CNN". The search strategy for Scopus Databases is "Article title, Abstract, Keywords: medical image classification & CNN & deep learning & convolutional neural network". "Records marked as ineligible by automation tools" means using the screening function of the database to set a number of screening conditions for the search results, so as to exclude some papers that do not fit the research scope. "Irrelevant studies" refers to studies with clearly unrelated topics or subjects, initially screened by reviewing titles and abstracts. "Low citation studies" refers to apply the low citation standard on an annual basis instead of across all literature, ensuring that highly cited works published each year are included within the research scope. "Low IF studies" refers the papers from journals with an IF less than 2. However, IF is not an absolute reference indicator, the number of citations of the papers themselves is also considered. If the IF value is below 2 but the paper has a convincing citations, it is retained. Additionally,

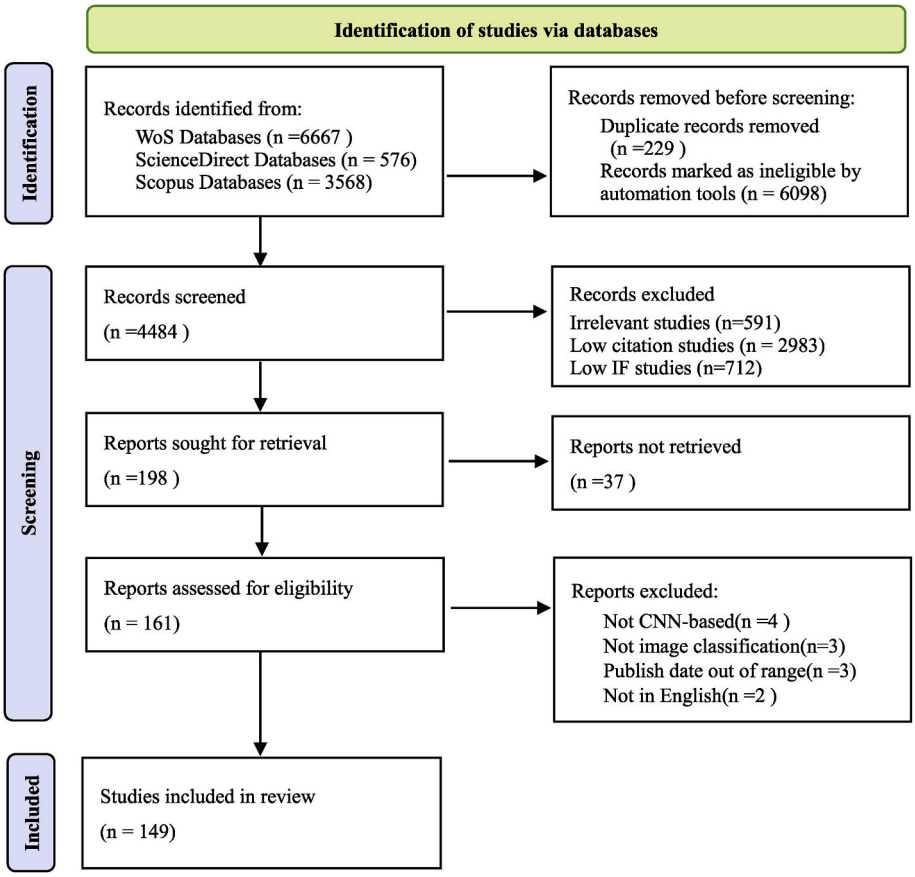


Fig. 1. The PRISMA flowchart for this review.

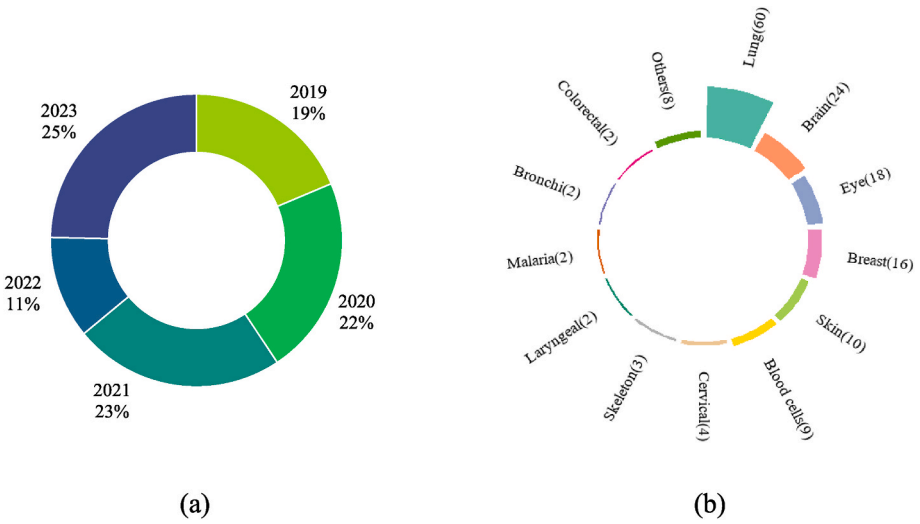


Fig. 2. Distribution of selected studies. (a) The studies' distribution by publish year. (b) The studies' distribution by body part.

while conference proceedings do not have IF values, papers from highly regarded conferences are also retained due to the significant influence of these conferences. "Reports not retrieved" refers to articles those we can not obtain the full text due to subscription or purchase requirements.

3. Evolution of CNN for medical image classification

Image classification is an important task in the field of computer vision. Over the last few years, CNN architectures have continued to evolve, demonstrating excellent performance not only in the computer vision tasks with natural images as classification objects, but also significantly improving the performance in medical image classification. According to our search for relevant literature on WoS, we found that the trend of CNN used in medical image classification has become more and more obvious in the past 10 years, as shown in Fig. 3. Indeed, CNN accounted for only 9 % of medical image classification methods in 2014, but by 2023 this figure has reached 54 %.

Due to the rapid iteration in this research field, we selected studies from the past five years instead of the past ten years for our review in order to focus on the latest technologies. Through detailed analysis and classification of the 149 included studies in the last five years, we have summarized several key research issues for CNN in medical image classification.

(1) Data preprocessing is a crucial step in medical image classification, as it can impact the model's performance, robustness, and generalization capability. Various preprocessing methods are currently employed, such as standardization and normalization, data augmentation, denoising, resizing, local region highlighting, Histogram Equalization (HE), etc.
(2) Utilizing pre-trained models on large-scale image datasets for transfer learning, especially by utilizing model parameters trained in natural images. This aids in accelerating model convergence and improving performance on medical image tasks.
(3) Improving the CNN architectures to improve model performance. Firstly, improving the existing CNN model can effectively improve the performance of the model in the medical image classification. Therefore, many variants of CNN have emerged that are applied in the medical field. Secondly, building new CNN structures to cope with the peculiarities of medical data. Thirdly, ensemble strategy is a good choice for optimizing models and improving performance, including the utilization of multiple CNNs and combination of CNNs with other methods.
(4) Medical image processing places a high demand on the explainability of decision making. Therefore, more and more studies are committed to developing techniques that explain the model's decisions, thereby increasing the trust of doctors and clinical decision-makers in the results produced by the model.

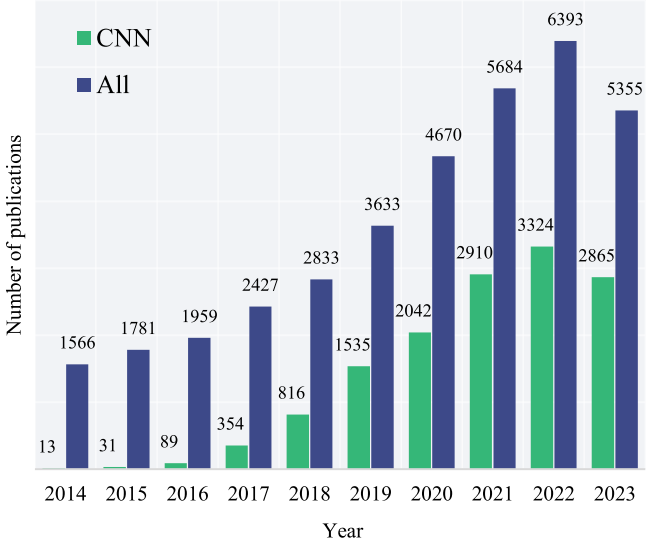


Fig. 3. Year-wise distribution of publications on medical image classification from WoS.

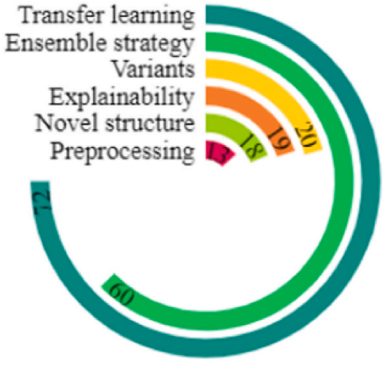


Fig. 4. The distribution of key research issues in selected studies.

As shown in Fig. 4, in our selected studies, there are 13 papers emphasize image preprocessing, 72 papers utilized transfer learning technology, 20 papers improved the existing CNN models, 18 papers proposed new structures, 60 papers used ensemble strategies, and 19 papers adopted explainability methods. It is worth noting that some literature focuses on two or more research issues simultaneously.

4. CNN based medical image classification

The first step in medical image classification is preprocessing which can highlight the key features and improve image quality. Medical image datasets are often very limited, and transfer learning has become the most effective strategy to deal with this problem. In the literature we selected, nearly half of the models use transfer learning strategies. Even newly proposed models or integrated models will consider pre-training on large-scale datasets to obtain initial weights. In this section, image preprocessing techniques and their impact on the model are first organized. Transfer learning methods are then introduced in detail, as well as fine-tuning strategies for different dataset sizes. Then, improvements for CNN structure are summarized, such as optimization based on existing models, new structures, and model ensemble strategies. Finally, existing interpretable methods are summarized and their role in understanding classification decisions is analyzed. Fig. 5 shows the CNN-based medical image classification process, covering all methods reviewed in this article.

4.1. Medical image preprocessing

The structure of medical images differs from natural images in that they usually contain many irrelevant regions and artifacts. So it is necessary to conduct preprocessing to enhance image quality, highlight crucial features, reduce noise, and meet model input requirements. This, in turn, aims to improve the model's feature extraction effect, generalization, and robustness. Commonly used preprocessing methods like normalization, HE, Contrast Limited Adaptive Histogram Equalization (CLAHE), denoising, resampling, local region highlighting and segmentation are widely performed in the surveyed studies.

A. Preprocessing for X-ray Images

X-ray images are usually grayscale images with a large contrast range. The main goal of preprocessing is to enhance the contrast of the region of interest, suppress noise, and improve the clarity of the image. Therefore, image enhancement methods such as HE, filters, and sharpening are often used for X-ray images. Three well-known enhancement techniques, CLAHE, high-frequency filtering and unsharp masking, were applied in conjunction with EfficientNetB0 in Ref. [22]. The real-time testing obtained an accuracy of 96.74 %. The use of these preprocessing techniques was intended to enhance the quality of musculoskeletal radiographic images, thereby improving the accuracy of deep learning models in medical diagnostics. Qi et al. [23] introduced a local phase-based image enhancement method to improve the diagnosis of COVID-19. The proposed method highlighted edges and important structures in X-ray images, enhancing visibility of diseased areas. Both the enhanced images and the original X-ray data, were used as inputs to the proposed CNN architecture. Finally, the CNN model achieved accuracies of 95.57 % and 94.44 % on two datasets, respectively. For COVID-19 detection, Avci et al. [24] used the Linear Discriminant Analysis (LDA) and Gray Level Co-occurrence Matrix as the feature reduction methods in the preprocessing stage. Gray Level Co-occurrence Matrix extracted texture features which contain the spatial relationship information of the image grayscale values. Then, LDA further reduced the dimension and selected these features to maximize the separability of different categories and improve classification performance. The accuracy increased to 99.94 % when this preprocessing method was applied.

In addition, some preprocessing methods have been proposed to address the problem of imbalanced data sets. A data generator [25] for data augmentation with meta-information was introduced to resample the dataset. The experiment demonstrated that this method significantly enhanced the MobileNet-V2 model's performance in classifying lung diseases, with an average AUC of 0.811 and an accuracy exceeding 90 %. Although the resampling technique required higher computational resources, it effectively addressed the issue of data imbalance which was crucial for improving classification performance. Various preprocessing steps included dataset balancing, medical experts' image analysis and data augmentation were employed in Ref. [26] to address the imbalanced dataset problems. The outcomes of the experiments revealed an impressive overall accuracy of 99.5 %. However, the medical experts' image analysis method used in this study required medical experts to manually analyze and screen images, which increased the steps and workload of the preprocessing process. In Ref. [27], a two-stage deep learning framework was designed to deal with the significant class imbalance encountered during the training of small lesion detectors.

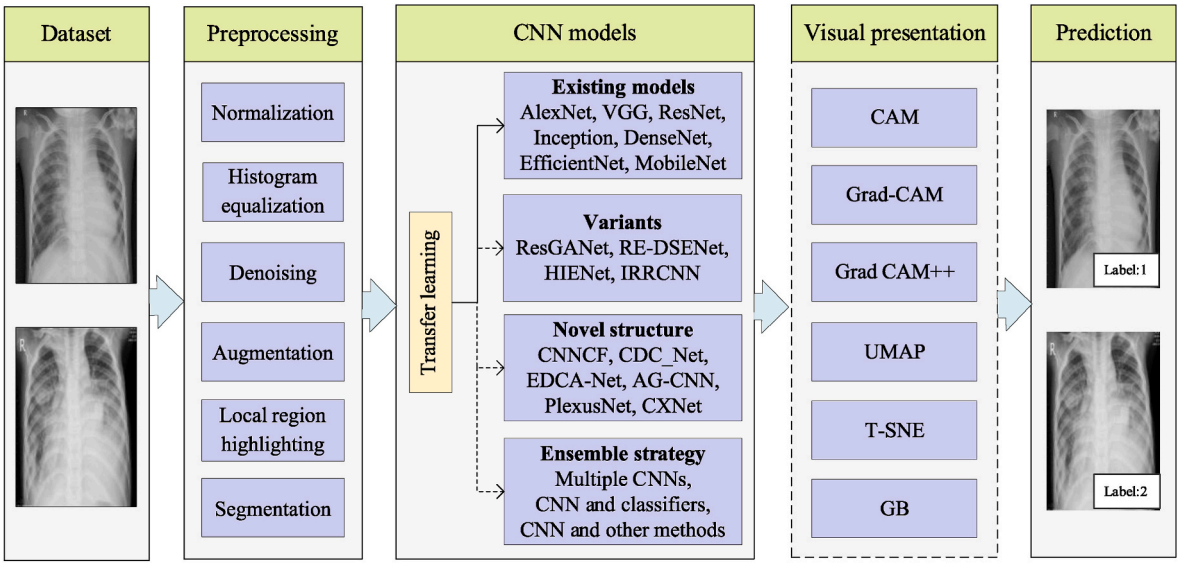


Fig. 5. General flow of CNN-based medical image classification. Dashed lines indicate actions that may be performed, some studies adopted it, others did not.

Initially, a deep cascade (DC) was trained, consisting of long sequences of decision trees. This process aimed to reduce the number of background samples reaching the final stage. The remaining samples were then input to a CNN, benefiting from both the rebalance and hard mining carried out by the DC. The DC-CNN was evaluated on two classification problems with significant imbalances: microcalcification detection and microaneurysm detection. It demonstrated superior performance in both cases compared to CNNs trained using commonly employed methods for mitigating class imbalance. Additionally, the DC-CNN was approximately 10 times quicker than the traditional CNN during testing. The research presented in this article showed that the combination of oversampling and hard mining exhibited the best performance. Specifically, data augmentation techniques such as flipping and rotating are more effective than other transformations like affine and warping. Simple undersampling of the majority class was less effective because it led to the loss of valuable information and inadequate representation of heterogeneity within the majority class.

B. Preprocessing for MRI Images

MRI images have high resolution, which facilitates detailed observation of fine structures and can clearly show the differences between different soft tissues. For MRI images, the most common preprocessing techniques are intensity normalization and contrast enhancement to improve image visualization and facilitate subsequent analysis. To address the data heterogeneity issue, a preprocessing method based on intensity normalization and adaptive contrast enhancement for MRI data was introduced in Ref. [28]. The implementation of this preprocessing step has notably mitigated the dataset heterogeneity stemming from various scanner technologies and acquisition protocols. Additionally, intensity normalization has proven beneficial in mitigating the substantial in-homogeneity observed in gliomas' tissues. In addition, image segmentation and extracting relevant features for MRI images are also crucial for the classification task.

C. Preprocessing for Histopathology Images

Histopathology images are generally high-resolution color images and are mainly used for morphological analysis of cells and tissues. However, image colors can vary significantly due to different staining methods. Color normalization [29] methods can eliminate color differences between different stains. In addition, by segmenting different stained areas, useful structural information can be extracted, which helps improve the performance of classification tasks.

D. Preprocessing for CT Images

CT images are generated based on density differences and thus have high contrast and resolution. Normalization, smooth filtering, contrast adjustment, and local regional highlighting are commonly used for CT image preprocessing. Singh et al. [30] proposed an intensity normalization technique that create some sharp edges and shapes around the anomalous region in CT scans. Specifically, intensity values below and above certain threshold points were set to constants, creating higher contrast around anomalous areas, making it easier for CNNs to detect these areas. Experimental results demonstrated that the proposed normalization technique, referred to as S3DCNN, surpasses the performance of the deeper 3D VGGNet variants. However, the process of selecting threshold points can be tedious and time-consuming, especially for large datasets. If the threshold points are not chosen properly, important information may be lost during the intensity normalization process. In order to improve the detection and classification accuracy of acute bleeding regions in CT brain images, literature [31] applied image thresholding in the preprocessing stage. It enhanced the accuracy and performance of CNN classification by emphasizing the pixel intensities that play a crucial role in feature discrimination. After applying threshold segmentation, the model's F1 score in binary classification tasks increased from 0.706 to 0.952. This preprocessing method enhanced image features, by which the model's classification accuracy was improved particularly in detecting subtle bleeding areas. To achieve optimal performance with this method, adjustments were required on different datasets to determine the best pixel threshold. On the basis of these common preprocessing techniques, using the image fusion technology, the classification accuracy can be improved more effectively. Gautam et al. [32] created two duplicates of the input images to remove some contrast abnormalities. In the first copy, contrast adjustment was implemented to improve the visualization of the target region. In the second copy, a 3×3 average filter was applied to produce a filtered image for pixel smoothing. Subsequently, both images were fused using a quadtree-based multi-focus image fusion method. This approach resulted in an improved image which can be easily classified by the newly introduced 13-layer CNN architecture. It demonstrated high accuracy in experiments on two different datasets. In the first experiment, the classification accuracy reached 98.33 % and 98.77 %; in the second experiment, the classification accuracies were 92.22 % and 93.33 %, respectively. It was proven that although contrast adjustment and image fusion add extra computation, they effectively improved the classification accuracy and other performance metrics of deep learning models.

E. Preprocessing for US Images

Ultrasound images have low contrast and resolution and contain a lot of speckle noise. When the acoustic properties of the target tissue and the surrounding tissue are similar, there will be blurred boundaries. Therefore, commonly used preprocessing methods include denoising, contrast enhancement, image smoothing, and edge detection. When selecting a preprocessing method, it is necessary to consider the specific application and objectives. If more structural information needs to be retained, methods such as adaptive filtering and bilateral filtering can be selected. In the case of low contrast, HE and gamma correction can significantly improve the visual effect [33].

F. Preprocessing for Retinal Images

The morphology, distribution, diameter, and branching characteristics of the vessel structures in the retinal fundus images provide important clinical information, and the color changes and brightness distribution of the images can reveal the states of different tissues. For retinal images, the most common preprocessing methods include denoisising, normalization, HE, CLAHE, segmentation, etc. To enhance retinal fundus images, a pragmatic approach [34] was employed, incorporating both HE and CLAHE. Following the image processing-based enhancement, CNN-based classification was performed. The results showed that the model's classification performance has significantly improved, achieving an accuracy of 97 %, a sensitivity of 94 %, and a specificity of 98 %. The implementation of HE and CLAHE techniques was simple and computationally fast, making them suitable for large-scale data processing, but the additional preprocessing steps and computational costs need to be considered.

G. Preprocessing for Skin Lesion Images

The edge features, size, symmetry, color features, and texture features in the skin lesions images are important indicators to assessing the nature of the lesions. Therefore, the commonly used preprocessing techniques include image enhancement, normalization, segmentation, feature extraction and so on. Aiming at the challenges of accurately classifying various types of skin lesions, Al-masni et al. [35] segmented skin lesion boundaries from complete dermoscopy images using a deep learning full-resolution convolutional network. By feeding the convolutional network with segmented skin lesions instead of utilizing entire dermoscopy images, more notable and representative features can

be extracted, resulting in an enhancement of the classification performance for skin lesion diagnosis. For example, in the ISIC 2016 test dataset, the F1 scores for benign and malignant cases of the Inception-ResNet-V2 model increased by 2.72 % and 4.71 %, respectively.

Table 1 presents the preprocessing methods used for different data types, in terms of tasks, CNN architecture, data types, preprocessing methods, classification performance, questions solved, as well as advantages and disadvantages.

4.2. Existing CNN models with transfer learning for medical image classification

By using pre-trained models previously trained on large datasets, we can directly apply their corresponding structures and weights to the problem we are dealing with [36]. This is referred to as transfer learning, which involves transferring pre-trained models to address the particular problems at hand, as shown in Fig. 6. Medical image classification tasks often encounter challenges such as data scarcity, class imbalance, and complex structures in specific domains. Transfer learning proves to be an effective approach in addressing these issues and improving the performance of medical image classification models [37]. The critical aspects of transfer learning involve selecting the appropriate pre-trained model and adopting an effective transfer strategy. The selected studies on these two issues will be reviewed below.

4.2.1. The selection of pre-trained models for transfer learning

Pre-trained models are typically selected from deep learning models trained on ImageNet, a dataset containing 1.2 million images belonging to 1000 classes. It is crucial to choose a model that performs well on the pre-training task and learns universal features. The selection of the pre-trained model can impact the performance of transfer learning, as favorable initial weights can expedite the model's learning process on the target task. Many researchers have conducted comparative studies on the transfer learning performance for different classification tasks and data modality.

A. Pre-trained Models Selected for X-ray Images

The most common research method in transfer learning involves using multiple transfer learning-based CNN models to compare which ones are more suitable for specific classification tasks or image modality. To overcome the difficulty of great demand of data, eight models pre-trained in ImageNet dataset were evaluated in Ref. [38]. These transfer learning models were proved to be computationally efficient and accurate for classifying chest X-ray images, and DenseNet121 achieved the best accuracy of 98.69 % in four classes classification task. Although these pre-trained models performed well on training data, overfitting might occur with new, unseen data, which could limit the model's ability to generalize. Apostolopoulos et al. [39] evaluated the performance of five state-of-the-art CNN models, named VGG19, MobileNet-V2, Inception, Xception, Inception-ResNet-V2. These CNN models were pre-trained in ImageNet and used to detect COVID-19. These models proved to be effective, and MobileNet-V2 achieved the best accuracy of 96.78 %. The limitation of this study was that comparisons were only made for specific tasks. AlexNet, VGG, Inception-V3, ResNet and DenseNet121 were evaluated in Ref. [40] to classify abnormality for frontal chest radio graphs. The result showed that pre-trained CNNs can accurately and effectively classify abnormal images. Among them, ResNet18 achieved the highest accuracy, reaching 96.64 %. In order to ensure the classification capability of CNN model, EfficientNetB0, VGG16 and Inception-V3 were evaluated for COVID-19 detection through transfer learning [41]. As a result, EfficientNetB0 model achieved the highest overall accuracy of 92.93 %. In this study, training time was reduced, but research was conducted with a limited dataset, which could be strengthened as more data becomes available. In order to enhance the efficiency and accuracy of detecting COVID-19 from chest X-ray images, Zebin et al. [42] implemented transfer learning using pre-trained VGG16, ResNet50, and EfficientNetB0 models as feature extractors. The results showed that the EfficientNetB0 model achieved the best performance with accuracy of 96.8 %. Transfer learning method also reduced training time, quickly improved accuracy and obtained better generalization in this study. But it required careful adjustment of parameters and training strategies, which can be complex and time-consuming.

The primary issue addressed by using transfer learning in Ref. [43] was improving the performance of the pneumonia detection model in the context of limited labeled data. Pre-trained AlexNet, DenseNet121, Inception-V3, ResNet18, and GoogLeNet were combined to detect pneumonia from chest X-ray images. The outputs of five models were aggregated into a prediction vector, and then the overall accuracy was improved by leveraging the strengths of each individual model. The results showed that the accuracy of combined model was 96.39 %, which was better than that of any single pre-trained model. This study indicated that combined multiple pre-trained models enhanced the model's robustness and accuracy. However, it could be computationally expensive, requiring significant resources for both training and inference. To overcome the challenges of training data scarcity, and improve model's performance on tuberculosis X-ray image classification tasks [44], pre-trained ResNet50, Inception-V3, and VGG19 were used to build an integrated model. The integrated model obtained the best overall accuracy of 86.42 % as the ResNet50 at 50 rounds of training, Inception-V3 at 10 rounds of training, and VGG19 at 50 rounds of training. This approach greatly reduced training time and the need for computing resources. Its disadvantage was that the model still needs to be tuned for the specific task to achieve optimal performance.

To assess the effectiveness of using transfer learning for feature extraction in image classification [45], utilized the state-of-the-art CNN called MobileNet to automatically classify lung diseases from X-ray images. The study compared the model's performance in feature extraction and classification tasks when trained from scratch on X-ray images, using ready-made features, and applying fine-tuning strategies. The results showed that the method trained from scratch achieved a seven-class classification accuracy of 87.66 % and a binary classification accuracy of 99.18 %, outperforming transfer learning techniques. Moreover, the fine-tuning strategy yielded better results than using ready-made features. Therefore, although transfer learning techniques could reduce training time and computational resources, the features extracted through transfer learning were less effective compared to those extracted from networks trained from scratch on X-ray images. In the diagnosis of COVID-19 [46], the EfficientNetB4 model based on transfer learning was used to improve the efficiency and accuracy of the model, as well as to reduce training time and computational resource consumption. The EfficientNetB4 model showed excellent performance in the automatic diagnosis of COVID-19, achieving an average accuracy of 99.62 % in binary classification tasks and 96.70 % in multi-class classification tasks. In this study, transfer learning not only significantly improved the performance of the model but also reduced training time and computational resource consumption. However, this study still faces significant challenges, as the existing datasets are not robust enough, limiting the model's generalization ability and accuracy. In order to improve the performance of deep learning models in medical image classification tasks, research [47] selected EfficientNetB7, which had shown excellent performance in multiple image classification tasks, as the base model. It utilized the weights from the ImageNet dataset and fine-tuned them on a specific dataset for pneumonia image classification. Through this transfer learning approach, different types of pneumonia were effectively classified, achieving an accuracy of 99.81 % on the X-ray dataset. However, due to the high complexity of the EfficientNetB7 model, training and inference time were relatively long, requiring high-performance computing resources, which may not be suitable for resource-limited environments. Facing challenges related to

Table 1
A summary of preprocessing methods used for different data types.

Studies/ Author	Tasks	CNN architecture	Data types	Preprocessing methods	Classification performance (Only report the best)	Questions solved	Advantages	Disadvantages
Narayan et al. [22]	Bone classification	EfficientNetB0	X-ray	CLAHE, high-frequency filtering and unsharp masking	Accuracy: 96.74 %	To enhance image quality.	Higher accuracy.	It needs to be carefully tuned according to the data set and task requirements.
Qi et al. [23]	COVID-19 diagnosis	AlexNet, ResNet50, SonoNet64, Xception, Inception-V4 and EfficientNetB4	X-ray	Local phase-based image enhancement	Accuracy: 95.57 % (Dataset1)Accuracy: 94.44 % (Dataset2)	To highlight key features and edges.	Enhancing COVID-19 detection.	It is applicable only to certain datasets.
Avci et al. [24]	COVID-19 detection	New CNN with 28 layers	X-ray	LDA and Gray Level Co-occurrence Matrix	Accuracy: 99.94 %	Feature selection and dimension reduction.	Better classification performance.	High computational cost.
Souid et al. [25]	Lung diseases classification	MobileNet-V2	X-ray	Resampling with a data generator for augmentation	Accuracy: 90 %	To address class imbalance.	Significant improvement in model performance.	It requires higher computational resources.
Reshi et al. [26]	COVID-19 detection	New CNN with 38 layers	X-ray	Dataset balancing, medical experts' image analysis, and data augmentation	Accuracy: 99.5 %	To address class imbalance.	Higher accuracy.	High dependency on manual effort.
Bria et al. [27]	Microcalcification and microaneurysm detection	DC-CNN based on VGGNet	X-ray	A DC of long sequences of decision trees that conduct rebalance and hard mining	Accuracy: 80.19 % (Microcalcification) Accuracy: 80.99 % (Microaneurysm)	To address class imbalance.	Better performance and faster inference process.	It is a complex multi-stage pipeline.
Mzoughi et al. [28]	Gliomas brain tumor classification	New multi-scale 3D CNN with 11 layers	MRI	Intensity normalization and adaptive contrast enhancement	Accuracy: 96.46 %	To address the data heterogeneity issue.	It can lead to an accurate classification.	It is a time-consuming preprocessing.
Sun et al. [29]	Uterine cancer diagnosis	HIENet	Histopathology Images	Standard score normalization and color normalization	Accuracy: 84.5 %	To reduce the impact of color differences on model performance.	Better classification performance.	Useful information related to color changes may be lost.
Singh et al. [30]	Acute brain hemorrhage classification	New Shallow 3DCNN	CT	Intensity normalization	Accuracy: 97 %	To create sharp contrast around anomalous areas.	Enhancing detection of subtle regions.	The process of selecting threshold points can be tedious and time-consuming.
Ker et al. [31]	Acute brain hemorrhages diagnosis	New 3D CNN	CT	Local region highlighting using image thresholding	F1 score: 95.2 %	To highlight critical features.	Better performance in detecting small lesions.	It needs dataset-dependent threshold adjustments.
Gautam et al. [32]	Stroke classification	New P_CNN with 13 layers	CT	Contrast adjustment, pixel smoothing , quadtree-based multi-focus image fusion	Accuracy: 98.77 %	To remove some contrast abnormalities.	Better classification performance.	Higher computational cost.
Hemanth et al. [34]	Diabetic retinopathy detection and classification	New CNN model	Retinal images	HE and CLAHE	Accuracy: 97 %	To enhance retinal fundus images.	Simple and computationally fast.	Additional preprocessing steps and computational costs need to be considered.
Al-masni et al. [35]	Skin lesions classification	Inception-V3, ResNet50, Inception-ResNet-V2, DenseNet201	Skin lesion images	Segmentation using full resolution convolutional network	Accuracy: 89.28 % (ResNet50)	To extract more notable and representative features.	Better classification performance.	Higher computational cost.

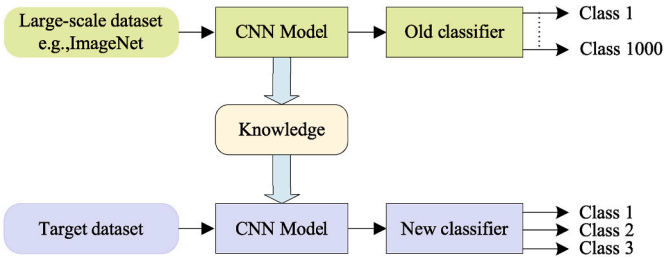


Fig. 6. Schematic diagram of transfer learning.

the scale and quality of currently available public COVID-19 datasets, study [48] identified the appropriate CNN model through a preliminary comparative study of several popular CNN models. The selected VGG19 model was pre-trained on a large-scale dataset and then fine-tuned extensively with appropriate parameters. It achieved a high level of accuracy in detecting COVID-19 across all three types of lung imaging modes: 86 % for X-rays, 100 % for US, and 84 % for CT scans. The research indicated that most deeper networks struggle to train effectively when data is limited.

B. Pre-trained Models Selected for MRI Images

Study [49] used the transfer learning method to overcome the problem of limited data. In this study, pre-trained AlexNet, VGG16, ResNet18, ResNet34, and ResNet50 were used to classify five classes of brain MRI images. Their performance was compared, and ResNet50 model obtained the highest classification accuracy of 95.23 %. The advantage of this study was that these models were developed with a small number of images, and it was the first study to classify five classes of brain diseases with the deep learning method. But this study did not perform well when using a single Alzheimer subject because the database used contained few subjects in each class. For brain tumors classification, pre-trained VGG16, VGG19, ResNet50 and DenseNet121 architectures were employed to extract features in Ref. [50], with three fully connected layers as the classifier. As a result, ResNet50 obtained the highest performance of 99.02 %, and DenseNet121 obtained the second-best performance of 98.91 %. The benefits of using transfer learning included the ability of these models to be trained with little data and less time. While the disadvantage was that the original MR image needed to undergo a lot of preprocessing in order to adapt to the input requirements of the deep learning model, which increased the workload of data preparation. To overcome this problem [51], proposed a multistage feature extraction and tandem method for early diagnosis of brain tumors. Pre-trained Inception-V3 and DenseNet201 were used as backbones. The proposed method achieved testing accuracies of 99.34 % with Inception-V3 and 99.51 % with DenseNet201, demonstrating the highest performance in brain tumor detection. In addition to obtaining higher accuracy, transfer learning method made the models more efficient on small datasets. However, pre-trained models may overfit new datasets. This problem can be worse if the new dataset differs significantly from the pre-trained dataset.

The existing classification methods of brain tumors have some problems, such as low precision and insufficient generalization ability of models, which can not provide reliable auxiliary diagnosis support in practical application. So a multistage brain tumor classification method based on deep CNN and extensive data enhancement techniques was proposed in Ref. [52], which used a pre-trained VGG model to extract features and performed multistage classification. The classification accuracy of the model was significantly improved, with an accuracy of 90.67 % for Radiopaedia dataset and 94.58 % for Brain tumor dataset. The advantage of this method was that it could effectively solve the problem of insufficient data and improved the accuracy. Its disadvantage was that it relied on a lot of computing resources, which affected the optimization and performance improvement of the model. Applying deep learning on small datasets and training CNNs from scratch is a challenging task. To address this issue, Swati et al. [53] proposed a block-wise fine-tuning strategy based on transfer learning using a pre-trained VGG19 model. This method achieved a notable accuracy of 94.82 % for brain tumor detection, which was significantly higher than many traditional approaches. Moreover, the model proposed was more versatile as it requires minimal preprocessing of 2D MR images and does not rely on handcrafted features. But there are reasons to worry about the process of fine-tuning VGG19 models can be computationally expensive and time-consuming, often requiring high-end GPUs and extensive memory resources, which may not be accessible in all clinical settings. To achieve an accurate and automated classification system for brain tumors, the pre-trained GoogLeNet model was used to extract the features of brain MRI images by transfer learning [54]. It showed an accuracy of 98 % on Figshare dataset, superior to all the most advanced methods. The application of transfer learning demonstrated its significant advantages in medical image classification, particularly in improving classification accuracy and generalization capability with limited data. However, transfer learning also presented challenges such as dependency on pre-trained models, potential overfitting, and the need for fine-tuning, which needed to be addressed in practical applications.

C. Pre-trained Models Selected for Histopathology Images

A lot of previous work for breast cancer image classification existed with lack of accuracy [55], presented a computer-aided diagnosis method using deep neural networks and transfer learning in order to overcome these challenges. Pre-trained ResNet18, ShuffleNet, and Inception-V3 were employed and compared, as a result, ResNet18 obtained the best average accuracy of 99.7 % for binary classification and 97.81 % for multi-class classification. The advantage of this research was that the accuracy was enhanced, while the disadvantage was that these models cannot handle more complex datasets to diagnose the breast cancer images. For breast cancer classification [56], transfer learning combined with structured filter pruning was used to improve model performance and efficiency, as well as deal with data constraints problem. Pre-trained deep CNN models such as VGG19, ResNet34, and ResNet50 were applied in this study. As a result, the classification performance of these models was significantly improved, the number of parameters and floating-point operations was significantly reduced after pruning. ResNet50 obtained the best accuracy of 92.07 %. In conclusion, transfer learning combined with structured filter pruning showed significant advantages in breast cancer classification tasks, such as model generalization was improved and classification performance was enhanced. Nevertheless, it was also accompanied by a certain increase in complexity and computational cost. Pre-trained models usually extract underlying features from natural images.

The most deep learning-based image classification methods have a large number of parameters and are not suitable for use in Internet of Medical Things (IoMT). Therefore, study [57] introduced a lightweight model based on transfer learning, ‘MobileNet-SVM’, for the automatic classification of breast cancer histology images. It achieved an accuracy of 91 % and an F1-score of 91.35 % on the test dataset. The MobileNet-SVM model not only had high accuracy but also required low computational speed and time, making it an ideal choice for IoMT imaging devices. However, despite its high computational efficiency, the MobileNet-SVM model was specifically designed for lightweight applications. Therefore it may not be able to compete with larger, more complex deep learning models when handling very complex or large-scale datasets. To improve the detection accuracy of breast cancer risk presence, the study [58] attempted to use the Inception-ResNet-V2 deep learning model based on transfer learning to detect breast cancer risk markers. This model was pre-trained on the ImageNet dataset and then fine-tuned, retaining the weights of the initial layers while only retraining the last few layers to adapt to the specific tasks of the breast

cancer dataset. Experimental results indicated that this method could effectively extract image features and, when applied to breast cancer detection tasks, achieved an accuracy of 91 %, demonstrating superior performance compared to traditional methods. However, its drawback was that the Inception-ResNet-V2 model structure was complex, requiring substantial computational resources and memory, which ordinary devices may find challenging to support.

D. Pre-trained Models Selected for CT Images

Study [59] investigated how deep transfer learning from non-medical images could enhance the performance of systems designed for classifying the malignancy of lung nodules. The authors employed 11 types of pre-trained CNN architectures, including VGG16, VGG19, MobileNet, Xception, Inception-V3, ResNet50, Inception-ResNet-V2, DenseNet169, DenseNet201, NASNetMobile, and NASNetLarge. These models were trained on the ImageNet dataset and repurposed as feature extractors through transfer learning to process nodular images in the LIDC and IDRI dataset. Among these CNN models, ResNet50 demonstrated the best performance, achieving an Area Under the Curve (AUC) of 93.1 % and an accuracy of 88.41 %, illustrating the effectiveness of transfer learning in enhancing classification performance. Transfer learning also significantly reduced the time and computational resources required to train deep learning models from scratch. This enabled the system to quickly adapt to classifying lung nodules, although the imbalance between malignant and benign nodules in the dataset could affect the training effectiveness and ultimate classification performance. To save time for training, a simple 2D deep learning framework [60], developed by transfer learning, was proposed to diagnose COVID-19 pneumonia from CT images. Each of the pre-trained VGG16, ResNet50, Inception-V3, and Xception was used as a backbone. Among the four pre-trained models, ResNet50 demonstrated outstanding diagnostic performance, with a sensitivity of 99.58 %, specificity of 100.00 %, and accuracy of 99.87 %, surpassing the performance of the other three pre-trained models. Although this method provided a simple and rapid model, and used less time to train by using pre-trained CNN models, it was prone to overfitting because the testing and training data originate from the same data source. In Ref. [47], pre-trained EfficientNetB7 was used for pneumonia classification with CT dataset. It obtained an accuracy of 99.88 % on the CT dataset. This model handled both X-ray and CT scan imaging data, demonstrating its applicability and robustness in different imaging modalities. In Ref. [48], the VGG19 model was selected for classify COVID-19 not only with X-ray dataset but also with CT dataset, and achieved a high level of accuracy for CT scans.

E. Pre-trained Models Selected for Microscopic Images

Since well-labeled datasets for a specific disease such as leukemia were very scarce, literature [61] adopted VGG16 pre-trained on large-scale datasets and migrated its knowledge to the leukemia detection task, enabling good performance with fewer labeled data. The experimental results showed that the pre-trained model achieved an accuracy of 84 % on independent test datasets. This paper showed that the pre-trained model could automatically extract high-quality features, reducing the need for large-scale labeled datasets. The shortcoming was that features of the pre-trained model may not be fully applicable to leukemia detection task and requires complex fine-tuning. Since collecting large amounts of biomedical data is difficult, transfer learning was used in Ref. [62]. The pre-trained VGG network was chosen because it captures fine image features essential for malaria diagnosis. The transfer learning-based VGG model achieved a classification accuracy of 93.1 % in identifying malaria-infected cells, demonstrating the potential of transfer learning in medical image analysis. Transfer learning method in this research reduced the reliance on domain expert knowledge for developing efficient feature extraction and classification algorithms, which could be costly in terms of expertise and time. But in the process of transfer learning, layers must be adjusted and configured. For example, the last few layers of the VGG model were replaced with an SVM to handle the classification of multidimensional features in this research.

F. Pre-trained Models Selected for US Images

For breast lesions classification, in order to obtain better performance of classification, 6 CNN architectures, namely AlexNet, ZFNet, VGG, ResNet, GoogLeNet and DenseNet, were systematically analyzed with transfer learning from ImageNet [63]. The experiments showed that deeper networks with fewer parameters obtain better results, and the DenseNet proved to be more suitable for breast lesions classification task. The application of transfer learning from the large-scale annotated ImageNet to classify breast lesions significantly enhanced the performance of various CNN architectures. Some models such as AlexNet, ZFNet, and VGG16, met the curse-of-dimensionality problem that may lead to overfitting. In Ref. [48], the VGG19 model was selected for classify COVID-19 with US dataset, and achieved an accuracy of 100 %.

G. Pre-trained Models Selected for Retinal Images

For diabetic retinopathy classification, transfer learning was used to enhance the accuracy of classification [64]. Five pre-trained models including AlexNet, GoogLeNet, Inception-V4, Inception-ResNet-V2 and ResNeXt50 were utilized. The experimental outcomes demonstrated that this method detected diabetic retinopathy with a higher accuracy, and the ResNeXt50 achieved the highest accuracy of 97.53 % among all the pre-trained models. Due to the high resolution dataset, the execution time for the classification increased.

H. Pre-trained Models Selected for Skin Lesion Images

In [65], in order to deal with the problems such as small scale of medical image dataset, difficulty in labeling, and data imbalance, a three-level fusion technique, MSM-CNN, was introduced in the task of classifying skin lesions. It integrated outcomes from three different networks and cropped images at multiple sizes. It explored how image size impacts skin lesion classification through transfer learning, achieving an average accuracy of 96.4 %. Employing a multi-scale and multi-network integration transfer learning approach enhanced the ability to capture both detailed and global features of skin lesions. This method elevated model performance, mitigated issues stemming from data imbalance, particularly in categories with small sample sizes. At the same time, this method also increased the model's complexity and computational demands, potentially slowing down the inference process. This is particularly challenging in resource-constrained environments.

The selection of model for transfer learning depends on the complexity of the target task, the size of the target dataset, computing resources, and domain differences. The specific choice depends on the specific task. For X-ray images, since they usually have relatively simple feature structures (such as bones, lung contours, etc.), pre-trained shallow models or lightweight models usually perform well for this modality, such as DenseNet121, ResNet18, MobileNet, EfficientNet, etc. Especially when the target dataset is small, transfer learning can significantly improve the generalization ability of the model. For MRI and CT images that contain three-dimensional structural information, since the grayscale value distribution of different tissues and organs of the human body is similar, the task usually involves fine boundaries or tissue segmentation. Such tasks require transfer learning models to capture more subtle features. Pre-trained larger and deeper models or 3D CNNs are more suitable for such tasks, such as VGG, ResNet50, DenseNet201, GoogLeNet, etc.

Table 2 shows some studies for medical image classification with

Table 2
Detailed implementation of transfer learning models.

Studies/Author	Tasks	Pre-trained CNN models	Target datasets	Data modality	Accuracy (Only report the best)	Questions solved	Advantages	Disadvantages
Kamal et al. [38]	COVID-19 detection	VGG19，Inception-V3，ResNet50, ResNet50-V2，MobileNet，MobileNet-V2，DenseNet121，NasNetMobile	Collected from Dr. Joseph Cohen and Chest X-Ray dataset	X-ray	DenseNet121: 98.69 %	To deal with small dataset.	Less requirement for data.	Overfitting problem may occur, which limits the model's ability to generalize.
Apostolopoulos et al. [39]	COVID-19 detection	VGG19, MobileNet-V2, Inception, Xception, Inception-ResNet-V2	Collected from public resources	X-ray	MobileNet-V2: 96.78 %	To evaluate the performance of state-of-the-art CNN models proposed over the recent years for medical image classification.	These models with transfer learning was proved effective.	Comparisons were only made for specific tasks.
Tang et al. [40]	Abnormality classification for frontal chest radio graphs	AlexNet, VGGNet16/19, Inception-V3, ResNet18/50, DenseNet121	Collected from four public datasets	X-ray	ResNet18: 96.64 %	To evaluate various deep CNN.	Better initialization of the model parameters and faster training.	The labels of the training images were text mined from the radiological report using NLP instead of manual ground-truth labels.
Gaur et al. [41]	COVID-19 detection	EfficientNetB0, VGG16, Inception-V3	COVID-19 radiography and Actualmed-COVID-chest Xray-dataset	X-ray	EfficientNetB0: 92.93 %	To ensure the classification capability of model.	Less training time.	The observations are from a limited amount of data set.
Zebin et al. [42]	COVID-19 detection	VGG16, ResNet50, EfficientNetB0	COVID-19 Image Data Collection	X-ray	EfficientNetB0: 96.8 %	To enhance the efficiency and accuracy.	Higher accuracy, less training time, and better generalization.	It requires careful adjustment of parameters and training strategies, which can be complex and time-consuming.
Chouhan et al. [43]	Pneumonia detection	Combined AlexNet, DenseNet121, Inception-V3, ResNet18 and GoogLeNet	Collected from GuangzhouWomen and Children's Medical Center	X-ray	96.40 %	To improve the performance in the context of limited labeled data.	Better robustness and higher accuracy.	Higher computational cost.
Hernández et al. [44]	Tuberculosis classification	Combined ResNet50, Inception-V3 and VGG19	Collected from a program	X-ray	86.42 %	To overcome the challenges of training data scarcity.	Less training time and computing resources.	The model still needs to be tuned for the specific task.
Apostolopoulos et al. [45]	COVID-19 detection and classification	MobileNet-V2	Collected from public resources	X-ray	99.18 % (Binary classification) 87.66 % (Multi-class classification)	To assess the effectiveness of using transfer learning.	Less training time and computational resources.	The features extracted through transfer learning are less effective than those extracted from networks trained from scratch on X-ray images.
Marques et al. [46]	COVID-19 detection	EfficientNetB4	Collected from public resources	X-ray	99.62 % (Binary classification) 96.7 % (Multi-class classification)	To improve the efficiency and accuracy of the model, and reduce training time and computational resource consumption.	Higher accuracy, less training time and computational resource consumption.	The model's generalization ability and accuracy are limited as the existing datasets are not robust enough.
Moussaid et al. [47]	Pneumonia classification	EfficientNetB7	1.MOMA Dataset2. MosMedData	X-ray, CT	99.81 % (Dataset1) 99.88 % (Dataset2)	To improve the performance.	Higher accuracy, and good applicability and robustness in different	It may not be suitable for resource-limited environments.

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Table 2 (*continued*)

Studies/Author	Tasks	Pre-trained CNN models	Target datasets	Data modality	Accuracy (Only report the best)	Questions solved	Advantages	Disadvantages
							imaging modalities.	
Horry et al. [48]	COVID-19 detection	VGG19	1.COVID-19 Image Data Collection and NIH Chest X-Ray2. COVID-19-CT Dataset3.POCOVID-Net Dataset	X-ray, CT, US	86 % (Dataset1) 100 % (Dataset2)84 % (Dataset3)	To deal with the scale and quality of currently available public COVID-19 datasets.	Higher accuracy.	The deep network struggle to train effectively when data is limited.
Talo et al. [49]	Brain disease classification	AlexNet, VGG16, ResNet18, ResNet34, ResNet50	Harvard Medical School database	MRI	ResNet50: 95.23 %	To overcome the issue of limited number of data.	Smaller number of brain MRI data.	The database used contains few subjects in each class.
Polat et al. [50]	Brain tumor classification	VGG16, VGG19, ResNet50, DenseNet21	Figshare dataset	MRI	ResNet50: 99.02 %	To classify brain tumors in MR images using transfer learning networks.	Little data and less time spent on training.	The original MR image needs to undergo a lot of preprocessing, which increases the workload of data preparation.
Noreen et al. [51]	Brain tumor classification	Inception-V3, DenseNet201	Figshare dataset	MRI	DenseNet201: 99.51 %	To solve the problem of pre-trained models extract features from natural images.	Higher accuracy and more efficient on small data set.	There is a risk of overfitting especially when the new data set differs significantly from the pre-trained data set.
Sajjad et al. [52]	Brain tumor classification	VGG19	1.Radiopaedia dataset2.Brain tumor dataset	MRI	90.67 % (Dataset1) 94.58 % (Dataset2)	To deal with the issues like low precision and insufficient generalization ability.	Higher accuracy and more efficient on small data set.	Need more computing resources.
Swati et al. [53]	Brain tumor classification	VGG19	CE-MRI	MRI	94.82 %	To deal with the problem of small datasets.	Higher accuracy and more generic.	The process of fine-tuning VGG19 model can be computationally expensive and time-consuming.
Deepak et al. [54]	Brain tumor classification	GoogLeNet	Figshare dataset	MRI	98 %	To achieve an accurate and automated classification system for brain tumors.	Higher accuracy and generalization capability with limited data.	Dependency on pre-trained models, potential overfitting, and the need for fine-tuning.
Aljuaid et al. [55]	Breast cancer classification	ResNet18, ShuffleNet, Inception-V3	BreakHis	Histopathology images	ResNet18: 99.7 % (Binary classification) ResNet18: 97.81 % (Multi-class classification)	To overcome the lack of accuracy.	Better accuracy.	The models cannot handle more complex datasets to diagnose the breast cancer images.
Choudhary et al. [56]	Breast cancer classification	VGG19, ResNet34, ResNet50	IDC dataset	Histopathology images	ResNet50 : 92.07 %	To improve model performance and efficiency, as well as deal with data constraints problem.	Better model generalization and classification performance.	Accompanied by a certain increase in complexity and computational cost.
Ogundokun et al. [57]	Breast cancer classification	MobileNet	BreakHis	Histopathology images	91 %	To build a lightweight model for breast cancer classification.	Higher accuracy, less computational resource and time.	It may not be able to handle very complex or large-scale datasets.
Humayun et al. [58]	Breast cancer classification	Inception-ResNet-V2	Collected from an open source library	Histopathology images	91 %	To improve the detection accuracy of breast cancer risk presence.	Higher accuracy.	The training process requires substantial computational resources and memory.

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Table 2 (*continued*)

Studies/Author	Tasks	Pre-trained CNN models	Target datasets	Data modality	Accuracy (Only report the best)	Questions solved	Advantages	Disadvantages
da Nóbrega et al. [59]	Lung nodule malignancy classification	VGG16, VGG19, MobileNet, Xception, Inception-V3, ResNet50, Inception-ResNet-V2, DenseNet169, DenseNet201, NASNetMobile, NASNetLarge	LIDC and IDRI	CT	ResNet50: 88.41 %	To explore how transfer learning enhance the performance of lung nodules classification systems.	Better performance, less training time, and computational resources.	It cannot deal with the problem of imbalance data.
Ko et al. [60]	COVID-19 detection	VGG16, ResNet50, Inception-V3, Xception	Collected from three public datasets	CT	ResNet50: 99.87 %	To minimize the training time.	It provides a simple and rapid model, and uses less time to training.	The models may exhibit generalization and overfitting issues because the testing and training data originate from the same data source.
Abhishek et al. [61]	Leukemia detection and classification	VGG16	Collected from public resources	Microscopic blood smears	84 %	In response to the lack of well-labeled data sets.	It reduces the need for large-scale labeled data sets.	The pre-trained model may not be fully applicable to leukemia detection task and requires complex fine-tuning.
Vijayalakshmi et al. [62]	Malaria detection	VGG19	Collected from two public datasets	Microscopic images	93.10 %	To deal with small dataset.	It reduces the reliance on domain expert knowledge.	Layers must be adjusted and configured.
Cao et al. [63]	Breast lesion classification	AlexNet, ZFNet, VGG, ResNet, GoogLeNet, DenseNet	Collected dataset	US	DenseNet: 87.5 %	To obtain better performance.	Better performance.	AlexNet, ZFNet, and VGG16 meet the curse-of-dimensionality problem that easy to overfitting.
Tariq et al. [64]	Diabetic retinopathy classification	AlexNet, GoogLeNet, Inception-V4, Inception, ResNet-V2, ResNeXt50	Collected from public resources	Retinal Images	ResNeXt50: 97.53 %	To enhance the accuracy of classification.	Higher accuracy.	Using a high resolution ultimately elevates the size of the dataset, which may increase the execution time for the classification.
Alsubait et al. [66]	Human protein atlas single cell classification	CSPNet, BoTNet, ResNet	HPA	Protein atlas single cell images	CSPNet: 95 %	To improve the accuracy and efficiency.	Higher accuracy, less training time, and strong generalization capabilities.	The effectiveness of transfer learning heavily depends on the quality and suitability of the pre-trained model.
Mahbod et al. [65]	Skin lesion classification	Combined EfficientNetB0, EfficientNetB1 and SeResNeXt-50	ISIC	Skin lesion images	96.30 %	To deal with the problems such as small scale of medical image data set, difficulty in labeling, and data imbalance.	Better performance, and it mitigates issues stemming from data imbalance particularly in categories with small sample sizes.	More complexity and computation, slower inference process.

transfer learning method, in terms of tasks, pre-trained CNN models, target datasets, data type, accuracy, questions solved, as well as advantages and disadvantages of the study.

4.2.2. *Transfer strategy for transfer learning*

Transfer learning requires the design of a reasonable strategy for transferring previously acquired knowledge to the target task. This involves deciding which layers need fine-tuning or training and how weights should be adjusted to adapt to the target task [36]. When reusing a pre-trained model on the target task, the original classifier is first removed, a suitable new classifier is added, and the model is then fine-tuned. Depending on the similarity between the source and target

dataset and the size of the target dataset, fine-tuning strategies typically adhere to three guidelines [67]: first, if the target dataset is large and there is a lot of computational power, the entire network can be trained from scratch. Second, if the dataset is small and the number of parameters is large, more layers should be frozen to avoid overfitting. On the contrary, if the dataset is large and the number of parameters is small, the model can be improved by training more layers for the new task, since overfitting is not an issue. Third, if the dataset is small and there is high similarity between source and target data, convolutional layers should be frozen, with only the classification layer trained. The three transfer strategies are shown in Fig. 7. Table 3 shows some transfer strategies used for medical image classification tasks, all listed models are pre-trained on ImageNet dataset.

4.2.3. Advantages and limitations of transfer learning

Transfer learning shows significant advantages in medical image classification. First, it addresses the problem of small medical datasets. Pre-trained models on large-scale datasets have already learned many general features, such as edges and textures, which are often applicable to medical images, thus improving the model's generalization ability. Second, transfer learning greatly reduces training time. The pre-trained model has already learned many low-level features during the pre-training phase, and it only needs to learn high-level features specific to medical images in the fine-tuning phase, avoiding the extensive computational resources and time required for training from scratch. Third, transfer learning effectively tackles the issue of overfitting. By providing a relatively stable initial set of weights, transfer learning makes models more robust when trained on small datasets, reducing the likelihood of overfitting—a common challenge in medical images. Finally, transfer learning helps handle complex image features. Pre-trained models in transfer learning have learned complex visual features from large amounts of data, enabling them to adapt to the diverse structures in medical images, such as different types of lesions, organs, and tissues, thereby enhancing classification accuracy.

However, transfer learning technology also has some limitations that need to be addressed. First, there is a large domain gap between medical images and natural images. Medical images usually have different feature distributions, such as the grayscale and texture features of X-ray, CT, and MRI images, which differ significantly from the color and object features of natural images. This domain gap means that the features learned from natural images are not always applicable to medical images. Some existing research has attempted to pre-train models on medical image datasets [68–72], and the results have shown that models pre-trained on medical image datasets are more effective than those pre-trained on natural images. However, due to the limited scale of medical image datasets, such studies are still relatively rare. Second, extensive fine-tuning increases model complexity and computational cost. The model requires more extensive fine-tuning if there is a large difference between the source and target datasets. In some cases, partial retraining of the model may be necessary, which reduces the time and resource advantages of transfer learning. Third, model architectures have limitations. Most pre-trained models are designed for 2D natural image classification, whereas many medical images, such as CT and MRI scans, are 3D images, making it difficult for transfer learning models to effectively handle the rich spatial information in 3D medical images. Finally, there is a lack of domain specialization. The specific characteristics of lesions or organs in medical images may require specialized model architectures or domain knowledge, and the general features in transfer learning models may not be sufficient for these tasks. It requires further optimization with domain expertise.

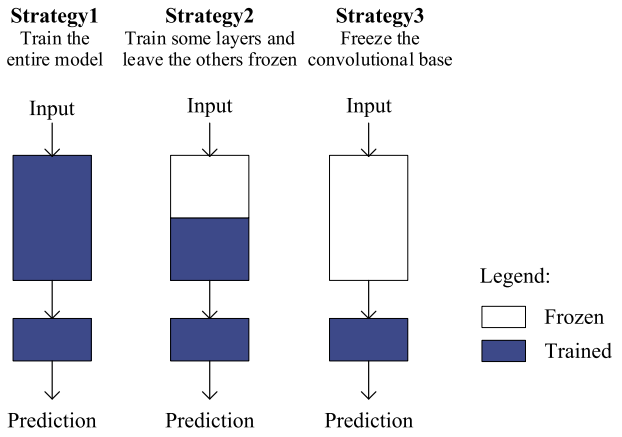


Fig. 7. Fine-tuning strategies [67].

4.3. CNN architectures for medical image classification

4.3.1. Variants of CNN for medical image classification

A lot of optimization work has been done based on classic CNN architectures, resulting in many different CNN variants. These improvements include optimizing the convolution kernel size and channel number, using depth-separable convolution and point-wise convolution, and introducing other modules, etc. Adoption of these methods has led to improved classification performance and efficiency.

A. Variants of ResNet

The ResNet architecture is the most popular research object, based on which the most variants have been produced. In Ref. [73], a Gliomas classification model based on ResNet with integration of pyramid dilated convolution was built in order to improve the classification accuracy. Initially, dilated convolution blocks with varying dilatation rates were amalgamated with traditional convolution blocks to create a pyramid dilated convolution module. Subsequently, the pyramid dilated convolution module was positioned at the base of the ResNet network to augment its receptive field. Experimental results illustrated that the proposed method achieved a classification accuracy of 80.11 %, surpassing the performance of classical deep learning methods. Although this model achieved high accuracy, it also faced two limitations: first, it only can handle 2D MRI scans; second, it relies on selecting images displaying clear tumor presence and necessitates expert assistance to manually label the training dataset, a process that is highly labor-intensive. To explore the task of image classification for colorectal cancer detection [74], different variants of ResNet, specifically ResNet-18 and ResNet-50 architectures, were improved and applied. By modifying the output layers of these architectures, colorectal cancer images were classified as benign or malignant. The study showed that ResNet-50 performed better than ResNet-18 in terms of accuracy, sensitivity, and specificity across various test sets. However, ResNet-50 had more layers, requiring more training time and computational resources. To explore lightweight deep learning models for COVID-19, a lightweight ResNet8 model was built by modifying the filter size and channel based on ResNet [75]. Compared with other existing models, the proposed model achieved an accuracy of 98.7 % in binary classification tasks and 98.3 % in multi-class classification tasks. This model performed well under different testing protocols, showing good reliability and robustness. Despite various improvements, the generated synthetic images may not completely cover all variations in real data, which could lead to overfitting in some cases. Due to the limited training data for medical images, simply applying models that perform well on natural images to the medical imaging field does not yield satisfactory results. Therefore [76], constructed a general backbone network that can enhance feature representation capabilities and simultaneously improve the performance of various medical computer vision tasks. In this research, a Group Attention Stacking module was proposed, resulting in a new ResNet variant, ResGANet. This new model achieved

Table 3
Detailed transfer strategies using for medical image classification tasks.

Studies/Author	Tasks	Pre-trained CNN models	Source datasets	Target datasets (Modality)	Size of target datasets	Fine-tuning strategy
Mahbod et al. [65]	Skin lesion classification	Combined EfficientNetB0, EfficientNetB1 and SeResNeXt-50	ImageNet	ISIC(Skin lesion images)	12927	Training from scratch (**Strategy1**)
Ogundokun et al. [57]	Breast cancer classification	MobileNet	ImageNet	BreakHis(Histology scans images)	2462	Training from scratch (**Strategy1**)
Kamal et al. [38]	COVID-19 detection	DenseNet121	ImageNet	Collected from Dr.JosephCohen [29] and Chext X-Ray dataset(X-ray)	760	Training part of convolutional layers (**Strategy2**)
Zebin et al. [42]	COVID-19 detection	EfficientNetB0	ImageNet	COVID-19 Image Data Collection (X-ray)	802	Training part of convolutional layers (**Strategy2**)
Chouhan et al. [43]	Pneumonia detection	Combined AlexNet, DenseNet121, Inception-V3, ResNet18 and GoogLeNet	ImageNet	Guangzhou Women and Children's Medical Center Pediatric Database (X-ray)	5232	Training part of convolutional layers (**Strategy2**)
Abhishek et al. [61]	Leukemia detection and classification	VGG16	ImageNet	Collected from clinical test (Microscopic blood smears)	1250	Training part of convolutional layers (**Strategy2**)
Vijayalakshmi et al. [62]	Malaria detection	VGG19	ImageNet	Collected from Tagore Medical College & Hospital, and MAMIC (Microscopic blood smears)	2600	Training part of convolutional layers (**Strategy2**)
da Nóbrega et al. [59]	Lung nodule malignancy classification	ResNet50	ImageNet	LIDC/IDRI(CT)	1018	Training classifier only (**Strategy3**)
Deepak et al. [54]	Brain tumor classification	GoogLeNet	ImageNet	Figshare(MRI)	3064	Training classifier only (**Strategy3**)
Swati et al. [53]	Brain tumor classification	VGG19	ImageNet	CE-MRI(MRI)	3064	Propose a **block-wise fine-tuning strategy**

state-of-the-art performance with fewer parameters and less computation, and it had good generalization ability. However, a drawback was that due to the introduction of multiple groups of attention mechanisms, the implementation and debugging of ResGANet may be more complex than that of traditional ResNet. To facilitate rapid and accurate screening for COVID-19, a multi-task prior attention residual learning framework has been proposed in Ref. [77]. It coupled two 3D-ResNets through a novel prior attention mechanism and designed a prior attention residual learning module that was stacked to form a deeper model. This framework significantly improved the performance of COVID-19 screening, achieving state-of-the-art results. The model framework was simply designed and could easily construct deep models by stacking blocks of prior attention mechanisms, but it had high computational costs and memory demands due to the use of 3D convolutional neural networks.

Improvements to ResNet include adding a pyramid dilated convolution module, lightweight design, and adding an attention mechanism. The pyramid dilated convolution module enables the network to extract features at different scales simultaneously, expands the model's receptive field. Thus the ability to capture local and global features is improved. Lightweight design is a means that most medical image processing models need to consider, such as using smaller convolution kernels and fewer channels to reduce models' parameters. Lightweight design models can run efficiently in resource-constrained environments. Two types of attention mechanisms are mentioned. One is the group attention mechanism which combines channel attention and spatial attention. By enhancing the feature representation capabilities of specific channels and spatial locations, the model improves the richness of feature expression while ensuring computational efficiency. The other is the prior attention mechanism. It utilizes prior knowledge to guide the model to focus on key lesion areas and reduce the interference of background or irrelevant areas. It is particularly suitable for the subtle distinction between COVID-19 and other lung diseases. The goals of these two attention mechanisms are different. The former aims to enhance the feature representation capabilities of the model, while the latter focuses on improving the discrimination ability of the model.

B. Variants of GoogLeNet

To improve the interdependence of remote features, a Long-Range unit was proposed and embedded in the Inception-V3 network in Ref. [78], enabling the network to capture local features and subsequently derive long-distance features through computation, effectively combining both local and long-range dependent features. The introduced method integrated correlations between long-range patches, enhancing the detection of Diabetic Retinopathy (DR). However, the incorporation of long-range dependencies added complexity and computational load to the model, which can make the training process more resource-intensive and time-consuming compared to traditional methods. Traditional manual feature extraction methods are often affected by individual cell differences and complex textures, while using CNNs alone may lose domain knowledge. Therefore [79], proposed a cell classification algorithm that combined Inception-V3 and manual features. It concatenated features extracted by Inception-V3 with manually extracted features, integrated them through a fully connected layer, and finally executed classification using a Softmax function. The experimental results showed that this method achieved an accuracy of 98.23 % in binary classification experiments and 94.68 % in seven-class classification experiments on the Herlev dataset. This research significantly improved the accuracy and robustness by leveraging the efficient architecture of Inception-V3 and feature concatenation methods, but it also increased the complexity of the algorithm, requiring more design and debugging work.

There are two directions of improvement in GoogLeNet: global feature extraction and manual feature utilization. The main goal of the Long-Range Unit is to enhance the model's ability to perceive global features. It can capture the global dependencies between different lesions in the image without destroying the original network structure. This structure is very suitable for identifying scattered and tiny lesion areas, such as diabetic retinopathy detection. And such a structure can be applied to most CNN models, not just GoogLeNet. The method of combining the features extracted by CNN with the manually extracted features is to solve the limitations of using deep learning or manual feature extraction methods alone. It has been proven that this method

performs better than using deep learning methods alone.

C. Variants of DenseNet

To tackle challenges like high resolution (gigapixel) of pathological images and limited annotation capabilities, DenseNet was optimized by introducing atrous dense connection and wider densely connected layers [80]. Convolutions with different dilation rates were adopted for multi-scale feature extraction. This optimization improved the classification accuracy of deep learning networks by removing error-marked patches in training sets, but it could be computationally expensive due to the integration of atrous convolutions and dense connections. To improve the classification accuracy of thoracic diseases [81], proposed a classification model named RE-DSENet. This model used DenseNet121 as the base network and classified 14 types of thoracic diseases using a multi-label classification approach. The model employed the Focal Loss function to address the issues of data imbalance and significant differences in texture features between different diseases, thereby enhancing the model's accuracy. Moreover, RE-DSENet reduced training time by 22.7 % compared to the base model, DenseNet121. One drawback of this method is that it does not consider the relationships between thoracic diseases, then future research is needed to explore how to integrate the correlations and dependencies between labels into the network learning to further enhance the algorithm's diagnostic capability.

The improvement of DenseNet can be divided into three aspects: multi-scale feature extraction, lightweight design and data imbalance solutions. The use of dilated convolution can improve DenseNet's ability to extract multi-scale features. This improvement expands the receptive field of convolution without increasing the number of parameters and captures features of different scales. The wider dense connection layer improves the model's feature representation ability by increasing the growth rate of DenseNet and reduces the number of network layers to reduce computational complexity. In addition, Focal Loss can handle the imbalance problem of different types of diseases in samples. It reduces the contribution of simple samples to the overall loss and increases the weight of difficult-to-classify samples by dynamically adjusting the loss weights of different samples.

D. Variants of UNet

A novel FractalCovNet architecture, derived from the UNet architecture, was proposed in Ref. [82], which could quickly and accurately diagnose COVID-19 cases from chest X-ray and CT scan images. The proposed approach comprised UNet architecture with fractal blocks and achieved an accuracy of 99 %. The study provided a low-cost and time-efficient method for diagnosing COVID-19, which is very beneficial for medical personnel dealing with a large number of cases. However, the performance of the model was constrained by the limited image data available for training the deep learning model. Faced with the problem of insufficient medical imaging data [83], enhanced the performance of machine learning models by integrating human visual attention and addressed the problem of overfitting in multitask learning models. The study employed a novel deep CNN architecture, known as the multitask UNet model, for saliency prediction and disease classification tasks in chest X-rays. The proposed model outperformed existing state-of-the-art models in both saliency prediction and disease classification tasks. Moreover, the model effectively mitigates the problem of overfitting, particularly in situations with limited datasets. However, the implementation and optimization of the model are complex and may require substantial computational resources.

The improvements to UNet mainly include the introduction of the Fractal module and the multi-task learning architecture. The Fractal module allows for the construction of a deeper neural network structure, which helps to extract more complex image features. The use of Fractal gives UNet a dual U-shaped structure, adding an extension path to the original UNet and allowing the model to extract richer contextual information from features at different levels. The multi-task learning architecture shares features between different tasks by sharing the encoder part of UNet, thereby improving the learning ability of each task. This architecture can effectively alleviate the overfitting problem and further improve the robustness and generalization ability of the model.

E. Variants of VGGNet

Traditional machine learning algorithms often perform poorly when processing complex pathological images. To improve diagnostic accuracy and efficiency [29], proposed a deep learning model called HIENet, which was based on the VGG16 architecture and incorporated two key attention modules: the position attention block and the channel attention block. These modules were designed to capture the relationships between different local regions in the input images and to learn the weights of the channels. The newly proposed model achieved an accuracy of 93.53 % in binary classification tasks and an average accuracy of 76.91 % in four-class tasks. By incorporating attention mechanisms, the HIENet model could effectively capture local features in images, enhancing diagnostic accuracy and sensitivity. However, due to the small size of the dataset, the model has a higher misclassification rate in some complex cases, particularly in distinguishing some cases of endometrial cancer and complex hyperplasia.

F. Variants of MobileNet

To improve the accuracy and efficiency of COVID-19 detection, study [84] selected the MobileNet-V2 model from multiple CNN structures for further improvements. The authors introduced new layers, such as bottleneck residual blocks and depthwise separable convolutions, to reduce the number of parameters and enhance the model's computational efficiency. The improved MobileNet-V2 model performed excellently in COVID-19 detection, achieving an accuracy of 98 % on the test dataset, and also showed excellent training time performance, with shorter compilation time. The challenge faced by this study is that the model requires a large amount of annotated data for training to ensure its accuracy and robustness. Since most hospitals and clinics do not have sufficient resources to deploy large deep learning models, it is very necessary to develop a lightweight and efficient deep learning model for the task of intracranial hemorrhage classification on CT scans.

G. Variants of FCN

In [85], the authors proposed a Lightweight-Fully Convolutional Network which was built using a combination of depth-wise separable convolutional layers and pointwise convolutional layers, as well as the inverted residual connections along with linear bottlenecks. While maintaining low computational demands, this model showed comparable performance in terms of accuracy and AUC. It also had smaller memory usage, fewer parameters, and shorter inference times, making it suitable for deployment in resource-limited clinical settings. Although the overall performance is good, it has lower sensitivity to certain types of bleeding, which may be due to the limited number of positive samples for these categories in the training data.

H. Variants of SqueezeNet

In [86], a lightweight convolutional neural network based on the SqueezeNet model was adopted to efficiently classify COVID-19 and other types of lung diseases under limited computational resources. The authors added batch normalization layers to improve training convergence and reduce overfitting, as well as transposed convolution layers and skip connections to integrate spatial and feature information. The improved SqueezeNet model performed excellently in terms of accuracy, sensitivity, specificity, precision, and F1 score. Moreover, the model's

classification time was shorter, significantly enhancing efficiency compared to more complex CNN designs. However, the limited number of existing COVID-19 datasets may affect the model's generalization ability.

I. Variants of SENet

Due to the uneven intensity distribution and blurred or missing boundaries of gastric tumors and lymph nodes in CT scans, and the highly variable shape of gastric tumors, developing automated solutions is very challenging. To address these challenges [87], proposed a 3D multi-attention-guided multitask learning network based on SENet, incorporating Visual Attention and Adaptive Spatial Attention modules to extract multi-scale features. For gastric tumor segmentation and lymph node classification tasks, the proposed method outperformed existing state-of-the-art algorithms on a CT image dataset, and also performed well on the MICCAI 2017 liver tumor segmentation challenge dataset. The complementary information extracted from different scales and dimensions effectively enhanced the segmentation and classification performance, and the bidirectional visual attention and adaptive spatial attention modules effectively addressed the issues of blurred boundaries and variable shapes. A challenge faced by this research is the high computational complexity of the 3D network, which demands substantial computational resources.

J. Variants of CapsNet

Limited medical image data is one of the biggest obstacles to using deep learning models. Therefore, study [88] was dedicated to researching a model that does not require preprocessing and can successfully classify within small datasets. The authors selected the CapsNet model and made several improvements, including adding convolutional layers and using the adaptive momentum estimation optimizer. This model achieved a high accuracy of 96.86 % on test data, outperforming other deep learning models such as CNNs and transfer learning methods. Even with limited data, the improved model can achieve high classification accuracy, and the CapsNet does not require extensive preprocessing. However, the architecture of CapsNet is complex, and its training is more challenging compared to simpler CNN models.

K. Other Variants

To enable rapid detection of COVID-19, a model based on Deep Convolutional Neural Network (DCNN) was proposed in Ref. [89], combined with a Multi-objective Adaptive Differential Evolution algorithm to optimize the hyperparameters of the CNN. The proposed MADE-CNN model outperformed existing machine learning models across various performance metrics, achieving an accuracy of up to 94.48 %. Sensitivity and specificity were also improved, reaching 93.83 % and 94.53 %, respectively, especially when dealing with imbalanced data. However, the performance of the model was somewhat dependent on the quality of the X-ray images, with low-quality images potentially affecting the classification results. The [90] aimed to leverage advanced deep learning models to enhance the accuracy and efficiency of breast cancer detection and classification. It proposed an Inception Recurrent Residual Convolutional Neural Network model. This model integrated Inception, Residual, and Recurrent convolutional layers, allowing it to capture more complex patterns in the data. The results showed that the model achieved a binary classification test accuracy of 99.05 % and a multi-class classification test accuracy of 98.59 % on the 2015 Breast Cancer Classification Challenge dataset. The model demonstrated superior classification performance compared to existing methods while using fewer network parameters. However, the integration of multiple network architectures increased the model's complexity.

Apart from improving the CNN structure itself, some studies have also proposed a general application framework that can be applied to any CNN model. For example, a deep CNN framework called DeTraC, comprising Decompose, Transfer and Compose, was validated as effective for AlexNet, VGG19, ResNet, GoogLeNet, and SqueezeNet [91]. The DeTraC framework could be summarized as simplifying the local structure of the data distribution with class-decomposition layer of DeTraC, training with sophisticated gradient descent optimization method, and refining the final classification with the class composition layer. Experimental results showed that DeTraC with VGG19 was the best combination, achieving an accuracy of 97.35 %. The DeTraC framework demonstrated superior performance, addressing significant challenges such as data irregularities and limited annotated datasets. However, its complexity and computational requirements must be considered for practical deployment.

All the CNN variants involved in this study are shown in Table 4.

4.3.2. Novel structures CNN for medical image classification

When building new CNN architectures, some techniques are often used and integrated with CNN. Such as residual connection, attention mechanism, and various optimization algorithms, etc.

A. Adopt Residual Connection

Residual technique is the most commonly used method. Classic CNNs generate classification results through hierarchical abstraction of original images, but these abstract features are relatively insensitive to the position and orientation of objects, and the lack of this spatial information limits further improvements in image classification accuracy. Therefore [69], introduced a deep learning framework based on Deep Residual Networks and Dilated Convolution for diagnosing and detecting childhood pneumonia. This method had overcame the issues of overfitting and degradation in deep models with its residual structure, and addressed the loss of feature space information due to increased model depth using dilated convolution. It can accurately identify image regions that most indicative of pneumonia, achieving a recall rate of 96.7 % and an F1 score of 92.7 % on the test dataset, significantly outperforming traditional CNN models. Moreover, this method effectively resolved issues related to low image resolution and partial occlusion of inflammatory areas. However, due to the depth of the model, its prediction efficiency is low in practical applications, necessitating further optimization of the network structure to enhance speed. In order to improve the accuracy and sensitivity of COVID-19 diagnosis, a modular CNNCF was designed in Ref. [92]. The CNNCF had two stages, each stage incorporated multiple residual blocks to preserve features across different layers, preventing model degradation. This novel model demonstrated outstanding performance in automatically identifying COVID-19 cases in both X-ray data and CT data. However, the performance of the model depended on high-quality labeled data, and the generalization ability between different data sets needs to be further verified. For breast cancer histopathology image classification, a self-training learning method with residual learning, named Histopathology-RDNet, was proposed in Ref. [93]. This approach was developed to address the challenge of requiring a large number of labeled images to train deep learning models effectively. The new CNN architecture had 64 layers divided into stacked-layers and shortcut connections, which obtained a higher accuracy of 96.3 % than much more complex models. The model contained many layers and different convolution filter sizes, resulting in a large amount of computation and longer training and inference time. In Ref. [94], CDC Net that combined residual network thoughts and dilated convolution was proposed for COVID-19 and other chest tract disorders classification, which showed significant performance as compared to VGG19, ResNet50, and Inception-V3. The proposed model was capable of detecting five different chest diseases, reducing the burden on doctors to use multiple diagnostic methods. It was trained and tested on multiple publicly available benchmark datasets, enhancing its generalization ability. However, variations between different datasets may lead to

Table 4
A summary of variants based on existing CNN models.

Backbones	Studies/ Author	Tasks	Improved architecture	Accuracy (Only report the best)	Questions solved	Advantages	Disadvantages
ResNet	Lu et al. [73]	Gliomas classification	ResNet with Pyramid dilated convolution resnet model	80.11 %	To improve classification accuracy.	Increase the receptive field of the original network and obtain higher accuracy.	It only can handle 2D MRI scans, and relies on selecting images displaying clear tumor presence and necessitates expert assistance to manually label the training dataset, a process that is highly labor-intensive.
	Sarwinda et al. [74]	Colorectal cancer detection	ResNet18/50 with modified output layer	ResNet50: 80 %	To explore the task of image classification for colorectal cancer detection.	Better accuracy, sensitivity, and specificity.	It requires more training time and computational resources.
	Karakanis et al. [75]	COVID-19 detection	ResNet8 with changed filter size and channel	98.7 % (Binary classification) 98.3 % (Multi-class classification)	To explore lightweight deep learning models for COVID-19.	Better performance, reliability and robustness.	There is a risk of overfitting.
	Cheng et al. [76]	Several diseases classification	ResGANet: ResNet with group attention block	81.66 % (Dataset1) 80 % (Dataset2)	To deal with limited training data.	Well performance with fewer parameters and less computation, and good generalization ability.	The implementation and debugging of ResGANet may be more complex due to the introduction of multiple groups of attention mechanisms.
	Wang et al. [77]	COVID-19 classification	3D-ResNet with attention	93.30 %	To facilitate rapid and accurate screening for COVID-19.	Well performance, and simple model framework.	It needs high computational costs and memory demands.
GoogLeNet	Luo et al. [78]	Diabetic retinopathy classification	Inception-V3 embedded with the proposed Long-Range unit	93.50 %	To improve the interdependence of remote features in retinal fundus photographs.	Combine both local and long-range dependent features, which enhances the detection of DR.	It's more resource-intensive and time-consuming to train the model due to the introduction of long-range unit.
	Dong et al. [79]	Cervical cell classification	Inception-V3 with artificial features	98.23 %	To combine the advantages of CNN and manual feature extraction.	Better accuracy and robustness.	It increases the complexity of the algorithm.
DenseNet	Li et al. [80]	Pathological image classification	DenseNet with atrous convolution and wider densely connected layers	94.10 % (Dataset1)92.05 % (Dataset2) 97.63 % (Dataset3)	To address the challenges such as the high resolution (gigapixel) of pathological images and the lack of annotation capabilities.	Improve the accuracy of deep learning networks by removing error-marked patches in training sets.	It can be computationally expensive due to the integration of atrous convolutions and dense connections.
	Zhao et al. [81]	Thorax diseases classification	RE-DSENet: DenseNet121 with multi-atlas segmentation module and classification module	85.7 %	To improve the classification accuracy of thoracic diseases.	Higher accuracy, less training time.	It does not consider the relationships between thoracic diseases.
UNet	Munusamy et al. [82]	COVID-19 detection	FractalCovNet: UNet with fractal blocks	99 % (Dataset1) 99 % (Dataset2)	To quickly and accurately diagnose COVID-19 cases.	It is a low-cost and time-efficient method.	The performance is constrained by the limited image data available for training.
	Zhu et al. [83]	Disease classification on chest X-ray	MT-UNet: UNet with classification heads to the bottleneck and the ending part of the architecture	67 %	To deal with the problem of insufficient medical imaging data.	It obtains better performance,and effectively mitigates the problem of overfitting.	It is complex and may requires substantial computational resources.
VGGNet	Sun et al. [29]	Endometrium disease classification	HIENet: VGG16 with position attention and channel attention	93.53 ± 0.81 % (Binary classification) 76.91 ± 1.17 % (Multi-class classification)	To improve diagnostic accuracy and efficiency.	Higher diagnostic accuracy and sensitivity.	Higher misclassification rate in some complex cases.
MobileNet	Akter et al. [84]	COVID-19 detection	MobileNet-V2 with depth-separable convolution filters instead of the usual convolution filters	98 %	To improve the accuracy and efficiency of COVID-19 detection.	Better performance, and shorter compilation time.	It requires a large amount of annotated data for training.

(*continued on next page*)

Table 4 (*continued*)

Backbones	Studies/ Author	Tasks	Improved architecture	Accuracy (Only report the best)	Questions solved	Advantages	Disadvantages
FCN	Lankireddy et al. [85]	Intracranial hemorrhages classification	Light-FCN: FCN with a combination of depth-wise separable convolutional layers and pointwise convolutional layers, as well as the inverted residual connections along with linear bottlenecks	92.90 %	To deal with insufficient resources in hospitals and clinics.	Low computational demands, comparable performance, smaller memory usage, fewer parameters, and shorter inference times.	It has lower sensitivity to certain types of bleeding.
SqueezeNet	Polsinelli et al. [86]	COVID-19 detection	SqueezeNet with a Transpose Convolutional Layer to the last Custom Fire Module	85.03 %	To efficiently classify COVID-19 under limited computational resources.	Good performance, and shorter classification time.	The model's generalization ability may be limited by insufficient existing COVID-19 datasets.
SENet	Zhang et al. [87]	Lymph node classification	SENet-50 with multi-attention	80.5 ± 5.3 %	To deal with the uneven intensity distribution and blurred or missing boundaries of gastric tumors and lymph nodes in CT scans, and the highly variable shape of gastric tumors.	Better performance, and it can address the issues of blurred boundaries and variable shapes.	High computational complexity of the 3D network.
CapsNet	Baydilli et al. [88]	White blood cells classification	WBCaps: Capsule Network with a convolutional layer in front of the model	96.86 %	To deal with limited medical image data.	Higher accuracy , and it does not require extensive preprocessing.	Its training is more challenging due to the complex architecture.
CNN	Singh et al. [89]	COVID-19 detection	MADE-based CNN: CNN with Multi-objective Adaptive Differential Evolution to tune hyper-parameters	94.48 %	To enable rapid detection of COVID-19.	Better performance especially with imbalanced data.	The performance of model depends on the quality of X-ray images.
GoogLeNet, ResNet, RCNN	Alom et al. [90]	Breast cancer classification	IRRCNN: An improved hybrid DCNN architecture based on inception, residual networks, and RCNN architecture	99.05 % (Binary classification) 98.59 % (Multi-class classification)	To enhance the accuracy and efficiency.	Better performance, and fewer parameters.	The integration of multiple network architectures increased the model's complexity.
AlexNet, VGG, ResNet, GoogLeNet, SqueezeNet	Abbas et al. [91]	COVID-19 detection	AlexNet, VGG19, ResNet, GoogLeNet, and SqueezeNet with DeTraC model that includes Decompose, Transfer, and Compose stages	VGG19: 93.1 %	To proposed a general application framework that can be applied to any CNN model.	Superior performance, and it can address issues like data irregularities and limited annotated datasets.	It has complexity and high computational requirements.

inconsistencies in model performance.

B. Adopt Attention Mechanism

The attention mechanism is also a commonly used technique. Most deep learning models designed for medical image classification only perform well on specific diseases, and their generalization is poor. Considering this challenge, an evolutionary attention-based network named EDCA-Net was designed with an effective and robust structure in Ref. [95]. First, the feature maps in the densely connected attentional network were automatically channel-wise weighted. Then two types of evolution, intra- and inter-evolution, were introduced to enhance the model's performance and generalizability. The intra-evolution phase optimized the weights of the model, while the inter-evolution phase enabled two instances to exchange training experiences throughout the training process. EDCA-Net outperformed existing state-of-the-art methods on three datasets, and the combination of internal and external evolution significantly enhanced the model's learning ability and generalization ability. However, the introduced evolutionary mechanisms and dense connection patterns may increase the complexity of the model and the demand for computational resources. Existing medical image analysis mostly focuses on image-level classification, with less attention paid to pixel-level fine-grained lesion area classification. For membranous nephropathy classification, a dimension-driven multi-path attention residual network named DDMARN [96] was introduced based on the characteristics of scattered distribution of pathological areas. DDMARN introduced channel attention mechanism which helps to effectively obtain the multi-scale features, resulting in better expression of the depth features. The proposed CNN model was proven to be very competitive in the classification tasks. The proposed model had only 700K parameters, significantly reducing computational complexity. Although pixel-level classification reduced classification complexity, it also leaded to the loss of some spatial context information, which may affect classification performance in certain cases. A new CNN architecture with a combination of squeeze and excitation and bottleneck layers for diabetic retinopathy classification was proposed in Ref. [97]. Two sub-networks consisting of excitation and bottleneck were created. If the two sub-networks classify the same label as output, the image will be classified with that label. The model achieved an accuracy of 98.7 %. One disadvantage of this method was that the architecture is complex, which may require high computational resources. To address the issue of low accuracy in traditional methods for detecting diabetic macular edema [98], proposed a new convolutional neural network algorithm that incorporated lesion attention modules. The new model architecture included multiple convolutional layers, pooling layers, and fully connected layers, utilizing

the lesion attention module to highlight the lesion areas, thereby improving classification accuracy. Experimental results showed that this model outperformed traditional deep learning models in diabetic macular edema detection. However, the performance of this model highly depended on the quality and diversity of the training data, which may result in varying performance across different datasets. To explore the integration of the attention mechanism into CNNs for glaucoma detection in fundus images [99], introduced an attention-based CNN for glaucoma detection, named AG-CNN. The model consisted of three subnets: the attention prediction subnet, the pathological area localization subnet, and the classification subnet. The attention prediction subnet collected ophthalmologists' attention maps through simulated eye-tracking experiments, generating attention maps for glaucoma detection. The pathological area localization subnet used these attention maps to locate pathological areas in fundus images. The classification subnet performed binary classification of glaucoma based on the localized pathological areas. Experimental results showed that the AG-CNN model achieved an accuracy of 95.3 %, sensitivity of 95.4 %, and specificity of 95.2 % in glaucoma detection, significantly outperforming existing deep learning methods. The innovation of this model lied in combining doctors' attention maps with pathological area localization, effectively enhancing detection performance, but it also introduced computational complexity and data dependency issues.

C. Adopt Optimization Algorithms

Some optimization algorithms are also used in CNN construction. A novel CNN model with Federated Learning was proposed for classifying skin disease in Ref. [100]. The Federated Learning technique was explored to address data privacy issues. With this technique, the accuracy of the model reached 94.15 %, which was higher than that of VGG16 and AlexNet. This method employed a federated learning framework, ensuring that data does not leave local devices, which not only protected user privacy but also enhanced the model's performance in detecting multiple types of skin diseases. Its drawback was that the federated learning framework increased the complexity of model training and inference. An improved model incorporating deep CNN and particle swarm optimization (PSO) techniques was designed [101] to optimize the problem of medical data classification from CT images. The PSO algorithm was used to obtained better image segmentation, and deep CNN was used to extract features and classify images. This approach, combining PSO and CNN, had achieved ebxcellent performance in medical image classification tasks. Especially, PSO showed the advantages of robust parameter control, computational efficiency, and easy implementation in image classification tasks. However, the framework combining PSO and CNN increased the complexity of model training and inference, potentially requiring higher computational resources.

D. Use the Neural Architecture Search

Neural Architecture Search (NAS) proves to be a valuable tool for optimizing the network design process for image classification and hyperparameters [102]. It is predominantly employed to ascertain microarchitecture and optimize floating point operations or inference efficiency [103]. Aiming at five clinical classification problems, Eminaga et al. [104] used the NAS method to design a simple yet effective model named PlexusNet to address the issues of complexity and scale in state-of-the-art convolutional neural network models. This model achieved a performance that was comparable to ResNet18, EfficientNetB0 and EfficientNetB1. It is clear that NAS can achieve a favorable trade-off between model complexity and performance in medical image classification tasks. But the performance of PlexusNet may be slightly lower than that of the most advanced SOTA models on some datasets.

E. Design Lightweight and Simple CNN Structures

Lightweight and simple CNN structures may show more satisfactory results than complex structures in medical image classification. Fine-tuning existing deep learning networks for medical image classification often results in overfitting and low transfer efficiency. In order to overcome such limitations, a 6-layers network, CXNet-m1, that was much shorter, thinner, but more powerful than pre-trained models, was proposed in Ref. [105]. It employed a novel loss function called sin-loss, which is capable of extracting discriminative information from misclassified and indistinguishable images. Additionally, the convolutional kernels of CXNet-m1 were optimized to enhance classification accuracy. Experimental results demonstrated that the lightweight CXNet-m1 model with sin-loss function achieved superior accuracy, recall rate, F1-score, and AUC value. This illustrated that designing a proper CNN can be more effective than fine-tuning existing deep networks, but the model's performance still heavily depends on the scale and quality of the training data. In order to overcome the problems of time-consuming and low accuracy of traditional methods, Nahiduzzaman et al. [106] dedicated to build lightweight convolutional neural networks for COVID-19 classification. They obtained a new model named ChestX-ray6 with 26 layers. This model had an accuracy of 80 % in detecting six diseases and an accuracy of 97.94 % in binary classification. Compared to other transfer learning models such as VGG19, ResNet50, DenseNet121 and MobileNetV2, the ChestX-ray6 model performs well in classification criteria, number of parameters, and processing time. However, it also had a high dependency on the dataset. Aiming to improve the accuracy and speed of detection and classification, Alyafeai et al. [107] proposed a lightweight CNN model with 28 layers for cervical cancer classification. The proposed method utilized a YOLO algorithm designed with an improved GoogLeNet model for real-time cervix region detection, followed by the classification of cervical tumors using features automatically extracted by a lightweight convolutional neural network model. This classification model achieved an AUC score of 82 % and operated 20 times faster than existing models. Its lightweight architecture made it suitable for deployment on mobile devices, enhancing its applicability in resource-limited environments. However, there was still room for improvement in classification accuracy.

In [108], an 8-layer CNN model, COVINet, was introduced for predicting COVID-19 from chest X-ray images. The model aimed to address the issue of small X-ray image datasets. Experiment results showed that the model achieved an accuracy of 97 % in binary classification, 90 % in three-class classification, and 85 % in four-class classification. Compared to VGG16 and AlexNet, COVINet performed slightly worse in binary and four-class classification but performed equally well in three-class classification. The model effectively addressed the issue of small dataset by incorporating data augmentation methods. The only drawback was the lower AUC value in four-class classification, indicating limited performance in distinguishing complex categories. The available MRI image databases are relatively small in scale, which limits the performance of machine learning models in classifying brain tumors [109]. was dedicated to developing a simpler CNN architecture that can perform comparably to more complex models while requiring fewer resources, making it suitable for deployment in clinical diagnostics and mobile platforms. In this research, a new CNN architecture with 22 layers was presented. It consisted of input, two main blocks, classification block, and output. The model could effectively classify brain tumors from MRI images, achieving a maximum accuracy of 97.28 %. It not only had a lightweight architecture and high accuracy but also had advantages in terms of speed. However, the model still faced challenges in generalization capability and complex classification scenarios. A multi-layer neural network EyeDeep-Net was developed for diagnosis of various eye problems from retinal fundus images in Ref. [110]. The main purpose of the design was to optimize model performance while making it suitable for deployment in resource-limited environments. The EyeDeep-Net model consisted of 14 convolutional layers, 2 fully connected layers, and 1 classification layer. It employed various techniques such as batch normalization, max pooling, and dropout to enhance

performance. This method achieved a testing accuracy of 76.04 %. It featured a lightweight and efficient architecture, making it suitable for deployment on mobile platforms. Additionally, the use of data augmentation and batch normalization enhanced the model's robustness and generalization capability on unseen data. Despite its strong performance, the model still faced challenges in distinguishing very similar classes of retinal diseases.

A summary of novel CNN structures is shown in Table 5, the main ideas and number of layers are described.

4.3.3. Ensemble CNNs for medical image classification

The combination and diversity of multiple models contribute to the accuracy of predictions, and the ensemble size and model selection have a crucial impact on the performance [111]. However, when choosing an ensemble approach, efficiency and memory become important factors to consider. In order to achieve the best trade-off between performance and efficiency and memory, it is essential to choose good models and appropriate ensemble strategy. The former represents which models can be combined, and the latter represents how to combine them [112].

A. Ensemble of Two or More CNNs

For alleviating the dependency of deep learning model on the data, the MobileNet-V2 and ResNet18 were combined based on the suggested weight factor in acute lymphoblastic leukemia classification [113], with a better performance than recent transfer learning-based techniques. The framework combined the computational efficiency of MobileNetV2 with the high accuracy of ResNet18. It not only demonstrated exceptionally high classification accuracy across multiple datasets but was also faster than methods based on transfer learning, as the second classifier was executed during the testing phase only when necessary. However, this framework required extensive hyperparameter tuning and optimization, making the training process complex and time-consuming. Aiming at medical images containing small important objects, a novel architecture combined ResNet50-V2 and modified FPN was presented [114]. This combination enabled the model to explore various resolutions of the image without losing details of small objects. When compared to Xception and ResNet50-V2, the proposed model achieved a higher accuracy of 98.49 %. By combining the advantages of FPN and ResNet50-V2, the model was capable of handling images at different scales, demonstrating strong robustness. Despite this, the model's generalization capability on different datasets still needs further validation. In Ref. [115], an automated system that can efficiently locate and classify eye disease was proposed, in which the DenseNet100 was used as a feature extractor and the CenterNet was used as a classifier. Experimental results showed that the model achieved an average accuracy of 98.1 % on the IDRiD dataset. This combined approach utilized DenseNet-100 to process low-intensity and noisy data images, effectively extracting features. By employing CenterNet as a one-stage detector, it ensured efficient computation. However, the complex computation involved is a major challenge for this combined approach.

An ensemble framework with exceptional capacity to extract deep features employing a backbone feature concatenation approach was designed for COVID-19 identification [116]. Pre-trained models included DenseNet, VGG16, and Inception-V3 were utilized to extract features from chest X-ray images. Then, the features from these models were concatenated at the final convolutional layer. The proposed method achieved an overall accuracy of 96.33 % in multi-class scenarios and an accuracy of 98.67 % in binary classification scenarios. The multi-head self-attention network and deep feature cascading method improved the computational efficiency and performance, showing significant advantages compared to the best performance of single models. However, in practical applications, it was still necessary to overcome challenges related to dataset size and computational resources. In Ref. [117], the DenseNet201, Resnet50-V2 and Inception-V3 were combined to detect COVID-19 from chest X-ray images. A modified weighted average ensembling method was employed, whereby greater weight was given to models demonstrating superior performance and lower validation errors. The ensemble models obtained a classification accuracy of 91.62 %. A major advantage was that ensembling effectively combines the strengths of different models, providing a robust framework for variations in medical image data. However, this ensemble method may incur higher computational costs and require more advanced hardware setups due to the complexity of training multiple models. In order to deal with the problems such as small scale of medical image dataset, difficulty in labeling, and data imbalance, a multi-scale multi-CNN ensemble consisting of EfficientNetB0, EfficientNetB1 and SeReNeXt5 was designed and evaluated for skin lesion analysis in Ref. [65]. This multi-scale and multi-network integration transfer learning approach enhanced the ability to capture both detailed and global features of skin lesions. It achieved the best average accuracy of 96.4 %. This method elevated model performance, mitigated issues stemming from data imbalance, particularly in categories with small sample sizes. At the same time, this method increased the complexity and computational overhead of the model, which might lead to a slower inference process, especially in resource-constrained environments. To overcome the challenges of training data scarcity, improve model's performance on tuberculosis X-ray image classification tasks [44], pre-trained ResNet50, VGG19, and Inception-V3 were used in an ensemble algorithm, with an overall accuracy of 85 %. This approach greatly reduced training time and the need for computing resources. It's disadvantages were that the model still needs to be tuned for the specific task to achieve optimal performance. Aiming at the problem of screening efficiency in diabetic retinopathy tasks, an identification and grading framework were explored using different ensembles of CNN models [112]. Inception-V3, Xception, Inception-ResNet-V2 as well as their different combinations were designed to construct the identification system. ResNet60, DenseNet169, DenseNet201, as well as their different combinations were designed to construct the grading system. The relationship between the number of CNN models in ensemble and the size of dataset was also explored. Results showed that the identification model performed best with an accuracy f 97.67 %, a sensitivity of 97.5 % and a specificity of 97.7 %. Meanwhile, the grading model achieved an accuracy f 96.5 %, a sensitivity of 98.1 % and a specificity of 98.9 %. This method delivered reproducible and consistent detection results with high sensitivity and specificity. However, its performance relied on the labeling experience of ophthalmology experts. Traditional methods for diagnosing COVID-19 have limitations such as high false negative rate. Therefore, a novel framework of cascaded deep learning classifiers was proposed to enhance the performance of these methods was proposed in Ref. [118]. Eleven pre-trained CNN models were compared firstly, then selected the three best performing models for integration to classify COVID-19 and pneumonia diseases. Experiments showed that the cascade of VGG16, ResNet50-V2 and DenseNet169 obtained the highest accuracy of 99.9 %. The cascaded architecture allowed for the simultaneous use of different deep learning models, optimizing the performance for each disease classification. While the cascaded framework was flexible, it required careful selection and tuning of models, which can be challenging without sufficient expertise.

Some studies have used four or more CNN models to build ensemble models for medical image classification. Aiming at the problem of how to develop a high-performance and interpretable deep learning model on small datasets to detect acute intracranial hemorrhage and classify its five subtypes, an ensemble model was introduced in Ref. [119]. VGG16, ResNet50, Inception-V3, and Inception-ResNet-V2 were combined to construct a ensemble model. Four ensemble models were used to predict the final classification probabilities through unweighted averaging method. This approach performed well on a relatively small training dataset, demonstrating high sensitivity and specificity. Additionally, the use of attention maps enhanced the model's transparency, making its predictions more understandable to clinicians. However, the data in this study were all sourced from a single institution, which may raise

Table 5
A summary of novel CNN architectures.

Studies/Author	Tasks	CNN architecture	Classification performance	Questions solved	Advantages	Disadvantages
Liang et al. [69]	Pediatric pneumonia classification	CNN with **residual** structure and dilated convolution (51 layers)	F1 score: 92.7 %Recall: 96.7 %	To address the lack of spatial information in extracted features.	Better performance, and it effectively resolve issues related to low image resolution and partial occlusion of inflammatory areas.	Low computational efficiency.
Liang et al. [92]	COVID-19 Detection	CNNCF with two stages, each stage consisted of multiple **residual** blocks	F1 score: 96.72 % Specificity: 99.33 %	To improve the accuracy and sensitivity.	Better performance, and it can deal with multi-modal data.	The performance of the model depends on high-quality labeled data.
Eltoukhy et al. [93]	Breast cancer detection	Histopathology-RDNet with **residual** learning technique (64 layers)	Accuracy: 96.3 %	To address the challenge of requiring a large number of labeled images to train deep learning models.	Higher accuracy.	It needs a large amount of computation and longer training and inference time.
Malik et al. [94]	COVID-19 detection	CDC_Net with **residual** structure and dilated convolution (17 layers)	Accuracy: 99.39 %	To classify COVID-19 and other chest tract disorders.	Good performance and generalization ability.	Variations between different datasets may lead to inconsistencies in model performance.
Zhu et al. [95]	Four diseases classification	EDCA-Net with evolutionary **attention-based** network	Accuracy: 99.44 % (Dataset1)Accuracy: 97.45 % (Dataset2) Accuracy: 99.94 % (Dataset3)Accuracy: 99.94 % (Dataset4)	To deal with poor generalization.	Better performance, good learning ability and generalization ability.	More complex and computational resources demand.
Zhang et al. [96]	Membranous nephropathy classification	DDMARN with a dimension-driven multi-path **attention residual structure** (36 layers)	Accuracy: 96.23 ± 0.17 %	In order to pay more attention to pixel-level fine-grained lesion area.	Good performance, less parameters, and low computational complexity.	It leads to the loss of some spatial context information.
Das et al. [97]	Diabetic retinopathy classification	CNN with **squeeze and excitation** and bottleneck layers	Accuracy: 98.7 %	To efficiently classify diabetic retinopathy.	Higher accuracy.	The architecture is complex, which may require high computational resources.
Saini et al. [98]	Diabetic macular edema classification	LCNN with **lesion-attention** module (25 layers)	Accuracy: 99.54 ± 1.23 %	To improve the accuracy.	Better performance.	The performance highly depends on the quality and diversity of the training data.
Li et al. [99]	Glaucoma detection	AG-CNN with an **attention** prediction subnet, a pathological area localization subnet and a glaucoma classification subnet	Accuracy: 95.3 %	To explore the integration of the attention mechanism into CNNs for glaucoma detection in fundus images.	Better performance.	It introduces computational complexity and data dependency issues.
Hossen et al. [100]	Skin diseases classification	CNN with **Federated Learning** (7 layers)	Accuracy: 94.11 %	To address data privacy issues.	Better performance, and it protects user privacy.	Higher complexity of training and inference.
Sreelakshmi [101]et al.	COVID-19 classification	CNN with **PSO technique**	Accuracy: 98.91 %	To optimize the problem of medical data classification from CT images.	Better performance.	Higher complexity of training and inference.
Eminaga et al. [104]	Colorectal cancer, endoscopic images, pneumonia, diabetes retinopathy, dermatologic images classification	PlexusNet built by **NAS**	Accuracy: 90.7 % (Dataset1)Accuracy: 95.3 % (Dataset2)	To address the issues of complexity and scale in state-of-the-art CNN models.	Better performance, and it achieves a favorable trade-off between model complexity and performance.	It may not perform very well on some datasets.
Xu et al. [105]	Anomaly detection on Chest X-Rays	**Lightweight** CXNet-m1 with hierarchical architecture and a novel loss function (6 layers)	Accuracy: 93.6 %	To overcome the problems of overfitting and low transfer efficiency.	More lightweight architecture, and better performance.	The model's performance heavily depends on the scale and quality of the training data.
Nahiduzzaman et al. [106]	Lung diseases classification	ChestX-Ray6 with **lightweight** architecture (8 layers)	Accuracy: 97.94 % (Binary classification) Accuracy: 80 % (Multi-class classification)	To overcome the problems of time-consuming and low accuracy.	Better performance, less parameters and processing time.	It also had a high dependency on the dataset.
Alyafeai et al. [107]	Cervical cancer classification	CNN with **lightweight** architecture (26 layers)	AUC: 82 %	To improve the accuracy and speed of detection and classification.	Good AUC score, faster running speed.	Relatively low accuracy.
Umer et al. [108]	COVID-19 classification	COVINet with **simple** architecture (8 layers)	Accuracy: 97 % (Binary classification)	To address the issue of small datasets.	Good performance, and it effectively addresses	It performs limited performance in

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Table 5 (*continued*)

Studies/Author	Tasks	CNN architecture	Classification performance	Questions solved	Advantages	Disadvantages
			Accuracy: 90 % (Three classes classification) Accuracy: 85 % (Four classes classification)		the issue of small dataset.	distinguishing complex categories.
Badza et al. [109]	Brain tumor classification	CNN with **simple** architecture (22 layers)	Accuracy: 97.28 %	To address the issue of small datasets.	Lightweight architecture, high accuracy and speed.	It faces challenges in generalization capability and complex classification scenarios.
Sengar et al. [110]	Retinal diseases diagnosis	EyeDeep-Net with **simple** architecture (18 layers)	Accuracy: 76.04 %	To optimize performance while making it suitable for deployment in resource-limited environments.	Lightweight and efficient architecture, good robustness and generalization capability on unseen data.	It faces challenges in distinguishing very similar classes of retinal diseases.

concerns about the system's generalizability. For diabetic retinopathy detection and classification, existing computer vision-based methods cannot accurately encode complex features, resulting in particularly low classification accuracy at early stages. Therefore, a deep learning ensemble approach to improve the classification performance for different stages of diabetic retinopathy was proposed in Ref. [120]. Five CNN models: ResNet50, Inception-V3, Xception, DenseNet121, and DenseNet169, were stacked to generate an ensemble-based framework, to combine information from multiple models. When compared to the state-of-the-art methods on the Kaggle dataset, the ensemble framework performed better. However, the disadvantage of this ensemble method was that the integration of multiple models and hyperparameter tuning increased the model's complexity, requiring higher computational resources and time. A total of 15 pre-trained CNN architectures including EfficientNets (B0-B5), Inception-V3, ResNet50, SeResNet50, ResNext50, DenseNet121, Xception, Inception-ResNet-V2, NasNetLarge, and NasNetMobile, were evaluated in Ref. [121]. Subsequently, an ensemble method was constructed based on majority voting using the best CNN combination to further enhance classification performance. Experimental results demonstrated that the majority voting of five architectures namely EfficientNetB0, EfficientNetB3, EfficientNetB5, Inception-ResNet-V2, and Xception achieved the highest accuracy of 85 %. The ensemble method provided more robust diagnostic accuracy by reducing the variance in the final predictions, but it also increased the model's complexity, requiring more computational resources and time for training and hyperparameter tuning. In Ref. [43], an ensemble model of AlexNet, DenseNet121, Inception-V3, ResNet18 and GoogLeNet was proposed for pneumonia detection. The predictions generated by five neural networks were aggregated into a prediction vector, and a majority voting scheme was applied to determine the final prediction. The results showed that the accuracy of combined model was 96.39 % which was better than that of the single pre-trained model. This study indicated that combined multiple pre-trained models enhanced the model's robustness and accuracy. But it could be computationally expensive and required significant computational resources for both training and inference. The CNN models VGGNet, DenseNet, ResNet, TinyNet, EfficientNet, MobileNet, Xception and Transformer were compared in Ref. [122]. According to the performance on an osteoarthritis classification task, two ensemble strategies were evaluated: weight and specialist. The weight ensemble strategy was to divide the sum of all models' result by the number of models to get the final predicted value. The specialist ensemble strategy was to assign greater weight to the probability distribution value of the model exhibiting the highest accuracy for each class. The proposed weight ensemble and specialist ensemble achieved an accuracy of 77.05 % and 72.76 %, respectively, which showed a considerable improvement. Although the ensemble model significantly improved the accuracy of degenerative arthritis classification, the increase in model complexity was an undeniable drawback.

The integration of multiple single CNN structures is also an aspect of ensemble strategy research. A synergic deep learning model integrated multiple deep convolutional neural networks was proposed [123], to solve the challenges brought by intra-class variation and inter-class similarity in medical image classification. The DCNN component took ResNet50 as the initial structure, and enabled them to learn from each other. The image representations learned by each pair of DCNNs were concatenated to form the input of synergic network, then a fully connected structure was used to predict whether the input image pair belongs to the same class. The proposed ensemble model achieved higher classification accuracy than individual DCNNs in several medical image classification tasks. This approach of simultaneous learning by multiple DCNN components not only enhanced the diversity of feature representations but also reduced the risk of overfitting associated with a single model. However, the complexity of the synergic deep learning model was also increased. In order to investigate how CNN characterizes cataract layer-by-layer, deconvolution operation was employed in a hybrid global-local feature representation model [124]. In this method, each fundus image was cut into eight local 950*950 pixel images, and multiple AlexNet models were used to extract local and global features from these images and original image. Finally, the predictions from CNN classifiers were combined by majority voting approach. The model achieved higher accuracy across multiple datasets. Additionally, the dual-path architecture made the model more robust against variations in image quality and content. Addressing the issue that the network initialization weights of existing deep learning models have not been well optimized, a multilevel CNN based on the weight initialization and sliding window fusion was proposed [125]. The proposed model employed a novel network weight initialization method, which enhances the nonlinear modeling capability of existing deep learning models. It also utilized a sliding window-based classifier fusion method, which sorts the classification probabilities and applies a sliding window to flexibly select and combine classification results, thereby improving the final classification accuracy. For multiple disease datasets, it achieved a higher average accuracy compared to traditional machine learning. Although this method increased the model's complexity and dependence on data, its excellent performance in various medical image classification tasks demonstrated its broad application prospects.

As shown in Table 6, ResNet is the most popular basic structure for building ensemble models. It was used in 12 of the total 17 ensemble models. Obviously, residual structure is very useful for medical image feature extraction as it can use skip connections to retain the characteristics of different layers in forward propagation. DenseNet has been used 8 times, which shows that dense connection is also very suitable for medical image classification tasks. In addition, Inception was used in 7

Table 6
A summary of ensemble models of multiple CNNs architectures.

Studies/ Author	Tasks	CNN architectures	Classification performance	Questions solved	Advantages	Disadvantages
Das et al. [113]	Acute lymphoblastic leukemia classification	MobileNet-V2, ResNet18	Accuracy: 99.39 % (Dataset1)Accuracy: 97.18 % (Dataset2)	For alleviating the dependency of deep learning model on the data.	Higher accuracy and more faster.	It requires extensive hyperparameter tuning and optimization, making the training process complex and time-consuming.
Rahimzadeh et al. [114]	COVID-19 detection	ResNet50-V2, modified FPN	Accuracy: 98.49 %	To deal with medical images containing small important objects.	Higher accuracy and strong robustness.	The model's generalization capability on different datasets still needs further validation.
Nazir et al. [115]	Diabetic eye disease detection	DenseNet100, CenterNet(modified ResNet)	Accuracy: 98.10 %	To build an automated system that can efficiently locate and classify eye disease.	It can effectively extract features and ensure efficient computation.	It requires complex computation.
Ukwuoma et al. [116]	COVID-19 detection	DenseNet, VGG16, Inception-V3	Accuracy: 96.33 % (Binary classification) Accuracy: 98.67 % (Multi classification)	To develop an efficient and accurate identification approach for COVID-19.	Better computational efficiency and performance.	It is necessary to overcome challenges related to dataset size and computational resources.
Das et al. [117]	COVID-19 detection	DenseNet201, ResNet50-V2, Inception-V3	Accuracy: 91.62 %	To improve the accuracy.	It provides a robust framework for variations in medical image data.	It needs higher computational costs and more advanced hardware setups.
Mahbod et al. [65]	Skin lesion classification	EfficientNetB0, EfficientNetB1, SeResNeXt50	Accuracy: 96.3 %	To deal with the problems such as small scale of medical image data set, difficulty in labeling, and data imbalance.	Better performance, mitigate issues stemming from data imbalance particularly in categories with small sample sizes.	More complexity and computation, slower inference process.
Hernández et al. [44]	Tuberculosis classification	ResNet50, Inception-V3, VGG19	Accuracy: 86.42 %	To overcome the challenges of training data scarcity.	Less training time and computing resources.	The model still needs to be tuned for the specific task.
Zhang et al. [112]	Diabetic retinopathy identification and classification	Identification system: Inception-V3, Xception, Inception-Resnet-V2Grading system: ResNet60, DensNet169, DenseNet201	Accuracy: 97.67 % (Identification) Accuracy: 96.5 % (Grading)	To improve the screening efficiency.	Better performance.	Its performance relies on the labeling experience of ophthalmology experts.
Karar et al. [118]	COVID-19 and pneumonia classification	VGG16, ResNet50-V2, DenseNet169	Accuracy: 99.9 %	To deal with the limitations such as high false negative rate.	Better performance.	It requires careful selection and tuning of models.
Lee et al. [119]	Acute intracranial hemorrhage detection	VGG16, ResNet50, Inception-V3,Inception-ResNet-V2	Precision: 85.8 %	To develop a high-performance and interpretable deep learning model on small datasets.	It performs well on a small training dataset, achieves a high sensitivity and specificity and better interpretability.	The data are all sourced from a single institution, which may result in poor generalizability.
Qummar et al. [120]	Diabetic retinopathy detection	ResNet50, Inception-V3, Xception, DenseNet121, DenseNet169	Accuracy: 80.8 %	To improve the performance.	Better performance.	The integration of multiple models and hyperparameter tuning increased the model's complexity, requiring higher computational resources and time.
Gifani et al. [121]	COVID-19 detection	EfficientNetB0, EfficientNetB3, EfficientNetB5, Inception-ResNet-V2 , Xception	Accuracy: 85.2 %	To improve the performance.	More robust diagnostic accuracy.	It increases the model's complexity, requiring more computational resources and time for training and hyperparameter tuning.
Chouhan et al. [43]	Pneumonia detection	AlexNet, DenseNet121, Inception-V3, ResNet18, GoogLeNet	Accuracy: 96.4 %	To improve the performance in the context of limited labeled data.	Better robustness, higher accuracy.	Higher computational cost.
Lee et al. [122]	Degenerative arthritis classification	VGGNet, DenseNet, ResNet,TinyNet, EfficientNet, MobileNet, Xception, ViT	Accuracy: 77.05 %	To improve the accuracy.	Higher accuracy.	More complex.
Zhang et al. [123]	Skin lesion classification	Multiple ResNet50	Accuracy: 86.3 %	To solve the challenges brought by intra-class variation	Higher accuracy, and it enhances the diversity of feature	More complex.

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Table 6 (*continued*)

Studies/ Author	Tasks	CNN architectures	Classification performance	Questions solved	Advantages	Disadvantages
				and inter-class similarity in medical image classification.	representations and reduces the risk of overfitting.	
Xu et al. [124]	Cataract grading	Multiple AlexNet	Accuracy: 86.24 %	In order to investigate how CNN characterizes cataract layer-by-layer.	Higher accuracy, more robust.	More complex.
An et al. [125]	Breast mass, brain tumor tissue, medical database and Alzheimer's disease classification	Multiple CNNs	Accuracy: 98.2 % (Dataset1)Accuracy: 95.71 % (Dataset2) Accuracy: 100 % (Dataset3)Accuracy: 95.9 % (Dataset4) Accuracy: 95.1 % (Dataset5)	To address the issue that the network initialization weights of existing deep learning models have not been well optimized.	Higher accuracy, better robustness and stability.	It increases the model's complexity and dependence on data.

studies, including its various versions, VGG was used in 5 studies, Xception was used in 4 studies, and EfficientNet, Inception-Resnet-V2, AlexNet and MobileNet were used in fewer than 4 studies.

B. Ensemble of CNN and Other Methods

Adopt appropriate classifiers Various types of classifiers can be connected in CNN-based medical image classification architectures. In this case, the original classification layers of CNN will be replaced by the new classifiers.

SVM is the most popular alternative to classification layers. The advantages of using SVM as a classifier were investigated in Ref. [36] for the task of classifying brain CT images. A CNN model, the well-known AlexNet model, and a modified version of AlexNet with a SVM classifier were trained. The results showed that the proposed modified pre-trained model, 'AlexNet-SVM', improved the accuracy to 93.48 %. This performance was superior to both a CNN created from scratch and the original AlexNet. Although the application of the SVM classifier enhanced the performance of AlexNet, the variability of the data remains a significant obstacle to the application of CNNs in medicine. The SVM was used as a post-classification model and integrated with the pre-trained VGG19 neural network to enhance the classification accuracy of malaria infection in microscopy images [62]. This combination took advantage of the feature extraction power of VGG19 and the classification strength of SVM when processing nonlinear data. The classification accuracy of the VGG19-SVM model reached 93.1 %, which was superior to that of traditional CNN models. SVM was able to handle high-dimensional data well in this study and provided better generalization ability on unseen data. However, this integration added computational complexity and required significant resources for training. In order to construct a lightweight model that suitable for the IoMT, the MobileNet and SVM was combined in Ref. [57]. The MobileNet-SVM model performed well in processing the histological images of breast cancer. It achieved an accuracy of 91 % and an F1-score of 91.35 % on the test dataset. However, because it was specifically designed for lightweight applications, despite its high computational efficiency, it may not be able to compete with larger, more complex deep learning models when handling very complex or large-scale datasets. A deep learning approach combining EfficientNet and SVM was proposed to enhance the accuracy of breast cancer diagnosis in Ref. [126]. Firstly, patches were extracted from whole slide images, and feature vectors were extracted using the EfficientNet model. Then, EfficientNet was fine-tuned by adding a global average pooling layer and a SVM classifier. Under this framework, EfficientNetB0, EfficientNetB2, EfficientNetB5, and EfficientNetB7 were compared. Eventually, the integration of EfficientNetB5 and SVM achieved the best performance, with accuracy of 99.49 % and 96.99 % in binary and multi-class tasks, respectively. This method reduced computational costs and complexity by selecting discriminative sub-images, achieving efficient computation. However, its generalization ability on other datasets needs further validation. For blood cancer classification task, an effective and computationally efficient approach that combines MobileNet-V2 and SVM was presented in Ref. [127]. This method incorporated depth-wise separable convolution, a tunable multiplier, and an inverted residual bottleneck structure, leading to a system that was both faster and more computationally efficient. Additionally, the use of SVM enhanced the system's overall performance by optimizing the hyperplane location. The proposed model achieved the highest accuracy of 99.39 %, 98.21 %, and 97.73 % in the ALLIDB1, ALLIDB2, and ASH databases, respectively. Although the performance improvement was significant, the introduction of SVM increased the model complexity and computational cost. To address the limitation of small-scale medical image databases, a combination of CNN features with SVM was adopted in Ref. [128] for medical images classification. This combined model achieved an overall classification accuracy of 95.82 %, surpassing the state-of-the-art methods. When training data was limited, the SVM classifier outperformed the softmax classifier in handing CNN features. Additionally, the CNN-SVM approach required fewer computations and less memory compared to transfer learning-based classification methods. The drawback of this method was that the performance of the SVM model depends on hyperparameter tuning, requiring numerous experiments to determine the optimal parameter settings.

To develop an accurate, efficient, and time-efficient method for detecting positive COVID-19 patients, the ensemble deep convolutional neural network model 'CoVNet-19' was proposed [129]. This model integrated DenseNet121 and VGG-19 for feature extraction, while an SVM classifier was utilized in the final classification stage due to its robustness and efficiency in managing high-dimensional feature spaces and classifying complex and nonlinear data. The CoVNet-19 model demonstrated high accuracy in classifying chest X-rays, achieving a binary classification accuracy of 99.71 % and a ternary classification accuracy of 98.28 %. Employing an SVM as the final classifier contributed to precise and robust classification outcomes, although training and fine-tuning multiple DCNNs and the SVM required significant computational resources. For cervical cells classification task, a new framework [130] was implemented that combines strong features derived from the Gray-Level Co-occurrence Matrix and Gabor filters with abstract features from the hidden layers. These fused features were then input into an SVM for classification. This strong feature CNN-SVM model performed excellently in both binary and seven-class tasks, with accuracies of 99.3 % and 93.75 %, respectively. Moreover, the model demonstrated high classification accuracy and robustness in multiple iterations and various dataset expansion experiments. But it was necessary to conduct dimensionality reduction through the Principal Components Analysis method because of the high dimensionality after feature fusion. Research [131] was dedicated to addressing the issues of data imbalance

and overfitting in medical diagnostic technology. This study employed the deep learning models ResNet50 and VGG16 for feature extraction. Subsequently, the Quantum Fruit Fly Algorithm technique (QFFA) was applied for feature selection to overcome local optima traps and enhance exploitation. The features selected by the QFFA technique were then applied to an SVM model for effective classification of medical diseases. This model outperformed existing models in several performance metrics, achieving a sensitivity of 99.26 % and an accuracy of 99.04 %. It effectively handled data imbalance issue and avoided overfitting. Additionally, the integration of QFFA technology, ResNet50 and VGG16 models, as well as the SVM classifier had also resulted in increased training time for the model. A novel automated method for COVID-19 detection was developed that utilizes parallel fusion and optimization of deep learning models [132]. The pre-trained AlexNet and VGG16 were employed. This study involved extracting features using a parallel fusion technique, followed by the selection of optimal features through an entropy-controlled firefly optimization approach. Finally, the chosen features were classified using Multiclass Support Vector Machine (MC-SVM). The MC-SVM performed well on multiple evaluation metrics, with an accuracy as high as 98 %. Additionally, compared to other classifiers, the error rate of MC-SVM was lower. However, the performance of MC-SVM highly depended on the methods of feature selection and fusion. To improve the accuracy and efficiency of COVID-19 classification, a three-step hybrid ensemble model was proposed in Ref. [133]. This hybrid ensemble model employed an improved AlexNet for feature extraction, the ReliefF algorithm for feature selection, and an SVM classifier for classification. The model achieved superior performance, with a classification accuracy of 98.642 $\pm$ 0.398 %. Additionally, it significantly outperformed other models in terms of classification time, such as SqueezeNet, Inception-V3, ResNet50, ResNet101, and VGG19. Like other ensemble models, this model increased the system's complexity due to the integration of multiple components. Two different DCNN models were utilized in Ref. [134], each with distinct architectures, in order to extract more discriminative features from images. After integrating features from the image metadata and DCNNs, an SVM classifier was trained to learn the distribution boundaries of each class. The proposed method achieved a binary classification accuracy of 86.17 % and a multi-class accuracy of 91.05 % for the DR classification task, demonstrating higher accuracy than other methods reported in the literature. However, the method faced difficulties in distinguishing certain categories such as grade 0 and grade 1, grade 3 and grade 4, leading to potential misclassification.

In four studies, a forward feed network-based neural network Extreme Learning Machine (ELM) [135] was embedded in CNN model to classify the features extracted by CNN. In Ref. [136], a parallel convolutional neural network was used for extracting features, followed by the application of the ELM technique for classifying diabetic retinopathy. This method attained accuracies of 91.78 % and 97.27 % on two distinct datasets. Moreover, in terms of classifier performance metrics, model parameters, layers, and prediction time, the proposed method surpassed other contemporary state-of-the-art models. Another key finding of the study was that the proposed framework exhibited stability across both large and small datasets, as well as balanced and imbalanced datasets. The only limitation of this method was that it has a higher misclassification rate for moderate categories, which may affect the overall classification performance. Study [137] proposed an ensemble framework that combines a lightweight parallel convolutional neural network with the ELM algorithm to improve the detection accuracy of pulmonary diseases. This method utilized the feature extraction capabilities of CNN and the rapid training and efficient classification abilities of ELM to enhance overall classification performance. In the classification task of multiple categories (17 types of pulmonary diseases), the CNN-ELM model achieved an overall accuracy of 90.92 %. Since ELM did not involve back-propagation, its training was faster than typical neural networks, allowing the use of ELM as a classifier to enable the model to process large amounts of data in a short time. However, integrating the weights of hidden layer in the ELM with the CNN model was a challenging task that requires specific technical to handle it. In order to improve the diagnostic efficiency of COVID-19, a smart recognition system for seven types of lung diseases based on ML technology has been proposed in Ref. [138]. In this system, a lightweight CNN was used to extract characteristic features from the raw pixel values of the CXR images. Then the Pearson Correlation Coefficient was employed to identify the best feature subset. Finally, the ELM was adopted to perform the classification task to assist faster learning and reduce computational complexity. The proposed CNN-PCC-ELM model achieved an accuracy of 96.22 % with an AUC of 99.48 % for eight class classification. The training time of the model was only 445 s, and the testing time was only 0.0156 s. However, there was also a problem of feature redundancy. In the initially extracted 512 features, a large number of redundant and irrelevant features needed to be further filtered. In Ref. [139], a new COVID-19 identification method was proposed to enhance classification performance. In this method, DenseNet-201 was used for feature extraction, followed by a Firefly algorithm for selecting the most optimal learning features. Then a new parallel high index technique was proposed to fuse two optimal vectors. The outcome was then passed on to an ELM for final classification. The results indicated an average classification accuracy of 94.76 %. Compared to Naive Bayes (NB), Linear Support Vector Machine (LSVM), Weighted K Nearest Neighbor (KNN), and Ensemble Trees, the performance of ELM was the best. The ELM classifier demonstrated the advantage of fast learning speed, capable of completing training and classification in a short time. Additionally, the integration of multiple methods also increased computational costs in this study.

An Orthogonal SoftMax Layer (OSL)-based model for Acute Leukemia detection was proposed in Ref. [37]. The proposed model utilized ResNet18 for deep feature extraction, followed by efficient OSL-based classification. The integration of OSL with ResNet18 aimed to enhance classification performance by ensuring orthogonality among weight vectors. The results showed that the model achieved classification accuracies of 99.39 %, 98.21 %, and 97.50 % on the ALLIDB1, ALLIDB2, and ASH datasets, respectively. Not only did the model achieve high accuracy and recall on different datasets, but the use of OSL also enhanced computational efficiency and reduced generalization errors. However, this model was specifically designed for small medical datasets, and its application to larger datasets might be somewhat limited. In response to the problem of small datasets of medical images, a deep convolutional neural network with Hierarchical Spiking Neural Network (HSNN) [140] was proposed, in which the CapsNet model with nesterov-accelerated adaptive moment estimation optimizer was used as feature extractor and the HSNN approach based on shark smell optimization algorithm was used as classifier. The effectiveness of the DCNN-HSNN technique was validated on both HIS2828 and ISIC2017 skin disease image datasets, with accuracies of 96.23 % and 94.6 % respectively. Although this method has demonstrated high accuracy across multiple datasets, the complexity of the model structure cannot be ignored, which may increase computational costs and implementation difficulty. In order to establish an automatic detection framework of COVID-19, a multi-level feature extraction-based approach was introduced in Ref. [141]. First, the conventional image descriptors, i.e., Scale-Invariant Feature Transform and Global Image Shape Transformation were utilized to extract, select, and fuse local and global features. Then the CNN extracted the deep features from the fused features. Finally, the Long Short-Term Memory (LSTM) RNN was used to classify the deep features. The proposed approach obtained an accuracy of 98.94 % on the SARS-CoV-2 CT scan dataset and an accuracy of 83.03 % on the SIRM Covid-19 CT scan and chest X-ray dataset. This deep learning method reduced training complexity and time by using fewer input parameters and layers. Moreover, it exhibited excellent overall performance when the data was imbalanced. However, overfitting still existed under certain training data split ratios. For exploring the application of CNN in laryngeal cancer diagnosis from histopathological

images, a three-stages method named LCDC-AOADL was presented [142]. The Inception-V3 was used for feature extraction, the Deep Belief Network (DBN) was applied for classification, and the Aquila Optimization Algorithm was utilized for hyperparameter tuning. All these methods combine to improve accuracy and efficiency of laryngeal cancer diagnosis. The experimental findings demonstrated that the LCDC-AOADL technique surpassed other recent methods in detection performance, achieving a maximum accuracy of 96.02 %, precision of 92.10 %, recall of 91.87 %, and an F-score of 91.86 %. In this study, multiple complex models and algorithms were involved, resulting in more resources and time for its implementation.

Combinations of different classifiers with CNN have also been explored in some studies. In Ref. [61], two classifiers were explored in combination with five different CNNs respectively. Pre-trained MobileNet, DenseNet121, ResNet152-V2, VGG16, Xception, and Inception-V3 were employed as feature extractors, and SVM and RF were used as classifiers. The best accuracy of 84 % was obtained by the combination of VGG16 with SVM. SVM performed well when there were clear separation boundaries and had good robustness, but it led to high computational complexity for large datasets, which limited its scalability. In order to optimize the detection of COVID-19, the features extracted from ResNet101 were fed to KNN and SVM separately in Ref. [143]. Both combinations achieved very high detection accuracy and sensitivity. Specifically, ResNet101 combined with SVM achieved an accuracy of 99.46 %, and ResNet101 combined with KNN achieved an accuracy of 99.86 %. In COVID-19 and pneumonia detection task, VGG16, VGG19, DenseNet160, DenseNet200, and sCNN those with key point detection methods, were connected to Softmax, RF, SVM, XG Boost respectively [144]. This study was designed to optimize the efficiency of using chest X-ray images to detect COVID-19 and pneumonia. The experiment results showed that the VGG19 model combined with RF as the classification layer obtained the highest accuracy of 96.6 %. The challenge of the method was that the combination of multiple models and algorithms increased the complexity of the system. In order to study the application of the latest deep learning techniques and machine learning algorithms in the classification of breast pathological images, seven CNN models for feature extraction and four classifiers formed 28 hybrid architectures in Ref. [145]. The authors also evaluated and ranked 28 hybrid architectures using Scott Knott statistical testing and Borda Count voting, and found that the hybrid architecture employed DenseNet 201 for feature extraction and the Multilayer Perceptron (MLP) as classifier emerged as the best-performing model, achieving accuracies of 99 % and 93.93 % on the FNAC dataset and BreakHis dataset respectively. This hybrid architecture demonstrated high classification accuracy across multiple datasets and various magnification factors, indicating strong generalization capabilities and robustness. However, it was important to note that MLP required careful tuning of multiple hyperparameters, which increased the difficulty of model optimization. Within a automatic COVID-19 classification frameworks, fifteen CNN architectures were chosen as feature extractor [146]. The extracted features were subsequently sent into Decision Tree (DT), RF, XG Boost, AdaBoost, Bagging classifier and LightGBM. According to the experimental results, the combination of DenseNet121 and Bagging tree classifier obtained the highest accuracy of 99 %, and the combination of ResNet50 and LightGBM achieved the second-best accuracy of 98 %. The study successfully demonstrated that combining deep learning models with powerful machine learning classifiers can achieve high accuracy in automatically detecting medical images of COVID-19. In practical applications, classifiers can be selected based on computational resources and specific needs. However, such combinations may increase computational costs and training time. Facing the uncertainty present during the COVID-19 detection, a deep uncertainty-aware transfer learning framework was proposed in Ref. [147]. In this research, eight different machine learning and statistical modeling techniques were used to classify COVID-19 cases from features extracted by VGG16, ResNet50, DenseNet121, and Inception-ResNet-V2. The method combined ResNet50 and SVM obtained the best results on both CT dataset and X-ray dataset. The study also found that predictive uncertainty estimates are much higher for CT images compared to X-ray images, indicating that more caution is needed when using machine learning to predict COVID-19. The features acquired by CNN model were concatenated with hand crafted features in Ref. [148], in order to improve the accuracy of brain tumor detection. Various classifiers: LR, DT, KNN, ensemble, LDA and SVM were utilized to predict the glioma from the fusion features of VGG19 and hand crafted features. Experimental results showed that the KNN classifier performed best on the BRATS 2015 dataset with an accuracy of 99.82 %, the ensemble classifier performed best on the BRATS 2016 dataset with an accuracy of 100 %, and the ensemble classifier again performed best on the BRATS 2017 dataset with an accuracy of 99.82 %. In this study, the fusion of manual and deep learning features significantly improved classification accuracy. Moreover, the ensemble strategy has a broad applicability and suitable for various types of MRI images. However, the feature fusion and optimization process is complex and may increase computational complexity and time. Research [59] evaluated the effect of transfer learning on the performance for lung nodule malignancy classification. 11 types of pre-trained CNN architectures were employed to extract features. Then NB, KNN, MLP, RF, SVM linear and SVM Radial Basis Function (RBF) were evaluated combined with these 11 CNN models. 66 hybrid architectures were built, in which feature extractor ResNet50 combined with SVM RBF classifier performed the best with accuracy of 88.41 % and AUC of 93.1 %. Because the deep extractor contained a large number of features, the classification time of the models was relatively long. The combinations of pre-trained GoogLeNet with SVM and KNN were proposed in Ref. [54] to achieve an accurate and automated classification system for brain tumors. The results demonstrated that the performance was significantly improved when using the SVM or KNN as classifier connected after GoogLeNet. Specifically, the SVM classifier obtained an accuracy of 97.8 $\pm$ 0.2 %, and the KNN classifier obtained an accuracy of 98.0 $\pm$ 0.4 %. Additionally, the system can still maintain high performance with fewer training samples. However, there was a potential risk of overfitting when using less training data.

Deep fully convolutional neural network usually has millions of parameters and inadequate number of training samples leading to overfitting and poor generalization. To deal with these challenges, a novel DenseNet based FCN architecture was designed for segmentation and classification was proposed in Ref. [149] for cardiac diagnosis. In this ensemble framework, four machine learning techniques—RF, SVM, MLP and Gaussian Naive Bayes (GNB)—were employed as a ensemble classifier, and the majority voting method finalized the disease prediction. It achieved a perfect accuracy of 100 %, ranking the first place in the automated cardiology diagnosis task in the ACDC-2017 Challenge. This study improved the accuracy of disease diagnosis by combining multiple classifiers, reducing the risk of error from a single model. Although the model's parameters and memory efficiency were high, the introduction of DenseNet and Inception modules, as well as the integration of multiple classifiers, had increased the computational complexity and training time.

Table 7 provides a detailed summary of the methods of adding a classifier. Some of the studies employed different CNN models and classifiers, we report the best combination and its performance in the table.

Models fusion A novel correlation learning mechanism combined CNN with Artificial Neural Network (ANN) was proposed in Ref. [150]. In this mechanism, CNN can find the most suitable filters in pooling and convolutional layers with the help of ANN. Consequently, the primary neural classifier exhibited accelerated learning and achieved enhanced efficiency. CNN and Transformer were combined to form a highly robust yet efficient hybrid model, named MedViT [151]. This proposed architecture integrated with the locality of CNN and the global connectivity of Transformer, showed superior robustness and generalization capabilities when compared to state-of-the-art studies. For breast cancer

Table 7
Detailed description of the combination methods those CNNs are used as feature extractor and other methods are used as classifier.

Studies/Author	Tasks	CNN Models	Classifiers	The best combination	Accuracy(Only report the best)	Questions solved	Advantages	Disadvantages
Dawud et al. [36]	Brain hemorrhage classification	AlexNet	SVM	/	93.48 %	To investigate the advantages of using SVM as a classifier.	Higher accuracy.	The variability of the data remains a significant obstacle to the application of CNNs in medicine.
Vijayalakshmi et al. [62]	Malaria detection	VGG19	SVM	/	93.1 %	To enhance the classification accuracy.	SVM is able to handle high-dimensional data well and provides better generalization ability on unseen data.	It adds computational complexity and requires significant resources for training.
Ogundokun et al. [57]	Breast cancer classification	MobileNet	SVM	/	91 %	To construct a lightweight model that suitable for the IoMT.	Better performance and higher computational efficiency.	It may not be able to handle very complex or large-scale datasets.
Ahmad et al. [126]	Breast cancer classification	EfficientNetB0/ EfficientNetB2/ EfficientNetB5/ EfficientNetB7	SVM	EfficientNetB5+SVM	99.49 %(binary classification) 96.99 %(multi classification)	To enhance the accuracy.	Less computational costs and lower complexity.	Its generalization ability on other datasets needs further validation.
Das et al. [127]	Blood cancer detection	MobileNet-V2	SVM	/	99.39 % (Dataset1)98.21 % (Dataset2)97.73 % (Dataset3)	To create an effective and computationally efficient approach for blood cancer classification task.	Better performance, faster and more computationally efficient.	The introduction of SVM increases the model complexity and computational cost.
Deepak et al. [128]	Brain tumor classification	CNN	SVM	/	95.82 %	To address the limitation of small-scale medical image databases.	Better performance, fewer computations and less memory requirements.	The performance of the SVM model depends on hyperparameter tuning, requiring numerous experiments to determine the optimal parameter settings.
Kedia et al. [129]	COVID-19 detection	VGG19 and DenseNet121	SVM	/	99.71 % (Binary classification) 98.28 % (Multi classification)	To develop an accurate, efficient, and time-efficient method for detecting positive COVID-19 patients.	It obtained precise and robust classification outcomes.	It requires significant computational resources to train and fine-tune multiple DCNNs and the SVM.
Jia et al. [130]	Cervical cell classification	CNN	SVM	/	99.3 % (Binary classification) 93.75 % (Multi classification)	To improve the performance.	High classification accuracy and robustness.	It needs dimensionality reduction through the Principal Components Analysis method which is complex.
Nijaguna et al. [131]	Breast cancer, diabetes, and Erythemato-Squamous disease classification	ResNet50 and VGG16	SVM	/	99.04 %	To address the issues of data imbalance and overfitting in medical diagnostic technology.	It effectively handles data imbalance issue and avoids overfitting.	It needs more training time.
Khan et al. [132]	COVID-19 detection	AlexNet and VGG16	SVM	/	98 %	To construct an automated method for COVID-19 detection.	Better performance and lower error rate.	The performance highly depends on the methods of feature selection and fusion.
Jin et al. [133]	COVID-19 detection	AlexNet	SVM	/	98.642 ± 0.398 %	To improve the accuracy and efficiency of COVID-19 classification.	Superior performance and efficiency.	It increases the system's complexity due to the integration of multiple components.

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Table 7 (*continued*)

Studies/Author	Tasks	CNN Models	Classifiers	The best combination	Accuracy(Only report the best)	Questions solved	Advantages	Disadvantages
Li et al. [134]	Diabetic retinopathy classification	Two DCNNs	SVM	/	86.17 % (Binary classification) 91.05 % (Multi classification)	To improve the performance.	Higher accuracy.	There is a risk of potential misclassification in distinguishing certain categories.
Nahiduzzaman et al. [136]	Diabetic retinopathy identification	PCNN	ELM	/	91.78 % (Dataset1) 97.27 % (Dataset2)	To improve the performance.	It has advantages in performance metrics, model parameters, layers, and prediction time, and exhibits stability across both large and small datasets, as well as balanced and imbalanced datasets.	It has a higher misclassification rate for moderate categories.
Nahiduzzaman et al. [137]	Lung diseases classification	CNN	ELM	/	90.92 %	To improve the detection accuracy of pulmonary diseases.	Higher accuracy, faster training.	It is a challenging task to integrate the weights of hidden layer in the ELM with the CNN model.
Nahiduzzaman et al. [138]	Lung diseases detection	CNN	ELM	/	96.22 %	To improve the diagnostic efficiency of COVID-19.	Faster learning and lower computational complexity.	There is also a problem of feature redundancy.
Khan et al. [139]	COVID-19 detection	DenseNet201	ELM	/	94.76 %	To enhance classification performance.	Better performance and faster training.	It increases the computational costs.
Das et al. [37]	Acute leukemia classification	ResNet18	OSL	/	99.39 % (Dataset1)98.21 % (Dataset2)97.50 % (Dataset3)	To enhance classification performance.	Higher accuracy and recall on different datasets, better computational efficiency and lower generalization errors.	Its application to larger datasets may be somewhat limited.
Jenifer et al. [140]	Skin lesion classification	CapsNet	HSNN	/	96.23 % (Dataset1)94.6 % (Dataset2)	To address the problem of small datasets of medical images.	Higher accuracy.	The model structure is complex.
Naeem et al. [141]	COVID-19 detection	CNN	LSTM	/	98.94 % (Dataset1)83.03 % (Dataset2)	In order to establish an automatic detection framework of COVID-19.	Good performance, lower training complexity and less training time. And it exhibits excellent overall performance when the data was imbalanced.	Overfitting still exists under certain training data split ratios.
Alrowais et al. [142]	Laryngeal cancer detection and classification	Inception-V3	DBN	/	96.02 %	For exploring the application of CNN in laryngeal cancer diagnosis.	Higher accuracy and efficiency.	More resources and time for implementation.
Abhishek et al. [61]	Leukemia detection and classification	MobileNet/DenseNet121/ ResNet152-V2/VGG16/ Xception/Inception-V3	SVM/RF	VGG16+SVM	84 %	In response to the lack of well-labeled data sets.	SVM performs well when there are clear separation boundaries and has good robustness.	SVM may lead to high computational complexity for large datasets.
Nawaz et al. [143]	COVID-19 detection	ResNet101	KNN/SVM	ResNet101+KNN	99.86 %	To optimize the detection of COVID-19.	Better performance.	/
Bhattacharyya et al. [144]	COVID-19 detection	DenseNet169/ DenseNet201/VGG16/ VGG19	Softmax/RF/ SVM/XG Boost	VGG19+RF	96.6 %	To optimize the efficiency of using chest X-ray images to detect COVID-19 and pneumonia.	Higher accuracy.	More complex.
Zerouaoui et al. [145]	Breast cancer classification	DenseNet201/Inception-V3/ Inception-ReseNet-V2/ MobileNet-V2/ResNet50/ VGG16/VGG19	MLP/SVM/ DT/KNN	DenseNet201+MLP	99 % (Dataset1) 93.93 % (Dataset2)	In order to study the application of the latest deep learning techniques and machine learning algorithms in the classification of breast pathological images.	Good accuracy, strong generalization capability and robustness.	The MLP requires careful tuning of multiple hyperparameters.

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Table 7 (*continued*)

Studies/Author	Tasks	CNN Models	Classifiers	The best combination	Accuracy(Only report the best)	Questions solved	Advantages	Disadvantages
Kassani et al. [146]	COVID-19 detection	MobileNet/DenseNet121/ DenseNet201/Xception/ ResNet50/ResNet152/ Inception-V3/Inception-ResNet-V2/VGGNet16/ VGGNet19/NASNetLarge/ NASNetMobile/ResNet50-V2/ResNet101-V2/ ResNet152-V2	DT/RF/ XGBoost/ AdaBoost/ Bagging tree/ LightGBM	DenseNet121+Bagging tree	99 %	To optimize the accuracy and efficiency of the COVID-19 detection.	Higher accuracy.	The combinations of CNN and machine learning classifiers may increase computational costs and training time.
Shamsi et al. [147]	COVID-19 detection	VGG16/ResNet50/ DenseNet121/Inception-ResNet-V2	KNN/SVM/ RBF SVM/ GP/RF/MLP/ Adaboost/ NB	ResNet50+SVM	87.9 % (Dataset1) 98.6 % (Dataset2)	To deal with the uncertainty present during the COVID-19 detection.	Good performance.	It needs more caution when using machine learning to predict COVID-19.
Saba et al. [148]	Brain tumor detection	VGG19	LR/DT/ KNN/ Ensemble/ LDA/SVM	Dataset1: VGG19+KNNDataset2: VGG19+ensembleDataset3: VGG19+ensemble	99.82 % (KNN) (Dataset1)100 % (ensemble) (Dataset2)99.82 % (ensemble) (Dataset3)	In order to improve the accuracy of brain tumor detection.	Higher accuracy, and it has a broad applicability.	The feature fusion and optimization process is complex and may increase computational complexity and time.
da Nóbrega et al. [59]	Lung nodule malignancy classification	VGG16/VGG19/MobileNet/ Xception/Inception-V3/ ResNet50/Inception-ResNet-V2/DenseNet169/ DenseNet201/ NASNetMobile/ NASNetLarge	NB/MLP/ SVM/KNN/ RF/SVM RBF	ResNet50+SVM RBF	88.41 %	To evaluate the effect of transfer learning on classification performance.	Better performance.	The classification time is relatively long because the deep extractor contains a large number of features.
Deepak et al. [54]	Brain tumor classification	GoogLeNet	SVM/KNN	GoogLeNet + SVM	98 ± 0.4 %	To achieve an accurate and automated classification system for brain tumors.	It can achieve high performance especially with fewer training samples.	There is a potential risk of overfitting when using less training data.
Khened et al. [149]	Cardiac diagnosis	DenseNet and FCN	RF, SVM, MLP and GNB	/	100 %	To deal with the challenges of overfitting and poor generalization of deep fully convolutional neural network.	Higher accuracy and lower error rate.	Higher computational complexity and more training time.

histopathology images classification, the CNN and LSTM were used to construct a hybrid model [152]. Both modules' outputs were merged through element-wise multiplication. The accuracy of this hybrid model reached 99 % for binary classification and 92.5 % for multi-class classification. A WOA-based CNN was proposed for skin cancer detection [153]. The enhanced version of Whale Optimization Algorithm (WOA) was employed to optimize selection of weights and biases of CNN, thus minimizing the error between the network's output and the desired output. In Ref. [154], a CNN architecture was evolved using Genetic Algorithm (GA), departing from conventional methods that rely on trial and error or predefined common structures. GA was employed to search for an optimal CNN structure that yielded superior results. This method had not only achieved high-precision of Glioma tumors grading, but also demonstrated remarkable success in classifying various types of brain tumor images. EfficientNet and Transformer were combined in Ref. [155] for tuberculosis detection from X-ray images. The pre-trained EfficientNet extracted the 2D features firstly, then the sequential feature maps were fed into the Vision Transformer, followed by propagation to customized layers to classify tuberculosis and other pneumonia categories. Aiming at COVID-19 classification on Chest X-ray images, the Choquet integral was introduced to integrated into DenseNet121, ResNet50, Inception-V3 and DCNN respectively [156]. The Choquet integral was a mathematical operator used for information fusion and decision-making, which helped to enhance the model's capability to prioritize relevant features and efficiently amalgamate them for decision-making. The accuracies of four CNN models those applied integrated Choquet were improved respectively. A three-stage hybrid architecture including density estimation, feature exploitation, and classification was proposed in brain tumors classification task [157]. In this architecture, the neural Autoregressive Distribution Estimation was trained to find appropriate joint distribution in density estimation stage, then two CNNs independently extracted features from both the raw image data and the output generated by the autoregressive model. The findings indicated that the hybrid CNN-NADE achieved high classification performance, especially in scenarios where medical image availability was limited. In Ref. [158], a Faster R-CNN was proposed to identify laryngeal neoplasms under the "small data" regime. It's architecture consisted of three parts, namely a backbone module, a Region Proposal Network (RPN), and an R-CNN detector module. ResNet50 was chosen as the backbone, extracting features from input images. RPN produced a series of region proposals, representing areas potentially containing objects of interest. Then, the R-CNN head utilized both the image features and region proposals to provide a category prediction. In Ref. [159], VGG16, DenseNet161, and ResNet50 were respectively combined with LSTM units, where the CNN extracted features from the entire image and the LSTM was responsible for generating sequential labels for a given dermoscopy image. The model also incorporated an attention module which interacted with the LSTM to define which features will be input to the latter.

Feature fusion For multi-label chest X-ray images classification, the visual vectors generated by ConvNeXt network were combined with semantic vectors encoded by language model BioBert [160]. This integration facilitated the mapping of disparate feature representations into a unified metric space, with semantic vectors serving as prototypes for each class. Subsequently, the metric relationship between images and labels was analyzed at both image and disease category levels. The experimental average AUC score attained 0.826, surpassing that of the comparison models. A novel parallel bi-branch model named Trans-CNN Net, which combined Transformer modules and CNN modules, was proposed in Ref. [161] for COVID-19 detection. Employing a cross-fusion principle, a bi-directional feature fusion architecture where features from both branches were fused reciprocally was designed. Additionally, the parallel structures of branches were fused using a feature fusion module, resulting in a model capable of extracting features at different scales. Experimental results on the COVIDx-CT dataset demonstrated a classification accuracy of 96.7 %, surpassing typical CNN networks like ResNet152 (95.2 %) and Transformer networks like Deit-B (75.8 %). For automatic Hepatocellular Carcinoma diagnosis, the features extracted by five CNN models were combined with powerful textural features obtained by classical approaches [162], which proved to be very efficient in many situations.

4.4. Exploration of explainability

Explainable deep learning-based classification is crucial in computer-aided medicine system, as it aids in understanding the rationale behind the decisions made by the deep learning model. In recent years, aiming to unveil the inner workings of deep learning models and overcome their black-box nature, various methods have been studied such as Class Activation Mapping (CAM), Gradient Weighted Class Activation Mapping (Grad-CAM), Grad-CAM++, T-distributed Stochastic Neighbor Embedding (T-SNE), Uniform Manifold Approximation and Projection (UMAP), and Guided Backpropagation (GB).

A. CAM, Grad-CAM, Grad-CAM++

CAM [163] generates a class-related heat map by re-weighting the classifier weights with the original feature map. This map intuitively highlights the parts that have the greatest impact on the model's decision. This visualization method allows medical professionals to understand how the model makes a specific prediction, thereby increasing their transparency and trust in the model. In Refs. [81,119], CAM was used to generate a category heat map by finding the weight corresponding to each pair of feature maps, and then finding their weighted sum. In the heat map, the lesion area marked by experts and the prediction area can be displayed at the same time, which is used to prompt professionals to pay attention to this area and achieve the role of auxiliary diagnosis.

Grad-CAM [164] is based on gradient information and uses the gradient of the convolutional layer to determine the impact of each feature map on the final classification. CAM can only be applied to CNN architectures with global average pooling layers, which requires a specific network structure. While the Grad-CAM can be applied to any CNN architecture and does not require a specific global pooling layer. It is visualized through gradient calculation, so it is more flexible and applicable to a wider range of CNN architectures. As a result, Grad-CAM is the most commonly used visualization method in medical image processing, such as studies [42,61,75,92,94,106,114,116,137,159,161,165,166] all used Grad-CAM to obtain the class activation maps which displayed the region of interest of the model in the image.

Grad-CAM++ [167] is an improved version of Grad-CAM, which aims to address the limitations of Grad-CAM when dealing with multiple objects or overlapping objects. By improving gradient weighting and introducing higher-order derivatives, it is able to better capture the complex relationship between target categories and different regions of the image. Therefore, Grad-CAM++ is particularly suitable for complex scenes with multiple objects and provides more explanatory visualization results. For a COVID-19 detection task, Grad-CAM++ was applied to aid clinical interpretation and increase acceptance among medical professionals [156].

B. T-SNE

T-SNE [168] is a nonlinear dimensionality reduction technique. It constructs a probability distribution by calculating the similarity between data points in a high-dimensional space, and attempts to reconstruct this distribution in a low-dimensional space. It uses Gaussian distribution to measure the similarity of high-dimensional data, and uses t distribution to process points in low-dimensional space. After T-SNE dimensionality reduction, the high-dimensional features extracted by CNN can be clearly displayed in 2D or 3D space, helping researchers to intuitively identify the distribution and clustering of different categories

in the data. This method retains the local structure of the data, but lacks in maintaining the global structure. As a widely-used technique for visualizing high-dimensional data, T-SNE feature visualization was utilized to ascertain the presence of crucial or sparse information within the fused feature set in Ref. [141].

C. UMAP

Unlike T-SNE, the goal of UMAP [169] is to minimize the difference in probability distribution between high-dimensional space and low-dimensional space by optimizing a similarity measure that spans high-dimensional space and low-dimensional space. It maintains both local and global structures, so it has an advantage in preserving global structures. In medical image classification, the features extracted by CNN are often of high dimensionality. UMAP effectively projects high-dimensional features into 2D or 3D space by maintaining the local and global structure of the data, allowing researchers to observe the distribution between different patient groups and disease states. This visualization method can help identify potential patterns and anomalies and provide support for clinical decision-making. Okyaz et al. [104] utilized UMAP method to indicate that globally pooled features from test images are tightly clustered while adhering to the respective image classes.

D. GB

GB [170] is an improved backpropagation technique that aims to generate more meaningful feature maps by modifying the gradient propagation process. Similar to traditional backpropagation, GB also calculates the gradient of the model output relative to the input image. But only the positive gradient and the gradient through the ReLU activation function are retained. In other words, it sets the negative gradient to zero. This process helps to emphasize those input features that contribute to the model's decision and ignore those features that have no effect on the decision. GB provides higher interpretability for CNN models by enhancing gradient information. Compared with Grad-CAM, the feature maps generated by GB have higher spatial resolution and can provide more detailed local information. This makes it particularly valuable in medical image analysis, because medical images often contain complex details and tiny abnormal areas. In Ref. [29], a CNN for classifying histopathological images of endometrial diseases was proposed, incorporating CAM and GB visualization method.

Table 8 presents the widely used explainable methods, in terms of CNN architecture, explainable methods, explaination mechanism, as

Table 8
A summary of explainable methods.

Studies/Author	CNN architecture	Explainable methods	Explanation mechanism	Advantages	Disadvantages
Zhao et al. [81]Lee et al. [119]	RE-DSENetEnsemble models	CAM	It generates a class-related heat map by re-weighting the global average pooling layer with the original feature map. This map intuitively highlights the parts that have the greatest impact on the model's decision.	It is simple, efficient, with clear category relevance, and suitable for lightweight models.	Low resolution, limited feature capture. It relies on the global average pooling layer.
Zebin et al. [42] Abhishek et al. [61] Karakanis et al. [75] Liang et al. [92] Malik et al. [94] Nahiduzzaman et al. [106] Rahimzadeh et al. [114] Ukwuoma et al. [116] Nahiduzzaman et al. [137] Barata et al. [159] Fan et al. [161] Mukhtorov et al. [165] Alghamdi et al. [166]	VGG16, ResNet50, EfficientNetB0 VGG16 ResNet8 CNNCF CDC_Net Lightweight CNN Ensemble models Ensemble models CNN-ELM Ensemble models Trans-CNN ResNet152 VGG16, ResNet18, DenseNet121	Grad-CAM	It uses the gradient of the convolutional layer to determine the impact of each feature map on the final classification.	It is more flexible and applicable to a wider range of CNN architectures.	Sensitive to noise, inaccurate positioning, limited resolution.
Deepanshi et al. [156]	Ensemble models	Grad-CAM++	It improves gradient weighting and introduces high-order derivatives based on Grad-CAM.	It has accurate fine-grained positioning, stronger category correlation and is suitable for complex multi-target scenarios.	High computational complexity, slow speed. It is still limited to the convolutional layer.
Naeem et al. [141]	CNN-LSTM	T-SNE	It maps high-dimensional data into two-dimensional or three-dimensional space by minimizing distribution differences while preserving the local structure of similar data points.	It is efficient high-dimensional data visualization that suitable for displaying local similarities of high-dimensional features.	High computational overhead. It not suitable for large-scale data and lacks in maintaining the global structure.
Eminaga et al. [104]	PlexusNet	UMAP	Similar to T-SNE, but it uses manifold learning to map high-dimensional data to a low-dimensional space by optimizing local and global structures and building a graph using nearest neighbor relationships.	It has better preservation of global structure and can handle larger datasets.	Sensitive to noise. And parameter selection is sensitive to results.
Sun et al. [29]	HIENet	GB	It is an improved back-propagation technique that generates visualizations to show how significantly an input contributes to a particular output.	It has higher spatial resolution, provides more detailed local information, and is suitable for complex networks.	Sensitive to noise, and difficult to distinguish different categories.

Table 9
Medical image datasets for various diseases.

Disease	Dataset	Sources	Data types	Classes	Size	Description
Lung disease	COVID-19 Radiography Database [171]	Chowdhury et al.	X-ray	4 classes: normal, COVID-19, viral pneumonia, lung opacity	21165	It is created by six different public sub-databases. 3487 images before augmentation and 21165 images after augmentation.
	Chest X-ray14 dataset [172]	National Institutes of Health Clinical Center	X-ray	14 classes: atelectasis, cardiomegaly, effusion, infiltration, mass, nodule, pneumonia, pneumothorax, consolidation, edema, emphysema, fibrosis, pleural_thickening, hernia	112120	It consists of 112,120 frontal-view chest X-rays from 30,805 patients.
	Chest X-Ray Images (Pneumonia) [173]	Guangzhou Women and Children's Medical Center	X-ray	2 classes: normal, pneumonia	5856	It consists of 1583 normal images and 4273 pneumonia images.
	Cohen [174]	Dr. Cohen at the University of Montreal	X-ray and CT	4 classes: normal, COVID-19, viral pneumonia, bacterial pneumonia	Regularly updated	It is a constantly updated public dataset used for healthy vs pneumonia, bacterial vs viral vs COVID-19 pneumonia, prognostic/severity predictions.
	SARS-CoV-2 CT scan dataset [175]	Eduardo Soares from the Lancaster University	CT	2 classes: COVID-19, Non-COVID-19	2482	It contains 1252 CT scans that are positive for SARS-CoV-2 infection (COVID-19) and 1230 CT scans for patients non-infected by SARS-CoV-2.
	COVIDx-CT dataset [176]	Hayden Gunraj	CT	3 classes: COVID-19, pneumonia, normal	431205	It comprises of 363699 training cases, 33725 validation cases and 33781 test cases.
	COVID-CTset [114]	Rahimzadeh et al.	CT	2 classes: COVID-19, normal	63849	It contains the full original CT scans of 377 persons, 15589 and 48260 CT scan images belonging to 95 COVID-19 and 282 normal persons, respectively.
	Large COVID-19 CT scan slice dataset [177]	Maede Maftouni	CT	3 classes: COVID-19, CAP, normal	17104	It is created from seven public datasets, and consists of 7593 COVID-19 images from 466 patients, 6893 normal images from 604 patients, and 2618 CAP images from 60 patients.
Skin lesion	DermNet image library [178]	DermNet NZ	Skin disease image	Various skin diseases	More than 20000	The images are in JPEG format with resolutions vary from image to image, but overall these are not extremely high resolution imagery.
	HIS 2828 [179]	Bar et al.	Skin disease image	4 classes: nervous tissue, connective tissue, epithelial tissue, muscular tissue	2828	It includes 1026 images under nervous tissue (NT) class, 484 images under connective tissue (CT) class, 804 images under epithelial tissue (ET) class, and 514 images into muscular tissue (MT) class.
	ISIC 2016 [180]	International Skin Imaging Collaboration	Dermoscopy image	2 classes: benign, melanoma	1279	It contains 900 training images and 379 test images.
	ISIC 2017 [181]	International Skin Imaging Collaboration	Dermoscopy image	2 classes: benign, melanoma and nevus of seborrheic keratosis	2750	It contains 2000 training images, 600 test images and 150 validation images.
	ISIC 2018 [182]	International Skin Imaging Collaboration	Dermoscopy image	7 classes: benign, seborrheic keratosis, basal cell carcinoma, actinic keratosis, dermatofibroma, vascular lesion, melanoma	11720	It contains 10015 training images, 1512 test images and 193 validation images.
Diabetic retinopathy	Kaggle DR 2015 competition [183]	Kaggle	Retinal fundus image	5 classes: no DR, mild, moderate, severe, PDR	34984	It contains a left and right field provided for every subject.
	APTOS 2019 [184]	Kaggle	Retinal fundus image	5 classes: no DR, mild, moderate, severe, PDR	3662	It comprises of 1805 negative, 370 mild, 999moderate, 193 proliferative and 295 severe DR.
	Messidor [185]	Decencière et al.	Retinal fundus image	5 classes: no DR, mild, moderate, severe, PDR	1200	It comprises of 400 images without dilation and 800 images with pupil dilation.
	EyePACS datasets [186]	EyePACS	Retinal fundus image	5 classes: no DR, mild, moderate, severe, PDR	88702	It includes 35126 training samples and 53576 test samples. But the data contains a large number of images noise and it's distribution is extremely unbalanced.
Brain tumor	Figshare [187]	Figshare	MRI	3 classes: glioma, meningioma, pituitary	3064	It contains 1426 images with glioma, 708 and 930 images corresponding to meningioma and pituitary tumor.
	BRATS 2015 [188]	Bjoern et al.	MRI	2 classes: low-grade gliomas, high-grade gliomas	274	It consists of 220 high grade gliomas and 54 low grade gliomas MRIs. The four MRI modalities are T1, T1c, T2, and T2FLAIR.
	BRATS 2017 [189]	Bjoern et al.	MRI	2 classes: low-grade gliomas, high-grade gliomas	285	It distribution is similar with BRATS 2015 with four MRI modalities as T1, T1ce, T2, and Flair for each scan.

(*continued on next page*)

Table 9 (*continued*)

Disease	Dataset	Sources	Data types	Classes	Size	Description
	Brain MRI Images for Brain Tumor Detection [190]	Navoneel Chakrabarty	MRI	2 classes: cancerous, non-cancerous	253	It consists of 155 images of malignant cancer and 98 images of benign non-cancerous tumors.
Brain hemorrhage	RSNA Intracranial Hemorrhage dataset [191]	RSNA	CT	5 classes: epidural hemorrhage, intraparenchymal hemorrhage, intraventricular hemorrhage, subdural hemorrhage, subarachnoid hemorrhage	21744	It has 21744 CT scans in total, out of which only 7652 had hemorrhages.
	Aminu Kano Teaching Hospital, Nigeria [36]	Aminu Kano Teaching Hospital	CT	2 classes: hemorrhage, nonhaemorrhage	12635	The images are originally of size 1024*1024*1 pixels, and 12635 images were obtained after augmentation.
Leukemia	ALLIDB1 [192]	Labati et al.	Microscopic image	2 classes: healthy, acute lymphoblastic leukemia	108	It contains 49 ALL affected and 59 healthy images.
	ALLIDB2 [192]	Labati et al.	Microscopic image	2 classes: healthy, acute lymphoblastic leukemia	260	It contains 130 ALL affected and 130 healthy images.
	C_NMC_2019 dataset [193]	Mourya et al.	Microscopic image	2 classes: healthy, acute lymphoblastic leukemia	15135	It contains 10661 training images, 1867 validation images and 2586 test images.
	ASH dataset [37]	ASH	Microscopic image	2 classes: acute myelogenous leukemia, no acute myelogenous leukemia	80	It contains 40 AML and 40 Non-AML images.
Breast cancer	BreakHis dataset [194]	Pathological Anatomy and Cytopathol-ogy, Parana, Brazil	Histopathology image	2 classes: benign, malignant	7909	It consists of 2480 benign and 5429 malignant images collected from 82 patients using different magnifying factors.
	Breast Histopathology Images [195]	Paul Mooney	Histopathology image	2 classes: invasive ductal carcinoma, no invasive ductal carcinoma	277524	It contains 198738 IDC negative and 78786 IDC positive patches of size 50x50, which were extracted from the original dataset consisted of 162 whole mount slide images of Breast Cancer.
	Digital Database for Screening Mammography (DDSM) [196]	University of South Florida	Mammographic image	2 classes: normal, abnormal	2620	It contains 695 cases of nomal and 1925 cases of abnormal.
Cervical disease	Intel & Mobile ODT dataset [197]	Intel and MobileODT companies	Cervigram image	3 classes: type 1, type 2, type 3	8015	It contains 1241 type 1, 4348 type 2, 2426 type 3 images.
	NCI Guanacaste project dataset [107]	American National Cancer Institute	Cervigram image	2 classes: cancer, no cancer	44000	It was collected from 7000 patient visits.
	Herlev dataset [79]	Herlev University Hospital	Cervical cells image	7 classes: superficial squamous epithelial, intermediate squamous epithelial, columnar epithelial, mild dysplasia, moderate dysplasia, severe dysplasia, carcinoma in situ	917	It contains 74, 70, 98, 182, 146, 197, 150 images for seven classes respectively.
Malaria disease	National Institutes of Health(NIH) malaria dataset [198]	National Institutes of Health	Cell image	2 classes: infected, non-infected	27558	It was collected from 150 P.falciparum-infected and 50 healthy patients, with equal instances of parasitized and uninfected cells.
White blood cells	BCCD database [199]	Mooney et al.	Microscopic image	4 classes: eosinophil, neutrophil, lymphocyte, monocyte	12444	It contains two subsets: 367 images before augmentation, and 12444 images after augmentation.
Alzheimer's Disease	ADNI [125]	Dr. Michael W. Weiner	MRI, PET	4 classes: MCI, AD, early MCI, late MCI	/	It contains 485 elderly controls, 551 MCI, 437 AD, 300 early MCI and 150 late MCI.
Cardiac disease	ACDC-2017 dataset [149]	University Hospital of Dijon, France	MRI	5 classes: NOR, MINF, DCM, HCM, ARV	150	It comprises 150 patients evenly distributed across five distinct classes. Training set contains 100 patients and test set contains 50 patients.
	2015 Kaggle data science bowl cardiac challenge data [149]	The National Heart, Lung, and Blood Institute	MRI	Not applicable	1140	It contains 500 patients for training, 200 for validation and 440 for final testing.

well as advantages and disadvantages.

5. Public medical image datasets

Algorithm models and datasets are the two most crucial foundations for medical image classification based on deep learning methods. A large amount of data is required to train and test the proposed models when studying any deep learning method or algorithm. High-quality datasets can help models accurately extract image features for efficient diagnosis and prognosis of various diseases. Earlier, we discussed the research on CNN models and their optimization strategies, and in this section, we will focus on datasets. With the advancement of related research and the accumulation of medical image data, numerous datasets for different diseases and data types have been made publicly available. This section provides an overview of open-source datasets mentioned in the literature, along with relevant descriptions. Some studies also use the private datasets those come from local hospitals or laboratories, making them inaccessible to the public.

In the included literature, the most commonly studied twelve diseases or medical image classification tasks are selected, and the relatively large-scale and representative public datasets are compiled. These twelve diseases or medical image classification tasks are: lung disease, skin lesion, diabetic retinopathy, brain tumor, brain hemorrhage, leukemia, breast cancer, cervical disease, malaria disease, white blood cells, Alzheimer's disease and cardiac disease classification. A total of 38 public datasets are compiled and shown in Table 9, with detailed description of the data type, the number and names of categories, and the size of the dataset.

6. Open research challenges

Medical image classification can assist healthcare professionals in the automatic diagnosis and grading of many diseases. CNN-based medical image classification has developed rapidly and gained popularity in the past decade. Despite CNN has demonstrated excellent performance in computer vision tasks with natural images as classification targets, there are still many challenges in successfully apply it to the field of medical image classification.

Lack of large-scale datasets Deep learning models are typically complex with a large number of parameters, requiring a significant amount of data to train the model and learn useful parameters. Complex models inferred from a limited amount of data often exhibit overfitting and perform poorly on any other datasets. This approach is highly undesirable as it only learns the characteristics of the training data. However, most problems in medical image analysis lack large-scale publicly available datasets. On one hand, this is because medical image data requires labeling by clinical experts, which requires high labor and time costs. On the other hand, medical image sharing is more difficult than sharing natural images due to privacy protection regulations and other issues.

Low data quality Due to the data collected from different institutions and devices, the quality of the collected images is uneven, and there is a lack of certain quality standards, resulting in underutilization of some datasets. Additionally, medical image data suffers from class imbalance issues. In actual medical image datasets, normal samples often far outnumber abnormal samples. For example, the RSNA Intracranial Hemorrhage dataset [85] used for the intracranial hemorrhage detection task contains a total of 21,744 CT scans, out of which only 7652 scans show hemorrhages. The data imbalance problem makes the model to tend to predict the majority class and ignore the minority class, especially in medical image classification where the minority class is often the more important pathological sample. Training CNN networks with imbalanced data can significantly reduce classification performance [200].

Poor model generalization ability The low generalization ability of CNN models in medical image classification exist in multiple aspects. First, the diversity and heterogeneity of medical image data makes a model performs well on a specific type of data but performs poorly on another type of data. Medical images come from different imaging equipment (such as CT, MRI, X-ray machines), hospitals and patients, which results in significant differences in image resolution, contrast, noise level and imaging angle. Even the same type of disease may show up in different shapes, sizes and locations in images from different patients. Therefore, the features learned by the model on the training data may not generalize effectively on different data sets. Secondly, the problems of small data set and data imbalance make the model perform well on training data, but cannot effectively classify when faced with unseen small sample. Limited training samples make it difficult for the CNN model to capture enough features and may overfit the training data. This overfitting problem reduces the generalization ability of the model, especially when dealing with rare lesions in the real world. Thus, it can easily lead to missed diagnosis or misdiagnosis. Thirdly, image noise and artifacts leads to insufficient model generalization ability in the real world. Medical images often contain various noises or artifacts (such as motion artifacts, reflections from metal implants, etc.), which can interfere with the model's feature extraction. Since the image quality in training data is usually higher than the actual data in clinical applications, the model may be insensitive to noise and perform poorly on data that is more affected by noise. Finally, the heterogeneity and inconspicuousness of lesions make the model have weak generalization ability when faced with highly variable lesions. The same type of disease may appear completely different in medical images in different patients, especially in the early stages of the disease where lesion features may be very faint or indistinguishable. This heterogeneity makes it difficult for CNN to learn universal features applicable to various situations from limited data.

Small lesion issue Medical images often encounter the challenge of very small lesion areas, such as microaneurysms in retinal images and microcalcifications with typical diameters less than 20 pixels in breast X-ray images [201]. Locating and detecting small lesions across the entire image space is a unique challenge in medical image classification, distinct from other image classification tasks.

Lack of explainability In medical image processing, understanding how deep learning models make decisions is crucial so that healthcare professionals can explain the decision outcomes to patients and make the most appropriate decisions for diagnosis and treatment. Deep learning approaches in medical image processing are evolving towards an end-to-end manner [1]. Although this approach improves recognition efficiency and accuracy, the black-box nature of deep learning models prevents healthcare professionals from understanding their internal workings. To some extent, this hinders the application of deep learning in medical clinical settings.

Limited computational resources First, high-resolution and large-scale medical image processing requires powerful computing hardware support. Medical images (such as CT, MRI, PET, etc.) are usually high-resolution, and some images (such as 3D images) require processing of large amounts of volumetric data. As the image size increases, the computational workload and memory usage of the convolutional layer will grow rapidly, resulting in significantly longer training and inference times. Second, deep networks lead to computational complexity. In order to improve the classification accuracy of the model, the network layers need to deepen, such as ResNet and DenseNet. These deep networks have millions or even hundreds of millions of parameters. The training process requires more computing resources and time. The memory and storage requirements are also large. The computational efficiency of the model may become low, especially when facing complex medical image classification tasks. Finally, the limitations of mobile devices and edge computing make it difficult to deploy the model. In some medical scenarios, especially in the fields of telemedicine and mobile health monitoring, CNN models need to be deployed on mobile devices or edge devices. However, the computing power, storage capacity, and energy consumption of mobile devices and edge computing

devices are limited, making it difficult to support the operation of large CNN models. Due to hardware limitations, models may need to be compressed or simplified, but this usually sacrifices model accuracy.

7. Future research directions

Literature research on medical image classification based on CNN is relatively abundant, however, certain areas still require further extensive investigation. Based on the challenges mentioned in the previous chapter, we summarize future research directions in this section to provide guiding principles for further studies.

Adopt effective image preprocessing methods Medical images vary in size and are prone to noise and artifacts [48], making preprocessing steps indispensable. Adopting effective preprocessing steps is crucial for improving model performance. Normalization, HE, image augmentation, local region highlighting and segmentation are all proved to be very effective preprocessing methods. Normalization and HE can enhance the contrast in images, making details more apparent and helping models to better recognize and classify important features. They also reduce variability in images from different sources by ensuring a consistent intensity distribution, thereby improving the model's robustness and accuracy. Image augmentation techniques, such as rotation, scaling, translation, and flipping, etc., can generate more diverse training data by applying various transformations to existing images. This not only prevents model overfitting but also enhances the model's generalization ability. Local region highlighting and segmentation method improves the accuracy of the classification by extracting the region of interest. In addition, the data resampling technology shows excellent performance in solving the class imbalance problem, and further research could explore resampling techniques in greater depth.

Optimize transfer learning strategies Transfer learning method hold promise for medical image classification as it requires less training time and is less complex, especially when dealing with limited data. Firstly, features obtained from different transfer learning models can be combined to develop hybrid models [65], which has been proven to be highly effective. Moreover, while using pre-trained models from large-scale natural image datasets has become a popular approach in medical image classification, there are significant differences between medical images and natural images in terms of data size, features, and task specifications, leading to limited performance improvements with transfer learning methods [202]. The more similar the pre-trained dataset is to the target task dataset, the more significant the performance improvement from transfer learning [72]. Therefore, further exploration and research on transfer learning on large-scale medical image datasets are crucial for medical image classification.

Design suitable CNN architectures Achieving higher accuracy is always a primary goal in medical image classification tasks. On one hand, larger and deeper CNN models often achieve better accuracy but can impact diagnostic efficiency and require more significant demands on datasets and computational resources. Moreover, classification accuracy does not always improve linearly with model depth. Once a certain depth is reached, further depth may not enhance accuracy. Furthermore, the nature of small-sample medical images prohibits excessively large and deep models. Therefore, designing a CNN model with the optimal balance between dataset size, model complexity, and classification accuracy to improve model generalization and robustness is a key focus for future exploration.

Combined with different approaches The integration of CNN models with other machine learning methods and optimization algorithms can leverage the strengths of both CNNs and other methods to construct more effective solutions for medical image classification. For instance, the transfer learning model is used as the feature extractor and the traditional machine learning method is used as the classifier [37,54,146,148,149]. Federated learning techniques can address data privacy concerns [100]. CNN models based on federated learning classify images more accurately and preserve privacy. Efficient and robust models can be built by combing Transformer and CNN [151], leveraging CNN's local features and Transformer's global connectivity. Optimization algorithms like WOA [153], GA [154] and may be other optimization algorithms can be employed to search for the optimal architecture of CNN, jointly optimizing all model parameters in the deep neural network pipeline.

Utilize automatic machine learning technology The automatic machine learning technology of neural network models can reduce human interference and quickly and accurately achieve a balance between model complexity, efficiency and accuracy. For example, NAS, which automatically designs a high-performance neural network architecture to solve specific tasks in the search space based on a customized search strategy, is a useful tool for optimizing the image classification and hyperparameter tuning process [104]. Building CNN models based on NAS will be a hot topic in future research.

Research on unsupervised learning technology Since high-quality, annotated medical image data is very scarce, unsupervised learning can mine potential features in the data without a large amount of annotated data, which provides potential for solving the difficulty of medical image annotation. There are two main potential unsupervised learning technologies, self-supervised learning and autoencoders. Self-supervised learning pre-trains the CNN model by generating pseudo labels, and then fine-tunes it with small-scale annotated data. For example, pseudo labels are generated through tasks such as image rotation, shearing, and reconstruction. The model learns image features through these tasks, so that it can learn effectively without annotation. Existing self-supervised learning methods include SimCLR [203] and MoCo [204]. Encoders are a typical unsupervised learning method that can learn compressed representations of images and extract key features from them. Variational autoencoders [205] and sparse autoencoders [206] have been used for anomaly detection in medical images, such as detecting abnormal lung images or early cancer lesion areas. However, unsupervised learning technology also faces some challenges. On the one hand, medical images contain many complex biological structures and lesions, and the distribution of these features is highly uneven, making it difficult for the model to capture useful patterns in unsupervised learning. On the other hand, unsupervised learning models usually lack clear evaluation criteria, especially in the field of medical image classification. How to measure the performance of the model is an urgent problem to be solved.

Adopt multi-modal data fusion Medical images can come from a variety of different imaging technologies, and the images provided by each imaging technology contain different information. Single-modal images often cannot fully reflect complex pathological conditions, while multi-modal data can provide complementary information from different perspectives to help the model better understand the condition. multi-modal data fusion can combine different types of medical information such as imaging data, genetic data, and pathological data to help doctors make more accurate diagnostic decisions in complex cases. multi-modal data fusion methods include multi-modal fusion based on CNN, multi-modal fusion based on attention mechanism, and joint learning. The multi-modal fusion method based on CNN designs a multi-input CNN model, inputs images from different modalities at the same time, uses parallel convolutional networks to process the features of each modality separately, and finally fuses them at the feature layer. The multi-modal fusion method based on attention mechanism determines which modality information is more important in the current classification task through the attention mechanism, thereby improving the flexibility and robustness of the model. Joint learning methods learn the representation of cross-modal features in the encoding stage by sharing the encoder part, and then processe the output of each modality through independent decoders. This not only retains the characteristics of each modality, but also shares cross-modal information.

Improve model explainability Understanding the prediction process and obtaining its inference rules is the next important research topic. Feature visualization methods such as CAM, Grad-CAM, and Grad-

CAM++ proposed in the current literature show the sensitivity of features to a certain class, which is helpful to model interpretability. But these are all "post-hoc" techniques which have many disadvantages as described above. To overcome the intrinsic limitations of "post-hoc" explainability techniques, future research could explore "ante-hoc" interpretability or interpretability-by-design approaches. These methods incorporate explainability directly into the model structure, allowing for explanations to emerge naturally during the prediction process. For example, part-prototype models [207] are interpretable-by-design neural networks that classify images by identifying class-representative prototypical parts in the input. These models simulate human-like reasoning processes, providing faithful and transparent explanations for their predictions. Unlike "post-hoc" techniques, "ante-hoc" models offer a more reliable framework for decision-making, particularly in high-stakes fields such as healthcare, where trustworthy and interpretable predictions are indispensable.

Optimize computing efficiency Lightweight CNN models with smaller volume and computational load make neural network models more efficient [106,107]. Compared with complex deep networks, the requirements for device storage capacity and computing power are relatively lower. Additionally, the distributed training method can effectively increase the computing speed and reduce the demand for computing resources. It can be further explored in the future.

Through a systematic review of published literature in this field, it was found that although CNN has achieved good results in medical image classification, there is still much room for improvement and further exploration. We hope that this systematic literature review will also help other researchers conduct related research to improve deep learning-based medical image processing technology.

8. Conclusion

The latest literature of the application of CNNs in medical image classification is reviewed in this article, providing a comprehensive overview of existing work. We find that CNN deep learning models can be applied to various areas of medical disease diagnosis, such as lung diseases, skin diseases, diabetic retinopathy, brain tumors, brain hemorrhage, leukemia, breast cancer, and other diseases. Through an in-depth review of six main techniques used: data preprocessing, transfer learning, variants of CNN, novel structure, ensemble strategy, and explainability, conclusions can be drawn as follows: Firstly, due to the diverse formats and sizes of medical images, image preprocessing is an indispensable step in medical image analysis, helping standardize initialized data and improve model performance. Secondly, most existing methods for medical image classification rely on transfer learning which is an effective strategy for small datasets. Even though novel architectures are proposed, transfer learning on large datasets is still considered due to the scarcity of available image data. However, since transfer learning models are trained on natural images, they may not always yield the expected results. Thirdly, in terms of CNN structure, existing structures based on transfer learning methods can be directly adopted, while improved versions of CNN structures and proposed new structures are also viable options. Additionally, ensemble strategies are also a good choice for optimizing models and improving performance. Fourthly, model interpretability can help medical personnel understand the principles and processes of decision-making, increasing confidence in decisions among medical personnel and patients. This is an important research direction in medical image classification based on deep learning methods. Furthermore, we summary the commonly used public datasets for major research areas for reference of researchers. Finally, the main challenges faced by CNNs in medical image analysis are discussed, along with future directions for addressing these challenges.

Medical image analysis using deep learning is a vast field that help radiologists accurately predict lesion areas and further classify them into respective types. The CNN method is one of the most studied deep learning methods in medical image analysis. Current research still faces many challenges, also providing good opportunities and broad research space for the academic community. In order to push deep learning methods into clinical applications and smoothly integrate them with smart healthcare, we still have a lot of work to do.

CRediT authorship contribution statement

Chao Chen: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nor Ashidi Mat Isa:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition. **Xin Liu:** Writing – review & editing, Software, Resources, Investigation, Conceptualization.

Declaration of competing interest

None Declared.

References

[1] G. Litjens, T. Kooi, B.E. Bejnordi, A.A.A. Setio, F. Ciompi, M. Ghafoorian, J.A. Van Der Laak, B. Van Ginneken, C.I. Sánchez, A survey on deep learning in medical image analysis, Med. Image Anal. 42 (2017) 60–88.

[2] K. Muhammad, S. Khan, J. Del Ser, V.H.C. de Albuquerque, Deep learning for multigrade brain tumor classification in smart healthcare systems: a prospective survey, IEEE Transact. Neural Networks Learn. Syst. 32 (2) (2021) 507–522.

[3] M.M. Adnan, M.S.M. Rahim, A.R. Khan, A. Alkhayyat, F.S. Alamri, T. Saba, S. A. Bahaj, Automated image annotation with novel features based on deep ResNet50-SLT, IEEE Access (2023).

[4] N. Dong, Q.Y. Feng, M.D. Zhai, J.F. Chang, X.M. Mai, A novel feature fusion based deep learning framework for white blood cell classification, J. Ambient Intell. Hum. Comput. (2022).

[5] M. Micucci, A. Iula, Recent advances in machine learning applied to ultrasound imaging, Electronics 11 (11) (2022) 1800.

[6] S.S. Yadav, S.M. Jadhav, Deep convolutional neural network based medical image classification for disease diagnosis, Journal of Big Data 6 (1) (2019).

[7] T. Mahmood, A. Rehman, T. Saba, L. Nadeem, S.A.O. Bahaj, Recent advancements and future prospects in active deep learning for medical image segmentation and classification, IEEE Access (2023).

[8] Y. Guo, Y. Liu, A. Oerlemans, S. Lao, S. Wu, M.S. Lew, Deep learning for visual understanding: a review, Neurocomputing 187 (2016) 27–48.

[9] Y. LeCun, L. Bottou, Y. Bengio, P. Haffner, Gradient-based learning applied to document recognition, Proc. IEEE 86 (11) (1998) 2278–2324.

[10] A. Krizhevsky, I. Sutskever, G.E. Hinton, Imagenet classification with deep convolutional neural networks, Adv. Neural Inf. Process. Syst. 25 (2012).

[11] K. Simonyan, A. Zisserman, Very Deep Convolutional Networks for Large-Scale Image Recognition, 2014 arXiv preprint arXiv:1409.1556.

[12] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, A. Rabinovich, Going deeper with convolutions, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015.

[13] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016.

[14] G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017.

[15] A.G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, H. Adam, *Mobilenets: Efficient Convolutional Neural Networks for Mobile Vision Applications.* arXiv Preprint arXiv:1704.04861, 2017.

[16] M. Tan, Q. Le Efficientnet, Rethinking model scaling for convolutional neural networks, in: International Conference on Machine Learning, PMLR, 2019.

[17] N. Subramanian, O. Elharrouss, S. Al-Maadeed, M. Chowdhury, A review of deep learning-based detection methods for COVID-19, Comput. Biol. Med. (2022) 143.

[18] R. Kumar, P. Kumbharkar, S. Vanam, S. Sharma, Medical images classification using deep learning: a survey, Multimed. Tool. Appl. (2023).

[19] H. Hassan, Z.Y. Ren, H.S. Zhao, S.J. Huang, D. Li, S.H. Xiang, Y. Kang, S.F. Chen, B.D. Huang, Review and classification of AI-enabled COVID-19 CT imaging models based on computer vision tasks, Comput. Biol. Med. 141 (2022).

[20] T. Majtner and E.S. Nadimi. Comparison of Deep Learning-Based Recognition Techniques for Medical and Biomedical Images.

[21] M.J. Page, J.E. McKenzie, P.M. Bossuyt, I. Boutron, T.C. Hoffmann, C.D. Mulrow, L. Shamseer, J.M. Tetzlaff, E.A. Akl, S.E. Brennan, The PRISMA 2020 statement: an updated guideline for reporting systematic reviews, Br. Med. J. 372 (2021).

[22] V. Narayan, P.K. Mall, A. Alkhayyat, K. Abhishek, S. Kumar, P. Pandey, Enhance-net: an approach to Boost the performance of deep learning model based on real-time medical images, J. Sens. 2023 (2023).

[23] X. Qi, L.G. Brown, D.J. Foran, J. Nosher, I. Hacihaliloglu, Chest X-ray image phase features for improved diagnosis of COVID-19 using convolutional neural network, Int. J. Comput. Assist. Radiol. Surg. 16 (2) (2021) 197–206.

[24] I. Avci, A. Alzabaq, A new respiratory diseases detection model in chest X-ray images using CNN, Trait. Du. Signal 40 (1) (2023) 145–155.
[25] A. Souid, N. Sakli, H. Sakli, Classification and predictions of lung diseases from chest X-rays using MobileNet V2, Applied Sciences-Basel 11 (6) (2021).
[26] A.A. Reshi, F. Rustam, A. Mehmood, A. Alhossan, Z. Alrabiah, A. Ahmad, H. Alsuwailem, G.S. Choi, An efficient CNN model for COVID-19 disease detection based on X-ray image classification, Complexity (2021) 2021.
[27] A. Bria, C. Marrocco, F. Tortorella, Addressing class imbalance in deep learning for small lesion detection on medical images, Comput. Biol. Med. 120 (2020).
[28] H. Mzoughi, I. Njeh, A. Wali, M. Ben Slima, A. BenHamida, C. Mhiri, K. Ben Mahfoudhe, Deep multi-scale 3D convolutional neural network (CNN) for MRI gliomas brain tumor classification, J. Digit. Imag. 33 (4) (2020) 903–915.
[29] H. Sun, X.X. Zeng, T. Xu, G. Peng, Y.T. Ma, Computer-aided diagnosis in histopathological images of the endometrium using a convolutional neural network and attention mechanisms, Ieee Journal of Biomedical and Health Informatics 24 (6) (2020) 1664–1676.
[30] S.P. Singh, L.P. Wang, S. Gupta, B. Gulyas, P. Padmanabhan, Shallow 3D CNN for detecting acute brain hemorrhage from medical imaging sensors, IEEE Sensor. J. 21 (13) (2021) 14290–14299.
[31] J. Ker, S.P. Singh, Y.Q. Bai, J. Rao, T. Lim, L.P. Wang, Image thresholding improves 3-dimensional convolutional neural network diagnosis of different acute brain hemorrhages on computed tomography scans, Sensors 19 (9) (2019).
[32] A. Gautam, B. Raman, Towards effective classification of brain hemorrhagic and ischemic stroke using CNN, Biomed. Signal Process Control 63 (2021).
[33] Y. Zhang, C. Huneau, J. Idier, D. Mateus, Ultrasound image reconstruction with denoising diffusion restoration models, in: Deep Generative Models, Springer Nature Switzerland, Cham, 2024.
[34] D.J. Hemanth, O. Deperlioglu, U. Kose, An enhanced diabetic retinopathy detection and classification approach using deep convolutional neural network, Neural Comput. Appl. 32 (3) (2020) 707–721.
[35] M.A. Al-masni, D.H. Kim, T.S. Kim, Multiple skin lesions diagnostics via integrated deep convolutional networks for segmentation and classification, Comput. Methods Progr. Biomed. (2020) 190.
[36] A.M. Dawud, K. Yurtkan, H. Oztoprak, Application of deep learning in neuroradiology: brain haemorrhage classification using transfer learning, Comput. Intell. Neurosci. 2019 (2019).
[37] P.K. Das, B. Sahoo, S. Meher, An efficient detection and classification of acute leukemia using transfer learning and orthogonal softmax layer-based model, IEEE ACM Trans. Comput. Biol. Bioinf 20 (3) (2023) 1817–1828.
[38] K.C. Kamal, Z.D. Yin, M.Y. Wu, Z.L. Wu, Evaluation of deep learning-based approaches for COVID-19 classification based on chest X-ray images, Signal Image and Video Processing 15 (5) (2021) 959–966.
[39] I.D. Apostolopoulos, T.A. Mpesiana, Covid-19: automatic detection from X-ray images utilizing transfer learning with convolutional neural networks, Physical and Engineering Sciences in Medicine 43 (2) (2020) 635–640.
[40] Y.X. Tang, Y.B. Tang, Y.F. Peng, K. Yan, M. Bagheri, B.A. Redd, C.J. Brandon, Z. Y. Lu, M. Han, J. Xiao, R.M. Summers, Automated abnormality classification of chest radiographs using deep convolutional neural networks, Npj Digital Medicine 3 (1) (2020).
[41] L. Gaur, U. Bhatia, N.Z. Jhanjhi, G. Muhammad, M. Masud, Medical image-based detection of COVID-19 using deep convolution neural networks, Multimed. Syst. 29 (3) (2023) 1729–1738.
[42] T. Zebin, S. Rezvy, COVID-19 detection and disease progression visualization: deep learning on chest X-rays for classification and coarse localization, Appl. Intell. 51 (2) (2021) 1010–1021.
[43] V. Chouhan, S.K. Singh, A. Khamparia, D. Gupta, P. Tiwari, C. Moreira, R. Damasevicius, V.H.C. de Albuquerque, A novel transfer learning based approach for pneumonia detection in chest X-ray images, Applied Sciences-Basel 10 (2) (2020).
[44] A. Hernández, A. Panizo and D. Camacho. An Ensemble Algorithm Based on Deep Learning for Tuberculosis Classification.
[45] I.D. Apostolopoulos, S.I. Aznaouridis, M.A. Tzani, Extracting possibly representative COVID-19 biomarkers from X-ray images with deep learning approach and image data related to pulmonary diseases, J. Med. Biol. Eng. 40 (3) (2020) 462–469.
[46] G. Marques, D. Agarwal, I.D. Díez, Automated medical diagnosis of COVID-19 through EfficientNet convolutional neural network, Appl. Soft Comput. 96 (2020).
[47] A. Moussaid, N. Zrira, I. Benmiloud, Z. Farahat, Y. Karmoun, Y. Benzidia, S. Mouline, B. El Abdi, J.E. Bourkadi, N. Ngote, On the implementation of a post-pandemic deep learning algorithm based on a hybrid CT-scan/X-ray images classification applied to pneumonia categories, Healthcare 11 (5) (2023).
[48] M.J. Horry, S. Chakraborty, M. Paul, A. Ulhaq, B. Pradhan, M. Saha, N. Shukla, COVID-19 detection through transfer learning using multimodal imaging data, IEEE Access 8 (2020) 149808–149824.
[49] M. Talo, O. Yildirim, U.B. Baloglu, G. Aydin, U.R. Acharya, Convolutional neural networks for multi-class brain disease detection using MRI images, Comput. Med. Imag. Graph. 78 (2019).
[50] Ö. Polat, C. Güngen, Classification of brain tumors from MR images using deep transfer learning, J. Supercomput. 77 (7) (2021) 7236–7252.
[51] N. Noreen, S. Palaniappan, A. Qayyum, I. Ahmad, M. Imran, M. Shoaib, A deep learning model based on concatenation approach for the diagnosis of brain tumor, IEEE Access 8 (2020) 55135–55144.
[52] M. Sajjad, S. Khan, K. Muhammad, W.Q. Wu, A. Ullah, S.W. Baik, Multi-grade brain tumor classification using deep CNN with extensive data augmentation, Journal of Computational Science 30 (2019) 174–182.
[53] Z.N.K. Swati, Q.H. Zhao, M. Kabir, F. Ali, Z. Ali, S. Ahmed, J.F. Lu, Brain tumor classification for MR images using transfer learning and fine-tuning, Comput. Med. Imag. Graph. 75 (2019) 34–46.
[54] S. Deepak, P.M. Ameer, Brain tumor classification using deep CNN features via transfer learning, Comput. Biol. Med. 111 (2019).
[55] H. Aljuaid, N. Alturki, N. Alsubaie, L. Cavallaro, A. Liotta, Computer-aided diagnosis for breast cancer classification using deep neural networks and transfer learning, Comput. Methods Progr. Biomed. (2022) 223.
[56] T. Choudhary, V. Mishra, A. Goswami, J. Sarangapani, A transfer learning with structured filter pruning approach for improved breast cancer classification on point-of-care devices, Comput. Biol. Med. 134 (2021).
[57] R.O. Ogundokun, S. Misra, A.O. Akinrotimi, H. Ogul, MobileNet-SVM: a lightweight deep transfer learning model to diagnose BCH scans for IoMT-based imaging sensors, Sensors 23 (2) (2023).
[58] M. Humayun, M.I. Khalil, S.N. Almuayqil, N.Z. Jhanjhi, Framework for detecting breast cancer risk presence using deep learning, Electronics 12 (2) (2023).
[59] R.V.M. da Nóbrega, P.P. Rebouças, M.B. Rodrigues, S.P.P. da Silva, C. Dourado, V. H.C. de Albuquerque, Lung nodule malignancy classification in chest computed tomography images using transfer learning and convolutional neural networks, Neural Comput. Appl. 32 (15) (2020) 11065–11082.
[60] H. Ko, H. Chung, W.S. Kang, K.W. Kim, Y. Shin, S.J. Kang, J.H. Lee, Y.J. Kim, N. Y. Kim, H. Jung, J. Lee, COVID-19 pneumonia diagnosis using a simple 2D deep learning framework with a single chest CT image: model development and validation, J. Med. Internet Res. 22 (6) (2020).
[61] A. Abhishek, R.K. Jha, R. Sinha, K. Jha, Automated detection and classification of leukemia on a subject-independent test dataset using deep transfer learning supported by Grad-CAM visualization, Biomed. Signal Process Control 83 (2023).
[62] A. Vijayalakshmi, B.R. Kanna, Deep learning approach to detect malaria from microscopic images, Multimed. Tool. Appl. 79 (21–22) (2020) 15297–15317.
[63] Z.T. Cao, L.X. Duan, G.W. Yang, T. Yue, Q. Chen, An experimental study on breast lesion detection and classification from ultrasound images using deep learning architectures, BMC Med. Imag. 19 (2019).
[64] H. Tariq, M. Rashid, A. Javed, E. Zafar, S.S. Alotaibi, M.Y.I. Zia, Performance analysis of deep-neural-network-based automatic diagnosis of diabetic retinopathy, Sensors 22 (1) (2022).
[65] A. Mahbod, G. Schaefer, C.L. Wang, G. Dorffner, R. Ecker, I. Ellinger, Transfer learning using a multi-scale and multi-network ensemble for skin lesion classification, Comput. Methods Progr. Biomed. (2020) 193.
[66] T. Alsubait, T. Sindi, H. Alhakami, Classification of the human protein atlas single cell using deep learning, Applied Sciences-Basel 12 (22) (2022).
[67] P. Marcelino, Transfer learning from pre-trained models, Data Sci. 10 (2018) 23.
[68] C. Li, Y.Y. Yang, H. Liang, B.Y. Wu, Transfer learning for establishment of recognition of COVID-19 on CT imaging using small-sized training datasets, Knowl. Base Syst. (2021) 218.
[69] G.B. Liang, L.X. Zheng, A transfer learning method with deep residual network for pediatric pneumonia diagnosis, Comput. Methods Progr. Biomed. (2020) 187.
[70] A.H. Alharbi, H.A.H. Mahmoud, Pneumonia transfer learning deep learning model from segmented X-rays, Healthcare 10 (6) (2022).
[71] M. Varma, M. Lu, R. Gardner, J. Dunnmon, N. Khandwala, P. Rajpurkar, J. Long, C. Beaulieu, K. Shpanskaya, F.F. Li, M.P. Lungren, B.N. Patel, Automated abnormality detection in lower extremity radiographs using deep learning, Nat. Mach. Intell. 1 (12) (2019) 578–583.
[72] M.E.H. Chowdhury, T. Rahman, A. Khandakar, R. Mazhar, M.A. Kadir, Z. Bin Mahbub, K.R. Islam, M.S. Khan, A. Iqbal, N. Al Emadi, M.B. Reaz, M.T. Islam, Can AI help in screening viral and COVID-19 pneumonia? IEEE Access 8 (2020) 132665–132676.
[73] Z.Y. Lu, Y.Z. Bai, Y. Chen, C.Q. Su, S.S. Lu, T.M. Zhan, X.N. Hong, S.H. Wang, The classification of gliomas based on a Pyramid dilated convolution resnet model, Pattern Recogn. Lett. 133 (2020) 173–179.
[74] D. Sarwinda, R.H. Paradisa, A. Bustamam and P. Anggia. Deep Learning in Image Classification Using Residual Network (ResNet) Variants for Detection of Colorectal Cancer.
[75] S. Karakanis, G. Leontidis, Lightweight deep learning models for detecting COVID-19 from chest X-ray images, Comput. Biol. Med. 130 (2021).
[76] J.L. Cheng, S.W. Tian, L. Yu, C.R. Gao, X.J. Kang, X. Ma, W.D. Wu, S.J. Liu, H. C. Lu, ResGANet: residual group attention network for medical image classification and segmentation, Med. Image Anal. 76 (2022).
[77] J. Wang, Y.M. Bao, Y.F. Wen, H.B. Lu, H. Luo, Y.F. Xiang, X.M. Li, C. Liu, D. H. Qian, Prior-attention residual learning for more discriminative COVID-19 screening in CT images, IEEE Trans. Med. Imag. 39 (8) (2020) 2572–2583.
[78] X.L. Luo, W. Wang, Y. Xu, Z.H. Lai, X.P. Jin, B.B. Zhang, D.V. Zhang, A deep convolutional neural network for diabetic retinopathy detection via mining local and long-range dependence, Caai Transactions on Intelligence Technology (2023).
[79] N. Dong, L. Zhao, C.H. Wu, J.F. Chang, Inception v3 based cervical cell classification combined with artificially extracted features, Appl. Soft Comput. 93 (2020).
[80] Y.X. Li, X.P. Xie, L.L. Shen, S.X. Liu, Reverse active learning based atrous DenseNet for pathological image classification, BMC Bioinf. 20 (1) (2019).
[81] G.Z. Zhao, S. Shao, M. Yu, Key techniques for classification of thorax diseases based on deep learning, Int. J. Imag. Syst. Technol. 32 (6) (2022) 2184–2197.
[82] H. Munusamy, K.J. Muthukumar, S. Gnanaprakasam, T.R. Shanmugakani, A. Sekar, FractalCovNet architecture for COVID-19 chest X-ray image classification and CT-scan image segmentation, Biocybern. Biomed. Eng. 41 (3) (2021) 1025–1038.

[83] H.Z. Zhu, R. Rohling and S. Salcudean. Jointly Boosting Saliency Prediction and Disease Classification on Chest X-Ray Images with Multi-Task UNet.
[84] S. Akter, F. Shamrat, S. Chakraborty, A. Karim, S. Azam, COVID-19 detection using deep learning algorithm on chest X-ray images, Biology-Basel 10 (11) (2021).
[85] P. Lankireddy, C. Sindhura and S. Gorthi. A New Lightweight Architecture and a Class Imbalance Aware Loss Function for Multi-Label Classification of Intracranial Hemorrhages.
[86] M. Polsinelli, L. Cinque, G. Placidi, A light CNN for detecting COVID-19 from CT scans of the chest, Pattern Recogn. Lett. 140 (2020) 95–100.
[87] Y.T. Zhang, H.M. Li, J. Du, J. Qin, T.F. Wang, Y. Chen, B. Liu, W.W. Gao, G.L. Ma, B.Y. Lei, 3D multi-attention guided multi-task learning network for automatic gastric tumor segmentation and lymph node classification, IEEE Trans. Med. Imag. 40 (6) (2021) 1618–1631.
[88] Y.Y. Baydilli, Ü. Atila, Classification of white blood cells using capsule networks, Comput. Med. Imag. Graph. 80 (2020).
[89] D. Singh, V. Kumar, V. Yadav, M. Kaur, Deep neural network-based screening model for COVID-19-infected patients using chest X-ray images, Int. J. Pattern Recogn. Artif. Intell. 35 (3) (2021).
[90] M.Z. Alom, C. Yakopcic, S. Nasrin, T.M. Taha, V.K. Asari, Breast cancer classification from histopathological images with inception recurrent residual convolutional neural network, J. Digit. Imag. 32 (4) (2019) 605–617.
[91] A. Abbas, M.M. Abdelsamea, M.M. Gaber, Classification of COVID-19 in chest X-ray images using DeTraC deep convolutional neural network, Appl. Intell. 51 (2) (2021) 854–864.
[92] S. Liang, H.X. Liu, Y. Gu, X.H. Guo, H.J. Li, L. Li, Z.Y. Wu, M.Y. Liu, L.X. Tao, Fast automated detection of COVID-19 from medical images using convolutional neural networks, Commun. Biol. 4 (1) (2021).
[93] M.M. Eltoukhy, K.M. Hosny, M.A. Kassem, Classification of multiclass histopathological breast images using residual deep learning, Comput. Intell. Neurosci. 2022 (2022).
[94] H. Malik, T. Anees, M. Din, A. Naeem, CDC_Net: multi-classification convolutional neural network model for detection of COVID-19, pneumothorax, pneumonia, lung Cancer, and tuberculosis using chest X-rays, Multimed. Tool. Appl. 82 (9) (2023) 13855–13880.
[95] H.D. Zhu, J. Wang, S.H. Wang, R. Raman, J.M. Górriz, Y.D. Zhang, An evolutionary attention-based network for medical image classification, Int. J. Neural Syst. 33 (3) (2023).
[96] X.Y. Zhang, W. Li, C.Z. Gao, Y. Yang, K. Chang, Hyperspectral pathology image classification using dimension-driven multi-path attention residual network, Expert Syst. Appl. (2023) 230.
[97] S. Das, K. Kharbanda, M. Suchetha, R. Raman, D.E. Dhas, Deep learning architecture based on segmented fundus image features for classification of diabetic retinopathy, Biomed. Signal Process Control 68 (2021).
[98] D.J.B. Saini, R. Sivakami, R. Venkatesh, C.S. Raghava, P.S. Dwarkanath, T.M. K. Anwer, L.K. Smirani, S.H. Ahammad, U. Pamula, M.A. Hossain, A.N.Z. Rashed, Convolution neural network model for predicting various lesion-based diseases in diabetic macula edema in optical coherence tomography images, Biomed. Signal Process Control 86 (2023).
[99] L. Li, M. Xu, X.F. Wang, L. Jiang, H.R. Liu and I.C. Soc. Attention Based Glaucoma Detection: A Large-Scale Database and CNN Model.
[100] M.N. Hossen, V. Panneerselvam, D. Koundal, K. Ahmed, F.M. Bui, S.M. Ibrahim, Federated machine learning for detection of skin diseases and enhancement of Internet of medical Things (IoMT) security, Ieee Journal of Biomedical and Health Informatics 27 (2) (2023) 835–841.
[101] S. Sreelakshmi, V.S. Anoop, A deep convolutional neural network model for medical data classification from computed tomography images, Expet Syst. (2023).
[102] T. Elsken, J.H. Metzen, F. Hutter, Neural architecture search: a survey, J. Mach. Learn. Res. 20 (1) (2019) 1997–2017.
[103] B. Zoph, Q.V. Le, Neural architecture search with reinforcement learning, arXiv preprint arXiv:1611.01578 (2016).
[104] O. Eminaga, M. Abbas, J.N. Shen, M. Laurie, J.D. Brooks, J.C. Liao, D.L. Rubin, PlexusNet: a neural network architectural concept for medical image classification, Comput. Biol. Med. 154 (2023).
[105] S.J. Xu, H. Wu, R.F. Bie, CXNet-m1: anomaly detection on chest X-rays with image-based deep learning, IEEE Access 7 (2019) 4466–4477.
[106] M. Nahiduzzaman, M.R. Islam, R. Hassan, ChestX-Ray6: prediction of multiple diseases including COVID-19 from chest X-ray images using convolutional neural network, Expert Syst. Appl. (2023) 211.
[107] Z. Alyafeai, L. Ghouti, A fully-automated deep learning pipeline for cervical cancer classification, Expert Syst. Appl. 141 (2020).
[108] M. Umer, I. Ashraf, S. Ullah, A. Mehmood, G.S. Choi, COVINet: a convolutional neural network approach for predicting COVID-19 from chest X-ray images, J. Ambient Intell. Hum. Comput. 13 (1) (2022) 535–547.
[109] M.M. Badza, M.C. Barjaktarovic, Classification of brain tumors from MRI images using a convolutional neural network, Applied Sciences-Basel 10 (6) (2020).
[110] N. Sengar, R.C. Joshi, M.K. Dutta, R. Burget, EyeDeep-Net: a multi-class diagnosis of retinal diseases using deep neural network, Neural Comput. Appl. 35 (14) (2023) 10551–10571.
[111] H. Bonab, F. Can, Less is more: a comprehensive framework for the number of components of ensemble classifiers, IEEE Transact. Neural Networks Learn. Syst. 30 (9) (2019) 2735–2745.
[112] W. Zhang, J. Zhong, S.J. Yang, Z.T. Gao, J.J. Hu, Y.Y. Chen, Z. Yi, Automated identification and grading system of diabetic retinopathy using deep neural networks, Knowl. Base Syst. 175 (2019) 12–25.
[113] P.K. Das, S. Meher, An efficient deep convolutional neural network based detection and classification of acute lymphoblastic leukemia, Expert Syst. Appl. 183 (2021).
[114] M. Rahimzadeh, A. Attar, S.M. Sakhaei, A fully automated deep learning-based network for detecting COVID-19 from a new and large lung CT scan dataset, Biomed. Signal Process Control 68 (2021).
[115] T. Nazir, M. Nawaz, J. Rashid, R. Mahum, M. Masood, A. Mehmood, F. Ali, J. Kim, H.Y. Kwon, A. Hussain, Detection of diabetic eye disease from retinal images using a deep learning based CenterNet model, Sensors 21 (16) (2021).
[116] C.C. Ukwuoma, D.S. Cai, M.B. Bin Heyat, O. Bamisile, H. Adun, Z. Al-Huda, M. A. Al-antari, Deep learning framework for rapid and accurate respiratory COVID-19 prediction using chest X-ray images, Journal of King Saud University-Computer and Information Sciences 35 (7) (2023).
[117] A.K. Das, S. Ghosh, S. Thunder, R. Dutta, S. Agarwal, A. Chakrabarti, Automatic COVID-19 detection from X-ray images using ensemble learning with convolutional neural network, Pattern Anal. Appl. 24 (3) (2021) 1111–1124.
[118] M.E. Karar, E. El-Din Hemdan, M.A. Shouman, Cascaded deep learning classifiers for computer-aided diagnosis of COVID-19 and pneumonia diseases in X-ray scans, Complex & Intelligent Systems 7 (1) (2021) 235–247.
[119] H. Lee, S. Yune, M. Mansouri, M. Kim, S.H. Tajmir, C.E. Guerrieri, S.A. Ebert, S. R. Pomerantz, J.M. Romero, S. Kamalian, R.G. Gonzalez, M.H. Lev, S. Do, An explainable deep-learning algorithm for the detection of acute intracranial haemorrhage from small datasets, Nat. Biomed. Eng. 3 (3) (2019), 173-+.
[120] S. Qummar, F.G. Khan, S. Shah, A. Khan, S. Shamshirband, Z.U. Rehman, I. A. Khan, W. Jadoon, A deep learning ensemble approach for diabetic retinopathy detection, IEEE Access 7 (2019) 150530–150539.
[121] P. Gifani, A. Shalbaf, M. Vafaeezadeh, Automated detection of COVID-19 using ensemble of transfer learning with deep convolutional neural network based on CT scans, Int. J. Comput. Assist. Radiol. Surg. 16 (1) (2021) 115–123.
[122] S.M. Lee, N. Kim, Deep learning model ensemble for the accuracy of classification degenerative arthritis, Cmc-Computers Materials & Continua 75 (1) (2023) 1981–1994.
[123] J.P. Zhang, Y.T. Xie, Q. Wu, Y. Xia, Medical image classification using synergic deep learning, Med. Image Anal. 54 (2019) 10–19.
[124] X. Xu, L.L. Zhang, J.Q. Li, Y. Guan, L. Zhang, A hybrid global-local representation CNN model for automatic cataract grading, Ieee Journal of Biomedical and Health Informatics 24 (2) (2020) 556–567.
[125] F.P. An, Medical image classification algorithm based on weight initialization-sliding window fusion convolutional neural network, Complexity (2019) 2019.
[126] N. Ahmad, S. Asghar, S.A. Gillani, Transfer learning-assisted multi-resolution breast cancer histopathological images classification, Vis. Comput. 38 (8) (2022) 2751–2770.
[127] P.K. Das, B. Nayak, S. Meher, A lightweight deep learning system for automatic detection of blood cancer, Measurement (2022) 191.
[128] S. Deepak, P.M. Ameer, Automated categorization of brain tumor from MRI using CNN features and SVM, J. Ambient Intell. Hum. Comput. 12 (8) (2021) 8357–8369.
[129] P. Kedia Anjum, R. Katarya, CoVNet-19: a Deep Learning model for the detection and analysis of COVID-19 patients, Appl. Soft Comput. (2021) 104.
[130] A.D.Y. Jia, B.Z.Y. Li, C.C.W. Zhang, Detection of cervical cancer cells based on strong feature CNN-SVM network, Neurocomputing 411 (2020) 112–127.
[131] G.S. Nijaguna, J.A. Babu, B.D. Parameshachari, R.P. de Prado, J. Frnda, Quantum Fruit fly algorithm and ResNet50-VGG16 for medical diagnosis, Appl. Soft Comput. (2023) 136.
[132] M.A. Khan, M. Alhaisoni, U. Tariq, N. Hussain, A. Majid, R. Damasevicius, R. Maskeliunas, COVID-19 case recognition from chest CT images by deep learning, entropy-controlled firefly optimization, and parallel feature fusion, Sensors 21 (21) (2021).
[133] W.Q. Jin, S.Q. Dong, C.Z. Dong, X.D. Ye, Hybrid ensemble model for differential diagnosis between COVID-19 and common viral pneumonia by chest X-ray radiograph, Comput. Biol. Med. 131 (2021).
[134] Y.H. Li, N.N. Yeh, S.J. Chen, Y.C. Chung, Computer-Assisted diagnosis for diabetic retinopathy based on fundus images using deep convolutional neural network, Mobile Inf. Syst. (2019), 2019.
[135] G.-B. Huang, Q.-Y. Zhu, C.-K. Siew, Extreme learning machine: theory and applications, Neurocomputing 70 (1–3) (2006) 489–501.
[136] M. Nahiduzzaman, M.R. Islam, M.O.F. Goni, M.S. Anower, M. Ahsan, J. Haider, M. Kowalski, Diabetic retinopathy identification using parallel convolutional neural network based feature extractor and ELM classifier, Expert Syst. Appl. 217 (2023).
[137] M. Nahiduzzaman, M.O.F. Goni, R. Hassan, M.R. Islam, M.K. Syfullah, S. M. Shahriar, M.S. Anower, M. Ahsan, J. Haider, M. Kowalski, Parallel CNN-ELM: a multiclass classification of chest X-ray images to identify seventeen lung diseases including COVID-19, Expert Syst. Appl. 229 (2023).
[138] M. Nahiduzzaman, M.O.F. Goni, M.R. Islam, A. Sayeed, M.S. Anower, M. Ahsan, J. Haider, M. Kowalski, Detection of various lung diseases including COVID-19 using extreme learning machine algorithm based on the features extracted from a lightweight CNN architecture, Biocybern. Biomed. Eng. 43 (3) (2023) 528–550.
[139] M.A. Khan, N. Hussain, A. Majid, M. Alhaisoni, S.A.C. Bukhari, S. Kadry, Y. Nam, Y.D. Zhang, Classification of positive COVID-19 CT scans using deep learning, Cmc-Computers Materials & Continua 66 (3) (2021) 2923–2938.
[140] P.I.R. Jenifer, S. Kannan, Deep learning with optimal hierarchical spiking neural network for medical image classification, Comput. Syst. Sci. Eng. 44 (2) (2023) 1081–1097.

[141] H. Naeem, A.A. Bin-Salem, A CNN-LSTM network with multi-level feature extraction-based approach for automated detection of coronavirus from CT scan and X-ray images, Appl. Soft Comput. (2021) 113.
[142] F. Alrowais, K. Mahmood, S.S. Alotaibi, M.A. Hamza, R. Marzouk, A. Mohamed, Laryngeal cancer detection and classification using Aquila optimization algorithm with deep learning on throat region images, IEEE Access 11 (2023) 115306–115315.
[143] S. Nawaz, S. Rasheed, W. Sami, L. Hussain, A. Aldweesh, E.T. Eldin, U.A. Salaria, M.S. Khan, Deep learning ResNet101 deep features of portable chest X-ray accurately classify COVID-19 lung infection, Cmc-Computers Materials & Continua 75 (3) (2023) 5213–5228.
[144] A. Bhattacharyya, D. Bhaik, S. Kumar, P. Thakur, R. Sharma, R.B. Pachori, A deep learning based approach for automatic detection of COVID-19 cases using chest X-ray images, Biomed. Signal Process Control 71 (2022).
[145] H. Zerouaoui, A. Idri, Deep hybrid architectures for binary classification of medical breast cancer images, Biomed. Signal Process Control 71 (2022).
[146] S.H. Kassani, P.H. Kassani, M.J. Wesolowski, K.A. Schneider, R. Deters, Automatic detection of coronavirus disease (COVID-19) in X-ray and CT images: a machine learning based approach, Biocybern. Biomed. Eng. 41 (3) (2021) 867–879.
[147] A. Shamsi, H. Asgharnezhad, S.S. Jokandan, A. Khosravi, P.M. Kebria, D. Nahavandi, S. Nahavandi, D. Srinivasan, An uncertainty-aware transfer learning-based framework for COVID-19 diagnosis, IEEE Transact. Neural Networks Learn. Syst. 32 (4) (2021) 1408–1417.
[148] T. Saba, A.S. Mohamed, M. El-Affendi, J. Amin, M. Sharif, Brain tumor detection using fusion of hand crafted and deep learning features, Cognit. Syst. Res. 59 (2020) 221–230.
[149] M. Khened, V.A. Kollerathu, G. Krishnamurthi, Fully convolutional multi-scale residual DenseNets for cardiac segmentation and automated cardiac diagnosis using ensemble of classifiers, Med. Image Anal. 51 (2019) 21–45.
[150] M. Wozniak, J. Silka, M. Wieczorek, Deep neural network correlation learning mechanism for CT brain tumor detection, Neural Comput. Appl. 35 (20) (2023) 14611–14626.
[151] O.N. Manzari, H. Ahmadabadi, H. Kashiani, S.B. Shokouhi, A. Ayatollahi, MedViT: a robust vision transformer for generalized medical image classification, Comput. Biol. Med. 157 (2023).
[152] M.M. Srikantamurthy, V.P.S. Rallabandi, D.B. Dudekula, S. Natarajan, J. Park, Classification of benign and malignant subtypes of breast cancer histopathology imaging using hybrid CNN-LSTM based transfer learning, BMC Med. Imag. 23 (1) (2023).
[153] N. Zhang, Y.X. Cai, Y.Y. Wang, Y.T. Tian, X.L. Wang, B. Badami, Skin cancer diagnosis based on optimized convolutional neural network, Artif. Intell. Med. 102 (2020).
[154] A.K. Anaraki, M. Ayati, F. Kazemi, Magnetic resonance imaging-based brain tumor grades classification and grading via convolutional neural networks and genetic algorithms, Biocybern. Biomed. Eng. 39 (1) (2019) 63–74.
[155] L.T. Duong, N.H. Le, T.B. Tran, V.M. Ngo, P.T. Nguyen, Detection of tuberculosis from chest X-ray images: boosting the performance with vision transformer and transfer learning, Expert Syst. Appl. 184 (2021).
[156] I. Budhiraja Deepanshi, D. Garg, N. Kumar, Choquet integral based deep learning model for COVID-19 diagnosis using eXplainable AI for NG-IoT models, Comput. Commun. 212 (2023) 227–238.
[157] R. Hashemzehi, S.J.S. Mahdavi, M. Kheirabadi, S.R. Kamel, Detection of brain tumors from MRI images base on deep learning using hybrid model CNN and NADE, Biocybern. Biomed. Eng. 40 (3) (2020) 1225–1232.
[158] S.J. Fang, J. Fu, C. Du, T. Lin, Y. Yan, Identifying laryngeal neoplasms in laryngoscope images via deep learning based object detection: a case study on an extremely small data set, Irbm 44 (6) (2023).
[159] C. Barata, M.E. Celebi, J.S. Marques, Explainable skin lesion diagnosis using taxonomies, Pattern Recogn. 110 (2021).
[160] Y.F. Jin, H.J. Lu, W.J. Zhu, W.L. Huo, Deep learning based classification of multi-label chest X-ray images via dual-weighted metric loss, Comput. Biol. Med. (2023) 157.
[161] X.L. Fan, X.F. Feng, Y.Y. Dong, H.C. Hou, COVID-19 CT image recognition algorithm based on transformer and CNN, Displays 72 (2022).
[162] D.A. Mitrea, R. Brehar, S. Nedevschi, M. Lupsor-Platon, M. Socaciu, R. Badea, Hepatocellular carcinoma recognition from ultrasound images using combinations of conventional and deep learning techniques, Sensors 23 (5) (2023).
[163] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, A. Torralba, Learning deep features for discriminative localization, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016.
[164] R.R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-cam: visual explanations from deep networks via gradient-based localization, in: Proceedings of the IEEE International Conference on Computer Vision, 2017.
[165] D. Mukhtorov, M. Rakhmonova, S. Muksimova, Y.I. Cho, Endoscopic image classification based on explainable deep learning, Sensors 23 (6) (2023).
[166] H.S. Alghamdi, Towards explainable deep neural networks for the automatic detection of diabetic retinopathy, Applied Sciences-Basel 12 (19) (2022).
[167] A. Chattopadhay, A. Sarkar, P. Howlader, V.N. Balasubramanian, Grad-cam++: generalized gradient-based visual explanations for deep convolutional networks, in: 2018 IEEE Winter Conference on Applications of Computer Vision (WACV), IEEE, 2018.
[168] M. Wattenberg, F. Viégas, I. Johnson, How to use t-SNE effectively, Distill 1 (10) (2016) e2.
[169] L. McInnes, J. Healy, J. Melville, Umap: Uniform Manifold Approximation and Projection for Dimension Reduction, 2018 arXiv preprint arXiv:1802.03426.
[170] W. Nie, Y. Zhang, A. Patel, A theoretical explanation for perplexing behaviors of backpropagation-based visualizations, in: International Conference on Machine Learning, PMLR, 2018.
[171] M.E. Chowdhury, T. Rahman, A. Khandakar, R. Mazhar, M.A. Kadir, Z.B. Mahbub, K.R. Islam, M.S. Khan, A. Iqbal, N. Al Emadi, Can AI help in screening viral and COVID-19 pneumonia? IEEE Access 8 (2020) 132665–132676.
[172] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, R.M. Summers, Chestx-ray8: hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017.
[173] D.Z. Kermany, Kang; goldbaum, michael, in: M. Data (Ed.), Large Dataset of Labeled Optical Coherence Tomography (OCT) and Chest X-Ray Images, 2018.
[174] J.P. Cohen, P. Morrison, L. Dao, K. Roth, T.Q. Duong, M. Ghassemi, Covid-19 Image Data Collection: Prospective Predictions Are the Future, 2020 arXiv preprint arXiv:2006.11988.
[175] E. Soares, Plamen Angelov, Sarah Biaso, Higa Froes, Michele, and Kanda Abe, Daniel., SARS-CoV-2 CT-scan dataset: a large dataset of real patients CT scans for SARS-CoV-2 identification, medRxiv (2020).
[176] H.a.W. Gunraj, Linda, Wong, Alexander, COVIDNet-CT, in: F.i. Medicine (Ed.), A Tailored Deep Convolutional Neural Network Design for Detection of COVID-19 Cases from Chest CT Images, 2020.
[177] M. Maftouni, A.C.C. Law, B. Shen, Y. Zhou, N. Ayoobi Yazdi, Z. Kong, A Robust Ensemble-Deep Learning Model for COVID-19 Diagnosis Based on an Integrated CT Scan Images Database, 2021.
[178] I. Coulson, in: N.Z.D. Society (Ed.), DermNet, 2021.
[179] Z. Lai, H. Deng, Medical image classification based on deep features extracted by deep model and statistic feature fusion with multilayer perceptron, Comput. Intell. Neurosci. (2018) 2018.
[180] D. Gutman, N.C. Codella, E. Celebi, B. Helba, M. Marchetti, N. Mishra, A. Halpern, Skin lesion analysis toward melanoma detection: a challenge at the international symposium on biomedical imaging (ISBI) 2016. Hosted by the International Skin Imaging Collaboration (ISIC), arXiv preprint arXiv:1605.01397, 2016.
[181] N.C. Codella, D. Gutman, M.E. Celebi, B. Helba, M.A. Marchetti, S.W. Dusza, A. Kalloo, K. Liopyris, N. Mishra, H. Kittler, Skin lesion analysis toward melanoma detection: a challenge at the 2017 international symposium on biomedical imaging (isbi), hosted by the international skin imaging collaboration (isic), in: 2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018), IEEE, 2018.
[182] N. Codella, V. Rotemberg, P. Tschandl, M.E. Celebi, S. Dusza, D. Gutman, B. Helba, A. Kalloo, K. Liopyris, M. Marchetti, Skin Lesion Analysis toward Melanoma Detection 2018: A Challenge Hosted by the International Skin Imaging Collaboration (Isic), 2019 arXiv preprint arXiv:1902.03368.
[183] D. Emma, Diabetic Retinopathy Detection. Kaggle.
[184] S.D.M. Karthik, APTOS 2019 Blindness Detection. Kaggle.
[185] E. Decencière, X. Zhang, G. Cazuguel, B. Lay, B. Cochener, C. Trone, P. Gain, R. Ordonez, P. Massin, A. Erginay, Feedback on a publicly distributed image database: the Messidor database, Image Anal. Stereol. 33 (3) (2014) 231–234.
[186] R. Kiefer, Glaucoma Dataset: EyePACS-AIROGS-Light-V2. Kaggle.
[187] J. Cheng, brain tumor dataset. figshare. Dataset. (2017).
[188] E. Giacomello, D. Loiacono, L. Mainardi, Brain MRI tumor segmentation with adversarial networks, in: 2020 International Joint Conference on Neural Networks (IJCNN), IEEE, 2020.
[189] Z. Ji, Y. Shen, C. Ma, M. Gao, Scribble-based hierarchical weakly supervised learning for brain tumor segmentation, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2019: 22nd International Conference, Shenzhen, China, October 13–17, 2019, Proceedings, Part III 22, Springer, 2019.
[190] N. Chakrabarty, Brain MRI Images for Brain Tumor Detection, 2019.
[191] A.E. Flanders, L.M. Prevedello, G. Shih, S.S. Halabi, J. Kalpathy-Cramer, R. Ball, J. T. Mongan, A. Stein, F.C. Kitamura, M.P. Lungren, Construction of a machine learning dataset through collaboration: the RSNA 2019 brain CT hemorrhage challenge, Radiology: Artif. Intell. 2 (3) (2020) e190211.
[192] R.D. Labati, V. Piuri, F. Scotti, All-IDB: the acute lymphoblastic leukemia image database for image processing, in: 2011 18th IEEE International Conference on Image Processing, IEEE, 2011.
[193] S. Mourya, S. Kant, P. Kumar, A. Gupta, R. Gupta, in: T.C.I. Archive (Ed.), ALL Challenge Dataset of ISBI 2019 (C-NMC 2019), 2019.
[194] F.A. Spanhol, L.S. Oliveira, C. Petitjean, L. Heutte, A dataset for breast cancer histopathological image classification, IEEE Trans. Biomed. Eng. 63 (7) (2015) 1455–1462.
[195] A. Janowczyk, A. Madabhushi, Deep learning for digital pathology image analysis: a comprehensive tutorial with selected use cases, J. Pathol. Inf. 7 (2016) 29.
[196] M. Heath, K. Bowyer, D. Kopans, P. Kegelmeyer Jr., R. Moore, K. Chang, S. Munishkumaran, Current status of the digital database for screening mammography, in: Digital Mammography: Nijmegen, 1998, Springer, 1998, pp. 457–460.
[197] BenO, Intel & MobileODT Cervical Cancer Screening. Kaggle.
[198] M. Yebasse, K.J. Cheoi, J. Ko, Malaria disease cell classification with highlighting small infected regions, IEEE Access 11 (2023) 15945–15953.
[199] P.P. Banik, R. Saha, K.D. Kim, An automatic nucleus segmentation and CNN model based classification method of white blood cell, Expert Syst. Appl. 149 (2020).
[200] S. Li, W. Song, H. Qin, A. Hao, Deep variance network: an iterative, improved CNN framework for unbalanced training datasets, Pattern Recogn. 81 (2018) 294–308.

[201] B.D. Bozorg, J. Zhang, B. ter Haar Romeny, Retinal microaneurysms detection using local convergence index features, arXiv (2017) 1–14.
[202] M. Raghu, C.Y. Zhang, J. Kleinberg and S. Bengio. Transfusion: Understanding Transfer Learning for Medical Imaging.
[203] T. Chen, S. Kornblith, M. Norouzi, G. Hinton, A simple framework for contrastive learning of visual representations, in: International Conference on Machine Learning, PMLR, 2020.
[204] K. He, H. Fan, Y. Wu, S. Xie, R. Girshick, Momentum contrast for unsupervised visual representation learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020.
[205] D.P. Kingma, Auto-encoding variational bayes, arXiv preprint arXiv:1312.6114 (2013).
[206] Q.V. Le, Building high-level features using large scale unsupervised learning, in: 2013 IEEE International Conference on Acoustics, Speech and Signal Processing, IEEE, 2013.
[207] L.A. De Santi, F.I. Piparo, F. Bargagna, M.F. Santarelli, S. Celi, V. Positano, Part-prototype models in medical imaging: applications and current challenges, BioMedInformatics 4 (4) (2024) 2149–2172.

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Review

Developments in Deep Learning Artificial Neural Network Techniques for Medical Image Analysis and Interpretation

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Abstract: Deep learning has revolutionised medical image analysis, offering the possibility of automated, efficient, and highly accurate diagnostic solutions. This article explores recent developments in deep learning techniques applied to medical imaging, including convolutional neural networks (CNNs) for classification and segmentation, recurrent neural networks (RNNs) for temporal analysis, autoencoders for feature extraction, and generative adversarial networks (GANs) for image synthesis and augmentation. Additionally, U-Net models for segmentation, vision transformers (ViTs) for global feature extraction, and hybrid models integrating multiple architectures are explored. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) process were used, and searches on PubMed, Google Scholar, and Scopus databases were conducted. The findings highlight key challenges such as data availability, interpretability, overfitting, and computational requirements. While deep learning has demonstrated significant potential in enhancing diagnostic accuracy across multiple medical imaging modalities—including MRI, CT, US, and X-ray—factors such as model trust, data privacy, and ethical considerations remain ongoing concerns. The study underscores the importance of integrating multimodal data, improving computational efficiency, and advancing explainability to facilitate broader clinical adoption. Future research directions emphasize optimising deep learning models for real-time applications, enhancing interpretability, and integrating deep learning with existing healthcare frameworks for improved patient outcomes.

Keywords: artificial intelligence; artificial neural networks; medical image analysis; deep learning; image classification and pattern recognition

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1. Introduction

The use of imaging for diagnosis in healthcare is substantial, amounting to about USD 100 billion globally per year [1]. Mounting pressures on healthcare facilities and the market for imaging diagnosis have led to increasing demands for diagnostic excellence in the clinical setting due to the rising number of clinical images, greater image complexity, and faster results demanded by clinicians. As a result, the need for new technologies is centred on providing solutions that will increase the effectiveness of the clinical process, improving healthcare systems, and provide accurate diagnosis for patients while improving care quality. Therefore, there has been high demand for technologies that can aid in the automation of workflows associated with the use of medical imaging for diagnosis, leading to advances in the use of artificial intelligence (AI) methods such as deep learning to assist radiologists in analysing complex image datasets [2].

1.1. Deep Learning Overview

Deep learning is a subfield of machine learning that leverages artificial neural network (ANN) architecture to acquire knowledge from large datasets and perform intricate operations. One of the main advantages of deep learning techniques is their ability to mimic the information processing complexity of the human brain [2,3]. The field has been studied since the 1980s, but gained prominence in recent years. This is because of access to large datasets for model training, improved algorithm development, and increased processing power of microprocessors. The structure of an ANN is an interconnection of nodes (sometimes referred to as processing elements or neurons) that can span up to multiple layers, depending on the intricacy of the tasks and capabilities of hardware resources. Each node gathers information from the previous layer and then transmits to the subsequent layer based on the configured characteristics and set parameters. The values of the parameters used are normally initially randomised, but then iteratively updated during training based on a set learning rate [4]. A deep learning network extracts deeper and intricate information as its number of architectural layers increases, resulting in optimised performance with large datasets and training iterations. This in turn ensures precise recognition of patterns in the data [3].

1.2. Deep Learning in Medical Imaging, Classification, and Segmentation

Deep learning has seen extensive applications in medical image analysis. It has been used for different medical imaging modalities, including X-ray radiographs, computed tomography (CT), ultrasound (US), and magnetic resonance imaging (MRI) scans, to provide predictive diagnosis and treatment. Subtle and intricate patterns presented by these medical images are effectively identified by the adapted deep learning approach, thereby providing a means to automate the feature extraction process. Image feature extraction or selection is a process that can be performed manually by a qualified specialist, but this can be time-consuming and subjective. When deep learning models are properly trained, they have an ability to accurately identify lesion or tumours, examine membranes and tissues for differences, and undertake diverse medicine-related tasks, thereby providing accelerated diagnostic outcomes. Therefore, deep learning is emerging in the realm of medical image analysis assisting with diagnosis. A summary of deep learning applications is shown in Figure 1.

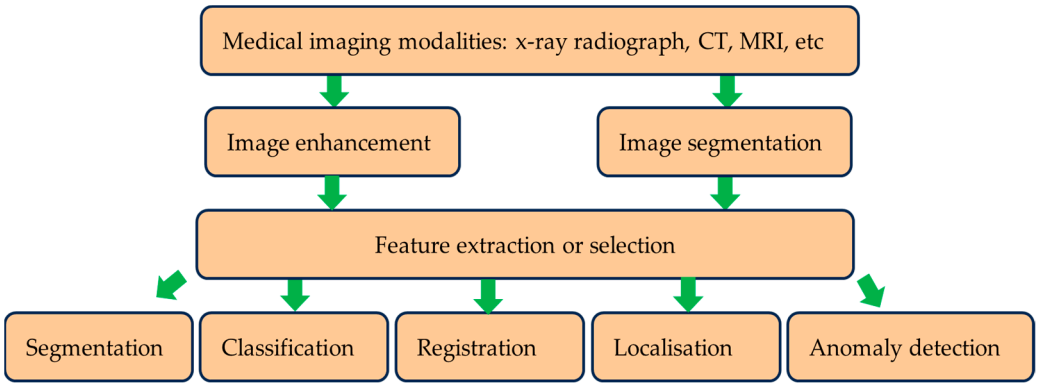


Figure 1. A summary of deep learning usage with medical images.

1.3. Challenges in Utilising Deep Learning in the Medical Field

Despite the significant advancements and capabilities of deep learning techniques in medical image analysis, there are certain limitations and challenges in their implementation and acceptance. For instance, most deep learning algorithms often lack explainability, i.e., they typically operate as black boxes [5]. In the medical field, where decision-making

processes are required for diagnosis and provision of treatment, the deep learning method of convolutional neural networks (CNNs), for instance, does not provide an insight into the manner by which it came up with a decision. Other challenges that applications of deep learning techniques can face include the following

- Overfitting during training. Overfitting causes poor accuracy when training deep artificial neural networks in recognising images not included during their training dataset (i.e., unseen or test images), even though the images in the training set could have been correctly recognised [6–10]. It has several possible causes that include insufficient training examples to facilitate adequate generalisation and excessive model parameters for the architecture. However, there are techniques that could be valuable to deal with overfitting. One technique is known as dropout [6,11–14], whereby some nodes in the architecture are temporarily left out during training. Another approach is to artificially extend the number of examples in the training dataset through a process known as data augmentation [15–17]. A comparison of different image data augmentation methods was reported in [18]. The technique has been applied to mammograms [19] and CT images [20]. Augmentation can be performed by manipulating the images through processes such as kernel filters, geometric transformation, random erasing, and mixing images. It can also be carried out through deep learning approaches such as adversarial training, neural style transformers, and generative adversarial networks [7]. An issue with performing data augmentation is selecting the best approach for a given set of images [21]. An exploration of the influence of different data augmentation techniques on the explainability of deep learning methods was reported in [22].
- Image annotation. Many deep learning algorithms are supervised, i.e., they require labelled images indicating their categories during their training phase. The labelling requires annotation of images by qualified medical practitioners. Because deep learning requires large datasets for training, this process can be time-consuming. However, there have been reports of automated and interactive image annotation methods that can assist with this operation, e.g., [23–28].
- Noisy images. Medical images can be noisy. Noise distorts the quality of images being used to train deep learning networks [29] and can reduce their ability to learn effectively.
- Interpretability. This is an area of great research interest to render decision-making by deep learning artificial neural networks more transparent, i.e., moving away from so-called black box behaviour to more interpretable decision-making. The issue of interpretability has been explored in many studies, e.g., [30–36]. Interpretability can be considered from multiple perspectives, e.g., user orientation for explanations provided, visualisation through graphs, charts rules etc, user comprehensibility through comprehensive reasoning, simplicity of explanations, local interpretation of a single datum and global interpretation of overall data, consistency in explanations, transparency in decision-making, and ethics and fairness in revealing bias and discrimination [31].
- Data sharing complexities and small datasets. Medical data gathered by a single institution may be insufficient to allow effective training of deep learning algorithms, and thus sharing across many institutions would be required. This, however, could be challenging due to regulatory, technical, and privacy concerns [37] and financial and time constraints, and limited availability of patients can limit the dataset. However, valuable techniques have been devised to address small-dataset problems in deep learning [38,39].

- Ethical issues. There are numerous ethical issues requiring careful consideration, e.g., patient anonymity, consent/assent, data handling and protection, data sharing, and vulnerable participants [40–43].
- Trust. There is an ongoing issue of relying on critical medical diagnostic results generated when the manner of their generation is not sufficiently transparent [4].
- Computational requirements and environmental issues. Training deep learning algorithms typically requires high computational capabilities and long durations. Many general-purpose computers do not have the means of delivering the required computational resources, and there is also the issue of the environmental aspects of using so much electricity to perform the required deep learning training [44,45].

This article explores deep learning techniques for medical image analysis, highlighting their roles in assisting diagnostic tasks. It examines CNNs for classification and segmentation, recurrent neural networks (RNNs) for sequential data, autoencoders for feature extraction, and generative adversarial networks (GANs) for data augmentation. It also discusses U-Net models for segmentation, vision transformers (ViTs) for long-range dependencies, and hybrid models that integrate multiple architectures. Different algorithms and their architectures are discussed. Associated mathematical models are presented to demonstrate the manner in which different imaging modalities operate. The study emphasises the transformative impact of deep learning in medical diagnostics and suggests future improvements in efficiency and interpretability. In the following sections, the materials and methods are explained and the results are discussed.

2. Materials and Methods

A systematic review was undertaken to explore the latest developments in deep learning ANNs for analysing medical images. As this field is growing, with many related articles published daily, a rapid review approach was undertaken [6]. The necessary constraints and criteria for inclusion were used when searching for literature that provided information about recent deep learning techniques used for medical image analysis. The following keywords were combined as part of the literature search: “deep learning” and “medical image analysis”. We also included imaging modalities and used “OR” and “AND” operators:

“deep learning” AND “medical image analysis”.
“deep learning” AND “medical image analysis” OR “CT”.
“deep learning” AND “medical image analysis” OR “MRI”.
“deep learning” AND “medical image analysis” OR “X-ray”.
“deep learning” AND “medical image analysis” OR “infrared thermal image”.
“deep learning” AND “medical image analysis” OR “ultrasound”.

The search was performed on different scientific databases, including Scopus, PubMed, and Google Scholar, and the inclusion and exclusion criteria are shown in Figure 2. The imaging modalities are shown in Figure 3.

We examined each article with a principal focus on the deep learning techniques deployed and the imaging modalities adopted. Articles that did not report either a deep learning algorithm or a medical imaging modality were excluded. The remaining articles were then filtered to include the deep learning techniques in whole or in part, as many works used a combination of deep learning techniques. As this is a wider review of deep learning methodologies, we have not looked critically into the clinical aspects of diseases to be diagnosed. The categories included were CNNs, RNNs, autoencoders, GANs, transfer learning (TL), ViTs, and hybrid models. We also considered studies that combined multiple deep learning algorithms, i.e., hybrid systems further refined our search to include the terms “hybrid”. For completeness, a summary of each method is included as part of the review.

As CNN architectures were widely deployed, mostly with other deep learning techniques, we further narrowed the search terms to include only publications that specifically used CNNs by updating the search terms to include ("convolutional neural network" AND "medical image analysis") OR ("CNN" AND "medical image analysis").

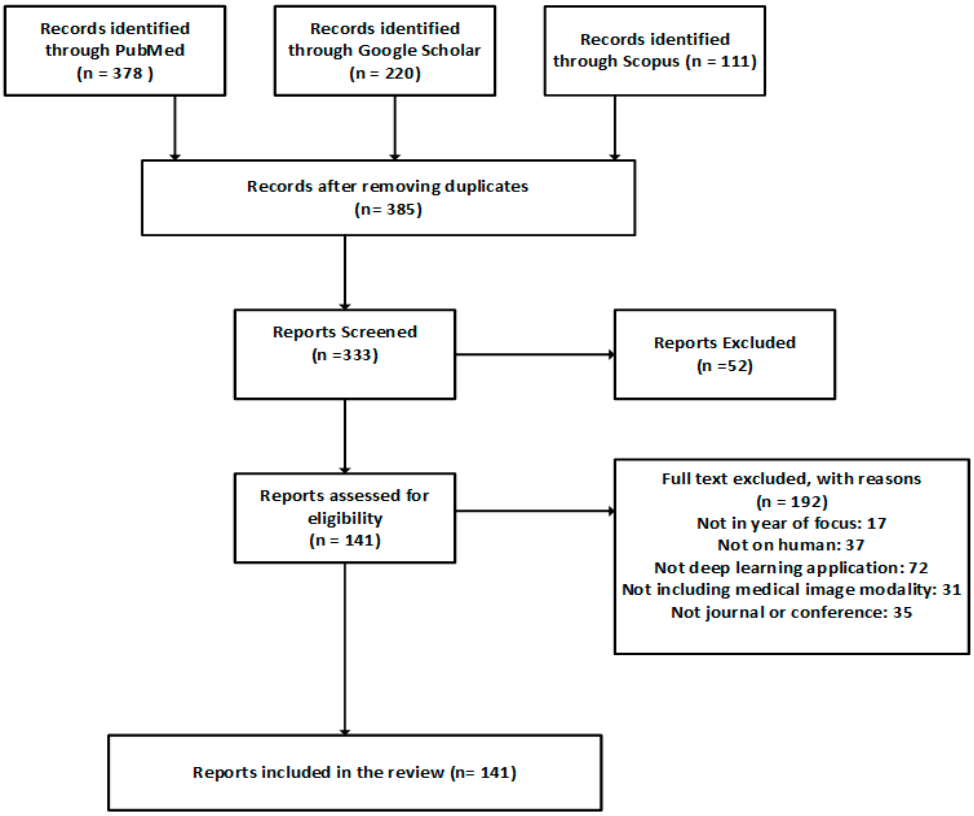


Figure 2. Inclusion and exclusion criteria of the systematic literature review using the PRISMA framework.

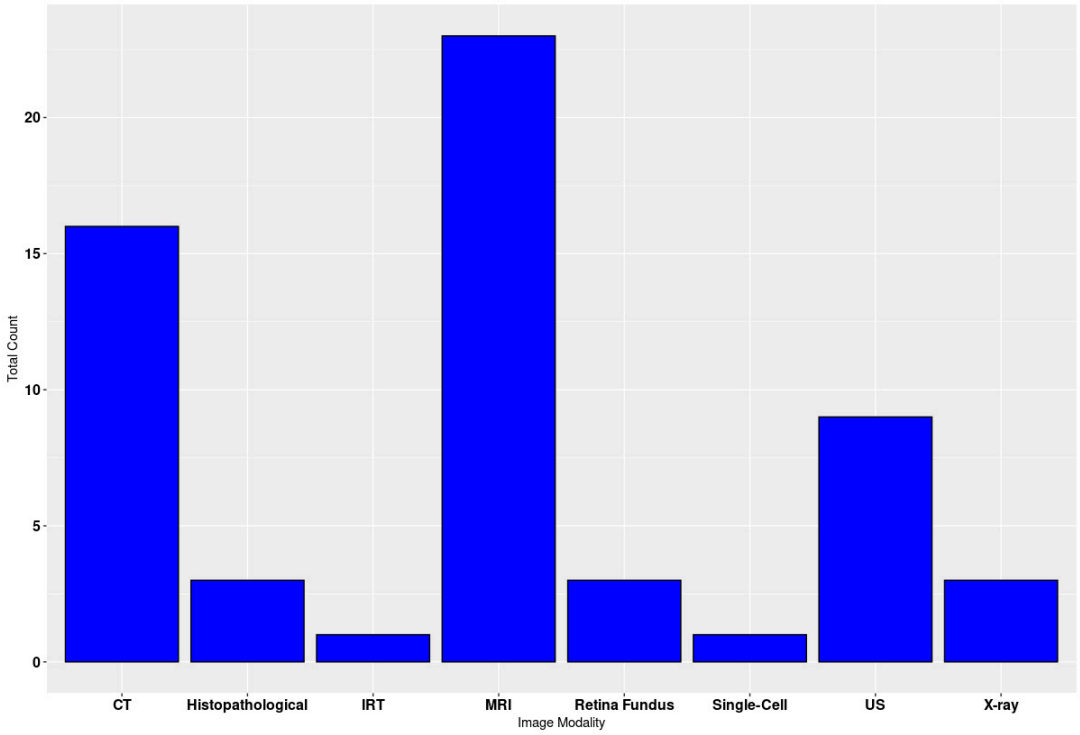


Figure 3. Count of imaging modalities on a section of articles included in the review.

Over 500 different studies were initially identified across the databases, with some duplicated across the databases, especially on PubMed and Google Scholar. After applying the exclusion criteria, this was refined to 141 articles being eventually reviewed in this study. The 141 articles were distributed over all the identified categories. We included a broad range of imaging modalities. The objective was to identify the deep learning techniques most suitable for medical imaging tasks, the processes employed to improve the monitoring and diagnosis performance of these techniques, and the optimisation performed.

3. Results

This section contains an overview of deep learning neural network concepts.

3.1. Convolutional Neural Networks

CNNs were the dominant technique for analysing medical images for diagnostic purposes [46]. It has gained immense popularity since 2012, when high-performance computing (HPC) became more accessible. This led to the ImageNet competition for different combinations of deep CNNs to achieve better diagnostic results when compared to human experts. CNNs are effective for several tasks, such as image segmentation, detection, registration, localisation, and classification [5]. They consist of numerous layers of convolutional filters with nonlinear activation functions combined with pooling layers, dropout layers, and fully connected layers. Their ability to extract complex spatial relationships and patterns in images has seen them used in various medical imaging modalities, such as magnetic resonance imaging (MRI), computed tomography (CT), X-rays, ultrasound (US), and histopathology, and more recently in infrared thermal (IRT) imaging [47,48]. Images associated with multiple conditions, e.g., bone fractures, cancer, liver diseases, pneumonia, and COVID-19, were segmented, classified, registered, and interpreted. The CNN architecture applied to the discrete Fourier-transformed infrared thermal images is shown in Figure 4. Although the study was a pilot, the model was effective in screening wrist fracture in children.

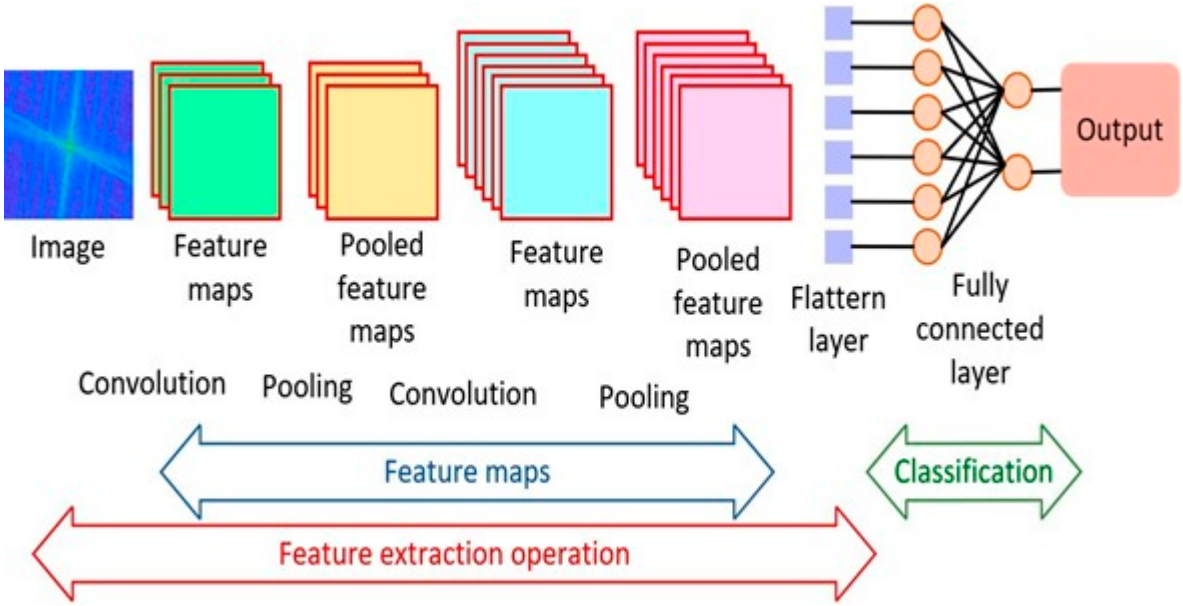


Figure 4. An example of a CNN used for the analysis of infrared thermal images [48].

The concept of convolution is simply a mathematical operation where a filter, also known as a kernel, is applied on an input image for feature extraction. The convolution

process is described mathematically next. Let the image input to the CNN be represented by a 3D matrix such that:

$$X \in \mathbb{R}^{H \times W \times D} \tag{1}$$

where H is the height of the image, W is the width of the image, and D is the depth of the image (representing the number of channels, which is usually three for an RGB image). When applied to a set of convolutional filters for each input, the output of the convolutional filter can be represented by:

$$Z_{i,j,k} = \sum_{h=0}^{H_k} \sum_{w=0}^{W_k} \sum_{d=0}^{D} X_{i+h,j+w,d}.K_{h,w,d,k} \tag{2}$$

where $Z_{i,j,k}$ is the output feature map for the filter k at position (i, j), K is the convolutional filter (kernel), and H_k and W_k are the height and width of the kernel, which are usually a sample of the size of the input. This process helps to identify local patterns such as the edges, shapes, and textures from the input image.

When the CNN algorithm is used for classification purposes, it is combined with other layers such as the activation layers, pooling layer, and the fully connected layer. The output from the convolutional filter is passed through an activation function $A_{i,j,k}$. If the rectified linear unit (ReLU) activation function is applied, the output from the convolutional layer becomes:

$$A_{i,j,k} = max\left(0,\ Z_{i,j,k}\right) \tag{3}$$

This operation helps to add nonlinearity by ensuring only positive values are retained, thereby helping to learn complex patterns from the image.

The pooling layer is then applied to the output of the activation layers. The effect of the pooling operation is to reduce spatial dimensions from the convolution operation and thus help the network to capture small translations in the image. This process also helps to reduce the computational complexity of the network. Assuming the max (maximum) pooling function is applied to the activation layer with a $p \times p$ window size, the output from the pooling layer is represented by:

$$A_{i,j,k} = \max_{h,w \in [1,p]} A_{i+h,j+w,k} \tag{4}$$

The fully connected layer is connected to the output of a flattened max pooling layer output. This layer takes a vector value and is a standard neural network, with each input connected to all the neurons in the next layer. The output of the fully connected (FC) layer is given by:

$$Z_l = W_l A_{l-1} + b_l \tag{5}$$

where A_{l-1} is the input into the FC layer and W_l and b_l represent the weights and biases of the FC layer. The final or output layer, which is also a vector, is mostly passed through a softmax function for a classification task. This function helps to convert the output classes into probabilities with the following expression:

$$\hat{y}_i = \frac{exp\ (Z_i)}{\sum_{j=1}^{C} exp\ (Z_j)} \tag{6}$$

where $\hat{y}_i$ represents the probability of predicting class i, and C represents the total number of classes being differentiated. The full CNN undergoes training to correctly learn the features of the image input. This training uses a loss function, usually a cross-entropy loss, when considering a classification task. The cross-entropy loss is given by:

$$L = -\sum_{i=1}^{C} y_i log\ (\hat{y}_i) \tag{7}$$

where $\hat{y}_i$ and y_i are the predicted probabilities for class i and the target label, respectively.

The gradient of the loss function is calculated with respect to the weights through a back-propagation formula:

$$\frac{\partial \mathrm{L}}{\partial \mathrm{W}} = \frac{\partial \mathrm{L}}{\partial \mathrm{Z}} \cdot \frac{\partial \mathrm{Z}}{\partial \mathrm{W}} \tag{8}$$

The weights are updated using the gradient descent, i.e.,

$$W_{new} = W_{old} - \eta \frac{\partial L}{\partial W} \tag{9}$$

where η $(0 < \eta \leq 1)$ represents the learning rate controlling the convergence rate.

Literature Review Findings for CNNs

Most studies have used a CNN individually, though there have been studies that applied a CNN in combination with another algorithm such as UNET and GANs for medical image diagnostics. However, in this section, the focus is on a CNN application on its own (other CCN-based algorithms, such as transfer learning, are discussed in Section 3.6). A lightweight CNN algorithm was used to detect COVID-19 from chest X-ray radiograph images [49]. In that study, their proposed CNN was inspired by the ResNet model [50], in which no layers were connected in sequence, creating a skip connection whereby neurons in a particular layer can be connected to another neuron further ahead. This arrangement helped to create a lightweight model that was effective in edge detection applications. Their study was compared with other CNN models such as CVDNet and deep GRU-CNN, and showed similar performance to models with reduced computational complexity. A deeper CNN in their work was proposed that used CT scans and X-ray radiographs for the detection of several pulmonary diseases, including COVID-19 and viral pneumonia [51]. They used varied datasets consisting of different imaging modalities for their work, which provided another dimension to the efficacy of CNNs in diagnostic imaging. They proposed a 26-layer deep CNN which was inspired by a wide residual network (WRN) [52]. It provided a faster training time when considering the deep nature of the architecture. This was achievable based on the sophistication of their hardware. Their model was effective in terms of the accuracy in detecting the different pulmonary diseases when compared to traditional methods. A CNN model was used to classify ultrasonic images of fatty liver [53]. A pertinent problem in that research was the similarity in the pathological ultrasonic images used for training the CNN algorithm. This sort of challenge might pose a problem for the CNN architecture, as it may struggle to extract distinct features for different pathological images. Therefore, there will be a need for deep convolutional layers, leading to computational complexities. However, they used pixel-level feature extraction as a preprocessing step and then proposed a CNN architecture comprising two convolutional layers, a pooling layer and a fully connected layer. They also tested the proposed method with a skip connection and improved its accuracy when compared to other algorithms such as VGGNet. The authors in [54] developed a CNN model to distinguish between benign and malignant tumours using radial endobronchial ultrasound (rEBUS) images. A total of 769 images were collected from hospitals in Taiwan for model training and internal validation, with an additional 300 and 92 images from two other hospitals used for external validation. The model was trained with image augmentation techniques to improve its robustness. Internal validation showed strong performance, with an AUC of 0.88, sensitivity of 0.80, and specificity of 0.75. For external validation, the AUC was 0.76 and 0.72 at the two external sites, with varying sensitivity and specificity.

After fine-tuning, performance improved, reaching AUC values of 0.78 and 0.82. The CNN also showed potential in identifying lung cancer subtypes, with moderate success for adenocarcinoma and squamous cell carcinoma, but limited accuracy for small cell lung cancer. Overall, the model proved promising for aiding rEBUS-based lung cancer diagnosis. The authors in [55] developed a deep convolutional neural network (DCNN) for the classification of ovarian cancer in the pelvic area using ultrasound images. They used data collected from ten hospitals in China between 2003 and 2019, including adult patients with adnexal (ovary-related) lesions and healthy controls. Validation of the proposed model was performed using three datasets: one internal and two external, all containing images from patients with either ovarian cancer or benign lesions. The DCNN showed high diagnostic performance, with area under the curve (AUC) scores of 0.911, 0.870, and 0.831 across the three datasets, respectively. They benchmarked their model against the diagnostic results of 35 radiologists, and the proposed DCNN model outperformed the 35 radiologists in accuracy for detecting ovarian cancer.

3.2. Recurrent Neural Networks

Recurrent neural networks (RNNs) are another class of neural network that have gained significant research consideration for modelling of sequential data. Their variability in length of their input and output also makes them suitable for natural language processing tasks [1]. An RNN has an internal memory unlike feed-forward networks, which helps to keep memory of the hidden states across different time stamps. In RNNs, there is a feedback loop between the outputs of the hidden layers [56]. This arrangement allows the RNNs to learn sequential patterns, making them suitable for tasks such as time-series prediction and video analysis [57]. RNN architecture normally suffers from the issue of vanishing gradients [53]. Their use transcends sequential and textual data. They can also be applied to imaging modalities, such as dynamic imaging in functional MRIs, with time-series characteristic information to monitor disease progress in a patient and to check how they respond to treatments. This functionality has been made popular by a variant of the network known as the long short-term memory network (LSTM). The LSTM works by introducing self-loops to allow for the flow of gradients for long durations [58]. The recurrent structure of the neural network is always enforced by the LSTM by introducing gating functions on the neurons in the hidden layer. The architecture of an RNN is shown in Figure 5.

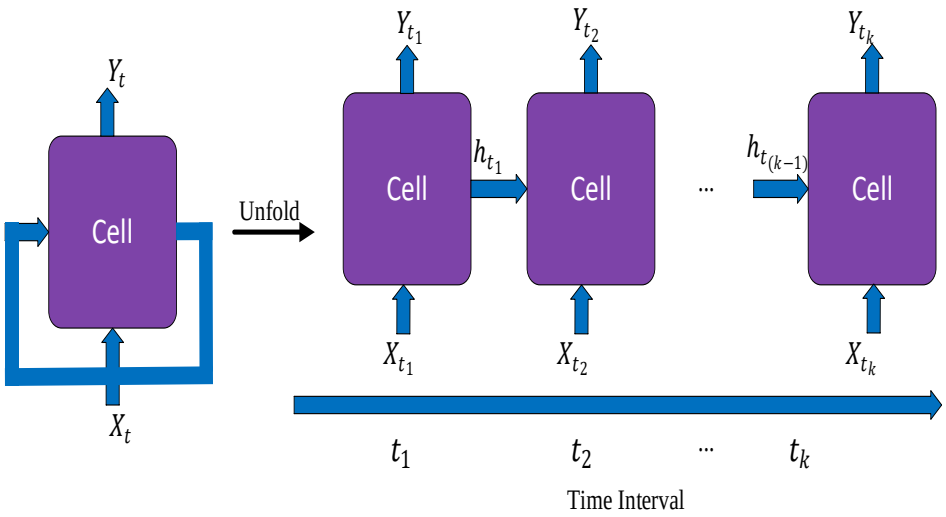


Figure 5. Structure of an RNN [53].

The mathematical representation of an RNN is described next. Let the sequential input to the RNN at time step t be x_t where $x_t \in \mathbb{R}^n$ is a vector, hence the input sequence becomes:

$$x_1, x_2, \ldots, x_T \tag{10}$$

where T represents the number of time steps in the sequence. For every time step t, a hidden state (h_t) is maintained. Each hidden state is updated as:

$$h_t = f(W_{hh}h_{t-1} + W_{xh}x_t + b_h) \tag{11}$$

where W_{hh} represents the hidden states or recurrent weights, W_{xh} is the weight of the connection between the input layer and the hidden layer, and b_h is the bias for the hidden state. The function f represents the activation function, typically the ReLU or the hyperbolic tangent. To obtain the output of the RNN at each time step, the output from each hidden state is multiplied by its associated weights and added to a chosen bias vector. This is represented as:

$$y_t = g\left(W_{hy}h_t + b_y\right) \tag{12}$$

where W_{hy} represents the weight matrix between the hidden state and the output and b_y represents the output bias vector. The function g is the activation function for the output, which could be task-dependent. The softmax activation function is mostly employed for classification tasks.

Literature Review Findings for RNNs

RNNs are widely used for machine learning or deep learning tasks that are time-related. For image diagnostics, they can be used to monitor disease progression and patient response to treatments based on different time stamps of when the images were taken. Most studies have based their classification tasks on an RNN, typically combining it with a CNN for the feature extraction stage, as shown in Table 1.

Table 1. Recurrent neural network applications.

Article	Imaging Modality	Task	CNN Feature Extraction	Disease/Body Part	Variant Used
[58]	MRI	Classification	Y	Alzheimer's	BGRU
[59]	Histopathological images	Classification	N	Breast cancer	None
[60]	MRI	Classification/segmentation	N	Brain tumour	LSTM
[57]	MRI	Segmentation	Y	Aorta	LSTM
[61]	MRI	Classification/localisation	Y	Knee ligament	LSTM
[62]	IRT	Classification	Y	Diabetes mellitus	LSTM
[56]	CT	Image denoising	N	Lungs	LSTM
[63]	MRI	Registration	Y	Brain cancer	LSTM

In one study, MR images were used for classification of Alzheimer's disease by analysing the longitudinal sequence of the MR images taken at different time steps to measure disease progression [58]. They combined a CNN for feature extraction and RNN for classification. For the RNN architecture, they cascaded three bidirectional gated recurrent units (BGRU) with the inputs from the CNN at multiple points, providing a longitudinal analysis. RNN architectures can suffer from vanishing gradients when the sequence of the images becomes too long, as they do not contain memory units to store sequences [64], hence the reason for developing variants such as LSTM and GRU. RNNs can also be used on their own without the addition of gated memory units. An RNN with slight modification was used in another study for the classification of breast cancer with

microscopic histopathological imaging [58]. The RNN architecture was gain-modulated using the honey badger algorithm (HBA) by updating the weights of the RNN during training. The capacity in which RNN was used in their study was for classification of different stages of breast cancer and not disease progression, hence why gated memory approaches were not used. RNNs can also be used for image segmentation tasks, especially in cases where time-series images of a part of the body are taken, requiring sequencing of the time frame for identification of anomalies. In another study, an RNN-based LSTM model was proposed together with U-Net to label aortic MR images [60]. This sort of task uses the capacity of the RNN architecture to segment temporal images, as most annotated images normally used for classification are static. The U-Net algorithm was combined with a CNN for feature extraction, with the sequencing part achieved with the RNN algorithm. The ability of a neural network to visualise several distinct images when compared to a human makes RNNs suitable for segmenting image labels for disease identification. RNNs can also been used to denoise diagnostic images. Images used for diagnosis can be susceptible to such interference as white and salt-and-pepper noise. An LSTM model was combined with particle swarm optimisation (PSO) algorithms to optimise the batch normalisation process of the RNN training to remove noise from CT images of the lungs [59]. The technique provided improved peak signal-to-noise ratio (PSNR) when compared to traditional noise removal techniques such as filtering-based and diffusion-based techniques.

3.3. Autoencoders

Autoencoders are unsupervised learning models used for dimensionality reduction and feature extraction with minimal distortion when their input is compared to their output [65]. They play an important role in the deep learning paradigm for medical image analysis [66]. They can help denoise or compress medical images and are useful in anomaly detection, where unusual patterns in images indicate potential medical issues. They can also be used as a semi-supervised deep learning model to produce annotated data in situations where there is a lack of substantial amount of annotated data available for training any deep learning network for tasks such as classification or segmentation [67]. Their architecture consists of an encoder and decoder structure with a latent space to store the value of the compressed data. Both the encoder and the decoder comprise a fully connected feed-forward neural network. The encoder converts the input image into a low-dimension compressed version, which is referred to as latent space or the encoder. The latent space contains only essential features of the input from the encoder and is kept as shallow as possible in terms of the number of neurons used to retain the compressed version of the input and computational efficiency. The encoder in turn transforms the latent space to a reconstruction of the input. A loss function is generally used during training to compare the input with its reconstruction [68]. The architecture of an autoencoder with an MR image as its input is shown in Figure 6.

The mathematical representation of the dimensionality reduction function of the autoencoder's encoder and decoder is described next. Let the input image to the encoder be represented by:

$$x \, \epsilon \, \mathbb{R}^n \tag{13}$$

The encoder in turn transforms the input x into a lower-dimension latent representation:

$$z \, \epsilon \, \mathbb{R}^d \tag{14}$$

where $d < n$.

The transformation function that produced the latent information is given as:

$$z = f_{enc}(x) = \sigma(W_{enc}x + b_{enc}) \tag{15}$$

where $W_{enc} \epsilon \mathbb{R}^{d \times n}$ is the encoder weight matrix, $b_{enc} \epsilon \mathbb{R}^{d \times n}$ represents the encoder's bias vector, and σ represents the selected activation function of the encoder, which is commonly ReLU or a sigmoid function. The latent representation z of the input x provides compressed information by only capturing the essential features of the input image, a feature referred to as bottleneck, in which the dimension of the latent feature is smaller than the input vector. This facilitates the autoencoder to learn a better way to efficiently represent the input vectors. The decoder network tries to transform the latent space z back to the reconstructed input $\hat{x}$ to match the input vector x. The decoder function can be represented thus:

$$\hat{x} = f_{dec}(z) = \sigma'(W_{dec}z + b_{dec}) \tag{16}$$

The loss function is used to minimize the difference between the input x and the reconstructed input $\hat{x}$. This difference is normally quantified by binary cross-entropy error (BCE) when the dataset is binary or mean squared error (MSE) for multi-class data.

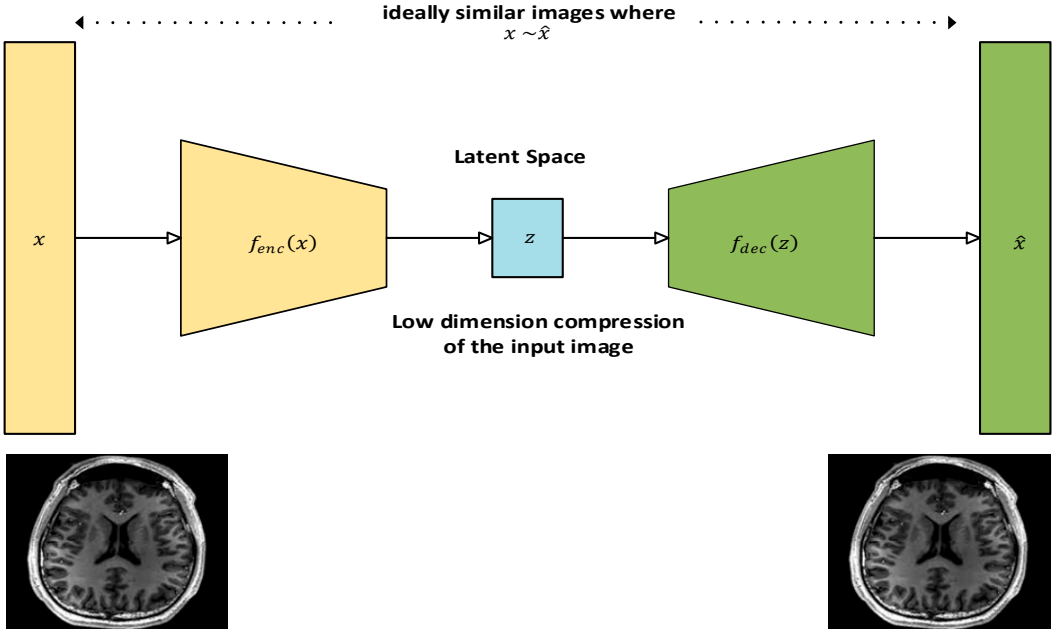


Figure 6. Autoencoder architecture reproducing an MR image.

The MSE loss is computed with the following function:

$$L(x, \hat{x}) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{x}_i)^2 \tag{17}$$

and the BCE loss is computed as follows:

$$L(x, \hat{x}) = -\sum_{i=1}^{n}[x_i log\ (\hat{x}_i) + (1 - x_i) log\ (1 - \hat{x}_i)] \tag{18}$$

Training of the autoencoder requires finding parameters for the weights and biases of the encoder and decoder such that the reconstruction loss is minimised over the training data. Gradient-based optimisation techniques such as stochastic gradient descent (SGD) or Adam can be used for this purpose. Table 2 shows the summary of autoencoder based techniques.

Table 2. Autoencoder-based techniques.

Article	Image Modality	Task	Disease/ Body Part
[67]	MRI	Augmentation/ segmentation	Brain
[69]	MRI	Denoising	Prostate
[70]	CT + others	Classification	Face
[71]	CT	Augmentation	Various
[72]	MRI and CT	Classification	Intracerebral haemorrhage
[73]	X-ray/digital histopathology	Anomaly detection	Various
[74]	Single-cell images	Classification	Myeloid leukaemia
[75]	None	Anomaly detection	None
[76]	CT	Classification	COVID-19
[77]	MRI	Denoising/classification	Autism/brain

Literature Review Findings for Autoencoders

Autoencoders are associated with both unsupervised and semi-supervised deep learning tasks. These tasks are normally preferred in the absence of the ample annotated datasets required for deep learning activities, especially in imaging analytics. Autoencoders are used to augment small annotated datasets, often before a segmentation task. For example, a method known as GenSeg was proposed that combines the generative aspects of autoencoders to generate a latent representation of tumour cells from a labelled health image and uses U-Net architecture to obtain the unique information of tumours present in the MR images. There is also other related research [69]. Noise reduction is another task that has benefited from the generative features of the autoencoder framework. The fusion of Bayes shrinkage fused wavelet transform (BSbFWT) was proposed for noise removal and an autoencoder block for generating a noiseless variant of an MR image of prostate cancer [72]. MR images can be prone to Gaussian and Rician noise, which is introduced during image capture by the MRI device and the imaging environment. The effectiveness of noise-reduced images was measured using several parameters, such as peak signal-to-noise ratio (PSNR), mean squared error (MSE), structural similarity index metric (SSIM), and mean absolute error (MAE), which outperformed numerous traditional filtering approaches presented in their work. Autoencoders can also be used in classification tasks [70,72,74,76]. Autoencoders were used to mesh a fully convoluted network for a classification task [73]. The authors proposed a convolutional mesh autoencoder (CMA) framework for the classification of syndromic craniosynostosis (SC) of three known SC variants—Muenke, Crouzon, and Apert disease—in infants and adults using 3D computed tomography (CT) images and others with very good percentages on evaluation metrics such as sensitivity, specificity, and accuracy when compared to normal subjects. These tasks are very critical, as late and inaccurate diagnosis might prove irreversible, causing permanent damage to the brain. As indicated, an autoencoder which is infused with four convolutional layers for encoding and decoding, was used for the construction of face models from the CT images. Autoencoders were also used to detect complex anomalies present in medical imaging [73,75,77]. In another study, autoencoders were applied to chest X-ray and digital pathology images [76]. Abnormalities that were barely visible, such as metastases in lymph nodes, always proved difficult to detect as they resemble normal images in pathological slides. The authors proposed a deep perpetual autoencoder that learnt the shared patterns of normal images and content similarities to abnormal ones and restored them correctly. For evaluation of their mode, they used the receiver operating characteristic (ROC), as it

integrates the classification performance of the normal and the abnormal class. Their model was also evaluated on non-medical images and performed well in comparison, indicating the suitability of their autoencoder for medical imaging purposes.

3.4. Generative Adversarial Networks

Generative adversarial networks (GANs) are groups of deep learning artificial neural networks that can be used for generating synthetic medical images, data augmentation, and improving image resolution. They are valuable for enhancing small datasets, which are common in medical imaging, and can also be used to create better training data for improving model performance. A GAN uses an unsupervised learning algorithm and can be used for mostly semi-supervised and unsupervised learning [78,79]. A GAN consists of two main parts—the generator (G) and the discriminator (D). The generator, which comprises a multilayer perceptron (LP), learns the data distribution of the input image and produces a similar image to the input, also known as "fake data". The job of the discriminator, which is also an MLP, is to discriminate between the generated image from the generator network and the input image. The result of this discrimination, which constitutes an error between the original input image and the generated image, is fed back to the generator input to make the generated image more realistic, i.e., closer to the original image. During training, the weights of the generator and discriminator are alternately updated, and the weight updates of the generator come from the discrimination error. Both networks are engaged in a competing optimisation process. This process continues until there is an equilibrium between the generator and the discriminator networks [1,80,81]. The architecture of a GAN is shown in Figure 7.

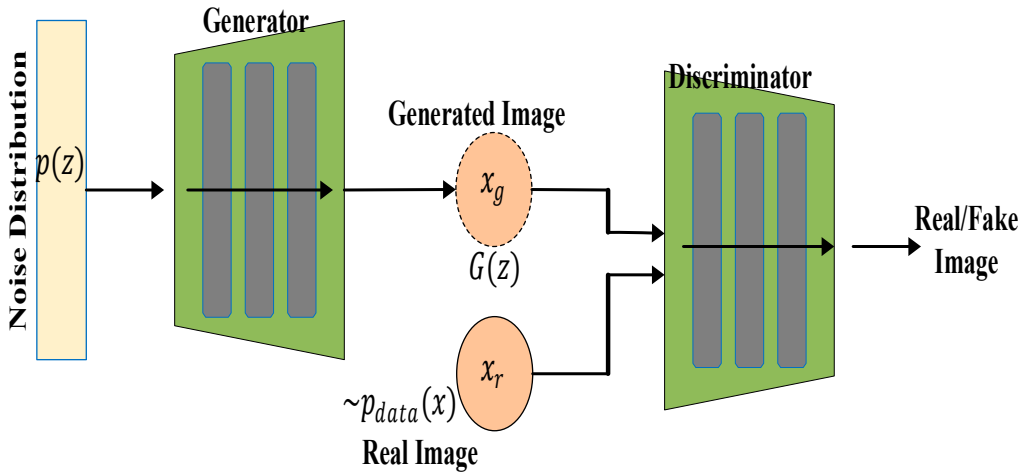


Figure 7. Structure of a GAN [79].

The generator (G) consists of an input noise vector z such that $z \sim p_z(z)$, where p_z is a prior distribution that could be Gaussian. The output of the generator, which could be an image, is represented by $G(z)$, and the objective is to maximize log $(D(G(z)))$. The input to the discriminator (D) is the image data sample x and the output of the generator network. The output to the discriminator is the probability that x is real or fake. This is represented by $D(x) \epsilon [0, 1]$, where 0 represents fake and 1 represents real. The goal of the discriminator network is to maximize $\log(D(x) + log(1 - D(G(z))))$. The discriminator loss $\mathcal{L}_D$ is given by:

$$\mathcal{L}_D = -\mathbb{E}_{x \sim p_{data}}[\log D(x)] - \mathbb{E}_{x \sim p_z}[\log(1 - (G(z)))] \quad (19)$$

where $-\mathbb{E}_{x\sim p_{data}}[\log D(x)]$ maximises the probability of correctly classifying the real data and $\mathbb{E}_{x\sim p_z}[\log(1-D(G(z)))$ minimises the probability of misclassifying the generated data. When both generator and discriminator are at a stable equilibrium, the combined minimax game function is defined as:

$$\underset{G}{\text{Min}}\ \underset{D}{\max}\ \mathbb{E}_{x\sim p_{data}}[\log D(x)] + \mathbb{E}_{x\sim p_z}[\log(1-(G(z)))] \quad (20)$$

Table 3 shows the summary of GAN-based techniques.

Table 3. GAN-based techniques.

Article	Imaging Modality	Task	Disease/Body Part	Variant Used
[79]	MRI/retina fundus	Image synthesis	-	-
[82]	MRI	Image resolution	Brain	Cycle-GAN
[83]	CT	Image synthesis	COVID-19	Enhanced vanilla
[84]	X-ray/CT	Image Denoising	Chest/thorax	CGAN
[85]	Various	Image resolution	Various	Enhanced vanilla
[86]	-	Image synthesis	Skin cancer	DCGAN
[87]	MRI/CT	Image synthesis	Head/neck	Vanilla GAN
[88]	MRI/CT	Image resolution	Bladder cancer	Enhanced vanilla
[89]	Retina fundus/MRI	Image resolution	Various	Vanilla GAN
[90]	CT/MRI	Translation	Thorax/brain	CGAN

Literature Review Findings for GANs

Deep learning technique such as GANs can be used in medical imaging in two different ways. The generative network in GANs has been used for image synthesis purposes to generate synthetic datasets for tasks where there are limited annotated datasets and using the discriminating networks of GANs for anomaly detection [79]. The quality of the generated synthetic images can be measured by the use of qualitative metrics such as Fréchet inception distance (FID), which is a measure of similarities between the representations of the generated and the real input images, structural similarity index measure (SSIM), which indicates the similarities of the structures (usually image contrast and brightness), and peak signal-to-noise ratio (PSNR), which is used to analyse the sensitivity of the generated image. PSNR is the most important metric when dealing with medical images. Quantitative metrics used for generated image quality include number of parameters (NoPs) to represent the total number of trainable parameters in the GAN and floating points of operations (FLoPs), which measure the cost of computation of the network [78]. Most studies that have used GANs have used them for image synthesis, image resolution, image translation, and image denoising. For example, a lightweight GAN (LEGAN) was proposed to generate high-fidelity images from MRIs and retina fundus images [79]. That technique boasts fewer parameters used in the training process and lower FID when compared to other variants of GAN such as CGAN, DCGAN, Pix2Pix, and so on. To achieve this, they used a two-stage GAN to create a coarse-to-fine paradigm, which is necessary in generating images with high sensitivity to the fine patterns of the original image. To lower the NoPs, redundancy in the convolutional kernel was eliminated by using the principal components of a normally fully ranked convolutional kernel for feature extraction. The resolution of MR images of the brain was improved by increasing the contrast using a GAN variant known as Cycle-GAN to aid in the segmentation of the images [85]. To achieve this, they used the image-to-image translation technique to create a high tissue contrast (HTC) of the real image. The attention block of the Cycle-GAN used in that study helped in focusing on a single tissue type and increasing the contrast within the tissue. Other studies have used GANs for medical image resolution, including [85]. For

denoising tasks, a GAN was used for denoising X-ray images using the CGAN variant of the GAN architecture [87]. The authors customised it to deal with spatially varying noise, which is often overlooked when dealing with medical images. To achieve this, the gradients of original images were merged with noisy images to obtain conditional information for the CGAN, thereby enhancing the contrast. The convolutional layers of the generator were used in full for better feature extraction. For improved consistency between the real and fake images, the reconstruction loss was combined with WGAN loss to create an objective loss for the network, obtaining remarkable PSNR and SSIM performance when compared to other state-of-the-art (SOTA) GAN architectures.

3.5. U-Net

U-Net architecture combines the best of CNN and encoder–decoder models, specifically for the purposes of segmenting medical images [91,92]. They have found application in major medical imaging modalities such as CT, X-ray, and MRI. U-Net's ability to exploit small, annotated data samples (based on its fully connected layers) by leveraging data augmentation and improved feature extraction made it a valuable technique for medical image segmentation [92]. The U-shaped architecture with skip connections help to delineate objects in images, making it highly effective in medical image analysis, particularly in tasks like tumour detection, organ delineation, and segmentation of medical images from various modalities, and variants have been widely embraced among the many different deep learning networks [93].

U-Net architecture is mainly composed of two paths. The first path is referred to as the contracting or encoder path. It uses a downsampling module that consists of several repeating convolutional blocks for semantic and contextual feature extraction. Each convolutional block has two successive 3×3 convolutions, ReLU activation functions, and a pooling layer. The pooling layer serves to increase the receptive field of the convolutional network with no extra burden of computing resources that might be introduced by an additional convolutional block. The second path of U-Net is the expansive or decoder path. It is saddled with the task of upsampling spatial resolutions of the feature maps from the contracting path, usually by a factor of two. During this operation, the dimensions of the features are reduced and a pixel-wise classification/resolution score is produced. The expansive path is made up of a 2×2 transposed convolutional layer (reversing the operation in the contracting path), which is followed by a 3×3 convolutional layer and a ReLU activation function. There is also a bottleneck layer that serves as a connection between the two paths. It is also comprised of two blocks of 3×3 convolutional layers and a ReLU activation function. The embedded skip connections in the bottleneck copy the output of each stage of the paths, helping to learn contextual and semantic representations in the deep and shallow layers, respectively [92,94]. The U-Net architecture is shown in Figure 8.

The encoders/contracting paths shown on the left-hand side of Figure 8 comprise several layers. Each layer l in the encoder is represented by the function:

$$f^l = \sigma\left(W^l * f^{l-1} + b^l\right) \tag{21}$$

where f^{l-1} represents input features from a previous layer, W^l, b^l is the convolutional weights and biases, and σ represents the activation function, which is ReLU in most cases.

The output of the encoder is normally max-pooled and can be represented as:

$$f^l_{pooled} = MaxPool\left(f^l\right) \tag{22}$$

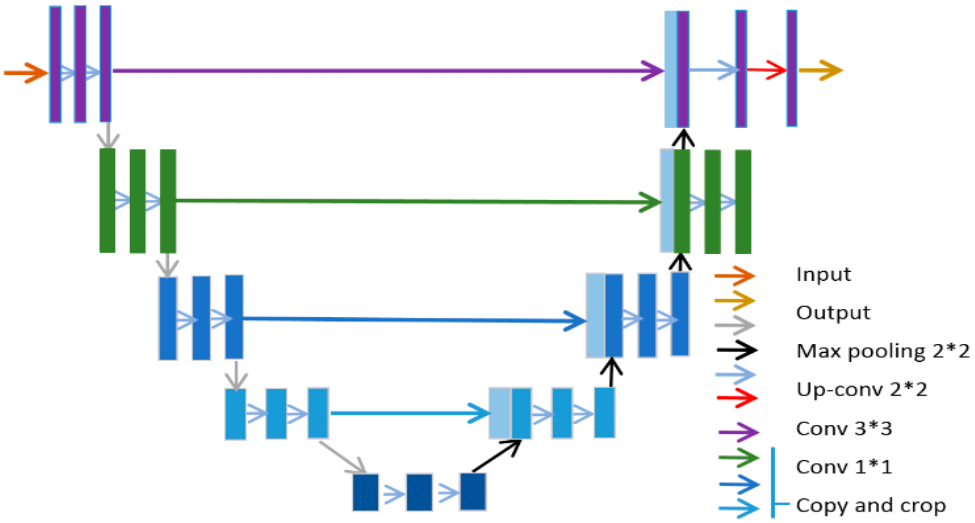


Figure 8. The architecture of U-Net [95].

The output is then passed through the bottleneck layer, which is another convolution function, represented by:

$$f_{bottleneck} = \sigma\left(W_b * f^L + b_b\right) \tag{23}$$

with L representing the number of layers in the encoder/contracting path. The upsampling taking place in the decoder is performed via a skip connection at every layer on the encoder. As such, each layer in the decoder upsamples the feature map, concatenates the corresponding encoder features, and applies convolution with the given function as:

$$f^l_{upsampled} = ConvTransponse\left(f^l_{bottleneck}\right) \tag{24}$$

$$f^l_{concat} = Concat\left(f^l_{upsampled}, f^{L-1}_{encoder}\right) \tag{25}$$

$$f^l = \sigma\left(W^l_{decoder} * f^l_{concat} + b^l_{decoder}\right) \tag{26}$$

At the output layer, there is a 1×1 convolutional layer that uses softmax activation to map the desired output segments:

$$y = Softmax\left(W_{out} * f^{L_{decoder}} + b_{out}\right) \tag{27}$$

During training of a U-Net, the loss function used is usually the cross-entropy loss, and it is normally applied pixel-wise.

Table 4 provides a summary of works that have implemented the U-Net segmentation.

Table 4. U-Net segmentation techniques.

Article	Imaging Modality	Disease/Body Part	Variant Used
[95]	MRI	Brain tumour	Enhanced U-Net
[96]	Various	Various	Enhanced U-Net
[97]	CT	Hepatocellular carcinoma	Enhanced U-Net
[98]	CT	Liver	Enhanced U-Net
[99]	MRI	Brain tumour	None
[100]	Colour fundus	Diabetic retinopathy	Enhanced U-Net
[101]	MRI	Various/musculoskeletal	Enhanced U-Net
[102]	MRI	Lower limb muscle	Attention U-Net/SCU-Net
[103]	MRI	Musculoskeletal	Various
[104]	Various	Various	Enhanced U-Net (U-Net++)
[105]	Ultrasound	Breast cancer	Enhanced U-Net (attention gate)

Literature Review Findings for U-Net

U-Net is mostly used for segmentation tasks, especially for segmentation of cancer for various imaging modalities. Over time, there have been a lot of modifications to the vanilla U-Net model as different researchers have tried to enhance different facets, ranging from the skip connections to modifying the convolutional layers with attention networks in a bid to increase segmentation quality or reduce computational burden. A multi-level feature assembly MFLA U-Net was reported that was integrated with a multi-scale information attention and pixel-vanishing attention mechanism [106]. This enhanced U-Net model was designed to boost segmentation performance. The model was tested on different medical imaging datasets with different modalities such as colonoscopy and dermoscopic images. Dice index coefficients were used as a metric to evaluate the effectiveness of the model in segmenting these images. It outperformed many state-of-the-art U-Net models on the datasets used for testing. A lightweight U-Net architecture was applied to a publicly available brain tumour dataset (BraTs) to segment brain tumours [99]. The focus of the study was mainly developing a low-resource U-Net framework that had a multimodal CNN encoder–decoder. Augmentation was excluded to reduce the computational demand on the network. The model achieved remarkable performance, with Dice coefficient values of up to 0.93 for specific segmented classes when compared to other U-Net models. An enhanced U-Net model with minimal parameters was reported [97]. The authors achieved this by developing a framework known as stack multi-connection simple reducing net, otherwise known as SRNet. This network used fewer convolution operation in the downsampling and upsampling processes, which in turn helped to reduce the total parameters of a vanilla U-Net algorithm by 20%. They also modified the original architecture by ensuring the convolutional layers were not stacked, helping to reduce information loss. Their model was also tested using the BraTs dataset. They obtained matching results with popular variants of the U-Net model, also using Dice coefficients as an evaluation tool. Others modified a vanilla U-Net for the purpose of improving the accuracy of segmentation [99]. They explored the weakness of U-Net models that only focus on contextual information and neglect other useful features of the channel. The developed HDA-ResUNet combined the best of attention mechanisms, U-Net, and dilated convolution. They evaluated their model on ISI and LiTS segmentation datasets and achieved than a conventional U-Net.

3.6. Transfer Learning

Deep learning techniques are known to be computationally intensive, owing to the large number of trainable parameters available in their networks. These parameters increase substantially as the network deepens. Also, the availability of large annotated medical image datasets is scarce, and this is a very important consideration in the use of deep learning for medical image analysis [107,108]. Transfer learning algorithms were developed to help solve such problems by providing a means to reparametrize an already-trained large deep learning network. These networks are CNN-based and trained on millions of images with different classes. The resulting learnt parameters are saved to be reused on other datasets. Some aspects of these networks are modified to suit new datasets. Transfer learning involves using pretrained deep learning models (e.g., ResNet, VGG, DenseNet, GogleNet, XceptionNet, AlexNet, Inception V3, and SqueezeNet) and fine-tuning them on medical images. Developing models when using transfer learning is performed in two stages: initialising the weights and fine-tuning. During initialisation of the weights, the weights of a previously trained model with a different dataset, i.e., AlexNet trained on ImageNet, as shown in Figure 9, are copied. When a new training dataset, such as medical images in this case, is used to train this model, the weights are updated. In the fine-tuning process, some of the CNN layers are frozen and thereby the

weights are not being updated. Another method of fine-tuning is freezing all the CNN layers barring the classification layer that is adjusted according to the requirements of the medical image data. This technique is highly effective when the available dataset is small or specific to a particular medical condition. It reduces training time and improves diagnostic accuracy [109]. Transfer learning can be categorised into inductive, transductive, and unsupervised learning based on data labels. These can also be categorised as homogeneous and heterogeneous based on how consistent the dataset features and labels are between the source and target domain [110,111].

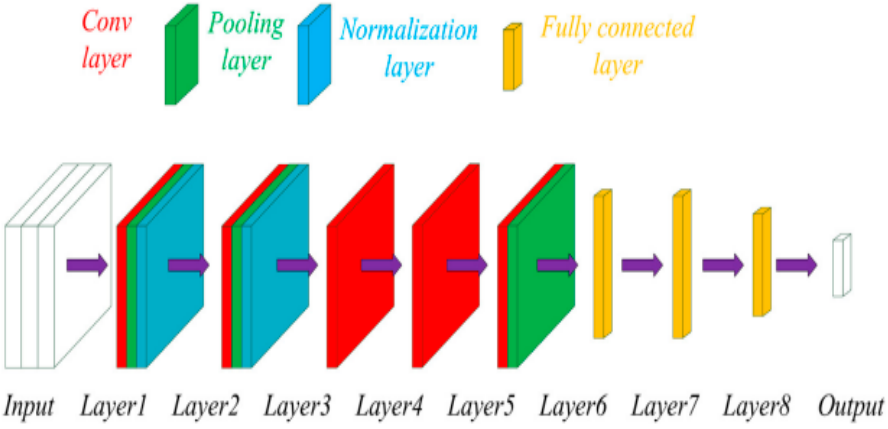


Figure 9. Structure of a transfer learning model—AlexNet [108].

The concept of transfer learning can be modelled mathematically as described next. The source domain be represented as:

$$D_s = \left\{ x_s^{(i)}, y_s^{(i)} \right\} \text{ for } {}_{i=1}^{N_s}, \tag{28}$$

where $x_s^{(i)} \in X_s$ represents the input data from the source domain, $y_s^{(i)} \in Y_s$ represents the corresponding labels, and N_s is the number of samples. The target domain is represented as:

$$D_T = \left\{ x_T^{(i)}, y_T^{(i)} \right\} \text{ for } {}_{i=1}^{N_T}, \tag{29}$$

where $x_T^{(i)} \in X_T$ represents the input data from the target domain and $y_s^{(i)} \in Y_s$ represents the corresponding labels. N_T is the number of samples, with the assumption that $X_s \neq X_T$ or $Y_s \neq Y_T$. The objective of the transfer learning function is to find a model $f_T(x_T; \varnothing_T)$ that minimises the loss in the target domain. If we represent the loss function by $\mathcal{L}_T$, then $\mathcal{L}_T$ can be defined as:

$$L_T = \frac{1}{N_T} \sum_{i=1}^{N_T} l(f_T(x_T^{(i)}; \varnothing_T),\ y_T^{(i)}) \tag{30}$$

where l is the specific loss algorithm, which can be cross-entropy or mean squared error and $\varnothing_T$ represents the target domain model parameters. The transfer learning process between the source and target domain takes two steps. We pretrain the model on the source domain using the objective loss function given by the equation:

$$L_s = \frac{1}{N_s} \sum_{i=1}^{N_s} l(f_s(x_s^{(i)}; \varnothing_s),\ y_s^{(i)}) \tag{31}$$

The learned parameters $\varnothing_s$ from the source domain are transferred to the target domain by freezing the base layers, where $\varnothing_s^{base}$ is fixed or by fine-tuning specific layers i.e., $\varnothing_T^{new}$.

The target domain overall model becomes:

$$f_T\left(x_T^{(i)}; \varnothing_T\right) = f_s^{base}\left(x_T; \varnothing_s^{base}\right) + f_T^{new}(x_T;\ \varnothing_T^{new}) \tag{32}$$

The learning process is then optimised by regularising the losses. Therefore, a combined loss function L is given by:

$$L = L_T + \lambda \cdot L_{regularisation} \tag{33}$$

where λ is the regularisation weight and $L_{regularisation}$ can be any term used for smooth transfer, e.g., L_2-norm.

Findings for Transfer Learning

Transfer learning (TL) techniques are mostly applied in the classification of medical images, as summarised in Table 5.

Table 5. Summary of transfer learning techniques.

Article	Imaging Modality	Disease/Body Part	TL Variant/Best Model
[112]	Histopathological images	Breast cancer	ResNet 50
[113]	MRI	Brain tumour	Improved ResNet 50
[114]	MRI	Alzheimer's	Various (EffiecientNet)
[115]	CT	Pulmonary nodules	Various (DenseNet)
[116]	X-ray/CT	COVID-19	Various (VGG 16)
[117]	MRI	Alzheimer's	Modified ResNET 18
[118]	Ultrasound	Thyroid	VGG-16

This is because the networks they are learning from have been pretrained for classification as well. Several transfer learning models, such as VGG16, DenseNet 121, and ResNet 50, were used for the binary classification tasks of X-ray radiographs and CT images of COVID-19 patients [118]. Most of the model parameters were frozen and the weights were not initialised. However, they were able to obtain very good results in terms of classification accuracy, with VGG16 performing best, with accuracy of 99%. Transfer learning models like those in [118] were compared and a custom CNN built from scratch to compare the efficacy of the TL process. The models were trained on different publicly available datasets with varying modalities, ranging from X-rays to CT scans of different diseases, including lung cancer and brain tumour. The initial convolutional layers of the TL networks used in that work was frozen and the weights of the top layers were updated. The models were trained on all the datasets, and improved accuracy was obtained with the TL models, with ResNet 50 showing the highest accuracy of 90% for the histopathological images. In another study, transfer learning techniques were trained on the ImageNet datasets [119]. The authors pretrained a novel hybrid DCNN model that combined convolutional layers with global average pooling for each layer with a skip connection in each of the convolutional layers. The 200,000 augmented unlabelled data were sourced from different repositories of biopsy breast cancer image datasets and were used to train the DCNN model. The pretrained DCNN model was then used to classify annotated skin cancer lesions as benign or malignant based on the weights from the novel pretrained model. They obtained improved accuracy when compared to other models tested on similar datasets. A transfer learning model (ResNet 18) was used for the diagnosis of different stages of Alzheimer's disease (AD) [120]. The model was fine-tuned by unfreezing all the layers, allowing it to update all it weights based on the publicly available DICOM MRI datasets used. High accuracy was obtained for the three different classes (in the range of 99%), and their model outperformed

other related works that classified AD. Qing Guan et al. [121] used an Inception-v3 TL model for the classification of papillary thyroid carcinomas (PTCs) and benign thyroid nodules using ultrasound images. A total of 2836 images from 2235 patients were used. The model was trained to crop nodule margins and make diagnostic predictions. The best performance was achieved using a 50-pixel margin and 384 × 384 image size. In the test group, Inception-v3 achieved sensitivity of 93.3% and specificity of 87.4%. Their proposed model was most accurate for nodules sized 0.5–1.0 cm.

3.7. Vision Transformers

Despite the significant successes recorded in enhancing the diagnostic accuracy of deep learning models such as CNN, RNN, and U-Net in the classification and segmentation of medical images, there remain some limitations. Their reliance on localised feature extraction leading to inductive bias and sequential operation makes them fall short when the medical imaging tasks requiring long-range dependency and global feature extraction [122]. Although initially designed for natural language processing tasks such as sentiment analysis [123], machine translation, and text summarisation [124], their ability to capture long range dependencies in image pixels helps to build a more robust segmentation–classification model. The vision transformer is a relatively new deep learning architecture that is increasingly being applied to medical imaging. Developed by Google in 2020, it performs segmentation or classification using the transformer architecture. The ViT creates a partition of the input images into multiple patches of 16 × 16 pixels and linearly embeds them. For the pixels to be suitable for the transformer architecture, they must be transformed into fixed-length vectors [125]. The self-attention mechanism represents the main feature of the ViT architecture, as this forms the basis of the interactions among the pixel patches. It also uses positional encoding to represent the spatial location of the image patches. Feed-forward layers placed after the self-attention layers are generally used by the model for final decisions [126]. Several large language models have been developed for medical image analysis [127]; however, we have focused on the legacy transformer model in our analysis. The original architecture of the vision transformer is shown in Figure 10.

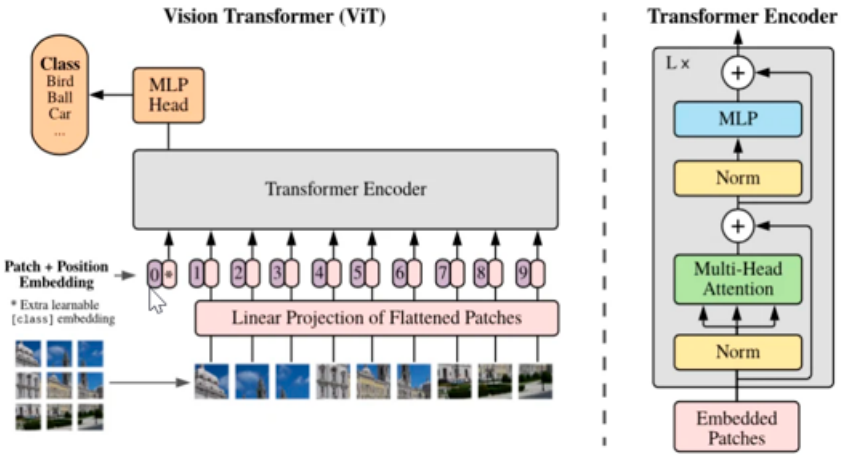


Figure 10. Vision transformer architecture [128].

If we represent the input to the ViT model by an image X, then X can be defined such that:

$$X \in \mathbb{R}^{H \times W \times C} \tag{34}$$

where H and W represent the height and width of the image, respectively, and C is the number of channels, which is usually three for RGB images. The input image is divided into non-overlapping patches of size P such that:

$$X_{patches} = \{x_{1,}x_{2,}\ldots,x_{N,}\} \tag{35}$$

where $N = \frac{H \times W}{P^2}$ is the total number of patches and each patch is flattened as:

$$x_1 \in \mathbb{R}^{P^2.C} \tag{36}$$

The patches are sent through the transformer encoder layers consisting of the multi-head attention layer, which computes the relationships between patches using the K, V, Q (keys, values, and queries) matrices, which are computed as:

$$K = Z^{(l-1)}W_K,\ V = Z^{(l-1)}W_V,\ Q = Z^{(l-1)}W_Q \tag{37}$$

where $Z^{(l-1)}$ is the positional patch embedding for each patch and W_K, W_V, and W_Q are the learnable weight projection matrices, which have the same size as embedding space. This is then passed to the feed-forward neural network (FFNN) and a classification head for a classification task [128].

Literature Review Findings for Vision Transformers

Several studies have used ViTs and their variants, including TNT, Swin, DeiT, and PVT [129], for classification, registration, and segmentation of medical images. Just like the convolution-based transfer learning models, learnable parameters from transformer models such as those can also be used for specific DL tasks. For example, a pretrained Swin transformer was used for the classification of breast cancer using publicly available breast X-ray images [130]. The dataset was resized to fit the Swin transformer input size and augmentation was also performed to improve generalisation, achieving very high classification rates based on the selected metrics for evaluation, with accuracy of 99.9% and precision of 99.8%. The results were compared with convolution-based TL algorithms (ResNet50 and VGG16), and the Swin transformer had superior performance. MR images of ischemic strokes were classified using ViT by [131]. The vanilla ViT or ViT base used in the study also had limited fine-tuning, and hyperparameters were adjusted to fit the datasets to be classified. They also augmented and resized the MR images as preprocessing steps before deploying the ViT model. They achieved an impressive accuracy score of 97.59% when compared to the VGG16 model used in a similar study, demonstrating the superiority of transformer models for classification tasks. Transformers were used for image registration tasks [132]. In that study, the authors developed the TransMorph algorithm, which is a hybrid transformer and convolutional network. The network leveraged the encoder–decoder architecture of transformers, but instead of the attention mechanism at the decoder, this was replaced with a convolutional network. For the transformer network, they used the Swin variant due to its ability to extract feature maps at different resolutions by merging patch layers, making it suitable for the image registration task. The algorithm was tested on different image-pair datasets comprising mainly MRI and CT modalities for registration purposes. They obtained very competitive results based on Dice score evaluations when compared to both traditional and other DL methods used for similar tasks. Chuantao Wang et al. [133] proposed a ViT model with a deep neural network called ConvTrans-Net for the segmentation of lymph node tumours using ultrasound images from Kaggle. The model works by concatenating the different feature vectors of the ultrasound images that are passed on to a multilayer perceptron, and the output of the multiple attention mechanism is passed on to a feed-forward layer for segmentation purposes. Their model was evaluated using the Jaccard similarity coefficient, precision, and recall, and was benchmarked with the NCP and WS-2017 models. They obtained a

Jaccard coefficient 85.21%, recall of 89.65%, and precision of 85.17 when compared to the benchmark models.

3.8. Hybrid Models

Most deployed deep learning methods use CNNs, as described in the preceding sections, with variants such as VGG, ResNet, LSTM, RNN, GAN, and GRU all being used for different medical image analysis. However, different authors have tried to combine the strengths of some of the models to deal with the weaknesses of the others by combining them, and this forms the basis of hybrid models [3]. Some studies have also combined different deep learning methodologies by focusing on the strengths of the methodologies. For example, a convolutional network's strong local feature extraction has been combined with the long-range dependencies of transformers when performing medical analysis.

3.8.1. Convolution-Based Hybrid Models

A hybrid model of convolution algorithms was developed comprising SegNet, MultResUNet, and the krill herd optimisation algorithm (KHO) to improve the segmentation of CT scans of liver lesions and RNA genome sequencing [134]. The SegNet framework provided the segmentation capacity of their model, utilising pixel-wise classification through the softmax layer. A CNN-based architecture was employed with MultiResUNet to handle the lesion segmentation, together with the SegNet framework. The hyperparameters of the models, when optimised through the KHO algorithm, helped to improve the segmentation process. The algorithm was tested with a publicly available LiTs dataset using evaluation metrics such as Dice coefficient, F1 score, and accuracy, and obtained better results, with F1 scores comparatively higher than the models used for comparison. A hybrid model of convolutional methods involving RestNet and U-Net (ResUNet) was developed for the segmentation of liver and tumours using CT images [135]. The model focused on improving the available models by providing improved image contrast and segmenting irregular tumour shapes and small tumours. The combination used the best of RestNet's residual connection and U-Net's encoder and decoder structure to enhance feature learning, segmentation precision, and efficiency. Various augmentation techniques such as rotation and reflection were also implemented to increase the variability in the dataset, and accuracy of 99.6% and a Dice coefficient of 99.2% were achieved. Yangyang Zhu et al. [136] developed a hybrid model for the classification of unexplained cervical lymphadenopathy (CLA) using ultrasound images of patients in an underdeveloped area of China. The CLA-HDM model was made up of three smaller models, each designed to handle a specific diagnosis task related to unexplained lymph node abnormalities (CLA). Model 1 checked whether the issue was benign (non-cancerous) or malignant (cancerous). Model 2 looked deeper into benign cases to decide between tuberculosis or a reactive condition. Model 3 examined malignant cases to determine if they were due to metastasis (spread from another cancer) or lymphoma. Each model had two input branches, one for greyscale ultrasound (BUS) images and another for colour Doppler (CDFI) images. The CDFI images were processed to emphasize important colour details. Then, both images were analysed by a deep learning model (ResNet-50). The efficacy of their model was evaluated using the area under the curve, which was above the 0.8 benchmark for each model. The authors in [137] developed an ensemble model of a custom-built convolutional neural network (5-CNN) and a transfer learning model using the pretrained VGG-19 architecture for the classification of thyroid disorders using ultrasound images of four categories: autoimmune, nodular, micro-nodular, and normal. The combined CNN-VGG method showed superior performance, achieving test accuracy of 97.35%, specificity of 98.43%, and sensitivity of 95.75%. It also demonstrated

strong predictive capabilities, with high positive and negative predictive values, and an area under the ROC curve of 0.96.

3.8.2. Convolution–Transformer-Based Hybrid Models

A hybrid model called TBConvl-NET that combines CNN, LSTM, and ViT for the segmentation of several diseases was developed using publicly available datasets of different modalities such as ultrasound and MRI [138]. The hybrid model targeted some well-known challenges in segmenting medical images, such as scale, texture, and shape of pathology. Due to the high computational resources required, they used a depth-wise separable convolution, thereby reducing computational overhead. Swin transformer blocks were used in the skip connections to help deal with the varying scales of the data and help preserve semantic information. They used the Dice index, accuracy, and Jaccard index to evaluate their model, compared the developed model with other hybrid segmentation models, and obtained improved results. A hybrid classification model that combines the transformer and the convolution model to improve the classification of skin lesion was developed using publicly available datasets [139]. It used the Swin-Unet architecture to perform image segmentation, leveraging the self-attention of the Swin transformer and robust hierarchical analysis of the UNet. This was combined with the Xception and ResNet 18 models for feature extraction to further improve on the image analysis. For hyperparameter tuning, a hybrid salp swam algorithm (HSSA) was used to obtain the optimal parameters, hence avoiding local minima during training. A gated recurrent unit (GRU) network was used for the eventual classification. Accuracy of 94.51% and 95.38% was achieved on the two datasets used in the study. The model performed better when compared to TL models like AlexNet and ResNet18. Laifa Yan et al. [140], proposed a framework focusing on the three-vessel view (3VV) that identifies three major heart vessels—the pulmonary artery, aorta, and superior vena cava—using ultrasound images. In the first step, a YOLOv5 (based on CNN architecture) model was used to detect these vessels and earmark the region of interest. In the second step, they used a modified DeepLabv3 model with a new attentional multi-scale feature fusion (AMFF) module to perform segmentation. Using a dataset of 511 images, the model achieved high accuracy, with Dice scores of over 85% for the pulmonary artery and aorta.

4. Discussion

In this article, we have provided a broad view of various deep learning methodologies used for medical imaging analysis. These included convolutional neural networks (CNNs), recurrent neural networks (RNNs), autoencoders, generative adversarial networks (GANs), U-Net architectures, vision transformers, and hybrid models. For each of these methodologies, some of the key findings were presented.

CNNs are extensively used for medical imaging tasks like disease detection, classification, and segmentation, and the most important contribution of the algorithms is the unrivalled feature extraction through convolutional layers, making them effective for tasks requiring spatial hierarchy. CNN models can be lightweight in terms of computing resources, as inspired by ResNet, while also balancing model performance. Deeper CNN architecture can be computationally expensive, but can be very useful in multi-disease detection, e.g., in the classification and detection of COVID-19 and viral pneumonia, achieving high accuracy due to advanced hardware. It can also be effective when the task includes pixel-level feature extraction. Examples can be seen in its use in the classification of ultrasound image classification of fatty liver. RNNs, particularly their variants like long short-term memory (LSTM) and gated recurrent units (GRUs), are suitable for sequential data tasks. Their ability to retain temporal dependencies is utilised in dynamic imaging

and disease progression monitoring. RNNs, which usually have convolutional layers, are effective for classification tasks for different imaging modalities, with MRI dominating. They have been applied in the detection of several diseases, such as Alzheimer's and breast cancer. LSTM variants of the RNN algorithm handle tasks like segmentation of temporal MR images and noise removal in diagnostic images. Some modifications have also been applied to RNN algorithm training, such as using the honey badger algorithm (HBA) for optimisation of the model training parameters.

Autoencoders are unsupervised models used for dimensionality reduction, anomaly detection, and data augmentation. Their encoder–decoder structure helps compress and reconstruct medical images with minimal distortion. Autoencoders are employed for image analysis tasks like segmentation and denoising and have been used with several imaging modalities for the analysis of different diseases. Enhancing autoencoders by fusion with wavelet transform can also help improve noise removal from medical images. They can also be combined with convolutional networks to improve disease classification. GANs are used for generating synthetic medical images, data augmentation, and improving image resolution or quality. The main components of a GAN comprise a generator for creating synthetic images and a discriminator for distinguishing between real and generated images. GANs can be very computationally intensive due to the dual deep convolutional networks. Research in this area is focused on reducing models' parameters and maintaining performance. Different variants of GANs have employed different image analysis tasks, with each variant tuned to fit the specific tasks.

U-Nets are specialised for medical image segmentation and have been widely adopted for tasks like tumour detection and organ delineation. The architecture combines encoder–decoder pathways with skip connections for improved feature representation. Different variants of the U-Net architecture also exist to enhance the vanilla model, making them suitable for specific imaging modalities while performing the main task of segmentation. Also, enhancements have been created to reduce model operating costs while retaining the segmentation process. More recent studies have been adding transformers to U-Nets to further enhance segmentation accuracy. Transfer learning leverages pretrained models (e.g., ResNet, VGG, DenseNet) on large datasets and fine-tunes them for medical imaging tasks. They are mostly used for classification tasks. Models like VGG16 and ResNet50 perform well for COVID-19 classification, achieving very high accuracy on different imaging modalities. Some transfer learning models are quite deep, with millions of parameters, making them very computationally intensive. Custom hybrid models pretrained on unlabelled medical datasets have also been used to improve classification accuracy for skin lesions and breast cancer images, mimicking the transfer learning ideology. They provide an option in terms of computing resources and domain-specific training and learning. ViTs are a relatively newer deep learning architecture, adapted for medical imaging tasks. They require global feature extraction and long-range dependencies. They divide images into patches and process them using self-attention mechanisms. ViT models can also be pretrained like most CNN models and learning transferred for newer task. Several variants have also been developed depending on the task. With larger training datasets, ViTs have been shown to outperform CNN models when used for tasks with fewer annotated data.

Hybrid models combine the strengths of multiple architectures (e.g., CNNs with RNNs or CNNs with transformers) to improve medical image analysis. CNN-based hybrids like ResUNet combine ResNet's feature extraction with U-Net's segmentation capabilities. They are used widely for cancer related image segmentation. Convo-transformer hybrids like TBConvl-NET integrate CNN, LSTM, and ViT for improved segmentation and classification of diseases. Hybrid models can also be optimised. Algorithms such as the hybrid salp

swarm algorithm (HSSA) could be used to obtain optimal parameters, thereby increasing model performance.

5. Conclusions

This review explored the transformative impact of deep learning algorithms on medical image analysis. Convolutional neural networks (CNNs) are widely used for feature extraction and disease classification across medical imaging modalities, achieving high accuracy with optimised computational costs. Recurrent neural networks (RNNs), including LSTM and GRU, enhance temporal analysis for disease progression monitoring. Autoencoders and generative adversarial networks (GANs) assist in data augmentation, denoising, and synthetic image generation. U-Net architectures improve segmentation for tumour detection and organ delineation. Vision transformers (ViTs) leverage attention mechanisms for superior classification and registration. Hybrid models combining CNNs, transformers, and optimisation techniques enhance performance, while transfer learning mitigates data scarcity, ensuring robust results across imaging applications. Together, these advancements underscore the versatility and efficiency of deep learning in medical diagnostics, paving the way for improved clinical outcomes and personalised healthcare solutions. Future advancements may focus on computational efficiency, integrating multimodal data, and enhancing interpretability for clinical adoption.

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References

1. Jeong, J.J.; Tariq, A.; Adejumo, T.; Trivedi, H.; Gichoya, J.W.; Banerjee, I. Systematic Review of Generative Adversarial Networks (GANs) for Medical Image Classification and Segmentation. *J. Digit. Imaging* **2022**, *35*, 137–152. [CrossRef] [PubMed]
2. Yu-Jen Chen, Y.; Hua, K.; Hsu, C.; Cheng, W.; Hidayati, S.C. Computer-aided Classification of Lung Nodules on Computed Tomography Images via deep Learning Technique. *OncoTargets Ther.* **2015**, *8*, 2015–2022. [CrossRef] [PubMed]
3. Zhu, Z. Advancements in Automated Classification of Chronic Obstructive Pulmonary Disease based on Computed Tomography Imaging Features Through Deep Learning Approaches. *Respir. Med.* **2024**, *234*, 107809. [CrossRef] [PubMed]
4. Sarmadi, A.; Razavi, Z.S.; Van Wijnen, A.J.; Soltani, M. Comparative Analysis of Vision Transformers and Convolutional Neural Networks in Osteoporosis Detection from X-ray Images. *Sci. Rep.* **2024**, *14*, 18007. [CrossRef]
5. Takahashi, S.; Sakaguchi, Y.; Kouno, N.; Takasawa, K.; Ishizu, K.; Akagi, Y.; Aoyama, R.; Teraya, N.; Bolatkan, A.; Shinkai, N.; et al. Comparison of Vision Transformers and Convolutional Neural Networks in Medical Image Analysis: A Systematic Review. *J. Med. Syst.* **2024**, *48*, 84. [CrossRef]
6. Srivastava, N.; Hinton, G.; Krizhevsky, A.; Sutskever, I.; Salakhutdinov, R. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. *J. Mach. Learn. Res.* **2014**, *15*, 1929–1958.
7. Shorten, C.; Khoshgoftaar, T.M. A Survey on Image Data Augmentation for Deep Learning. *J. Big Data* **2019**, *6*, 60. [CrossRef]
8. Tian, Y.; Zhang, Y. A Comprehensive Survey on Regularization Strategies in Machine Learning. *Inf. Fusion* **2022**, *80*, 146–166. [CrossRef]
9. Xiao, M.; Wu, Y.; Zuo, G.; Fan, S.; Yu, H.; Shaikh, Z.A.; Wen, Z. Addressing Overfitting Problem in Deep Learning-Based Solutions for Next Generation Data-Driven Networks. *Hindawi Wirel. Commun. Mob. Comput.* **2021**, *2021*, 8493795. [CrossRef]
10. Ying, X. An Overview of Overfitting and its Solutions. *IOP Conf. Ser. J. Phys. Conf. Ser.* **2019**, *1168*, 022022. [CrossRef]

11. Garbin, C.; Zhu, X.; Marques, O. Dropout vs. Batch Normalization: An Empirical Study of Their Impact to Deep Learning. *Multimed. Tools Appl.* **2020**, *79*, 12777–12815. [CrossRef]
12. Helmbold, D.P. Surprising Properties of Dropout in Deep Networks. *J. Mach. Learn. Res.* **2018**, *18*, 1–28.
13. Piotrowskia, A.P.; Napiorkowskia, J.J.; Piotrowska, A.E. Impact of Deep Learning-Based Dropout on Shallow Neural Networks Applied to Stream Temperature Modelling. *Earth Sci. Rev.* **2020**, *201*, 103076. [CrossRef]
14. Salehin, I.; Kang, D.-K. A Review on Dropout Regularization Approaches for Deep Neural Networks within the Scholarly Domain. *Electronics* **2023**, *12*, 3106. [CrossRef]
15. Xu, M.; Yoon, S.; Fuentes, A.; Park, D.S. A Comprehensive Survey of Image Augmentation Techniques for Deep Learning. *Pattern Recognit.* **2023**, *137*, 109347. [CrossRef]
16. Kumar, T.; Brennan, R.; Mileo, A.; Bendechache, M. Image Data Augmentation Approaches: A Comprehensive Survey and Future Directions. *IEEE Access* **2024**, *12*, 187536–187571. [CrossRef]
17. Yang, Z.; Sinnott, R.O.; Bailey, J.; Ke, Q. A Survey of Automated Data Augmentation Algorithms for Deep Learning-Based Image Classification Tasks. *Knowl. Inf. Syst.* **2023**, *65*, 2805–2861. [CrossRef]
18. Nanni, L.; Paci, M.; Brahnam, S.; Lumini, A. Comparison of Different Image Data Augmentation Approaches. *J. Imaging* **2021**, *7*, 254. [CrossRef]
19. Oza, P.; Sharma, P.; Patel, S.; Adedoyin, F.; Bruno, A. Image Augmentation Techniques for Mammogram Analysis. *J. Imaging* **2022**, *8*, 141. [CrossRef]
20. Sandfort, V.; Yan, K.; Pickhardt, P.J.; Summers, R.M. Data augmentation using generative adversarial networks (CycleGAN) to improve generalizability in CT segmentation tasks. *Sci. Rep.* **2019**, *9*, 16884. [CrossRef]
21. Wei, J.; Wang, Q.; Song, X.; Zhao, Z. The Status and Challenges of Image Data Augmentation Algorithms. *J. Phys. Conf. Ser.* **2023**, *2456*, 012041. [CrossRef]
22. Liu, X.; Karagoz, G.; Meratnia, N. Analyzing the Impact of Data Augmentation on the Explainability of Deep Learning-Based Medical Image Classification. *Mach. Learn. Knowl. Extr.* **2025**, *7*, 1. [CrossRef]
23. Mamat, N.; Othman, M.F.; Abdulghafor, R.; Alwan, A.A.; Gulzar, Y. Enhancing Image Annotation Technique of Fruit Classification Using a Deep Learning Approach. *Sustainability* **2023**, *15*, 901. [CrossRef]
24. Jayaraj, R.; Lokesh, S. Automatic Image Annotation Using Adaptive Convolutional Deep Learning. *Intell. Autom. Soft Comput.* **2023**, *36*, 481–497. [CrossRef]
25. Zhang, D.; Islam, M.M.; Lu, G. A Review on Automatic Image Annotation Techniques. *Pattern Recognit.* **2012**, *45*, 346–362. [CrossRef]
26. Galbusera, F.; Cina, A. Image Annotation and Curation in Radiology: An Overview for Machine Learning practitioners. *Eur. Radiol. Exp.* **2024**, *8*, 11. [CrossRef]
27. Yin, S.; Bi, J. Medical Image Annotation Based on Deep Transfer Learning. *J. Appl. Sci. Eng.* **2019**, *22*, 385–390.
28. Zhang, Y.; Chen, J.; Xiangxun, M.; Wang, G.; Bhatti, U.A.; Huang, M. Interactive Medical Image Annotation Using Improved Attention U-net With Compound Geodesic Distance. *Expert Syst. Appl.* **2024**, *237 Pt A*, 121282. [CrossRef]
29. Sriwong, K.; Kerdprasop, K.; Kerdprasop, N. The Study of Noise Effect on CNN-Based Deep Learning from Medical Images. *Int. J. Mach. Learn. Comput.* **2021**, *11*, 202–207. [CrossRef]
30. Huang, Z.; Li, F.; Wang, Z.; Wang, Z. Interpretability of Deep Learning. *Int. J. Future Comput. Commun.* **2022**, *11*, 34–39. [CrossRef]
31. Xu, B.; Yang, G. Interpretability Research of Deep Learning: A Literature Survey. *Inf. Fusion* **2025**, *115*, 102721. [CrossRef]
32. Antamis, T.; Drosou, A.; Vafeiadis, T.; Nizamis, A.; Ioannidis, D.; Tzovaras, D. Interpretability of Deep Neural Networks: A Review of Methods, Classification and Hardware. *Neurocomputing* **2024**, *601*, 128204. [CrossRef]
33. Marcinkevičs, R.; Vogt, J.E. Interpretable and Explainable Machine Learning: A Methods-Centric Overview with Concrete Examples. *WIREs Data Min. Knowl. Discov.* **2023**, *13*, e1493. [CrossRef]
34. Linardatos, P.; Papastefanopoulos, V.; Kotsiantis, S. Explainable AI: A Review of Machine Learning Interpretability Methods. *Entropy* **2021**, *23*, 18. [CrossRef]
35. Carvalho, D.V.; Pereira, E.M.; Cardoso, J.S. Machine Learning Interpretability: A Survey on Methods and Metrics. *Electronics* **2019**, *8*, 832. [CrossRef]
36. Jiang, L.; Xu, C.; Bai, Y.; Liu, A.; Gong, Y.; Wang, Y.-P.; Deng, H.-W. Autosurv: Interpretable deep learning framework for cancer survival analysis incorporating clinical and multi-omics datawas partially supported by previous studies. *Npj Precis. Oncol.* **2024**, *8*, 4. [CrossRef]
37. Balachandar, N.; Chang, K.; Kalpathy-Cramer, J.; Rubin, D.L. Accounting for Data Variability in Multi-Institutional Distributed Deep Learning for Medical Imaging. *J. Am. Med. Inform. Assoc.* **2020**, *27*, 700–708. [CrossRef]
38. Safonova, A.; Ghazaryan, G.; Stiller, S.; Main-Knorn, M.; Nendel, C.; Ryo, M. Ten Deep Learning Techniques to Address Small Data Problems with Remote Sensing. *Int. J. Appl. Earth Obs. Geoinf.* **2023**, *125*, 103569. [CrossRef]
39. Piffer, S.; Ubaldi, L.; Tangaro, S.; Retico, A.; Talamonti, A. Tackling the Small Data Problem in Medical Image Classification with Artificial Intelligence: A Systematic Review. *Prog. Biomed. Eng.* **2024**, *6*, 032001. [CrossRef]

40. D'Antonoli, T.A. Ethical Considerations for Artificial Intelligence: An Overview of the Current Radiology Landscape. *Diagn. Interv. Radiol.* **2020**, *26*, 504–511.
41. Safdar, N.M.; Banja, J.D.; Meltzer, C.C. Ethical Considerations in Artificial Intelligence. *Eur. J. Radiol.* **2020**, *122*, 108768. [CrossRef] [PubMed]
42. Chen, I.Y.; Pierson, E.; Rose, S.; Joshi, S.; Ferryman, K.; Ghassemi, M. Ethical Machine Learning in Healthcare. *Annu. Rev. Biomed. Data Sci.* **2021**, *4*, 123–144. [CrossRef] [PubMed]
43. Chau, M. Ethical, Legal, and Regulatory Landscape of Artificial Intelligence in Australian Healthcare and Ethical Integration in Radiography: A Narrative Review. *J. Med. Imaging Radiat. Sci.* **2024**, *55*, 101733. [CrossRef] [PubMed]
44. Chen, C.; Zhang, P.; Zhang, H.; Dai, J.; Yi, Y.; Zhang, H.; Zhang, Y. Deep Learning on Computational-Resource-Limited Platforms: A Survey. *Mob. Inf. Syst.* **2020**, *1*, 8454327. [CrossRef]
45. Pietrołaj, M.; Blok, M. Resource Constrained Neural Network Training. *Sci. Rep.* **2024**, *14*, 2421. [CrossRef]
46. Chlap, P.; Min, H.; Vandenberg, N.; Dowling, J.; Holloway, L.; Haworth, A. A review of Medical Image Data Augmentation Techniques for Deep Learning Applications. *J. Med. Imaging Radiat. Oncol.* **2021**, *65*, 545–563. [CrossRef]
47. Masumoto, R.; Eguchi, Y.; Takeuchi, H.; Inage, K.; Narita, M.; Shiga, Y.; Inoue, M.; Toshi, N.; Tokeshi, S.; Okuyama, K.; et al. Automatic Generation of Diffusion Tensor imaging for the Lumbar Nerve using Convolutional Neural Networks. *Magn. Reson. Imaging* **2024**, *114*, 110237. [CrossRef]
48. Shobayo, O.; Saatchi, R.; Ramlakhan, S. Convolutional Neural Network to Classify Infrared Thermal Images of Fractured Wrists in Pediatrics. *Healthcare* **2024**, *12*, 994. [CrossRef]
49. Chauhan, S.; Edla, D.R.; Boddu, V.; Rao, M.J.; Cheruku, R.; Nayak, S.R.; Martha, S.; Lavanya, K.; Nigat, T.D. Detection of COVID-19 Using Edge Devices by a Light-Weight Convolutional Neural Network from Chest X-ray Images. *BMC Med. Imaging* **2024**, *24*, 1. [CrossRef]
50. Wu, Z.; Shen, C.; Van Den Hengel, A. Wider or Deeper: Revisiting the ResNet Model for Visual Recognition. *Pattern Recognit.* **2019**, *90*, 119–133. [CrossRef]
51. Zhu, H.; Liu, Y.; Gao, X.; Zhang, L. Combined CNN and Pixel Feature Image for Fatty Liver Ultrasound Image Classification. *Comput. Math. Methods Med.* **2022**, *2022*, 9385734. [CrossRef] [PubMed]
52. Yu, K.; Tseng, Y.; Yang, H.; Liu, C.; Kuo, P.; Lee, M.; Huang, C.; Kuo, L.; Wang, J.; Ho, C.; et al. Deep learning with test-time augmentation for radial endobronchial ultrasound image differentiation: A multicentre verification study. *BMJ Open Respir. Res.* **2023**, *10*, e001602. [CrossRef] [PubMed]
53. Gao, Y.; Zeng, S.; Xu, X.; Li, H.; Yao, S.; Song, K.; Li, X.; Chen, L.; Tang, J.; Xing, H.; et al. Deep learning-enabled pelvic ultrasound images for accurate diagnosis of ovarian cancer in China: A retrospective, multicentre, diagnostic study. *Lancet Digit. Health* **2022**, *4*, e179–e187. [CrossRef]
54. Abdulahi, A.T.; Ogundokun, R.O.; Adenike, A.R.; Shah, M.A.; Ahmed, Y.K. PulmoNet: A Novel Deep learning based Pulmonary Diseases Detection Model. *BMC Med. Imaging* **2024**, *24*, 51. [CrossRef] [PubMed]
55. Zhang, K.; Sun, M.; Han, T.X.; Yuan, X.; Guo, L.; Liu, T. Residual networks of residual networks: Multilevel residual networks. *IEEE Trans. Circuits Syst. Video Technol.* **2017**, *28*, 1303–1314. [CrossRef]
56. Kim, J.; Hong, J.; Park, H. Prospects of Deep Learning for Medical Imaging. *Precis. Future Med.* **2018**, *2*, 37–52. [CrossRef]
57. Zhang, H.; Qie, Y. Applying Deep Learning to Medical Imaging: A Review. *Appl. Sci.* **2023**, *13*, 10521. [CrossRef]
58. Rajeev, R.; Samath, J.A.; Karthikeyan, N.K. An Intelligent Recurrent Neural Network with Long Short-Term Memory (LSTM) BASED Batch Normalization for Medical Image Denoising. *J. Med. Syst.* **2019**, *43*, 234. [CrossRef]
59. Yao, W.; Bai, J.; Liao, W.; Chen, Y.; Liu, M.; Xie, Y. From CNN to Transformer: A Review of Medical Image Segmentation Models. *J. Digit. Imaging Inform. Med.* **2024**, *37*, 1529–1547. [CrossRef]
60. Cui, R.; Liu, M. RNN-based Longitudinal Analysis for Diagnosis of Alzheimer's Disease. *Comput. Med. Imaging Graph.* **2019**, *73*, 1–10. [CrossRef]
61. Anbalagan, V.; Balasubramanian, V. HBO-GMRNN: Honey Badger Optimization Based Gain Modulated Recurrent Neural Network for Classification of Breast Cancer. *Biomed. Signal Process. Control.* **2024**, *91*, 105910. [CrossRef]
62. Amarneni, S.; Valarmathi, D.R.S. Diagnosing the MRI Brain Tumour Images Through RNN-LSTM. *E Prime Adv. Electr. Eng. Electron. Energy* **2024**, *9*, 100723. [CrossRef]
63. Zhu, K.; Chen, Y.; Ouyang, X.; White, G.; Agam, G. Fully RNN for Knee Ligament Tear Classification and Localization in MRI scans. *Electron. Imaging* **2022**, *34*, 227-1–227-6. [CrossRef]
64. Gulshan; Arora, A.S. Automated Prediction of Diabetes Mellitus Using Infrared Thermal Foot Images: Recurrent Neural Network Approach. *Biomed. Phys. Eng. Express* **2024**, *10*, 025025. [CrossRef]
65. Ayub, S.; Kannan, R.J.; Alsini, R.; Hasanin, T.; Sasidhar, C. LSTM-Based RNN Framework to Remove Motion Artifacts in Dynamic Multicontrast MR Images with Registration Model. *Wirel. Commun. Mob. Comput.* **2022**, *2022*, 5906877. [CrossRef]
66. Das, T.; Saha, G. Addressing Big Data Issues Using RNN Based Techniques. *J. Inf. Optim. Sci.* **2020**, *40*, 1773–1785. [CrossRef]

67. Zhang, Y. A Better Autoencoder for Image: Convolutional Autoencoder. In Proceedings of the ICONIP17-DCEC, Guangzhou, China, 14–18 October 2017; pp. 11–21. Available online: http://users.cecs.anu.edu.au/Tom.Gedeon/conf/ABCs2018/paper/ABCs2018_paper_58.pdf (accessed on 17 April 2025).
68. Baldi, P. Autoencoders, unsupervised learning, and deep architectures. In Proceedings of the ICML Workshop on Unsupervised and Transfer Learning, Edinburgh, UK, 26 June–1 July 2012; JMLR Workshop and Conference Proceedings. pp. 37–49.
69. Vorontsov, E.; Molchanov, P.; Gazda, M.; Beckham, C.; Kautz, J.; Kadoury, S. Towards Annotation-Efficient Segmentation Via Image-to-Image Translation. *Med. Image Anal.* **2022**, *82*, 102624. [CrossRef]
70. Wang, W.; Huang, Y.; Wang, Y.; Wang, L. Generalized autoencoder: A neural network framework for dimensionality reduction. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops, Columbus, OH, USA, 23–28 June 2014; pp. 490–497.
71. Juneja, M.; Kaur Saini, S.; Kaul, S.; Acharjee, R.; Thakur, N.; Jindal, P. Denoising of Magnetic Resonance Imaging Using Bayes Shrinkage Based Fused Wavelet Transform and Autoencoder Based Deep Learning Approach. *Biomed. Signal Process. Control.* **2021**, *69*, 102844. [CrossRef]
72. O' Sullivan, E.; Van De Lande, L.S.; Papaioannou, A.; Breakey, R.W.F.; Jeelani, N.O.; Ponniah, A.; Duncan, C.; Schievano, S.; Khonsari, R.H.; Zafeiriou, S.; et al. Convolutional Mesh Autoencoders for the 3-Dimensional Identification of FGFR-Related Craniosynostosis. *Sci. Rep.* **2022**, *12*, 2230. [CrossRef]
73. Wolf, D.; Payer, T.; Lisson, C.S.; Lisson, C.G.; Beer, M.; Götz, M.; Ropinski, T. Self-supervised Pre-training with Contrastive and Masked Autoencoder Methods for Dealing with Small Datasets in Deep Learning for Medical Imaging. *Sci. Rep.* **2023**, *13*, 20260. [CrossRef]
74. Chen, R.; Song, Y.; Huang, J.; Wang, J.; Sun, H.; Wang, H. Rapid Diagnosis and Continuous Monitoring of Intracerebral Hemorrhage with Magnetic Induction Tomography Based on Stacked Autoencoder. *Rev. Sci. Instrum.* **2021**, *92*, 084707. [CrossRef] [PubMed]
75. Shvetsova, N.; Bakker, B.; Fedulova, I.; Schulz, H.; Dylov, D.V. Anomaly Detection in Medical Imaging with Deep Perceptual Autoencoders. *IEEE Access* **2021**, *9*, 118571–118583. [CrossRef]
76. Elhassan, T.A.; Mohd Rahim, M.S.; Siti Zaiton, M.H.; Swee, T.T.; Alhaj, T.A.; Ali, A.; Aljurf, M. Classification of Atypical White Blood Cells in Acute Myeloid Leukemia Using a Two-Stage Hybrid Model Based on Deep Convolutional Autoencoder and Deep Convolutional Neural Network. *Diagnostics* **2023**, *13*, 196. [CrossRef] [PubMed]
77. Zhang, H.; Guo, W.; Zhang, S.; Lu, H.; Zhao, X. Unsupervised Deep Anomaly Detection for Medical Images Using an Improved Adversarial Autoencoder. *J. Digit. Imaging* **2022**, *35*, 153–161. [CrossRef]
78. Li, D.; Fu, Z.; Xu, J. Stacked-Autoencoder-Based Model for COVID-19 Diagnosis on CT Images. *Appl. Intell.* **2020**, *51*, 2805–2817. [CrossRef]
79. Zhang, H.; Chen, J.; Liao, B.; Wu, F.; Bi, X. Deep Canonical Correlation Fusion Algorithm Based on Denoising Autoencoder for ASD Diagnosis and Pathogenic Brain Region Identification. *Interdiscip. Sci. Comput. Life Sci.* **2024**, *16*, 455–468. [CrossRef]
80. Gao, J.; Zhao, W.; Li, P.; Huang, W.; Chen, Z. LEGAN: A Light and Effective Generative Adversarial Network for medical image synthesis. *Comput. Biol. Med.* **2022**, *148*, 105878. [CrossRef]
81. Singh, N.K.; Raza, K. *Medical Image Generation Using Generative Adversarial Networks: A Review*; Springer: Singapore, 2021; pp. 77–96.
82. Xun, S.; Li, D.; Zhu, H.; Chen, M.; Wang, J.; Li, J.; Chen, M.; Wu, B.; Zhang, H.; Chai, X.; et al. Generative adversarial networks in medical image segmentation: A review. *Comput. Biol. Med.* **2021**, *140*, 105063. [CrossRef]
83. Yi, X.; Walia, E.; Babyn, P. Generative adversarial network in medical imaging: A review. *Med. Image Anal.* **2019**, *58*, 101552. [CrossRef]
84. Hamghalam, M.; Wang, T.; Lei, B. High tissue contrast image synthesis via multistage attention-GAN: Application to segmenting brain MR scans. *Neural Netw.* **2020**, *132*, 43–52. [CrossRef]
85. Zhu, Q.; Ye, H.; Sun, L.; Li, Z.; Wang, R.; Shi, F.; Shen, D.; Zhang, D. GACDN: Generative adversarial feature completion and diagnosis network for COVID-19. *BMC Med. Imaging* **2021**, *21*, 154. [CrossRef] [PubMed]
86. Li, Y.; Zhang, K.; Shi, W.; Miao, Y.; Jiang, Z. A Novel Medical Image Denoising Method Based on Conditional Generative Adversarial Network. *Comput. Math. Methods Med.* **2021**, *2021*, 9974017. [CrossRef]
87. Ahmad, W.; Ali, H.; Shah, Z.; Azmat, S. A new generative adversarial network for medical images super resolution. *Sci. Rep.* **2022**, *12*, 9533. [CrossRef] [PubMed]
88. Mutepfe, F.; Kalejahi, B.K.; Meshgini, S.; Danishvar, S. Generative Adversarial Network Image Synthesis Method for Skin Lesion Generation and Classification. *J. Med. Signals Sens.* **2021**, *11*, 237–252. [CrossRef]
89. Touati, R.; Le, W.T.; Kadoury, S. A feature invariant generative adversarial network for head and neck MRI/CT image synthesis. *Phys. Med. Biol.* **2021**, *66*, 095001. [CrossRef] [PubMed]
90. Xiao, Y.; Chen, C.; Wang, L.; Yu, J.; Fu, X.; Zou, Y.; Lin, Z.; Wang, K. A novel hybrid generative adversarial network for CT and MRI super-resolution reconstruction. *Phys. Med. Biol.* **2023**, *68*, 135007. [CrossRef]

91. Mahapatra, D.; Bozorgtabar, B.; Garnavi, R. Image super-resolution using progressive generative adversarial networks for medical image analysis. *Comput. Med. Imaging Graph.* **2019**, *71*, 30–39. [CrossRef]
92. Uzunova, H.; Ehrhardt, J.; Handels, H. Memory-efficient GAN-based domain translation of high resolution 3D medical images. *Comput. Med. Imaging Graph.* **2020**, *86*, 101801. [CrossRef]
93. Ronneberger, O.; Fischer, P.; Brox, T. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Lecture Notes in Computer Science*; Springer Science + Business Media: Berlin, Germany, 2015; pp. 234–241. [CrossRef]
94. Azad, R.; Aghdam, K.; Rauland, A.; Jia, Y.; Avval, H.; Bozorgpour, A.; Karimijafarbigloo, S.; Cohen, J.P.; Adeli, E.; Merhof, D. Medical Image Segmentation Review: The Success of U-Net. *IEEE Trans. Pattern Anal. Mach. Intell.* **2024**, *46*, 10076–10095. [CrossRef]
95. Krithika Alias Anbudevi, M.; Suganthi, K. Review of Semantic Segmentation of Medical Images Using Modified Architectures of UNET. *Diagnostics* **2022**, *12*, 3064. [CrossRef]
96. Ehab, W.; Li, Y. Performance Analysis of UNet and Variants for Medical Image Segmentation. *arXiv* **2023**, arXiv:2309.13013.
97. Ding, Y.; Chen, F.; Zhao, Y.; Wu, Z.; Zhang, C.; Wu, D. A Stacked Multi-Connection Simple Reducing Net for Brain Tumor Segmentation. *IEEE Access* **2019**, *7*, 104011–104024. [CrossRef]
98. Wang, Z.; Zou, Y.; Liu, P.X. Hybrid dilation and attention residual U-Net for Medical Image Segmentation. *Comput. Biol. Med.* **2021**, *134*, 104449. [CrossRef]
99. Khan, R.A.; Luo, Y.; Wu, F. RMS-UNet: Residual Multi-Scale UNet for Liver and Lesion Segmentation. *Artif. Intell. Med.* **2022**, *124*, 102231. [CrossRef]
100. Kong, Z.; Zhang, M.; Zhu, W.; Yi, Y.; Wang, T.; Zhang, B. Data enhancement based on M2-Unet for liver segmentation in Computed Tomography. *Biomed. Signal Process. Control* **2022**, *79*, 104032. [CrossRef]
101. Chetty, G.; Yamin, M.; White, M. A low resource 3D U-Net based deep learning model for medical image analysis. *Int. J. Inf. Technol.* **2022**, *14*, 95–103. [CrossRef]
102. Huang, K.; Yang, Y.; Huang, Z.; Liu, Y.; Lee, S. Retinal Vascular Image Segmentation Using Improved UNet Based on Residual Module. *Bioengineering* **2023**, *10*, 722. [CrossRef]
103. Lin, Z.; Dall'ara, E.; Guo, L. A novel mean shape based post-processing method for enhancing deep learning lower-limb muscle segmentation accuracy. *PLoS ONE* **2024**, *19*, e0308664. [CrossRef]
104. Ding, T.; Shi, K.; Pan, Z.; Ding, C. AI-based automated breast cancer segmentation in ultrasound imaging based on Attention Gated Multi ResU-Net. *PeerJ Comput. Sci.* **2024**, *10*, e2226. [CrossRef]
105. Henson, W.H.; Li, X.; Lin, Z.; Guo, L.; Mazzá, C.; Dall'ara, E. Automatic segmentation of lower limb muscles from MR images of post-menopausal women based on deep learning and data augmentation. *PLoS ONE* **2024**, *19*, e0299099. [CrossRef]
106. Lin, Z.; Henson, W.H.; Dowling, L.; Walsh, J.; Dall'ara, E.; Guo, L. Automatic segmentation of skeletal muscles from MR images using modified U-Net and a novel data augmentation approach. *Front. Bioeng. Biotechnol.* **2024**, *12*, 1355735. [CrossRef] [PubMed]
107. Zhou, Z.; Rahman Siddiquee, M.M.; Tajbakhsh, N.; Liang, J. *UNet++: A Nested U-Net Architecture for Medical Image Segmentation*; Springer International Publishing: Berlin/Heidelberg, Germany, 2020; p. 3.
108. Garbaz, A.; Oukdach, Y.; Charfi, S.; El Ansari, M.; Koutti, L.; Salihoun, M. MLFA-UNet: A multi-level feature assembly UNet for medical image segmentation. *Methods* **2024**, *232*, 52–64. [CrossRef]
109. Kora, P.; Ooi, C.P.; Faust, O.; Raghavendra, U.; Gudigar, A.; Chan, W.Y.; Meenakshi, K.; Swaraja, K.; Plawiak, P.; Rajendra Acharya, U. Transfer learning techniques for medical image analysis: A review. *Biocybern. Biomed. Eng.* **2021**, *42*, 79–107. [CrossRef]
110. Yu, X.; Wang, J.; Hong, Q.; Teku, R.; Wang, S.; Zhang, Y. Transfer learning for medical images analyses: A survey. *Neurocomputing* **2022**, *489*, 230–254. [CrossRef]
111. Ayana, G.; Dese, K.; Abagaro, A.M.; Jeong, K.C.; Yoon, S.; Choe, S. Multistage Transfer Learning for Medical Images. *Artif. Intell. Rev.* **2024**, *57*, 232. [CrossRef]
112. Atasever, S.; Azginoglu, N.; Terzi, D.S.; Terzi, R. A comprehensive survey of deep learning research on medical image analysis with focus on transfer learning. *Clin. Imaging* **2022**, *94*, 18–41. [CrossRef]
113. Kim, H.E.; Cosa-Linan, A.; Santhanam, N.; Jannesari, M.; Maros, M.E.; Ganslandt, T. Transfer learning for medical image classification: A literature review. *BMC Med. Imaging* **2022**, *22*, 69. [CrossRef]
114. De Santana, M.A.; Pereira, J.M.S.; Silva, F.L.D.; de Lima, N.M.; de Sousa, F.N.; de Arruda, G.M.S.; de Lima, R.d.C.F.; da Silva, W.W.A.; Santos, W.P.d. Breast cancer diagnosis based on mammary thermography and extreme learning machines. *Res. Biomed. Eng.* **2018**, *34*, 45–53. [CrossRef]
115. Kumar, S.; Choudhary, S.; Jain, A.; Singh, K.; Ahmadian, A.; Bajuri, M.Y. Brain Tumor Classification Using Deep Neural Network and Transfer Learning. *Brain Topogr.* **2023**, *36*, 305–318. [CrossRef]
116. Archana, B.; Kalirajan, K. An Intelligent Magnetic Resonance Imagining-Based Multistage Alzheimer's Disease Classification using Swish-Convolutional Neural Networks. *Med. Biol. Eng. Comput.* **2024**, *63*, 885–899. [CrossRef]
117. Zhu, Y.; Jin, P.; Bao, J.; Jiang, Q.; Wang, X. Thyroid ultrasound image classification using a convolutional neural network. *Ann. Transl. Med.* **2021**, *9*, 1526. [CrossRef] [PubMed]

118. Kalusivalingam, A.K.; Sharma, A.; Patel, N.; Singh, V. Enhancing Diagnostic Accuracy in Medical Imaging through Convolutional Neural Networks and Transfer Learning Algorithms. *Int. J. Artif. Intell. Mach. Learn.* **2021**, *2*, 1–24.
119. Alzubaidi, L.; Al-Amidie, M.; Al-Asadi, A.; Humaidi, A.J.; Al-Shamma, O.; Fadhel, M.A.; Zhang, J.; Santamaría, J.; Duan, Y. Novel Transfer Learning Approach for Medical Imaging with Limited Labeled Data. *Cancers* **2021**, *13*, 1590. [CrossRef] [PubMed]
120. Odusami, M.; Maskeliūnas, R.; Damaševičius, R.; Krilavičius, T. Analysis of Features of Alzheimer's Disease: Detection of Early Stage from Functional Brain Changes in Magnetic Resonance Images Using a Finetuned ResNet18 Network. *Diagnostics* **2021**, *11*, 1071. [CrossRef]
121. Guan, Q.; Wang, Y.; Du, J.; Qin, Y.; Lu, H.; Xiang, J.; Wang, F. Deep learning based classification of ultrasound images for thyroid nodules: A large scale of pilot study. *Ann. Transl. Med.* **2019**, *7*, 137. [CrossRef]
122. Tian, D.; Jiang, S.; Zhang, L.; Lu, X.; Xu, Y. The role of large language models in medical image processing: A narrative review. *Quant. Imaging Med. Surg.* **2023**, *14*, 1108–1121. [CrossRef]
123. Ogunleye, B.; Sharma, H.; Shobayo, O. Sentiment Informed Sentence BERT-Ensemble Algorithm for Depression Detection. *BDCC* **2024**, *8*, 112. [CrossRef]
124. Bora, A.; Cuayáhuitl, H. Systematic Analysis of Retrieval-Augmented Generation-Based LLMs for Medical Chatbot Applications. *Make* **2024**, *6*, 2355–2374. [CrossRef]
125. Pu, Q.; Xi, Z.; Yin, S.; Zhao, Z.; Zhao, L. Advantages of transformer and its application for medical image segmentation: A survey. *Biomed. Eng. OnLine* **2024**, *23*, 14. [CrossRef]
126. Berroukham, A.; Housni, K.; Lahraichi, M. Vision Transformers: A Review of Architecture, Applications, and Future Directions. In Proceedings of the 2023 7th IEEE Congress on Information Science and Technology (CiSt), Agadir/Essaouira, Morocco, 16–22 December 2023; pp. 205–210. [CrossRef]
127. Van, M.; Verma, P.; Wu, X. On large visual language models for medical imaging analysis: An empirical study. In Proceedings of the 2024 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE), Wilmington, DE, USA, 19–21 June 2024; pp. 172–176.
128. Dosovitskiy, A.; Beyer, L.; Kolesnikov, A.; Weissenborn, D.; Zhai, X.; Unterthiner, T.; Dehghani, M.; Minderer, M.; Heigold, G.; Gelly, S.; et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv* **2021**, arXiv:2010.11929.
129. Han, K.; Wang, Y.; Chen, H.; Chen, X.; Guo, J.; Liu, Z.; Tang, Y.; Xiao, A.; Xu, C.; Xu, Y.; et al. A Survey on Vision Transformer. *IEEE Trans. Pattern Anal. Mach. Intell.* **2023**, *45*, 87–110. [CrossRef] [PubMed]
130. Tanimola, O.; Shobayo, O.; Popoola, O.; Okoyeigbo, O. Breast Cancer Classification Using Fine-Tuned SWIN Transformer Model on Mammographic Images. *Analytics* **2024**, *3*, 461–475. [CrossRef]
131. Abbaoui, W.; Retal, S.; Ziti, S.; El Bhiri, B. Automated Ischemic Stroke Classification from MRI Scans: Using a Vision Transformer Approach. *J. Clin. Med.* **2024**, *13*, 2323. [CrossRef] [PubMed]
132. Chen, J.; Frey, E.C.; He, Y.; Segars, W.P.; Li, Y.; Du, Y. TransMorph: Transformer for unsupervised medical image registration. *Med. Image Anal.* **2022**, *82*, 102615. [CrossRef]
133. Wang, C.; Zhang, J.; Liu, S. Medical ultrasound image segmentation with deep learning models. *IEEE Access* **2022**, *11*, 10158–10168. [CrossRef]
134. Ramamurthy, M.; Krishnamurthi, I.; Vimal, S.; Robinson, Y.H. Deep Learning Based Genome Analysis and NGS-RNA LL Identification with a Novel Hybrid Model. *Biosystems* **2020**, *197*, 104211. [CrossRef]
135. Rahman, H.; Bukht, T.F.N.; Imran, A.; Tariq, J.; Tu, S.; Alzahrani, A. A Deep Learning Approach for Liver and Tumor Segmentation in CT Images Using ResUNet. *Bioengineering* **2022**, *9*, 368. [CrossRef]
136. Zhu, Y.; Meng, Z.; Fan, X.; Duan, Y.; Jia, Y.; Dong, T.; Wang, Y.; Song, J.; Tian, J.; Wang, K.; et al. Deep learning radiomics of dual-modality ultrasound images for hierarchical diagnosis of unexplained cervical lymphadenopathy. *BMC Med.* **2022**, *20*, 269. [CrossRef]
137. Vasile, C.M.; Udriștoiu, A.L.; Ghenea, A.E.; Popescu, M.; Gheonea, C.; Niculescu, C.E.; Ungureanu, A.M.; Udriștoiu, Ș.; Drocaș, A.I.; Gruionu, L.G. Intelligent diagnosis of thyroid ultrasound imaging using an ensemble of deep learning methods. *Medicina* **2021**, *57*, 395. [CrossRef]
138. Iqbal, S.; Khan, T.M.; Naqvi, S.S.; Naveed, A.; Meijering, E. TBConvL-Net: A hybrid deep learning architecture for robust medical image segmentation. *Pattern Recognit.* **2025**, *158*, 111028. [CrossRef]
139. Obayya, M.; Saeed, M.K.; Alruwais, N.; Alotaibi, S.S.; Assiri, M.; Salama, A.S. Hybrid Metaheuristics with Deep Learning-Based Fusion Model for Biomedical Image Analysis. *IEEE Access* **2023**, *11*, 117149–117158. [CrossRef]
140. Yan, L.; Ling, S.; Mao, R.; Xi, H.; Wang, F. A deep learning framework for identifying and segmenting three vessels in fetal heart ultrasound images. *Biomed. Eng. OnLine* **2024**, *23*, 39. [CrossRef] [PubMed]

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3D TransUNet: Advancing Medical Image Segmentation through Vision Transformers

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***Abstract*— Medical image segmentation plays a crucial role in advancing healthcare systems for disease diagnosis and treatment planning. The u-shaped architecture, popularly known as U-Net, has proven highly successful for various medical image segmentation tasks. However, U-Net's convolution-based operations inherently limit its ability to model long-range dependencies effectively.**

To address these limitations, researchers have turned to Transformers, renowned for their global self-attention mechanisms, as alternative architectures. One popular network is our previous TransUNet, which leverages Transformers' self-attention to complement U-Net's localized information with the global context. In this paper, we extend the 2D TransUNet architecture to a 3D network by building upon the state-of-the-art nnU-Net architecture, and fully exploring Transformers' potential in both the encoder and decoder design. We introduce two key components: 1) A *Transformer encoder* that tokenizes image patches from a convolution neural network (CNN) feature map, enabling the extraction of global contexts, and 2) A *Transformer decoder* that adaptively refines candidate regions by utilizing cross-attention between candidate proposals and U-Net features. Our investigations reveal that different medical tasks benefit from distinct architectural designs. The Transformer encoder excels in multi-organ segmentation, where the relationship among organs is crucial. On the other hand, the Transformer decoder proves more beneficial for dealing with small and challenging segmented targets such as tumor segmentation. Extensive experiments showcase the significant potential of integrating a Transformer-based encoder and decoder into the u-shaped medical image segmentation architecture. TransUNet outperforms competitors in various medical applications, including multi-organ segmentation, pancreatic tumor segmentation, and hepatic vessel segmentation. It notably surpasses the top solution in the BrasTS2021 challenge. Code and models are available at `https://github.com/Beckschen/3D-TransUNet`.

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I. Introduction

Convolutional neural networks (CNNs), particularly fully convolutional networks (FCNs) [1], have risen to prominence in the domain of medical image segmentation. Among their various iterations, the U-Net model [2], characterized by its symmetric encoder-decoder design augmented with skip-connections for improved detail preservation, stands out as the preferred choice for many researchers. Building on this methodology, remarkable progress has been witnessed across diverse medical imaging tasks. These advancements encompass cardiac segmentation in magnetic resonance (MR) imaging [3], organ delineation using computed tomography (CT) scans [4]–[7], and polyp segmentation in colonoscopy recordings [8].

Despite CNNs' unparalleled representational capabilities, they often falter when modeling long-range relationships due to the inherent locality of convolution operations. This limitation becomes particularly pronounced in cases with large inter-patient variations in texture, shape, and size. Recognizing this limitation, the research community has been increasingly drawn to Transformers, models built entirely upon attention mechanisms due to their innate prowess in capturing global contexts [9]. In the realm of medical image segmentation, our prior work with TransUNet [10] stands as a testament to the potential of transformers. However, a pivotal observation from our research indicates that simply substituting a CNN encoder with a Transformer can lead to suboptimal outcomes. Transformers process inputs as 1D sequences and prioritize global context modeling, inadvertently producing features of low resolution. Directly upsampling such features fails to reintroduce the lost granularity. In contrast, a hybrid approach combining CNN and Transformer encoders seems more promising. It effectively harnesses the high-resolution spatial details from CNNs while also benefiting from the global context provided by Transformers.

In this study, we extend the original 2D TransUNet architecture to a 3D configuration, delving deeper into the strategic incorporation of Transformers in both encoding and decoding processes. This leap is rooted in the prowess of the nnU-Net framework, with a vision to surpass its established standards. Our 3D TransUNet unfolds through two primary mechanisms: Firstly, the Transformer Encoder tokenizes image patches from CNN feature maps, allowing a seamless fusion of global self-attentive features with high-resolution CNN features skipped from the encoding path, for enabling precise localization. Secondly, the Transformer Decoder redefines conventional per-pixel segmentation as a mask classification, framing prediction candidates as learnable queries. Specifically, these queries are progressively refined by synergizing cross-attention with localized multi-scale CNN features. In addition, we introduce coarse-to-fine attention refinement in the Transformer decoder where for each segmentation class, an initial candidate set is meticulously refined using attention mechanisms focused on the prediction's foreground, ensuring that each iterative refinement sets a new standard for the subsequent, culminating in continuously improved segmentation accuracy.

By integrating Transformers in the encoder and decoder components of the U-Net-like architectures, we show that our designs allow the framework to preserve the advantages of Transformers while enhancing medical image segmentation. Intriguingly, multi-organ segmentation, which leans heavily on global context information—such as the interplay among diverse abdominal organs—tends to gravitate towards the Transformer encoder design. Conversely, tasks like small target segmentation, such as tumor detection, generally benefit more from the Transformer decoder design. Our extensive experiments demonstrate the superior performance of our method compared to competing approaches across various medical image segmentation tasks. In summary, our contributions can be summarized as follows:

- We introduce a Transformer-centric medical image segmentation framework, incorporating self-attention within the sequence-to-sequence prediction context, applicable to both 2D and 3D medical image segmentation tasks.
- We thoroughly investigate the effects of integrating vision transformers into the encoder and the decoder of the u-shaped segmentation architectures, providing insights on tailoring designs to cater to distinct medical image segmentation challenges.
- We achieve the state-of-the-art results on various medical image segmentation tasks, and release our codebase to encourage further exploration in applying Transformers to medical applications.

II. Related Works

Combining CNNs with self-attention mechanisms. Various studies have attempted to integrate self-attention mechanisms into CNNs by modeling global interactions of all pixels based on the feature maps. For instance, Wang *et al.* designed a non-local operator, which can be plugged into multiple intermediate convolution layers [11]. Built upon the encoder-decoder u-shaped architecture, Schlemper *et al.* [12] proposed additive attention gate modules which are integrated into the skip-connections. Different from these approaches, we employ Transformers for embedding global self-attention in our method.

Transformers. Transformers were first proposed by [9] for machine translation and established state-of-the-arts in many NLP tasks. To make Transformers also applicable for computer vision tasks, several modifications have been made. For instance, Parmar *et al.* [13] applied the self-attention only in local neighborhoods for each query pixel instead of globally. Child *et al.* [14] proposed Sparse Transformers, which employ scalable approximations to global self-attention. Recently, Vision Transformer (ViT) [15] achieved state-of-the-art on ImageNet classification by directly applying Transformers with global self-attention to full-sized images. To the best of our knowledge, our TransUNet originally proposed in 2021 is the first Transformer-based medical image segmentation framework, which builds upon the highly successful ViT. Based on TransUNet, nnformer [16] improves the methodology by interleaving convolution with self-attention. In another development, CoTR [17] offers a more efficient self-attention in the Transformer encoder. Another Transformer architecture is the Swin Transformer [18] a hierarchical vision transformer that employs shifted windows to capture local and global information for efficient and scalable visual processing. Subsequent models such as SwinUNet [19] and SwinUETR [20] have been developed for medical image segmentation.

Mask classification for segmentation. DETR [21] is the first work that uses Transformer as a decoder with learnable object queries for object detection. In the context of recent advancements in transformers [22]–[27], a novel variation known as mask Transformers has emerged. This variant introduces segmentation predictions by employing a collection of query embeddings to represent object and its associated mask. Wang *et al.* [23] first develops mask Transformer with memory embedding, and Cheng *et al.* [24] further formulate the query update in a manner of DETR [21]. At the core of mask transformers lies the decoder, which is responsible for processing object queries as input and progressively transforming them into mask embedding vectors [24]–[27]. This process enables the model to effectively handle segmentation tasks and produce accurate results.

III. Method

Given a 3D medical image (*e.g.*, CT/MR scan) $\mathbf{x} \in \mathbb{R}^{D\times H\times W\times C}$ with the spatial resolution of $D\times H\times W$ and C number of channels. Our goal is to predict the corresponding pixel-wise labelmap with size $D\times H\times W$. The most common way is to directly train a CNN (*e.g.*, U-Net) to first encode images into high-level feature representations, which are then decoded back to the full spatial resolution. Our approach diverges from conventional methods by thoroughly exploring the attention mechanisms utilized in both the encoder and decoder phases of standard U-shaped segmentation architectures, employing Transformers. In Section III-A, we delve into the direct application of Transformers for encoding feature representations from segmented image patches. Following this,

in Section III-B, we elaborate on the implementation of the query-based Transformer, which serves as our decoder. The detailed architecture of TransUNet is then presented in Section III-C.

A. Transformer as Encoder

Image sequentialization. Following [15], we first perform tokenization by reshaping the input $\mathbf{x}$ into a sequence of flattened 3D patches $\{\mathbf{x}_i^p \in \mathbb{R}^{P^3 \cdot C} | i = 1, .., N\}$, where each patch is of size $P \times P \times P$ and $N = \frac{DHW}{P^3}$ is the number of image patches (*i.e.*, the input sequence length).

Patch embedding. We map the vectorized patches $\mathbf{x}^p$ into a latent d_{enc}-dimensional embedding space using a trainable linear projection. To encode the patch spatial information, we learn specific position embeddings which are added to the patch embeddings to retain positional information as follows:

$$\mathbf{z}_0 = [\mathbf{x}_1^p\mathbf{E}; \mathbf{x}_2^p\mathbf{E}; \cdots ; \mathbf{x}_N^p\mathbf{E}] + \mathbf{E}^{pos}, \tag{1}$$

where $\mathbf{E} \in \mathbb{R}^{(P^3 \cdot C) \times d_{enc}}$ is the patch embedding projection, and $\mathbf{E}^{pos} \in \mathbb{R}^{N \times d_{enc}}$ denotes the position embedding.

Each Transformer layer consists of Multihead Self-Attention (MSA) and Multi-Layer Perceptron (MLP) blocks (Eq. (2)(3)). Therefore the output of the ℓ-th layer can be written as follows:

$$\mathbf{z}'_\ell = \mathrm{MSA}(\mathrm{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, \tag{2}$$

$$\mathbf{z}_\ell = \mathrm{MLP}(\mathrm{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell, \tag{3}$$

where $\mathrm{LN}(\cdot)$ denotes the layer normalization operator and $\mathbf{z}_\ell$ is the encoded image representation.

B. Transformer as Decoder

1) Coarse candidate estimation: Traditional approaches, such as U-Net, predominantly view medical image segmentation as a per-pixel classification task. In this paradigm, each pixel is classified into one of the possible K categories, typically achieved by training a segmentation model with the per-pixel cross-entropy (or negative log-likelihood) loss.

Instead of considering individual pixels, our approach in this paper treats medical image segmentation as a mask classification problem. We introduce the concept of an "organ query", a d_{dec}-dimensional feature vector that represents each organ in the image. With a predefined set of N organ queries, our goal for an image comprising K segmentation classes is to segregate the image into N distinct candidate regions. Subsequently, we aim to assign the corresponding organ label to each region. Importantly, it is worth noting that the value of N does not have to align with the number of classes, as demonstrated in prior studies [22]. In fact, we intentionally set N to be significantly larger than K, to minimize the risk of false negatives. Assume the dimension of the object queries is d_{dec}, the coarse predicted segmentation map can be computed by the dot product between the initial organ queries $\mathbf{P}^0 \in \mathbb{R}^{N \times d_{dec}}$ and the embedding of the U-Net last block feature $\mathbf{F} \in \mathbb{R}^{D \times H \times W \times d_{dec}}$:

$$\mathbf{Z}^0 = g(\mathbf{P}^0 \times \mathbf{F}^\top), \tag{4}$$

where $g(\cdot)$ is sigmoid activation followed by a hard thresholding operation with a threshold set at 0.5. As shown in Eq. (4), $\mathbf{P}$ can be seen as the $1\times1\times1$ convolutional kernel in the standard U-Net segmentation head.

2) Transformer decoder: Figure 1 illustrates our customized 3D Transformer decoder designed specifically to refine organ queries, thereby enhancing the coarse prediction $\mathbf{Z}^0$. Similar to the structure seen in the Transformer encoder (detailed in Section III-A), the self-attention mechanism (*i.e.*, the MSA block) in each layer will enable the Transformer decoder to comprehensively engage with image features and capture organ query interrelations. Recognizing the rich localization in intermediate CNN features, which complements the Transformer's global image context, we refine organ queries in each decoder layer by integrating cross-attention with localized multi-scale CNN features.

Our strategy involves concurrent training of a CNN decoder and the Transformer decoder. In the t-th layer, the refined organ queries are denoted as $\mathbf{P}^t \in \mathbb{R}^{N \times d_{dec}}$. Simultaneously, an intermediate U-Net feature is mapped to a d_{dec}-dimensional feature space denoted as $\mathcal{F}$ to facilitate cross-attention computation. Notably, when the count of upsampling blocks aligns with the Transformer decoder layers, multi-scale CNN features can be projected into the feature space $\mathcal{F} \in \mathbb{R}^{(D_t H_t W_t) \times d_{dec}}$, where D_t, H_t, and W_t specify the spatial dimensions of the feature map at the t-th upsampling block. Moving to the $t+1$-th layer, organ queries are updated using cross-attention as follows:

$$\mathbf{P}^{t+1} = \mathbf{P}^t + \mathrm{Softmax}((\mathbf{P}^t\mathbf{w}_q)(\mathcal{F}\mathbf{w}_k)^\top) \times \mathcal{F}\mathbf{w}_v, \tag{5}$$

where the t-th query features undergo linear projection to form queries for the next layer using the weight matrix $\mathbf{w}_q \in \mathbb{R}^{d_{dec} \times d_q}$. The U-Net feature, $\mathcal{F}$, is similarly transformed into keys and values using parametric weight matrices $\mathbf{w}_k \in \mathbb{R}^{d_{dec} \times d_k}$ and $\mathbf{w}_v \in \mathbb{R}^{d_{dec} \times d_v}$. Note a residual path is used for updating $\mathbf{P}$ following previous studies [25]. Next, we will introduce how to incorporate a coarse-to-fine attention refinement to further enhance the accuracy of segmentation results.

3) Coarse-to-fine attention refinement: The value of coarse-to-fine refinement in medical image segmentation, particularly for small target segmentation, is well-established [4], [28], [29]. This technique employs a coarse mask from an initial stage to guide subsequent refinements. Here, to integrate a seamless coarse-to-fine refinement process within the Transformer decoder, we have incorporated a mask attention module [25]. This enhancement aims to ground the cross-attention within the foreground region based on the former coarse prediction for each category, to reduce the background noise and better focus on the region of interest. This improved attention map iteratively aids subsequent, finer segmentation stages.

Concretely, we start by setting the organ queries and the coarse-level mask prediction as $\mathbf{P}^0$ and $\mathbf{Z}^0$ (based on Eq. (4)) respectively, and then begin the iterative refinement process. At the t-th iteration, using the current organ query features $\mathbf{P}^t$ and coarse prediction $\mathbf{Z}^t$, we compute the masked cross-attention, which refines $\mathbf{P}^{t+1}$ for the subsequent iteration. This

Algorithm 1: Iterative coarse-to-fine refinement

Input : Parametric weight matrices $\mathbf{w}_q, \mathbf{w}_k, \mathbf{w}_v$;
Organ embedding $\mathbf{P}$, U-Net last feature $\mathbf{F}$;
The U-Net t-th layer feature $\mathcal{F}$;
Max number of iterations T;

Output: Fine segmentation map $\mathbf{Z}^T$, predicted class label $\hat{\mathbf{y}}$;

$t \leftarrow 0$;
$\mathbf{P}^0 \leftarrow \mathbf{P}$;
$\mathbf{Z}^0 \leftarrow g(\mathbf{P}^0 \times \mathbf{F}^\top)$;
repeat
  Update $\mathbf{P}^{t+1}$ according to Eq. (6);
  Update $\mathbf{Z}^{t+1} \leftarrow g(\mathbf{P}^{t+1} \times \mathbf{F}^\top)$;
  $t \leftarrow t + 1$;
until $t = T$;
Compute the class label $\hat{\mathbf{y}}$ by Eq. (8) and Eq. (9);

Return: $\mathbf{Z}^T$, $\hat{\mathbf{y}}$.

computation incorporates the existing coarse prediction $\mathbf{Z}^t$ into the affinity matrixn, as detailed in Eq.(5):

$$\mathbf{P}^{t+1} = \mathbf{P}^t + \text{Softmax}((\mathbf{P}^t\mathbf{w}_q)(\mathcal{F}\mathbf{w}_k)^\top + h(\mathbf{Z}^t)) \times \mathcal{F}\mathbf{w}_v, \tag{6}$$

where

$$h\left(\mathbf{Z}^t(i,j,s)\right) = \begin{cases} 0 & \text{if } \mathbf{Z}^t(i,j,s) = 1 \\ -\infty & \text{otherwise} \end{cases} \tag{7}$$

where i, j, s are the coordinate indices. This formula restricts the cross-attention mechanism to focus solely on the foreground, nullifying it for all other regions. By iteratively updating both the organ queries and the corresponding mask predictions, our Transformer decoder systematically refines the segmentation results across multiple iterations. A detailed description of this iterative process is outlined in Algorithm 1. The refinement cycle persists until the iteration count t reaches the maximum threshold T, which is equivalent to the number of layers in the Transformer decoder.

Fine segmentation decoding. After the final iteration, the updated organ queries $\mathbf{P}^T$ can be decoded back to the finalized refined binarized segmentation map $\mathbf{Z}^T$ by the dot product with U-Net's last block feature $\mathbf{F}$, following Eq. (4). To associate each binarized mask with one semantic class, we further use a linear layer with weight matrices $\mathbf{w}_{fc} \in \mathbb{R}^{d \times K}$ that projects the refined organ embedding $\mathbf{P}^T$ to the output class logits $\mathbf{O} \in \mathbb{R}^{N \times K}$. Formally, we have:

$$\mathbf{O} = \mathbf{P}^T\mathbf{w}_{fc} \tag{8}$$

$$\hat{\mathbf{y}} = \arg\max_{k=0,1,\ldots,K-1} \mathbf{O} \tag{9}$$

where k is the label index. The final class labels associated with the refined predicted masks $\mathbf{Z}^T$ is $\hat{\mathbf{y}} \in \mathbb{R}^N$.

C. TransUNet

Our 3D TransUNet is built upon the state-of-the-art nnU-Net architecture, with the aim of surpassing its established standards. We illustrate the overall framework of the proposed TransUNet in Figure 1. We instantiate our 3D TransUNet with three distinct configurations:

1) Encoder-only: **A CNN-Transformer hybrid encoder** is employed where CNN is first used as a feature extractor to generate a feature map for the input. Patch embedding is applied to feature patches instead of from raw images. For the decoding phase, we use a standard U-Net decoder. We choose this design since 1) it allows us to leverage the intermediate high-resolution CNN feature maps in the decoding path; and 2) we find that the hybrid CNN-Transformer encoder performs better than simply using a pure Transformer as the encoder. The Encoder-only model will be trained using a hybrid segmentation loss consisting of pixel-wise cross entropy loss and dice loss.

2) Decoder-only: In this configuration, we use a conventional CNN encoder for the encoding phase. As for the decoding phase, we use a **CNN-Transformer hybrid decoder** in the segmentation model. Initially, the organ queries $\mathbf{P}$ are set to zero. Before being processed by the Transformer decoder, they are augmented with learnable positional embeddings following Eq. (1). Then as aforementioned in Section III-B, $\mathbf{P}$ will be gradually refined conditioned on the U-Net features and be decoded back into the full-resolution segmentation map. We train the network with the Hungarian matching loss following previous works [21], [23] to update the organ queries throughout the decoding layers. This loss aims to match pairs between predictions and ground-truth segments. It combines pixel-wise classification loss and binary mask loss for each segmented prediction:

$$\mathcal{L} = \lambda_0(\mathcal{L}_{ce} + \mathcal{L}_{dice)} + \lambda_1\mathcal{L}_{cls}, \tag{10}$$

where the pixel-wise classification loss $\mathcal{L}_{ce}$ and $\mathcal{L}_{dice}$ denote binary cross-entropy loss and dice loss, respectively [30]. The classification loss $\mathcal{L}_{cls}$ is instantiated by the cross-entropy loss for each candidate region. λ_0 and λ_1 are the hyper-parameters for balancing the per-pixel segmentation loss and the mask classification loss.

We also employ deep supervision, applying the training loss to the output at each stage of the TransUNet decoder.

3) Encoder + Decoder: Here, we integrate both the Transformer encoder and the Transformer decoder into the 3D nnU-Net model. And then similar to the decoder-only model, here we also use the Hungarian matching loss to train the whole network.

Lastly, we would like to note that our method, though built upon the 3D nnU-Net, can be easily modified to fit 2D tasks by simply switching the backbone model and reducing all operations back to 2D.

IV. Experiments and Discussion

A. Dataset and Evaluation

Synapse multi-organ segmentation dataset[1]. We use the 30 abdominal CT scans in the MICCAI 2015 Multi-Atlas Abdomen Labeling Challenge, with 3779 axial contrast-enhanced abdominal clinical CT images in total.

Each CT volume consists of $85 \sim 198$ slices of 512×512 pixels, with a voxel spatial resolution of $([0.54 \sim 0.54] \times$

[1] https://www.synapse.org/#!Synapse:syn3193805/wiki/217789

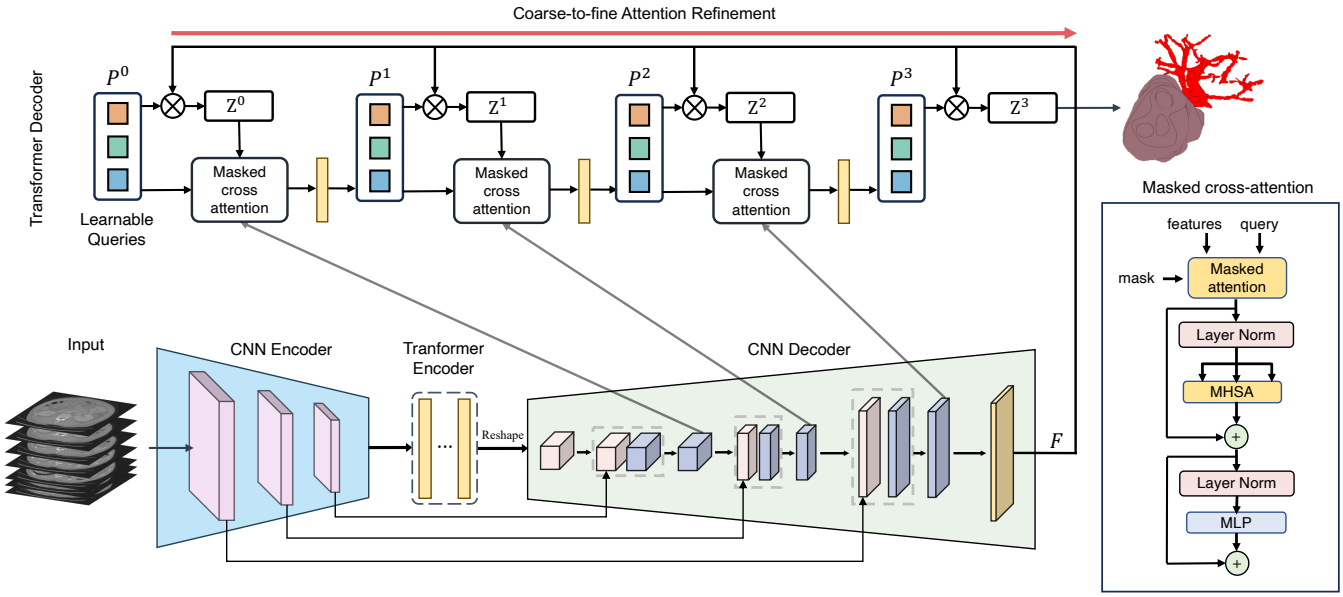


Fig. 1. Overview of 3D TransUNet. We develop two opted architectural components: 1) The Transformer encoder where a CNN encoder is firstly used for local image feature extraction, followed by a pure Transformer encoder for global information interaction; and 2) the Transformer decoder that reframes per-pixel segmentation as mask classification using learnable queries, which are refined through cross-attention with CNN features, and employs a coarse-to-fine attention refinement approach for enhanced segmentation accuracy.

$[0.98 \sim 0.98] \times [2.5 \sim 5.0])\text{mm}^3$. Following [31], we report the average DSC and average Hausdorff Distance (HD) on 8 abdominal organs (aorta, gallbladder, spleen, left kidney, right kidney, liver, pancreas, spleen, stomach with a random split of 18 training cases (2212 axial slices) and 12 cases for validation, following the split setting in [10].

BraTS2021 brain tumor segmentation challenge[2]. BrasTS2021 Challenge is the most recent and largest dataset for brain tumor segmentation. 1251 multi-parametric MRI scans were provided with segmentation labels to the participants. 4 contrasts are available for the MRI scans: Native T1-weighted image, post-contrast T1-weighted (T1Gd), T2-weighted, and T2 Fluid Attenuated Inversion Recovery (T2-FLAIR). Annotation were manually performed by one to four raters, with final approval from experienced neuro-radiologists. The labels include regions of GD-enhancing tumor (ET), the peritumoral edematous/invaded tissue (ED), and the necrotic tumor core (NCR). All MRI scans were pre-processed by co-registration to the same anatomical template, interpolation to isotropic 1mm^3 resolution and skull-stripping. The image sizes of all MRI scans and associated labels are $240\times240\times155$. In our experiments, we apply 5-fold cross-validation with the same data split used by the No.1 solution [32] in BraTS2021 challenge.

Medical Segmentation Decathlon (MSD) HepaticVessel[3]. The MSD HepaticVessel, a task of Medical Segmentation Decathlon [33], consists of 443 portal venous phase CT scans obtained from patients with a variety of primary and metastatic liver tumors. The corresponding target ROIs were the vessels and tumors within the liver. This data set was selected due to the tubular and connected nature of hepatic vessels neighboring heterogeneous tumors. We apply 5-fold cross-validation to evaluate the methods on this dataset.

Large scale pancreatic mass dataset. Our dataset of venous phase 2930 CT scans, is collected from a high-volume US hospital. To the best of our knowledge, it is one of the largest scale pancreatic tumor CT dataset. Pancreatic ductal adenocarcinoma (PDAC) is of the highest priority among all pancreatic abnormalities with a 5-year survival rate of approximately 10% and is the most common type (about 90% of all pancreatic cancers). The labels include Pancreas, PDAC and Cyst. The dataset is randomly split into a training of 2123 CT scans and a testing dataset of 807 CT scans. The model validation is conducted on a subset of training set. The training set includes 1017 PDACs, 462 Cyst, and 644 normal pancreases. The testing set includes 506 PDACs, 271 Cyst, 300 normal pancreases. The evaluation metrics include the dice score, the sensitivity and the specificity following the criterion in [34].

BraTS2023 brain metastases dataset. The BraTS-MET dataset [35] plays a crucial role in developing advanced algorithms for detecting and segmenting brain metastases, with the goal of ensuring seamless integration into clinical practice. This dataset comprises a variety of untreated brain metastasis mpMRI scans collected from multiple institutions [36]–[38] following standard clinical protocols. It is important to mention that our work exclusively utilizes the BraTS2023-MET version of the dataset. Currently, it consists of 238 cases, and we employ a default 5-fold cross-validation method in our analysis.

[2]http://braintumorsegmentation.org/
[3]http://medicaldecathlon.com/

TABLE I
IMPLEMENTATION DETAILS INCLUDING THE ARCHITECTURE HYPERPARAMETER, TRAINING SETTINGS, AND DATA AUGMENTATION.

	Synapse	MSD vessel	BraTS	Pancreas
category	multi-organ	organ&tumor	tumor	organ&tumor
crop size	40×224×192	64×192×192	128×128×128	40×224×192
batch size/gpu	2	2	2	2
lr	8e-2	3e-4	3e-4	3e-4
optimizer	sgd	adamw	adamw	adamw
lr schedule	cosine	cosine	cosine	cosine
downsample	[4, 5, 5]	[4, 5, 5]	[5, 5, 5]	[3, 5, 5]
num of query	n/a	20	20	20
C2F stage	3	3	3	3
augmentation	random rotation, scaling, flipping, white Gaussian noise, Gaussian blurring, adjusting brightness and contrast, simulation of low resolution, Gamma transformation			

B. Implementation Details

Training. In all our 3D experiments, we adhere to nnU-Net's prescribed data augmentation procedures to enhance the diversity of our training dataset. To facilitate effective training, we employ a batch size of 2 using 1 Nvidia RTX 8000 GPU. A comprehensive breakdown of our implementation details can be found in Table I, encompassing critical aspects such as architectural hyperparameters, training configurations, and data augmentation techniques customized for various datasets. We experiment with both 1-layer and 12-layer ViT for implementing the Transformer encoder. Specifically, the 12-layer ViT model is pretrained on the ImageNet21k dataset [39], with additional LayerScale [40]. The latent dimensions d_{enc} and d_{dec} are set as 768 and 192 respectively. For computing the Hungarian matching loss, λ_0 and λ_1 are set as 0.7 and 0.3.

In line with nnU-Net's pioneering adaptive architecture design, our TransUNet exhibits adaptability tailored to the characteristics of the data it processes. Similar to nnU-Net, our backbone architecture is inherently self-adaptive for different medical datasets, particularly in determining crucial factors such as the number of down-sampling layers and the allocation of channels at each stage. The details can be found in Table I. It is noteworthy that our TransUNet maintains compatibility with 2D image data, requiring minimal architectural modifications to accommodate various imaging scenarios.

Testing. Given a CT/MR scan, we do inference in a sliding-window manner. By leveraging the aggregation of all patches, we assign a probability vector to a voxel in position (i, j, s): $\sum_{n=1}^{N}(\mathbf{Z}_{n,ijs}^{T}) \in \mathbb{R}^{K}$, followed by an argmax to obtain hard prediction.

C. Analytical Study of Three Configurations

We performed a variety of ablation studies to thoroughly evaluate the proposed three configurations under the TransUNet framework, $i.e.$, Encoder-only, Decoder-only and Encoder+Decoder.

To assess how effective Transformer encoders are against CNN encoders, and likewise for decoders, we conducted comprehensive experiments of our 3D TransUNet model. This evaluation encompassed both Encoder-only and Decoder-only configurations and compared them to the baseline nnU-Net across multi-organ segmentation and hepatic vessel tumor segmentation tasks. The results of these comparisons on 3D TransUNet are summarized in Tables II and III.

For multi-organ segmentation, while the decoder-only design demonstrates a modest performance enhancement (87.63% compared to 87.33%), the encoder-only configuration, especially when employing the 12-layer ViT encoder initialized with pre-trained weights from ImageNet, achieves a significant 0.8% improvement in Dice score, reaching 88.11%. This advantage becomes even more pronounced when adopting a 2D U-Net backbone, with performance gains of 2.80% and 3.02% at resolutions of 224×224 and 512×512, respectively, as outlined in Table VIII.

As for the vessel tumor segmentation, the encoder-only design's performance improvement, while present, remains relatively subtle. Both the 1-layer and 12-layer ViT encoders yield comparable results (66.30% and 66.35%), slightly outperforming the baseline nnU-Net's score of 66.04%. In contrast, the decoder-only configuration exhibits a substantial increment, recording a gain of 1.63% (67.67% versus 66.04%).

To summarize, our results show the encoder-only design thrives in multi-organ segmentation, while the decoder-only configuration is more adept in vessel tumor segmentation. This distinction aligns with the inherent strengths of the Transformer encoder, which captures global context information such as the intricate relationships among various targets, making it especially effective for multi-organ tasks. Conversely, the Transformer decoder, tailored to refine small and hard targets, is aptly suited for tumor segmentation. However, a combined approach of the Transformer encoder and decoder (Encoder+Decoder) does not offer further enhancements for either multi-organ or hepatic vessel segmentation. The reason may be attributed to their strengths may overlap or counteract, preventing synergistic benefits in these specific tasks. As a result, our subsequent experiments will predominantly employ the Encoder-only model for segmenting multiple objects and the Decoder-only model for segmentation tasks targeting smaller and more challenging tumors or lesions.

D. Deep Analysis for Transformer Decoder

1) Number of organ/tumor queries: For successful training, the number of learnable queries must be at least equal to the number of classes ($i.e.$, each class must have at least one query). However, when we varied the number of queries in the segmentation process, we observed that the performance of our 3D TransUNet with the Decoder-only configuration remains largely unaffected by this parameter. A detailed summary of these findings is presented in Table IV.

2) Multi-scale CNN feature for updating queries: A defining characteristic of our Transformer decoder is its integration of multi-scale features from the CNN decoder, which are rich in localization details. These features play a pivotal role in progressively refining the learnable queries through the synergy of cross-attention with localized multi-scale CNN representations. Our experimentation, as summarized in Table V, encompasses the configuration of Decoder-only. The consistently observed performance enhancements, as compared to

TABLE II
COMPARISON ON THE SYNAPSE MULTI-ORGAN CT DATASET (AVERAGE DICE SCORE %, AND DICE SCORE % FOR EACH ORGAN). WE USE PRETRAINED MODEL FROM [15] FOR THE ENCODER WITH 12 TRANSFORMER LAYERS.

encoder 1-layer	encoder 12-layer	decoder	Aorta	Gallbladder	Kidney (L)	Kidney (R)	Liver	Pancreas	Spleen	Stomach	Avg. Dice
			93.04	78.82	84.68	88.46	97.13	81.50	91.68	83.34	87.33
✓			93.07	79.56	86.16	87.68	97.22	81.71	92.56	83.23	87.65
	✓		92.97	81.15	85.76	87.47	97.03	81.76	93.39	85.31	**88.11**
		✓	92.88	82.06	86.04	87.70	97.10	82.08	91.14	82.03	87.63
	✓	✓	92.67	81.66	85.29	87.76	97.34	82.69	91.90	85.59	**88.11**

TABLE III
ABLATION OF TRANSFORMER ARCHITECTURE ON MSD VESSEL DATASET WITH DICE SCORE METRICS (%). EXPERIMENTS ARE CONDUCTED IN FIVE-FOLD CROSS-VALIDATION.

encoder 1-layer	encoder 12-layer	decoder	Vessel	Tumor	Avg. Dice
			63.71	68.36	66.04
✓			63.47	69.12	66.30
	✓		63.67	69.02	66.35
		✓	64.41	70.94	**67.67**
✓		✓	64.58	69.89	67.24

TABLE IV
ABLATION OF NUMBER OF QUERIES ON MSD VESSEL DATASET WITH DICE SCORE METRICS (%). EXPERIMENTS ARE CONDUCTED IN FIVE-FOLD CROSS-VALIDATION.

Number of Queries	Vessel	Tumor	Avg. Dice
5	64.75	70.32	67.53
20	64.41	70.94	67.67
40	64.32	70.41	67.37

TABLE V
ABLATION OF TRANSFORMER DECODER ON MSD VESSEL DATASET WITH DICE SCORE METRICS (%). EXPERIMENTS ARE CONDUCTED IN FIVE-FOLD CROSS-VALIDATION.

Transformer decoder	Multi-scale	masked cross attention	Vessel	Tumor	Average
✓	✓	✓	64.41	70.94	67.67
✓	✓		64.37	70.71	67.54
✓			64.19	69.89	67.04
			63.71	68.36	66.04

TABLE VI
PERFORMANCE COMPARISON ON BRATS2023 FOR BRAIN METASTASIS CHALLENGE WITH LESION-WISE DICE SCORE METRICS (%). MASKED CROSS ATTENTION IS NOT USED FOR THE DECODER-ONLY MODEL FOR TRAINING STABILITY. EXPERIMENTS ARE CONDUCTED IN FIVE-FOLD CROSS-VALIDATION.

Method	ET	TC	WT	Avg. Dice
nnU-Net	54.90	58.67	55.75	56.44
Encoder-only	54.79	58.96	56.05	56.60
Decoder-only	56.80	61.12	60.09	59.34

TABLE VII
PERFORMANCE COMPARISON ON IN-HOUSE LARGE-SCALE PANCREATIC TUMOR SEGMENTATION DATASET.

Method	DSC				Sensitivity	Specificity
	Pancreas	PDAC	Cyst	Average		
nnU-Net	83.8	56.94	56.88	65.97	89.34	91.00
Encoder-only	83.77	58.38	57.98	66.71	91.71	84.66
Decoder-only	85.35	62.66	61.04	69.69	89.94	97.33

the baseline Transformer decoder—where the segmentation mask is computed by directly employing the dot product of the learned query and the last-layer CNN feature—underscore the indispensable nature of incorporating multi-scale CNN features in the query updating process.

3) Coarse-to-fine refinement in Transformer decoder: In each Transformer decoder layer, the coarse-to-fine refinement uses the predicted mask from the current iteration to constrain the cross-attention within the foreground region, therefore refining the organ queries at the next iteration. To demonstrate the effectiveness of this strategy, we have selected vessel tumor segmentation as a representative case study. This choice is motivated by the Transformer decoder's demonstrated proficiency in segmenting small targets, such as tumors or lesions. As illustrated in Table V, the integration of coarse-to-fine refinement (masked cross-attention) consistently yielded enhanced results. For a more intuitive understanding, we provide a qualitative example in Fig. 2, elucidating how this attention refines masks for intricate targets. From the first to the third iteration, the segmentation quality of the tumor has been significantly improved.

E. Generalization to Other Tumor/Lesion Datasets

We assess the generalizability of our 3D TransUNet across different imaging modalities and tasks, including brain metastases segmentation from MRI data (BraTS2023), pancreatic tumor (PDAC) and cyst segmentation. The outcomes are detailed in Table VI and Table VII. Our findings confirm that the Encoder-only variant offers modest enhancements, whereas the Decoder-only configuration substantially boosts performance for tumor/lesion segmentation. For example, in brain metastatic lesion segmentation, the Encoder-only model provides a marginal enhancement, whereas the Decoder-only model contributes to a significant Dice improvement of

TABLE VIII
COMPARISON ON THE SYNAPSE MULTI-ORGAN CT DATASET (AVERAGE DICE SCORE %, AND DICE SCORE % FOR EACH ORGAN).

Scale	Method	Avg. Dice	Aorta	Gallbladder	Kidney (L)	Kidney (R)	Liver	Pancreas	Spleen	Stomach
2D (224×224)	U-Net [2]	74.68	84.18	62.84	79.19	71.29	93.35	48.23	84.41	73.92
	Pyramid Attn [41]	73.08	82.57	56.25	75.78	70.51	93.46	50.02	83.95	72.13
	DeepLabv3+ [42]	76.35	82.00	62.85	78.89	75.24	93.96	57.75	86.57	73.57
	UNet++ [43]	76.65	86.93	63.69	77.86	68.29	93.91	59.23	87.81	75.49
	AttnUNet [12]	75.57	55.92	63.91	79.20	72.71	93.56	49.37	87.19	74.95
	TransUNet	**77.48**	87.23	63.13	81.87	77.02	94.08	55.86	85.08	75.62
2D (512×512)	U-Net [2]	81.34	89.69	69.98	83.08	74.13	95.10	67.73	90.50	80.51
	Pyramid Attn [41]	80.08	88.59	65.91	84.45	75.15	95.30	60.06	91.84	79.33
	DeepLabv3+ [42]	82.50	88.79	72.16	88.13	79.52	95.58	65.97	90.02	79.87
	UNet++ [43]	81.6	89.65	71.68	82.92	75.15	94.92	69.06	89.42	80.01
	AttnUNet [12]	80.88	89.46	67.09	83.83	75.98	95.28	68.48	88.63	78.26
	nnU-Net [44]	82.92	91.55	73.43	82.74	73.61	96.01	71.81	94.29	79.94
	TransUNet	**84.36**	90.68	71.99	86.04	83.71	95.54	73.96	88.80	84.20
3D	V-Net [30]	68.81	75.34	51.87	77.10	80.75	87.84	40.05	80.56	56.98
	DARR [31]	69.77	74.74	53.77	72.31	73.24	94.08	54.18	89.90	45.96
	nnU-Net [44]	87.33	93.04	78.82	84.68	88.46	97.13	81.50	91.68	83.34
	CoTr [17]	85.72	92.96	71.09	85.70	85.71	96.88	81.28	90.44	81.74
	nnFormer [16]	85.32	90.72	71.67	85.60	87.02	96.28	82.28	87.30	81.69
	VT-UNet [45]	70.72	78.25	44.76	77.51	78.16	91.63	45.18	82.20	68.04
	Swin UNETR [20]	73.51	82.94	60.96	80.41	71.14	91.55	56.71	77.46	66.94
	TransUNet	**88.11**	92.97	81.15	85.76	87.47	97.03	81.76	93.39	85.31

TABLE IX
PERFORMANCE COMPARISON ON MSD VESSEL DATASET WITH DICE SCORE METRICS (%). EXPERIMENTS ARE CONDUCTED IN FIVE-FOLD CROSS-VALIDATION.

Method	Vessel	Tumor	Avg. Dice
nnU-Net	63.71	68.36	66.04
nnFormer [16]	63.21	69.37	66.29
VT-UNet [45]	60.88	59.82	60.35
Swin UNETR [20]	57.65	58.31	57.98
TransUNet	64.10	70.60	**67.67**

2.9%. This trend is even more pronounced for PDAC and cystic lesion segmentation. Specifically, Encoder-only model enhances the PDAC segmentation from 56.94% to 58.38%, while Decoder-only model pushes it further to 62.66%. In the case of cystic lesions, Encoder-only model lifts nnU-Net's performance from 56.88% to 57.98%, with Decoder-only model further elevating it to 61.04%. Consequently, the aggregate metrics of average Dice, Sensitivity, and Specificity all witness marked improvements with Decoder-only model.

F. Comparison with State-of-the-arts

We compare our TransUNet to previous 2D and 3D state-of-the-art methods on multi-organ segmentation and hepatic vessel tumor segmentation in Table VIII and Table IX. With the 2D version built on the U-Net architecture and the 3D variant grounded in the 3D nnU-Net framework, our TransUNet consistently outperforms other state-of-the-art methods, underscoring its efficacy across diverse U-Net frameworks. As discussed above, leveraging the Transformer Encoder's ability to capture global organ relationships, we use the Encoder-only design for multi-organ segmentation. Conversely, given the Transformer Decoder's prowess in refining small targets, we opt for the Decoder-only setup for tumor segmentation. Specifically, we compare TransUNet against a spectrum of methodologies, including: 1) 2D techniques such as U-Net [2], DeepLabv3+ [42], and UNet++ [43], complemented by attention-augmented CNN methods like AttnUNet [12] and Pyramid Attn [41], evaluated across resolutions of 224×224 and 512×512; 2) 3D approaches like V-Net [30], DARR [31], and 3D nnU-Net [44], accompanied by cutting-edge Transformer-centric strategies including CoTR [17], nnformer [16], VT-UNet [45], and Swin UNETR [20]. As corroborated by the results in Table VIII, TransUNet not only surpasses traditional CNN-based self-attention models but also outperforms numerous state-of-the-art Transformer-oriented techniques. For example, when benchmarked against recent state-of-the-art Transformer-based methods like CoTr and nnformer, our TransUNet achieves approximately a 10% improvement in Dice scores for the challenging task of gallbladder segmentation and about a 3% enhancement in overall segmentation

Notably, as evidenced in Table X, our 3D TransUNet surpasses the top-ranked solution, nnU-Net-Large [32], from the BraTS2021 challenge, underscoring the robustness and efficacy of our proposed approach.

V. CONCLUSION

While U-Net has been successful, its limitations in handling long-range dependencies have prompted the exploration of Transformer as an alternative architecture, exemplified by our previously developed TransUNet, harnessing the combined strengths of U-Net and Transformers. Our study extends the TransUNet architecture to a 3D network, building upon nnU-Net's foundation. Leveraging Transformers' capabilities in both encoder and decoder design, we introduce 1) A

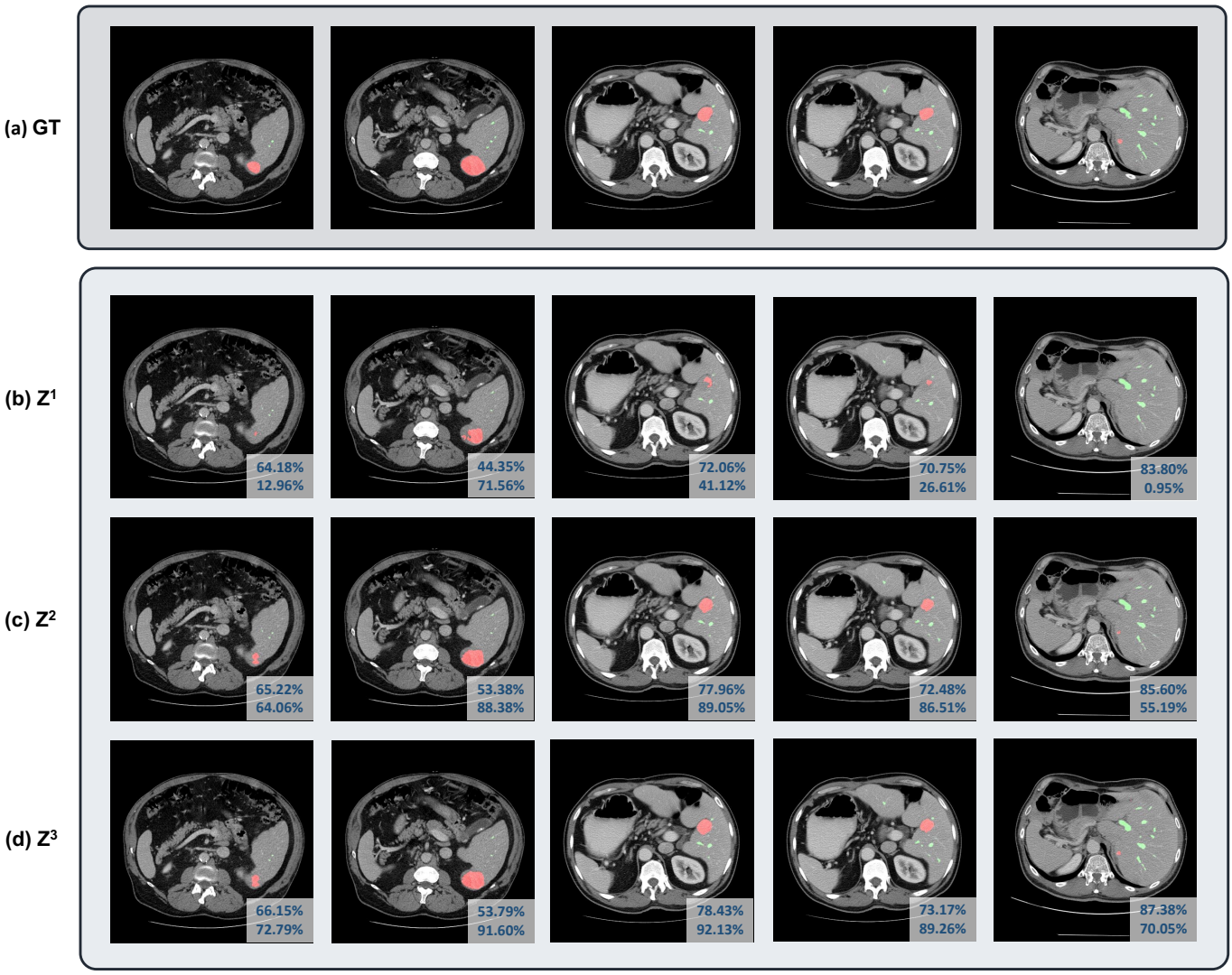


Fig. 2. Visualizations of outputs from different iterations during coarse-to-fine refinement: (a) Groundtruth. (b-d) the segmentation mask at the first to the third iteration. Different columns represent different samples from MSD Vessel Dataset. The dice coefficients of vessels and tumors are indicated in the first and the second row of the lower right corner of each image respectively.

TABLE X
PERFORMANCE COMPARISON ON BRATS2021 CHALLENGE FOR BRAIN TUMOR SEGMENTATION WITH DICE SCORE METRICS (%). EXPERIMENTS ARE CONDUCTED IN FIVE-FOLD CROSS-VALIDATION.

Method	ET	TC	WT	Avg. Dice
nnU-Net	88.05	91.92	93.79	91.25
AxialAttn [32]	87.23	91.88	93.21	90.77
nnUNet-Large [32]	88.23	92.35	93.83	91.47
TransUNet	88.85	92.48	93.90	**91.74**

Transformer encoder that tokenizes CNN feature map patches, facilitating a richer extraction of global contexts; and 2) A Transformer decoder designed to adaptively refine segmentation regions, capitalizing on cross-attention mechanisms between candidate proposals and U-Net features. Our investigations further illuminate that medical segmentation tasks have varying architectural preferences and offer insights into tailoring design configurations to suit these specific requirements. Empirical results showcase our 3D TransUNet's superior performance in diverse medical segmentation tasks. Notably, our method excels in multi-organ, pancreatic tumor, hepatic vessel, and brain metastases segmentation. Additionally, we achieved notable results in the BraTS2021 challenge.

REFERENCES

[1] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 3431–3440.

[2] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *International Conference on Medical image computing and computer-assisted intervention*. Springer, 2015, pp. 234–241.

[3] L. Yu, J.-Z. Cheng, Q. Dou, X. Yang, H. Chen, J. Qin, and P.-A. Heng, "Automatic 3d cardiovascular mr segmentation with densely-connected volumetric convnets," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2017, pp. 287–295.

[4] Y. Zhou, L. Xie, W. Shen, Y. Wang, E. K. Fishman, and A. L. Yuille, "A fixed-point model for pancreas segmentation in abdominal ct scans," in *International conference on medical image computing and computer-assisted intervention*. Springer, 2017, pp. 693–701.

[5] X. Li, H. Chen, X. Qi, Q. Dou, C.-W. Fu, and P.-A. Heng, "H-denseunet: hybrid densely connected unet for liver and tumor segmentation from ct volumes," *IEEE transactions on medical imaging*, vol. 37, no. 12, pp. 2663–2674, 2018.

[6] Q. Yu, L. Xie, Y. Wang, Y. Zhou, E. K. Fishman, and A. L. Yuille, "Recurrent saliency transformation network: Incorporating multi-stage visual cues for small organ segmentation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 8280–8289.

[7] X. Luo, J. Chen, T. Song, Y. Chen, G. Wang, and S. Zhang, "Semi-supervised medical image segmentation through dual-task consistency," *AAAI Conference on Artificial Intelligence*, 2021.

[8] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "Unet++: A nested u-net architecture for medical image segmentation," in *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*. Springer, 2018, pp. 3–11.

[9] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in neural information processing systems*, 2017, pp. 5998–6008.

[10] J. Chen, Y. Lu, Q. Yu, X. Luo, E. Adeli, Y. Wang, L. Lu, A. L. Yuille, and Y. Zhou, "Transunet: Transformers make strong encoders for medical image segmentation," *arXiv preprint arXiv:2102.04306*, 2021.

[11] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 7794–7803.

[12] J. Schlemper, O. Oktay, M. Schaap, M. Heinrich, B. Kainz, B. Glocker, and D. Rueckert, "Attention gated networks: Learning to leverage salient regions in medical images," *Medical image analysis*, vol. 53, pp. 197–207, 2019.

[13] N. Parmar, A. Vaswani, J. Uszkoreit, L. Kaiser, N. Shazeer, A. Ku, and D. Tran, "Image transformer," in *International Conference on Machine Learning*. PMLR, 2018, pp. 4055–4064.

[14] R. Child, S. Gray, A. Radford, and I. Sutskever, "Generating long sequences with sparse transformers," *arXiv preprint arXiv:1904.10509*, 2019.

[15] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *ICLR*, 2021.

[16] H.-Y. Zhou, J. Guo, Y. Zhang, X. Han, L. Yu, L. Wang, and Y. Yu, "nnformer: Volumetric medical image segmentation via a 3d transformer," *IEEE Transactions on Image Processing*, 2023.

[17] Y. Xie, J. Zhang, C. Shen, and Y. Xia, "Cotr: Efficiently bridging cnn and transformer for 3d medical image segmentation," in *Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part III 24*. Springer, 2021, pp. 171–180.

[18] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 10 012–10 022.

[19] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, and M. Wang, "Swin-unet: Unet-like pure transformer for medical image segmentation," in *European conference on computer vision*. Springer, 2022, pp. 205–218.

[20] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H. R. Roth, and D. Xu, "Swin unetr: Swin transformers for semantic segmentation of brain tumors in mri images," in *International MICCAI Brainlesion Workshop*. Springer, 2021, pp. 272–284.

[21] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *European conference on computer vision*. Springer, 2020, pp. 213–229.

[22] R. Strudel, R. Garcia, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 7262–7272.

[23] H. Wang, Y. Zhu, H. Adam, A. Yuille, and L.-C. Chen, "Max-deeplab: End-to-end panoptic segmentation with mask transformers," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 5463–5474.

[24] B. Cheng, A. Schwing, and A. Kirillov, "Per-pixel classification is not all you need for semantic segmentation," *Advances in Neural Information Processing Systems*, vol. 34, 2021.

[25] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 1290–1299.

[26] Q. Yu, H. Wang, S. Qiao, M. Collins, Y. Zhu, H. Adam, A. Yuille, and L.-C. Chen, "k-means mask transformer," in *European Conference on Computer Vision*. Springer, 2022, pp. 288–307.

[27] Q. Yu, H. Wang, D. Kim, S. Qiao, M. Collins, Y. Zhu, H. Adam, A. Yuille, and L.-C. Chen, "Cmt-deeplab: Clustering mask transformers for panoptic segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 2560–2570.

[28] Z. Zhu, Y. Xia, W. Shen, E. Fishman, and A. Yuille, "A 3d coarse-to-fine framework for volumetric medical image segmentation," in *2018 International conference on 3D vision (3DV)*. IEEE, 2018, pp. 682–690.

[29] L. Xie, Q. Yu, Y. Zhou, Y. Wang, E. K. Fishman, and A. L. Yuille, "Recurrent saliency transformation network for tiny target segmentation in abdominal ct scans," *IEEE transactions on medical imaging*, vol. 39, no. 2, pp. 514–525, 2019.

[30] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-net: Fully convolutional neural networks for volumetric medical image segmentation," in *3DV*, 2016.

[31] S. Fu, Y. Lu, Y. Wang, Y. Zhou, W. Shen, E. Fishman, and A. Yuille, "Domain adaptive relational reasoning for 3d multi-organ segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2020, pp. 656–666.

[32] H. M. Luu and S.-H. Park, "Extending nn-unet for brain tumor segmentation," *arXiv preprint arXiv:2112.04653*, 2021.

[33] M. Antonelli, A. Reinke, S. Bakas, K. Farahani, B. A. Landman, G. Litjens, B. Menze, O. Ronneberger, R. M. Summers, B. van Ginneken *et al.*, "The medical segmentation decathlon," *arXiv preprint arXiv:2106.05735*, 2021.

[34] Z. Zhu, Y. Xia, L. Xie, E. K. Fishman, and A. L. Yuille, "Multi-scale coarse-to-fine segmentation for screening pancreatic ductal adenocarcinoma," in *Medical Image Computing and Computer Assisted Intervention–MICCAI 2019: 22nd International Conference, Shenzhen, China, October 13–17, 2019, Proceedings, Part VI 22*. Springer, 2019, pp. 3–12.

[35] A. W. Moawad, A. Janas, U. Baid, D. Ramakrishnan, L. Jekel, K. Krantchev, H. Moy, R. Saluja, K. Osenberg, K. Wilms *et al.*, "The brain tumor segmentation (brats-mets) challenge 2023: Brain metastasis segmentation on pre-treatment mri," *arXiv preprint arXiv:2306.00838*, 2023.

[36] E. Oermann, K. Link, Z. Schnurman, C. Liu, Y. J. F. Kwon, L. Y. Jiang, M. Nasir-Moin, S. Neifert, J. Alzate, K. Bernstein *et al.*, "Longitudinal deep neural networks for assessing metastatic brain cancer on a massive open benchmark." *preprint*, 2023.

[37] J. D. Rudie, R. S. D. A. Weiss, P. Nedelec, E. Calabrese, J. B. Colby, B. Laguna, J. Mongan, S. Braunstein, C. P. Hess, A. M. Rauschecker *et al.*, "The university of california san francisco, brain metastases stereotactic radiosurgery (ucsf-bmsr) mri dataset," *arXiv preprint arXiv:2304.07248*, 2023.

[38] E. Grøvik, D. Yi, M. Iv, E. Tong, D. Rubin, and G. Zaharchuk, "Deep learning enables automatic detection and segmentation of brain metastases on multisequence mri," *Journal of Magnetic Resonance Imaging*, vol. 51, no. 1, pp. 175–182, 2020.

[39] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein *et al.*, "Imagenet large scale visual recognition challenge," *International journal of computer vision*, vol. 115, pp. 211–252, 2015.

[40] H. Touvron, M. Cord, A. Sablayrolles, G. Synnaeve, and H. Jégou, "Going deeper with image transformers," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 32–42.

[41] H. Li, P. Xiong, J. An, and L. Wang, "Pyramid attention network for semantic segmentation," *arXiv preprint arXiv:1805.10180*, 2018.

[42] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 801–818.

[43] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "Unet++: Redesigning skip connections to exploit multiscale features in image segmentation," *IEEE transactions on medical imaging*, vol. 39, no. 6, pp. 1856–1867, 2019.

[44] F. Isensee, P. F. Jaeger, S. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnu-net: a self-configuring method for deep learning-based biomedical image segmentation," *Nature methods*, vol. 18, no. 2, pp. 203–211, 2021.

[45] H. Peiris, M. Hayat, Z. Chen, G. Egan, and M. Harandi, "A robust volumetric transformer for accurate 3d tumor segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2022, pp. 162–172.